

Rice Congress 2010



Rice Research and Development Institute
Department of Agriculture

Rice Congress 2010

Selected Papers

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Rice Congress 2010

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The responsibility for this publication rests with the Department of Agriculture, Sri Lanka (DOASL)

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CONTENTS

Foreword	v
Rice Supply and Demand in Sri Lanka: The Future Outlook R.M. Herath, R.D.D. Rajapaksa, H.U. Warnakulasooriya and T.H.C.S. Perera	01
Promotion of Inappropriate Technology in Rice Cultivation D. Sumith de Z. Abeyesiriwardena	19
Rice Variety Improvement for the Wet Zone of Sri Lanka A.P. Bentota, G.A. Jinadasa, D.S. de Z. Abeyesiriwardena, S.C. Wanigasuriya, B.G.D.S. Weerasingha and N.P.S. de Silva	29
Potentials and Limitations of Hybrid Rice Production in Sri Lanka S.W. Abeysekera and S.N. Jayawardena	55
Identification of Candidate Genes for Salt Tolerance in Rice A.R.F. Zahara, D.M. de Costa and W.A.J.M. de Costa	69
Weedy Rice: A Threat to Direct Seeded Rice Culture in Sri Lanka A.S.K. Abeysekera, U.B. Wickrame, L. Nugaliyadde and Y.B. Iqbal	97
A Review of Rice Pest Management Research in Sri Lanka A.S. Pushpakumari, L. Nugaliyadde, H.L.K. Liyanage, K.S. Wickramasinghe and G.D.S.N. Chandrasena	113
Management of Rice Diseases in Sri Lanka During the Last Decade Y.J.P.K. Mithrasena and D.M.N. Dissanayake	137
Management of Rice Sheath Blight with Emphasis on Biological Control D.M. de Costa and D.M.N. Dissanayake	145
Meeting Challenges to Realize Potential Yield in Rice Cultivation in Sri Lanka W.M.W. Weerakoon, S.N. Jayawardena, K.G.D.S. Kahadewela, M.A.P.W.K. Malaviarachchi, S.C. Wanigasuriya and A.H.G. Mitrasena	159

Land Resource Planning for Achieving High Productivity in Rice Lands	185
W.M.A.D.B. Wickramasinghe, M.A.K. Munasinghe and K.M.A. Kendaragama	
Recent Advancement in Mechanization of Rice Production in Sri Lanka	195
M.H.M.A. Bandara, B.M.C.P. Balasuriya and Y.B. Iqbal	
Past Achievements and Future Directions of the Seed Paddy Sector in Sri Lanka	205
M.B.N.W.M. Attanayake, W.M.I. Weerasekara, D.B. Weeratunga and G.M.W. Chithral	
Climate Change and Rice Cultivation in Sri Lanka: Impacts, Adaptation and Mitigation Strategies	219
W.M.W. Weerakoon, W.A.J.M. de Costa and S.P. Nissanka	
Geo-informatics for Efficient Use of Resources in Rice Production	241
W.M.U.K. Rathnayake, Ranjith Premalal de Silva and N.D.K. Dayawansa	
Determination of Onset Time of Rains during Yala and Maha Seasons in Major Rice Growing Regions of Sri Lanka	253
R.D. Chithranayana and B.V.R. Punyawardena	
Achievements and Strategies for Improving Water Productivity in Rice Cultivation	263
W.M.U.K. Rathnayake, H.B. Nayakekorala and A.H.S. de Silva	
Productivity Improvement of Rice Lands through Soil Fertility Management	277
D.N. Sirisena, W.M.J. Bandara and A.S. Jayasena	

FOREWORD

The Department of Agriculture has engaged in rice research since its inception and made a significant contribution for the development of rice sector in Sri Lanka through generation of improved technology. Increase in production to achieve self-sufficiency in rice has been the national goal in early days and now the country reaching self-sufficiency so that time has come to see what is beyond self-sufficiency.

To date, the department has held four symposia to review the progress of rice research and development and to plan for future. The fifth symposium, 'Rice Symposium 2010' organized by the Rice Research and Development Institute of the Department of Agriculture was held in December 2010 to review the progress made by and the programmes of many stakeholders involved directly and indirectly in the rice sector during last decade with a view of recommending reforms considered necessary to uplift the rice sector as a strong component significantly contributing towards the overall development of the national economy.

Thirty four papers were presented in this symposium in 9 sessions under different themes to cover all aspects of rice sector in Sri Lanka. However, with the limitations we are bound to include only 18 selected papers in this book.

Editors

Rice Supply and Demand in Sri Lanka: The Future Outlook

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Abstract

Rice production in Sri Lanka has shown a remarkable growth during the last four decades. Annual production has reached the self-sufficiency level. This paper examines the changes in the demand and supply of rice and possible avenues to face the challenges of the next decade. Rice still behaves as a normal commodity and demand for rice is projected to increase by 12% by 2020 mainly resulting from population growth. In the production increase that occurred from 1990 decade to 2000 decade, 31% is attributed to area expansion and 63% to productivity improvement. The contribution of yield increase to production increase is higher in rainfed and minor irrigated areas than major irrigation areas. Yield and area response function analysis showed that, area and yield are responsive to lag year farm gate price of paddy and fertilizer prices. Real profits in the Dry Zone have been increasing while it fluctuated widely in the Wet Zone. Rice production during 2000 decade has grown at the highest growth rate (3.49%) annually compared to past three decades, which was resulting from area growth (2.06%) and yield growth (1.40%). Rice production is still increasing and rice supply is projected to increase by 41% by 2020 surpassing the demand. Maintaining of present growth of rice production is an attractive choice to policy makers. However, this would require continuation of favorable output and input price policies and technological development and dissemination process. A pragmatic approach is required to maintain the balance of supply and demand to sustain current growth of rice production ensuring attractive price for rice farmers. Promotion of rice based products and grain quality improvement and reducing the influence of wheat flour on demand for rice and crop diversification, cultivation of organic, traditional and high value rice and exploring the possibility of expansion of rice exports and maintaining buffer stocks can be used to a certain degree to maintain the balance of supply and demand.

Introduction

The world today is characterized by a prevailing mood of despair regarding the reducing of rice yield in Asia due to the impact of climate change. Rice production in the world will be thwarted as temperature increases in rice

growing areas with continued climate change. Rising temperatures in the past 25 years have already cut rice yield at several key growing locations by 10-20% (University of California, 2010). International organizations and concerned professionals are busy organizing seminars and conferences to

raise awareness regarding the lowering of rice production and to mobilize global resources to address this issue.

In Sri Lanka, the situation in rice production gives an ambition as well as despair. The ambition arises from our ability to achieve self sufficiency in rice production through combination of technology, services and government policies. Despair looms because of struggle to deal with the issue of rice surplus and to sustain present trend in rice production matching the demand and farmers' welfare in view of profitability and competitiveness with globalization process, threat of impact of climate change and depletion of natural resources. In the past, we were naturally concerned with achieving self sufficiency. We did not overlook the problems that are likely to arise from potential change of climate and depletion of natural resources.

It is in this global and national perspective we consider that the race to stay ahead of ensuring food security is not yet over. Rice continues to be the most important commodity in the Sri Lankan economy, being the staple food contributing 45% of the population's total calorie intake on average and as much as 60-65% of the households in the lowest income quartile. Rice is grown on nearly 1 million ha of the country's arable land annually and is a major source of livelihood of 1.5 million farm families.

Because of the political and economic importance of rice in the country, the rice sector has historically been

the central focus of government agricultural policy. Government interventions have been aimed to achieve several, often conflicting, objectives - stabilizing prices, raising farm incomes, providing rice at low prices to consumers and attaining rice self sufficiency in the pursuit of food security. Over the past four decades, however, the level and nature of these government interventions and the relative importance of policy objectives have changed in response to changes in the domestic factors affecting rice demand and supply, the macroeconomic environment, and the political-economic forces.

The purpose of this paper is to examine the changes in the demand and supply of rice in Sri Lanka during the past four decades, the problems likely to emerge in coming decade regarding the demand and supply of rice, and suggest possible policy alternatives to face the challenges of the next decade.

Trends In Demand for Rice Determinants of Demand for Rice

Demand for rice mainly depends on the level of per capita income, its own price, prices of substitutes (wheat flour) and urbanization rate and population growth. With increase in income, people have the tendency to substitute low cost source of energy such as coarse grain for rice. But at higher levels of income, rice becomes an inferior good. As incomes increase, people go for diversified diets and substitute rice for high value, quality

food with more protein and vitamin. A recent study has shown that the demand responses to income and price of the major substitute (wheat flour) are still positive (Herath *et al.*, 2007). A close look at the per capita consumption reported in the House Hold Income and Expenditure Surveys from 1980/81 to 2006/07 strengthen the above fact that per capita consumption of rice is still increasing (Table 1).

Urbanization is the other factor that affects the demand for rice. Urbanization reduces per capita rice consumption. For same level of income, the per capita consumption of rice is low by about 32% in urban areas compared to rural areas. This may be due to tendency towards diversifying staple food diet by urban people with the changing life styles, coupled with increasing per capita income and availability and accessibility to wheat flour based products in different forms in the market. However, rate of urbanization is still low and only about 22% of total population live in urban areas in the country.

Population growth is the major force behind the demand for rice. Population in Sri Lanka has been growing at a rate of 1.1% per annum and if this population growth rate is expected to continue till 2020, Sri Lankan population is expected to increase by about 11.5% in next 10 years.

Future Demand for Rice

Per capita consumption of rice decreased from 116 kg/year in 1980 to 103 kg/year in 1995 while per capita consumption of wheat flour increased from 28 kg/year in 1980 to 42 kg/year in 1995 (Table 1). However, per capita consumption of rice has started to increase again from 2001 onwards and it has reached a level of 108 kg/year by year 2007. Conversely, the per capita consumption of wheat flour has reduced during the same period, may be due to substitution effect between these two major foods.

Both cross price elasticity of rice and wheat, and income elasticity of rice have been found positive (Herath *et al.*, 2007). Apparently, per capita consumption of rice will further increase in coming years with gradual increase of per capita income, particularly incomes of the poorer segments of the country and decrease of ratio of rice price to the price of wheat flour, and the flexibility of the domestic rice industry to respond to domestic market needs. Diversification of domestic rice market has increased by making diverse brands of rice available to match the demands of the diverse consumer segments. Particularly, the time and effort needed to prepare rice has decreased with advances made in rice milling and marketing. Availability of rice flour based products such as string hopper flour and noodles that are ready for cooking at home have increased.

Table 1: Per capita consumption of rice and wheat flour in Sri Lanka; 1980 -2007.

Year	Rice (kg/person/year)	Wheat Flour (kg/person/year)
1980/81	116	28
1985/86	112	33
1990/91	110	31
1995/96	103	42
2001/02	104	38
2006/07	108	25

Source: House Hold Income and Expenditure Survey Reports

Per capita availability reported in food balance sheet published by the Department of Census and Statistics is also used as a proxy to examine the trend in per capita consumption of rice. Fig. 1 shows change in per capita availability of rice and wheat flour from 1970 to 2009. It is clear from figure that per capita availability has fluctuated over

time with slight increasing trend from 2004 strengthening the hypothesis of increase in per capita consumption of rice. With higher production recorded in 2008, per capita availability of rice shot up to 122 kg. However it dropped to 113 kg in 2009.

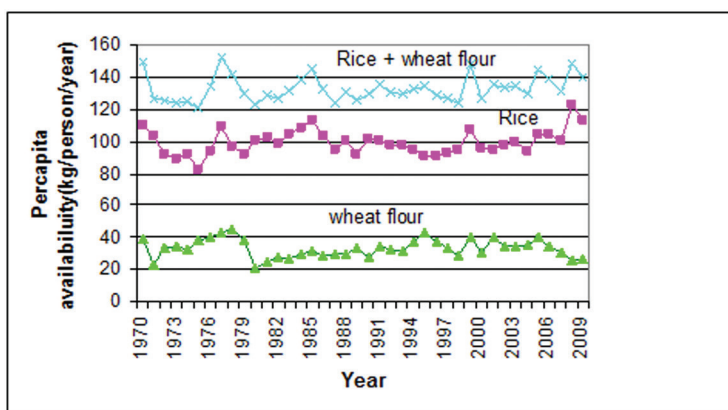


Figure 1: Per capita availability of rice and wheat flour during 1970-2009.

Source: Food balance sheets, Department of Census and Statistics

As projection on per capita income, price of wheat flour and urbanization rate are not available, previous studies have projected future demand for rice considering only the growth of population that brings major boost in demand for rice. Accordingly, population projections only are used

in predicting future demand in this study. Table 2 presents demand for rice estimated using the 115 kg/year/head as the safe consumption requirement. Accordingly the demand for rice will increase by nearly 12% and rice production must increase to 2.7 million tons by year 2020, from the present level of about 2.35 million tons.

Table 2: Future rice requirement at 115 kg per capita per year.

Year	Population (000')	Requirement at 115 kg per capita (000't)
2010	20675	2378
2011	20902	2404
2012	21132	2430
2013	21365	2457
2014	21600	2484
2015	21837	2511
2016	22078	2539
2017	22320	2567
2018	22566	2595
2019	22814	2624
2020	23065	2652

Trends in Paddy Production

Trends in paddy production, harvested area and average yield in Sri Lanka from 1970 to 2009 are illustrated in Fig. 2. At glance, it is apparent that there are three homogeneous phases of growth in area, yield and paddy production. These periods are i: 1970-1979, ii: 1980-1999, iii: 2000-2009. The first and the third periods are characterized by faster growth rates whereas the second period typifies slower growth. Piece wise regression analysis also strengthens the above hypothesis. However, since

rice congress is convened once in ten years it is appropriate to examine the changes in paddy production, yield and area in ten year periods. Accordingly, 40 year period of 1970-2009 is divided into 4 ten year sub periods. Ten year averages of harvested area, yield and production are presented in the Table 3. Dramatic changes in the Sri Lankan rice economy have occurred over the past four decades. The marked increases can be seen in ten year averages of harvested area, average yield, and paddy production between

decades 1970-1979 and 1980 -1989, and also between decades 1990-1999 and 2000-2009. However, there is no marked growth in above parameters between decades 1980 -1989 and 1990 -1999. In contrast to between 1980-

1989 and 1990-1999, harvested area, yield and production has increased during 2000-2009. Ten year averages of harvested area, yield and production has shot up to 0.77 million ha, 4.1t/ha and 3.1 million tons, respectively in 2000-2009.

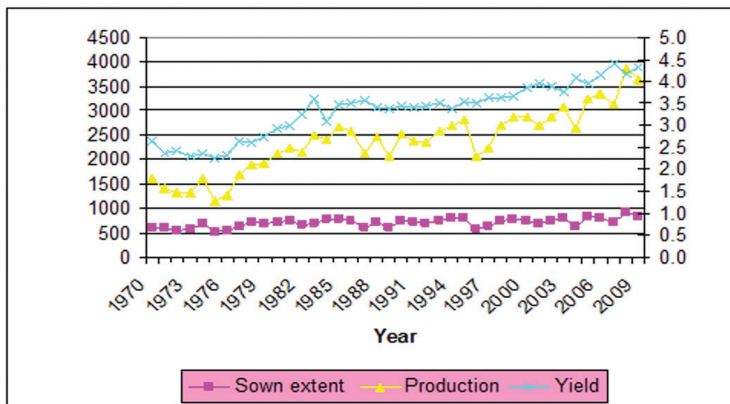


Figure 2: Trends in sown extent, production and yield of rice.

Table 3: Trends in rice production in Sri Lanka, 1970-2009 (10 year averages).

Periods	Harvested Area (000'ha)	Yield (t/ha)	Production (000' t)
1970-1979	610.4	2.5	1513.7
1980-1989	705.7	3.3	2334.0
1990-1999	716.9	3.5	2516.8
2000-2009	772.5	4.1	3136.0

Relative contribution of yield and area to production growth

Contribution of yield and area changes to production changes between the aforesaid four periods were decomposed using the following growth accounting formula which has been applied by Hazell (1982);

$$\Delta Q = \Delta \bar{A} \bar{Y}_1 + \bar{A}_1 \Delta \bar{Y} + \Delta \bar{A} \Delta \bar{Y} + E$$

Where ΔQ , $\Delta \bar{Y}$ and $\Delta \bar{A}$ are the differences in mean annual production, mean annual yield and mean annual area harvested, respectively while $\bar{A}_1$ and $\bar{Y}_1$ are the mean annual area and

mean annual yield for the first period. E denotes the sum of the computational error and any change in Q due to change in covariance between A and Y. Terms $\Delta\bar{A}\bar{Y}_1$, $\bar{A}_1\Delta\bar{Y}$, and $\Delta\bar{A}\Delta\bar{Y}$ represent the contribution of area change, yield change and the interaction between yield and area change, respectively. Of the production increase that occurred from 1970-1979 to 1980-1989 period, 63.1% was due to yield increase and only 28.6% was due to area expansion (Table 4). In the production change from 1980-1989 to 1990-1999, contribution of yield increase increased to 75.5% while area expansion's contribution decreased to 20.5% showing that adoption of varieties such as Bg 300

which were cultivated more than 21% of land area during this period became more and more important. In the production change from 1990-1999 to 2000-2009 too, yield increase dominated over area expansion. However, yield improvement's contribution decreased to 63.1% while area expansion's contribution increased to 31.5%. The implication is that area expansion mainly through increasing cropping intensity is also more important to production growth. Growth of harvested area, yield and production over each period and relative contribution of yield and area expansion to production growth is discussed in preceeding section.

Table 4: Source and their contribution to change in mean annual rice production between ten year periods during 1970 to 2009

Source of Mean Production Change	Change in Rice Production					
	1970-1979 to 1980 -1989		1980 -1989 to 1990 -1999		1990-1999 to 2000-2009	
	Absolute (000't)	%	Absolute (000't)	%	Absolute (000't)	%
Effect of area change ($\Delta A \square 1$)	235.11	28.66	37.39	20.45	195.08	31.51
Effect of yield change ($\square 1 \Delta \square$)	517.91	63.14	138.00	75.46	390.91	63.14
Effect of interaction between yield and area change ($\Delta \square \Delta \square$)	80.86	9.86	2.21	1.21	30.29	4.89
Error	-13.59	-1.66	5.28	2.89	2.82	0.46
Total change	820.29	100	182.88	100	619.12	100

Table 5 presents a similar decomposition of production change between three periods at a more disaggregated level i.e. by different types of irrigation. A striking feature is that the turning of negative change of mean annual paddy production in rainfed areas between 1980-1989 and 1990-1999 to a positive change from 1990-1999 to 2000-2009. In both cases between 1980-1990 and 1990-1999,

and 1990-1999 and 2000-2009 under major irrigation, contribution of area expansion was higher than contribution of yield increase. With respect to minor and rain fed areas yield increases dominated the contribution of area expansion in change of mean annual production while offsetting the negative contribution of area expansion under rainfed areas in case of between 1990-1999 to 2000-2009.

Table 5: Source and their contribution to change in mean annual irrigation wise rice production between ten year periods during 1980 to 2009.

Source of Mean Production Change	Change in Rice Production ('000 t)					
	1980 -1989 to 1990 -1999			1990-1999 to 2000-2009		
	Irrigation System			Irrigation System		
	Major	Minor	Rainfed	Major	Minor	Rainfed
Effect of area change ($\Delta A \Delta \bar{Y}_i$)	152.65 (81.22)	-1.57 (-13.36)	-117.57 (-127.32)	276.81 (52.18)	56.16 (44.44)	-33.22 (-100.03)
Effect of yield change($\bar{A}_i \Delta \bar{Y}$)	37.41 (19.90)	9.68 (82.36)	27.54 (29.83)	208.12 (39.23)	61.95 (49.02)	71.26 (214.60)
Effect of interaction between yield and area change ($\Delta \square \Delta \square$)	4.59 (2.44)	-0.03 (-0.28)	-5.21 (-5.65)	40.01 (7.54)	7.39 (5.85)	-4.50 (-13.55)
Error	-6.70 (3.57)	3.68 (31.28)	2.90 (3.14)	5.53 (1.04)	0.87 (0.69)	-0.34 (-1.01)
Total change	187.94 (100)	11.76 (100)	-92.34 (100)	530.46 (100)	126.37 (100)	33.21 (100)

Figures in parenthesis are the percentages

Growth rates of harvested area, yield and production over decades

Compound growth rates of harvested area, yield and paddy production within each ten year period were calculated to examine the growth of these variables over each period and how the area and yield contributed to growth of paddy production in each period. Paddy production was growing at a rate of 2.71% in 1970s (Table 6). In this period, about two-third of the growth in rice production was accounted for by increasing area at the rate of 1.82%, mainly through opening up of cultivation frontier due to irrigation expansion and highly favorable output prices. Only a moderate growth of yield that was achieved as yield potentials of old improved varieties have been already tapped, adoption of new-improved varieties were at initial stages, use of fertilizer was low, and unfavorable weather conditions prevailed. However, growth rate of paddy production in 1970s was not sustained into the 1980s and 1990s. In 1980s the yearly growth rate of paddy production fell to 0.35%. In contrast to 1970s, the growth of yield (1.62%) accounted for sustaining the growth of production at a rate of 0.35% while offsetting the decline of harvested area at a rate of 1.25%. Thus, growth in rice production became completely dependent on yield growth during this period. Wide spread of high yielding varieties and increased use of fertilizer and cultivation of new land under assured irrigation despite decline in sown area in unfavorable

environments especially in the Wet Zone have equally contributed to the growth of yield during this period. In 1990s paddy production was growing at a rate of 0.68%. Growth of yield (0.82%) has contributed to sustain the growth of paddy production offsetting the negative growth of harvested area. By the 1990s real paddy price has dropped quite sharply, adoption of high yielding varieties and expansion of irrigated area have leveled off. Rain fed areas contracted substantially as paddy production ceased to be profitable at the low real prices in these less favorable environments. Between 2000 and 2009 the yearly growth of rice production accelerated to 3.49% on the average and the country turned from being a net importer of 5 to 10% of its national requirements to being self sufficient. In contrast the acceleration of growth in rice production between 2000 and 2009 was achieved through productivity as well as expansion of sown area. Ten year average of harvested area increased up to 0.77 million ha as the annual growth rate increased sharply from -0.14% in 1990-1999 to 2.60% in 2000-2009. Growth of harvested area accounted for nearly 60% of production growth by mostly increasing cropping intensity up to 147%. Growth rate of yield increased from 0.82% in 1990-1999 to 1.4% in 2000-2009 accounting for nearly 40% of production growth in 2000-2009. Yield and area might have grown rapidly between 2000 and 2009 especially in latter part of 2000s due to highly favorable paddy prices and fertilizer prices and effective research, seed supply and extension programs.

Table 6: Compound growth rates (%) of harvested area, yield and rice production over decades from 1970 to 2009.

Periods	Harvested Area	Yield	Production
1970-1979	1.82	0.87	2.71
1980-1989	-1.25	1.62	0.35
1990-1999	-0.14	0.82	0.68
2000-2009	2.06	1.40	3.49

Factors affecting rice production, area and yield

Paddy production, area and yield depend on price factors such as paddy prices and fertilizer prices and non price factors such as technology and weather. Previous studies conducted by Ranaweera *et al.* (1990) and Samarathunga *et al.* (1984) concluded that paddy production is slightly responsive to prices. Ekanayake (2006) studied how the demand for fertilizer responds to price of fertilizer and farm gate price of paddy based on fertilizer demand functions fitted using aggregated data covering the period 1981-2005. His study concluded that the demand for fertilizer is inelastic to own price and farm gate price of paddy and demand for fertilizer is more responsive to farm gate price of paddy than its own price. While the computed own price elasticity of Urea, MOP and TSP was -0.15, -0.21, -0.07, respectively elasticity of Urea, MOP and TSP with respect to farm gate price of paddy was 0.33, 0.79, 0.41 in that order. However, later studies by

Weerasuriya and Gunaratne (2009) and Rajapaksha and Karunagoda (2008) concluded that paddy production in Sri Lanka is responsive to price of fertilizer. Further, Rajapaksha and Karunagoda (2008) showed that paddy production is more responsive to farm gate price of paddy than price of fertilizer.

This brief review pose a question whether supply response pattern of paddy to prices has changed with fertilizer subsidy introduced in 2005. Therefore, yield and area response functions were fitted to annual aggregate data for the period 1970-2009 using Nerlovian lag model to identify factors affecting rice area and yield (Table 7). While R² and F value of the area response function fall within inclusive range on goodness of fit of the function, yield response function has good overall statistical fit according to high adjusted R² and F values and has no serious problem of auto-correlation as implied by the Durbin-Watson D statistics which is closed to two. According to the

yield response function, yield is significantly responsive to lag price of N fertilizer, percentage area under high yielding varieties which captures the technological advancement, and favorable weather conditions. The estimated area response function shows

that paddy area is also responsive to lagged farm-gate price of paddy, lagged price of urea, and favorable weather conditions. According to this analysis, the increasing trend in area and yield may be attributed to price factor, technology and weather condition.

Table 7: Yield and area response functions of rice (1970-2009).

Independent Variable	Yield	Area (by irrigation type)		
		Major	Minor	Rain-fed
Constant	-0.9484 (-1.06)+	-7.498 (-1.07)	2.698 (0.29)	39.935* (7.42)
Farm-gate paddy price (Rs./kg) lagged by one year	-0.0693 (-0.95)	-0.1942 (-1.4)	0.3754** (2.03)	0.2841* (2.65)
Price of Urea (Rs./mt) lagged by one year	-0.0670* (-3.33)	-0.069** (-1.73)	-0.1884* (-3.55)	-0.1459* (-4.740)
Area under high yielding varieties (%)	0.5865* (4.80)	-	-	-
Area under irrigation (%)	0.0874 (0.42)	-	-	
Weather (dummy variable)	-0.0373* (-2.25)	0.0598** (1.73)	0.1379* (3.0)	0.0719* (2.70)
Time trend	-	0.0108* (3.22)	0.0051 (0.262)	-0.0135* (-5.23)
Adj-R2	0.859	0.767	0.351	0.854
F value	37.68	28.20	5.46	49.35
Durbin-Watson statistics	1.681	1.232	1.775	1.388

+ Figures in parenthesis are the t-values of coefficients, * indicates significant at 0.05 level, ** indicates significant at 0.1 level

Profitability in paddy cultivation

Profitability is also an important factor which determines paddy production in Sri Lanka. Farmers' interests on continuation of paddy cultivation and invest on paddy cultivation depend on profitability. Profitability of paddy cultivation in two environments namely irrigated Dry Zone and rain fed Wet Zone are presented in Tables 8 and 9. Cost of cultivation data pertaining to Maha season which is the major paddy growing season and published by the Socio Economics and Planning Centre of Department of the Agriculture were used. Anuradahapura, Polonnaruwa and Ampara districts were selected to represent the Dry Zone irrigated condition and the Wet Zone rain fed environment was represented by Kandy and Kalutara districts.

Change of nominal returns from paddy cultivation

Return to land and management (RLM) in the Dry Zone (DZ) increased from Rs. 3220/ha in 1999 to Rs. 88422/ha in 2008 during 1999-2009 period showing an increasing trend taken as a whole when exclude the declines observed in three seasons: 2002/03, 2005/06 and 2008/09. RLM in the Wet Zone (WZ) has fluctuated widely recording negative values for all the years except in Maha 2007/08. Year to year changes in RLM are due to fluctuation of farm gate paddy prices and yield from year to year. Return to family labor (RFL) is also an important economic parameter

which determines farmers' interest towards paddy cultivation. Return to manday of family labor during 1999 - 2009 fluctuated from Rs. 206 to Rs. 2330 in the DZ: Rs. 142 to Rs. 657 in the WZ. Return to unit of capital (RUC) was over 1 during the entire period in the DZ and WZ, except in 2003 Maha in the WZ. It is worth to note that RLM, RFL and RUC were comparatively high during last two years. This is due to high farm gate paddy prices prevailed during these years.

Change of real returns from paddy cultivation

In order to examine the change of profitability over time taking away the effect of inflation, nominal values were deflated using the Consumer Price Index (CPI) for all commodities published by the Department of Census and Statistics. CPI value for 1999 (75.3) was taken as 100 (base value). In the DZ real values of RLM and RFL showed increasing trend while in the WZ they fluctuated widely. While RUC in the DZ varied widely, in the WZ it showed a declining trend. Increases of RLM and RFL in real term in the DZ indicate that purchasing power of income earned from paddy cultivation has increased over the last 10 year period.

Table 8: Return from paddy cultivation under irrigated condition in the Dry Zone in Maha.

Year	Nominal Return				Real Return			
	Return to Family Labour, Land & Mgt (Rs./ha)	Return to Land & Mgt (Rs./ha)	Return to Family Labour (Rs./md)	Return to Unit of Capital (Rs.)	Return to Family Labour, Land & Mgt (Rs./ha)	Return to Land & Mgt (Rs./ha)	Return to Family Labour (Rs./md)	Return to Unit of Capital (Rs.)
2000	9515	3220	206.00	1.04	9515	3220	206.00	1.04
2001	19710	9383	379.67	1.71	18563	8838	357.59	1.61
2002	25441	15588	353.67	1.55	20989	12860	291.79	1.28
2003	21472	7126	397.33	1.67	16170	5366	299.23	1.26
2004	32077	16655	544.00	1.88	22832	11855	387.22	1.34
2005	38000	19178	702.67	1.99	24820	12526	458.95	1.30
2006	31081	13225	691.33	1.94	18286	7781	406.74	1.14
2007	47048	24726	1034.33	2.36	25164	13225	553.23	1.26
2008	114005	88422	2057.67	3.70	52640	40827	950.09	1.71
2009	115795	86221	2330.67	3.45	43624	32482	878.03	1.30

Mgt - management, md – man days

Table 9: Return from paddy cultivation under rainfed condition in the Wet Zone in Maha.

Year	Nominal Return				Real return			
	Return to Family Labour, Land & Mgt (Rs./ha)	Return to Land & Mgt (Rs./ha)	Return to Family Labour (Rs./md)	Return to Unit of Capital (Rs.)	Return to Family Labour, Land & Mgt (Rs./ha)	Return to Land & mgt (Rs./ha)	Return to Family Labour (Rs./md)	Return to Unit of Capital (Rs.)
2000	6805	-12167	142.50	1.38	6805	-12167	142.50	1.38
2001	7815	-7348	180.50	1.33	7360	-6920	170.00	1.25
2002	11038	-2228	248.50	1.44	9107	-1838	205.02	1.19
2003	-3041	-16032	160.00	0.92	-2290	-12074	120.49	0.69

2004	6666	-12176	205.50	1.21	4745	-8667	146.28	0.86
2005	591	-15180	200.00	1.02	386	-9915	130.63	0.67
2006	6135	-20273	221.50	1.22	3610	-11927	130.32	0.71
2007	10350	-11774	281.50	1.34	5536	-6297	150.56	0.72
2008	46191	16998	714.00	2.12	21328	7848	329.68	0.98
2009	37180	-2599	657.50	1.86	14007	-979	247.70	0.70

Mgt – management, md - man days

Present level of cost and return from rice cultivation

Table 10 presents current cost and returns from paddy cultivation. It is clear that rice cultivation in both environments is profitable at present. Further, profitability is high in the Dry Zone environment due to comparatively higher yield. High cost of cultivation in the DZ compared to the WZ is due to high input use such as fertilizer and labor with high wage rates. Of the total cost of cultivation in the DZ, labor constitutes the largest share which is 43%. Materials and power account for

34% and 23%, respectively. In the WZ 65% of the cost is on labor while 23% and 12% go for material and power, respectively. This shows that labor cost is the most significant component of cost of cultivation in the DZ as well as in the WZ environments. In the DZ about 44% of hired labor is used whereas in the WZ only about 34% of hired labor is used. At present there is tendency of using combine harvesters in the DZ as an alternative to increasing labor cost. By using combine harvesters cost of labor can be down by about 8%.

Table 10: Present level of cost and return from rice cultivation.

Item	Dry Zone	Wet Zone
Average yield (t/ha)	5.19	3.09
Farm gate price (Rs./kg)	31.24	30.38
Cost of cultivation (Rs./ha)	81,361.13	85,646.51
Gross return (Rs./ha)	161,443.77	96,777.54
Net return (Rs./ha)	80,082.64	11,427.55
Unit cost of production (Rs./kg)	15.89	27.87
Net return to land and management	103,898.63	56,505.18
Return to family labor (Rs./man day)	3,702.89	2,236.44

Return to unit of capital	3.26	2.43
Average holding size (ha)	0.97	0.38

Projection of rice production

A study conducted by Ranaweera *et al.* (1990) employed second degree polynomial functions to make future projection and show the trend in harvested area, production and yield in 15 years from 1975 to 1990 and concluded that these variables are approaching the maxima. However, the high growth rate experienced in 2000s suggest that paddy production, harvested area and average yield have not yet approached their upper limit

and are still in the form of increasing trend. The linear and second degree polynomial trend fitted to production, area harvested and national average yields strengthen the above hypotheses. Previous examination of paddy sector made projection of paddy production based on historical yield growth (Samarathunga *et al.*, 2000). Table 11 presents projected paddy production up to 2020 assuming that present growth rate of yield and harvested area will maintain up to 2020.

Table 11: Projected rice production up to 2020.

Year	Harvested Area (000' ha)	Yield (t/ha)	Production (000't)
2010	859.48	4.40	3782.38
2011	877.19	4.46	3914.34
2012	895.26	4.52	4050.91
2013	913.70	4.59	4192.24
2014	932.52	4.65	4338.50
2015	951.73	4.72	4489.86
2016	971.34	4.78	4646.51
2017	991.35	4.85	4808.62
2018	1011.77	4.92	4976.38
2019	1032.61	4.99	5150.00
2020	1053.89	5.06	5329.67

If it is assumed that the present growth of harvested area and yield will continue up to 2020 Sri Lanka will outpace the demand with domestic supply by 2011 and there will be a huge surplus of rice by year 2020. In spite of the optimistic projections three questions remain to be answered.

- Will it be possible to maintain price factors which affect the rice yield and area favorably?
- How can Sri Lanka handle the excess supply in a way that farm gate price of rough rice is not affected if the above optimistic projections are realistic?

Options for Next Decade

With the scenario presented above, two possible options are put forwarded for discussion in this section;

1. Maintaining rough rice production at present growth rate through increase in yield and area.

This option is most appealing to policy makers and even researchers. However, note should be taken that this requires maintaining price factors which affect yield and area favorably through managing excess production in addition to the technological breakthrough with an accelerated and well focused technology transfer process. Question that arises in this regard is:

Is it possible to manage excess production through export for niche markets, storing as a buffer stock, increasing the consumption of rice by popularization of rice based products prepared substituting rice flour for wheat flour, promoting for other purposes such as animal feed?

2. Producing rice only for requirement adjusting extent under rice cultivation during Yala season

This option directed towards producing rice only for requirement could be well justified if such program is planed properly. This requires an estimate of the extent to be cultivated during Yala season accurately based on the rice requirement considering the Maha production, commercial stock position, buffer stock requirement etc. If this option is adopted the rest of rice land can be utilized for cultivation of other field crops during Yala season based on suitability and cultivation of organic rice, traditional rice and quality rice for niche markets. This policy would require encouraging and organizing farmers introducing incentive and marketing channels for cultivation of other field crops.

Conclusions

Rice production in Sri Lanka has grown significantly over the last four decades especially in the latter part of this decade achieving nearly sufficiency level. The significant advances made in the latter part of this decade needs to be continued in order to meet the growing demand of the increasing population ensuring the food security in the country. In order to maintain these trends, existing favorable rice policies must be continued following pragmatic approaches while technological breakthroughs are being attempted. If the present trend of growth of rice production is maintained, there will be a substantial rice surplus in the country. Therefore, program for management of such a surplus or supply should receive utmost attention to avoid price falls detrimental to the farming community.

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Promotion of Inappropriate Technology in Rice Cultivation

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Abstract

Rice is grown as a food crop but not for medicinal or other purposes such as fodder in Sri Lanka. In order to improve productivity and grain quality in rice, use of improved and appropriate technology is imperative. Adoption of inappropriate technology may frustrate and discourage the farmer ultimately resulting in low productivity and production. Inappropriate technologies in rice cultivation are evident with respect to both cultivars as well as crop management practices. Promotion of traditional cultivars (TCs), unless for a specific purpose or introduced cultivars that are not adaptable to local conditions should be considered inappropriate. At the same time, whether promotion of Hybrid Rice in farmers' fields in Sri Lanka is appropriate is a question in the present context. Why promotion of TCs is considered inappropriate is discussed in detail as an example. TCs are not technically sound and economically viable under the present circumstances. Cultivation of improved cultivars is more environmentally friendly than that of TCs. No tillage practice except in places where tillage is not possible, use of effective microorganisms, prophylactic application of pesticides, use of various kinds of micro-nutrient solutions and hormones as foliar sprays, use of various plant extracts as pesticides and the system of rice intensification' (SRI) can be identified as inappropriate crop management technologies that are promoted in rice cultivation. Out of these, why SRI is considered an inappropriate technology is discussed in detail as an example. SRI which saves seed at the expense of labor is the least adaptable, technically not sound and practically not feasible method of stand establishment in rice with no yield or any other advantage over standard transplanting or broadcasting.

Key Words: Cultivar; Management practices, Promotion of inappropriate technology, Rice

Introduction

Rice (*Oryza sativa* L.) is the staple food of Sri Lankan's and it provides 40% of carbohydrate and 45% of protein daily requirements of the people in Sri Lanka. Depending on the degree of polishing it also provides varying amounts of fat, fiber, vitamins particularly vitamin B and minerals. Rice in Sri Lanka is grown as a food

crop but not for medicinal or other purposes such as fodder. In order to improve productivity and grain quality in rice, use of improved and appropriate technology is very imperative. Any technology to be promoted should be technically sound (efficient, effective and sustainable), practically feasible, economically viable, socially acceptable and environmentally friendly and technologies that do not

fulfill these requirements become inappropriate depending on the place and time. Appropriate technology at present in a country like China may not be appropriate in Sri Lanka and appropriate technology in Sri Lanka in the past may be even several years ago may not be appropriate at present.

Adoption of inappropriate technology may frustrate and discourage farmers due to yield and income losses, failures in achieving expectations and increased risk of cultivation (Abey Siriwardena and Dissanayake, 2008 & 2009). Large scale Adoption of inappropriate technology would result in gradual decrease in rice productivity and production in the country affecting both rice farmers and consumers and national economy at large. One example for such an inappropriate technology is the introduction of the rice cultivar TN1 from Taiwan during mid 1960's, which introduced the 'bacterial leaf blight' disease, a disastrous disease of rice, into Sri Lanka. Therefore, the duty of rice scientists in the country is not only to encourage and promote the use of appropriate technology but also to discourage the introduction and promotion of inappropriate technology in rice cultivation. Thus, the objectives of the present paper are to identify some of the inappropriate technologies that are detrimental for rice cultivation and to discuss two of them in detail as examples so that all the stake holders in rice production; officers who are involving in rice research and development, policy makers, farmers and the general public become aware of the gravity of the issue.

Inappropriate Technologies that are Promoted

Inappropriate technologies that are promoted in rice cultivation can be identified with respect to cultivars as well as crop management practices. Within cultivars, inappropriate technologies that need to be paid immediate attention are promotion of traditional cultivars (TCs) and promotion of introduced varieties that are not adaptable to local conditions. However, whether promotion of hybrid rice for cultivation in farmers' fields in Sri Lanka is an inappropriate technology is a question in this regard because hybrid rice technology in Sri Lanka as at present is not well developed to be adopted in farmers' fields. This should not be misunderstood to discourage hybrid rice research as a policy as research is the only way that can make hybrid rice an appropriate technology in rice in Sri Lanka in time to come. Adoption of hybrid rice should be a highly appropriate technology if hybrids with comparatively higher heterosis can be developed together with efficient hybrid rice seed production technologies. Why promotion of TCs is considered an inappropriate technology will be discussed in detail as an example. TCs are not technically sound and economically viable in the current context where high productivity is expected with increasing population.

No tillage practice as a blanket recommendation, use of effective microorganisms, prophylactic application of pesticides, use of various

kinds of micro-nutrient solutions and hormones as foliar sprays, use of various plant extracts as pesticides and system of rice intensification (SRI) are inappropriate crop management technologies. Some of them are without adequate research evidence of their efficacy. Out of these, why SRI is considered an inappropriate technology will be discussed in detail as an example. SRI which saves seed at the expense of labor is the least adaptable and acceptable method of stand establishment in rice cultivation because it has no yield or any other advantage over standard transplanting or broadcasting.

Promotion of traditional cultivars

At present, TCs are being promoted in rice cultivation by purporting attributes such as tolerance to pests and diseases, nutritional quality and palatability of cooked rice, suitability for organic farming and health benefits. However, this is far from the reality.

No TC except Murungakayan is tolerant to Leaf Blast, the major disease of rice and no TC is tolerant to Bacterial Leaf Blight, the other major disease of rice, whereas almost all the improved cultivars (ICs) are incorporated with genetic resistance identified from exotic genetic sources to both these diseases. Similarly, almost all the TCs except 'Heenati' group are susceptible to Brown Plant Hopper (BPH), a major pest in rice. Even the Heenati group shows very low level of tolerance to BPH. In addition, all the

TCs are susceptible while many ICs are resistant to Gall Midge, another frequent major pest in rice. Most of the ICs possess resistance to at least either Brown Plant Hopper or Gall Midge or both. A rare exception is 'Dahanala' a TC which possesses genetic resistance for rice Thrips, an important pest in seedling stage. This trait has not yet been possible to transfer successfully into ICs owing to inadequate research input. All the TCs as well as ICs with no exception have been found to be susceptible to Rice Bug, Leaf Roller and Stem Borer that are other major pests frequently damaging rice.

Varieties with high nutritional quality and palatability are available not only within TCs but also within ICs, if rice is milled and processed appropriately. Bw 272-6b is an IC with high protein, high iron, high fiber and high zinc having very high palatability of cooked rice. Cooked rice of Bg 360, an IC known as 'Keera Samba' already in the market, is highly palatable with high protein.

ICs are more suitable for organic farming than TCs because ICs are tolerant to at least few major pests, all the major diseases and lodging. In addition ICs are capable of giving much higher grain yields and hence profits than TCs at all levels of organic manure (Unpublished data at the Rice Research and Development Institute) as all the ICs have higher Harvest Index (grains: above ground biomass ratio) (0.5-0.55) than that of TCs (0.25-0.3).

Poor adaptability of TCs

TCs are lodging and low yielding. The percentage distribution of rice variety categories in Sri Lanka from 1950 to 2009 is shown in Fig. 1. During the 1950s, rice area in Sri Lanka was totally covered by the TCs and the average rice yield of the country during that period was less than 1t/ha (Abey Siriwardena, 2000). In contrast at present, more than 98% of the rice area is cultivated to ICs and the average rice yield in the country is about 4.2 t/ha. Thus the key factor that has contributed in increasing productivity of rice in the

country is the adoption of ICs in place of TCs due to higher adaptability of ICs than that of TCs.

Interestingly, in early 1990s TCs had been heavily promoted and area under TCs was increased up to about 10%, the highest area cultivated to TC after 1985 (Fig. 1). However, it could not be sustained for more than two years and area under TCs came back to less than 2% again. This clearly shows the poor adaptability and acceptability and thus the poor popularity of TCs among farmers when compared to ICs.

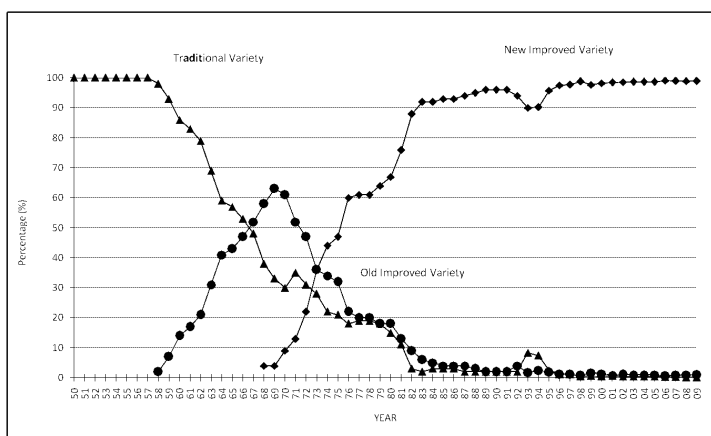


Figure 1: Percentage distribution of rice varietal categories in Sri Lanka, 1950-2009

Health properties

With respect to health properties, some TCs are said to possess medicinal properties according to ayurvedic medicinal practitioners but there is no scientific evidence to establish this. Also some TCs are reported to be low in glycemic index (GI) and high in anti-

oxidant properties in some recent news paper articles. The question is whether the full range of ICs has been analyzed for these characteristics. There are also ICs which are good in these attributes. The GI of an IC, recently developed 'Red Basmathi Type' rice, is as low as 54 and is considered a low GI variety (Baker IDI, Heart and Diabetes Institute, 2009). The anti-

oxidant capacity of Bw 272-6b, an IC, is as high as 2.242 r (Unpublished data at the Industrial Technology Institute, Sri Lanka). Thus, it is obvious that there are also ICs that are good in health properties. Therefore, there is no rationale for promoting only the TCs on health considerations. What is important is that desirable health characteristics if any in low yielding TCs should be transferred to high yielding ICs through breeding so that consumers will have the option of having them at affordable prices.

Is cultivation of TCs an appropriate technology?

Thus TCs are not technically sound and economically viable under the present circumstances. On the other hand, there are so many ICs of rice that are technically sound and economically viable for cultivation at present. Even the adoption of integrated pest management practices is comparatively easier with ICs as they are more tolerant to major rice pests and diseases than TCs. In addition, ICs have a higher fertilizer response than that of TCs at all levels of soil fertility. Therefore, it is justifiable to consider the use of TCs an appropriate technology in rice cultivation? However, TCs should be conserved for future breeding activities in which their desirable characteristics if any can be transferred to ICs when appropriate scientific methods become available in the future.

System of rice intensification (SRI)

System of rice intensification (SRI) has been first developed in Madagascar about 26 years ago. However, a method very similar to SRI but without using the name SRI had been promoted in Sri Lanka about 103 years ago (Amarasekara, 1907). SRI is a method of rice cultivation where young rice seedlings (about 10 day old) are transplanted at a comparatively wider spacing (25-30 cm or more) with only one plant/hill and no standing water until 45 days after planting with nourishment from compost rather than chemical fertilizer. SRI promoters claim that SRI is capable of giving a grain yield of 20 t/ha or even more only with organic manure and compost while naturally resisting pests and diseases, saving water by avoiding continuous flooding and saving seed paddy by about 80-90% (Uphoff, 2001; CIIFAD, 1999).

Underlying principle of SRI

Unlike wheat, barley, sorghum or maize, rice is a heavy tillering cereal where a range of stand establishment methods are adopted depending on the degree of making use of tillering. The range of stand establishment methods in rice with their seed rates and special features is shown in Fig. 2 which is self-explanatory (Abeyasiriwardena *et al.*, 2009). There are two extremes

in the range of stand establishment methods; high density broadcasting with no tillering at all is one extreme and SRI with maximum tillering is the other extreme.

In the SRI, rice crop is allowed to have as many effective tillers per plant as possible with the minimum number of mother plants or culms per unit land area. Therefore, tillering has to be promoted in SRI to achieve its potential yield and management practices specific to SRI have been developed (Uphoff and Fernandes, 2002) to promote heavy tillering keeping the

maximum possible uniformity among tiller panicles (Abey Siriwardena *et al.*, 2009). In SRI, individual panicles are relatively bigger but the number of panicles per unit land area is relatively low (Abey Siriwardena *et al.*, 2009; Nissanka and Bandara, 2004). The unique feature of SRI is that it saves seed by five and ten times over standard transplanting and standard broadcasting, respectively but at the expense of additional labor for manual transplanting, weed control and water management.

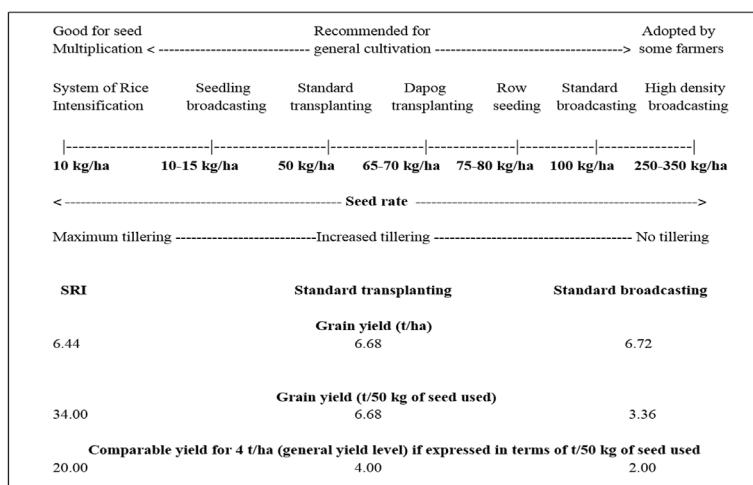


Figure 2: Range of stand establishment methods of rice with their seed rate, tillering behavior and grain yield in terms of t/ha and t/50 kg seed used for cultivation.

Grain yield

Grain yield of rice can be expressed on area basis as grain yield per unit area cultivated (t/ha) or on seed basis as grain yield per unit amount of seed used for cultivation (t/kg of seed). The

amount of seed required for standard transplanting of one hectare (50 kg of seed) is considered as the base when grain yield is expressed on seed basis (Abey Siriwardena *et al.*, 2009). In general, the most limiting resource in rice cultivation is used as the base in

expressing grain yield so that grain yield is generally expressed as t/ha in the present day context as the most limiting resource at present is not the seed but the land. However, as the unique feature of SRI is saving seed, grain yield under SRI may be expressed on seed basis to highlight the seed saving efficiency of SRI over the other stand establishment methods. Only 10 kg of seed/ha is used in SRI whereas 50 kg and 100 kg of seed/ha are used in the standard transplanting and broadcasting methods, respectively.

Abey Siriwardena *et al.* (2009) reported that when the grain yield of rice was expressed on area basis as t/ha, grain yield among SRI, standard transplanting and standard broadcasting were found to be 6.44 t/ha, 6.68 t/ha and 6.72 t/ha, respectively and they were comparable. However, when the grain yield was expressed on seed basis as t/50kg of seed used for cultivation, the grain yield of 34.00 t/50 kg of seed under SRI was highly attractive when compared to that of 6.68 t/50 kg of seed under standard transplanting and 3.36 t/kg of seed under standard broadcasting, irrespective of the variety (Fig. 2). In this context, for example, the yield level of 4 t/ha expressed on area basis (which is similar across SRI, standard transplanting and standard broadcasting) is equivalent to 20 t/50 kg of seed expressed on seed basis under SRI (Fig. 2). Interestingly, these yield levels under SRI expressed on seed basis are, however, comparable to the reported yield levels of 20 t or more (Uphoff, 2001) expressed on area basis under SRI.

Grain yield of rice expressed on area basis does not vary among stand establishment methods that range from high density broadcasting to SRI. However, SRI is expected to give five and ten times more grain yields than that of standard transplanting and standard broadcasting, respectively if the grain yield is expressed on seed basis as SRI uses only five and ten times less seed than that of standard transplanting and standard broadcasting, respectively. In general, the yield level of 4 t/ha is common in rice. This area basis yield level is equivalent to 20 t/50 kg of seed used for cultivation on seed basis under SRI. It appears that SRI promoters have misunderstood this point and they express seed basis yield level of 20 t/50kg of seed under SRI in area basis as 20 t/ha.

Reality of SRI

As already indicated, management practices for SRI has been developed to maximize tillering at the maximum rate possible. Thus, SRI is highly sensitive to environment as it depends on heavy and rapid tillering. None of the other stand establishment methods in rice depends on tillering of the crop to the degree that SRI depends on so that SRI is the least adaptable method of stand establishment in rice (Abey Siriwardena and Dissanayake, 2008 & 2009).

Standing water in the field suppresses tillering so that standing water is avoided during the vegetative phase of the crop under SRI. SRI performs well only without standing water. This point is misinterpreted by SRI

promoters saying that SRI saves water. The same amount of water can even be saved with any of the other stand establishment methods as well even with a yield improvement, if standing water is not retained in the field or complete drainage is assured. The reality is that, unlike SRI, all the other stand establishment methods perform equally well with and without standing water.

If there is no standing water in the field, weeds will become a serious problem but application of herbicides has to be avoided in SRI as it suppresses tillering. Therefore, labor-intensive manual weeding or mechanical weeding using rotary weeders is the only alternative with SRI. Thus, SRI requires 38-54% more labor than standard methods of stand establishment in rice (Rakotamalala, 1997) so that SRI is highly labor-intensive.

Due to low densities, transplanted fields are more tolerant to Brown Plant Hopper and Leaf Roller than broadcast sown fields but SRI has no special advantage over standard transplanting in this regard. The only advantage of SRI is that it saves seed over the other stand establishment methods. However, seedling broadcasting also saves seed as much as SRI does.

Use of compost in rice fields has long been recognized as a good complementary practice. However, whether application of compost alone could provide the nutrient requirement for high yields (20 t/ha) is questionable.

Providing the plant nutrient requirement in the form of compost alone is practically not feasible and if feasible it is very expensive. In addition, use of only compost in large quantities to satisfy the nutrient requirement of the rice crop in naturally rice growing low lying areas will lead to soil problems. Thus, the best practice is to supplement chemical fertilizer with organic matter.

Is SRI an appropriate technology ?

In short, SRI which saves seed at the expense of labor is the least adaptable method of stand establishment in rice with no yield or any other advantage over standard transplanting or broadcasting. A country like Sri Lanka should concern more on labor saving while reducing the risk of getting poor yield than on saving seed in rice cultivation. What should really matter is the return on investment. At present, the most limiting and expensive resources in rice cultivation are land and labor but not the seed. In short high yields are expected per unit land area while minimizing the labor cost. Thus, a method that can save labor with maximizing yield per unit land area while reducing the risk of getting low yield due to poor adaptability even at the expense of seed is the most appropriate for rice cultivation (Abeyasiriwardena and Dissanayake, 2008 & 2009).

Though SRI was re-introduced to Sri Lanka in 1999, hardly any area cultivated to SRI is found at present, even after 11 years of re-introduction.

Poor farmer acceptance of a technology is a clear signal of its unsuitability. SRI is identified as the latest in a family of unconfirmed field observations (UFOs) that has some resemblance to space UFO (Sinclair, 2004). Thus, is it justifiable to consider SRI an appropriate technology?

Conclusions

Adoption of inappropriate technology in rice cultivation should not be promoted. Adoption of traditional cultivars, introduced cultivars that are not adaptable to local conditions and hybrid rice at present level of development are inappropriate technologies with respect to cultivars. No tillage practice as a blanket recommendation, use of effective microorganisms, prophylactic application of pesticides, use of various kinds of micro-nutrient solutions and hormones as foliar sprays, use of various plant extracts as pesticides and system of rice intensification (SRI) are inappropriate crop management technologies.

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Rice Variety Improvement for the Wet Zone of Sri Lanka

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Abstract

Potential for further increase in rice production in the Wet Zone is high. Improving profitability through enhancing productivity and hence the sustainability of rice farming while maximally utilizing the available natural resources in the Wet Zone have become imperative when the population growth, food security and the national development are concerned. Climatic, edaphic, socio-economic and other constraints to rice production have been identified. The use of appropriate varieties is the key to overcome most of these constraints. Rice varieties presently used in the Wet Zone have been identified and the rice varieties suitable for cultivation in the different environments within the Wet Zone have been proposed in order to enhance the average productivity of rice to the expected level of 4.0 t/ha within next five years in the Wet Zone. Depending on the production constraints, the rice plant type suitable for the Wet Zone is defined so that the future rice improvement program should be focused on this to bridge the gap between potential yield and the realized yield at farmer level in the Wet Zone. The characteristics that were considered in defining the Wet Zone plant type were reduced tillering and large panicles, high density grains, loosely compacted canopy and thick and erect leaf characteristics, diverse crop growth duration, intermediate plant height and increased biomass with stronger culms, harvest index around 0.5, strong root system, no pre-harvest sprouting, high yield potential, high variety adaptability and stability over diverse environments, tolerance to abiotic stresses namely iron toxicity, salinity and submergence, tolerance to biotic stresses namely leaf blast, bacterial leaf blight, sheath blight, stem rot, brown plant hopper, gall midge, sheath mites and thrips and high grain quality.

Introduction

Classification of agro-ecological regions in the country clearly identified three basic agro-climatic zones namely Wet, Intermediate and Dry (Punyawardena *et al.*, 2003). Wet Zone (WZ) is further classified into Low Country Wet Zone (LCWZ), Mid

Country Wet Zone (MCWZ) and Up Country Wet Zone (UCWZ). Rice is not grown predominantly in the UCWZ and some agro-ecological regions in the MCWZ (WM1a and WM1b). It is grown predominantly in nine agro ecological regions out of 15 in the WZ and all the agroecological regions (24) in the Dry and Intermediate Zones in Sri

Lanka. LCWZ spreads across Colombo (100%) and part of Kalutara (95%), Galle (93%), Gampaha (95%), Kegalle (60%), Matara (60%) and Rathnapura (45%) districts. MCWZ spreads across part of Kandy (50%), Kegalle (35%), Rathnapura (15%) and Matara (10%) districts. UCWZ spreads across part of Nuwaraeliya (60%), Kegalle (5%),

Rathnapura (5%), Kandy (8%) and Matara (3%) districts. Agro-ecological regions and land area wise contribution of administrative districts to those agro-ecological regions and rainfall, land terrain and major soil groups in each agro-ecological region in the LCWZ and MCWZ are presented in Table 1 to further characterize the Wet Zone.

Table 1: Agro-ecological regions and land area wise contribution of administrative districts to these agro-ecological regions and rainfall, land terrain and major soil groups in each agro-ecological region in the LCWZ, MCWZ and UCWZ.

Agro-ecological Zone/Region	Area Wise Contribution of Administrative Districts (%)		Total Annual Rain-Fall (mm)	Land Terrain	Major Soil Groups
Low Country Wet Zone					
WL 1a	Matara	35	>3200	Rolling, undulating and hilly	Red Yellow Podzolic & Low Humic Gley soil
	Galle	37			
	Kalutara	60			
	Colombo	30			
	Gampaha	6			
	Rathnaputa	30			
	Kegalle	35			
WL 1b	Kalutara	20	>2800	Undulating and Rolling	Red Yellow Podzolic & Low Humic Gley soil
	Galle	3			
	Colombo	20			
	Gampaha	12			
WL2a	Matara	25	>2400	Rolling undulating and flat	Red Yellow Podzolic Low Humic Gley soil & Bog, Half Bog soil
	Galle	50			
	Kalutara	15			
	Colombo	15			
	Rathnaputa	15			

WL 2b	Gampaha	3	>2200	Steeply dissected, rolling and undulating	Red Yellow Podzolic, Reddish Brown Latasolic & Low Humic Gley soil
	Galle	3			
	Kegalle	25			
	Colombo	3			
WL3	Colombo	35	>1700	Rolling and undulating	Red Yellow Podzolic with Soft & Hard Laterite, Low Humic Gley and Regasol
	Gampaha	74			
Mid & Up Country Wet Zone					
WM 1a	Rathnapura	8	>3300	Mountainous steeply dissected hilly & rolling	Red Yellow Podzolic & Lithosol Soils
	Kegalle	18			
	Kalutara	5			
	Galle	7			
	Matara	10			
	Nuwara Eliya	5			
	Kandy	5			
WM 1b	Rathnapura	5	>2900	Steeply dissected hilly & rolling	Red Yellow Podzolic & Lithosol Soils
	Kandy	10			
WM 2a	Rathnapura	5	>2200	Steeply dissected hilly & rolling	Red Yellow Podzolic, Reddish Brown Latazolic, Low Humic Gley
	Kandy	10			
WM 2b	Kandy	15	>1800	Steeply, hilly & rolling	Reddish Brown Latazolic,Imature Brown Loam, Low Humic Gley & Red Yellow Podzolic Soils
	Kegalle	12			
WM 3a	Kegalle	5	>1600	Steeply, hilly & rolling	Low Humic Gley & Lithosol soils
	Kandy	5			
WM3b	Kandy	10	>1400	Hilly, rolling, undulating & steep	Reddish Brown Latazolic,Imature Brown Loam, Low Humic Gley
WU1	Rathnapura	5	>3100	Mountainous steeply dissected hilly & rolling	Red Yellow Podzolic, Mountain Regosols & Lithosol Soils
	Kegalle	5			
	Nuwara Eliya	10			
WU 2a	Nuwara Eliya	10	>2400	Steeply dissected, Hilly & Rolling	Red Yellow Podzolic Soils
	Kandy	4			

WU 2b	Nuwara Eliya 15 Kandy 4	>2200	Mountainous, Steeply dissected, Hilly and Rolling	Red Yellow Podzolic, Mountain Regosols & Lithosol Soils
WU3	Nuwara Eliya 15	>1800	Hilly & rolling	Red Yellow Podzolic Soils

Out of the total rice extent of 0.727 million ha in Sri Lanka, approximately 25% is in the WZ (Jayawardena and Peiris, 1980). Furthermore, out of the total rice extent in the WZ approximately 80% is in the LCWZ while rest is in the MCWZ. As about 20% of the national rice production is coming from the 25% of the total rice extent in the WZ, WZ seems to have deterred the attention of authorities in making a concerted effort to its improvement (Jayawardena and Peiris, 1980). In the WZ, inexpensive or green water is sufficiently available for successful rice crops in both Maha and Yala seasons so that despite many constraints to rice production, cropping intensity in the WZ is very close to that in the Low Country Intermediate and Dry Zones (LCI&DZ). Approximate cropping intensities in the WZ and LCI&DZ were 110% and 118% in 2007 and 141% and 147% in 2008, respectively. WZ is getting adequate rainfall during both seasons and the mountains and adjacent highlands of the rice tracts in the WZ are acting as reservoirs by storing water during rainy period and releasing it gradually and continuously through spring flows, interflows and stream flows to low lying rice tracks (Panabokke and Somasiri, 1980). This has highly favorable consequences on annual rice production and the national food security in the country. In both the

WZ and LCD&IZ comparatively larger area is cultivated in the Maha season. However, Yala cultivated area as a percentage of that of Maha is higher in the WZ (71.2%) than that in the LCD&IZ (65.4%) indicating comparatively higher availability of green water in the WZ and importance of the WZ with respect to national food security. Panabokke and Punyawardena (2000) reported that one of the special features of rice production that creates problems in Sri Lanka is the high year-to-year as well as seasonal variation in the quantity of production. This problem however, is much less in the WZ than in the LCD&IZ. In the WZ, Yala production is about 90.4% of Maha production whereas in the LCD&IZ, Yala production is about 82.2% of Maha production.

WZ is not too far behind with respect to the productivity of rice when compared to the LCD&IZ (Table 2). Yield difference between WZ and LCD&IZ is only about 1 t/ha. Within the WZ, rice yield in the LCWZ representing majority of the zone is lower than that in the MCWZ. Comparatively lower yield in the LCWZ may be attributed to comparatively higher temperature (Abeyasiriwardena, 1999) and poor drainage. Another interesting feature is that Maha yields are higher than Yala yields in the WZ and this is other way around in the LCD&IZ.

The same observation was also made by Jayawardena and Peris (1980). Comparatively high proportion of rice lands in the WZ particularly in the LCWZ has been abandoned or fallowed for long due to seawater intrusion, water logging, development of soil related problems, poor maintenance of drainage systems, prone to flooding etc. However, even in the lowest

potential fields there is no other crop to replace rice (Jayawardena and Peris, 1980). The solutions and the national level policies proposed to alleviate these problems were emphasized by Kendaragama and Bandara (2000). The lowest rice yield was recorded in the Kalutara district within the LCWZ, where most of the above problems are dominant.

Table 2: Grain yield (t/ha) of rice in different agro-climatic zones and seasons in Sri Lanka (2007-2009).

Season	Agro-climatic Zone			
	Wet Zone		Dry & Intermediate Zone	
	Low Country	Mid & Up Country	Average	
Yala	3.16	4.06	3.34	4.50
Maha	3.64	4.22	3.75	4.37
Average*	3.45	4.13	3.59	4.41

* Averages are weighted averages by land area cultivated.

Although Sri Lanka is at the verge of self-sufficiency in rice at present, improving profitability through enhancing productivity and hence the sustainability of rice farming while maximally utilizing the available natural resources in the WZ have become imperative when the population growth, food security and the national development are concerned. Potential for further increase in rice production in the WZ is high as indicated above. Thus, in this context, Identification of presently existing constraints to rice production and finding ways and means to overcome those constraints

utilizing improved technology is important. The easiest and the cheapest way to achieve this goal is the use of appropriate rice varieties. Though many rice varieties are available to cater for specific problems the frequent use of inappropriate rice varieties has been observed in the LCWZ. This paper discusses the constraints to further growth in rice production, selection of appropriate rice varieties adaptable for diverse environments and the rice plant type suitable for the WZ to be considered in the future variety improvement program in order to enhance the rice production in the WZ.

Constraints to Further Growth in Rice Production

Climatic constraints

About 90% of the rice lands in the WZ are cultivated under rain-fed condition. Annual rainfall in the WZ varies from 1600 to >3300 mm. Uncertain rainfall pattern of the zone especially in Yala season remarkably affects the rice cultivation. Excess water and unexpected rains at land preparation, crop establishment, heading, harvesting and processing adversely affect the rice production in this zone. This may be one of the reasons for low Yala yields in the zone. Furthermore, excess water causes yield losses as a result of frequent occurrence of submergence and flooding in the zone (Punyawardena *et al.*, 2003).

Relative humidity which varies from 60%-96% and temperature (minimum $18.3 \pm 0.49^{\circ}\text{C}$ and maximum $33.2 \pm 0.51^{\circ}\text{C}$) also result in comparatively higher respiration rates in the plants causing yield losses. The influence of high temperature and high humidity at flowering also cause sterility the famous phenomena of “Ehala pussa” during Yala season in the WZ (Abey Siriwardena *et al.*, 2002; Weerakoon *et al.*, 2008). This so-called Ehala Pussa has become a serious problem making entire rice tracts amounting to several hectares almost completely sterile (Abey Siriwardena *et al.*, 2002). Furthermore, low temperature during the reproductive

phase in rice in Maha season in the WZ has been reported to increase grain sterility adversely affecting the rice yield (Ota and Devasundarajah, 1966; Jayawardena and Peris, 1980). This phenomenon which is observed in Maha season is commonly referred as “Seethala Boll”.

During rainy period, the low light intensity and less sunshine hours are prevalent in the LCWZ due to high cloud cover percentage in the sky. The small narrow “yaya” type landscape further aggravates this situation due to shading effects from canopy structure of adjacent uplands.

Edaphic constraints

In the WZ, main great soil group is documented as Red Yellow Podsollic (Ultisol). Furthermore, Immature Brown Loam soils and Red Yellow Latasolic soils are also available in comparatively a lesser extent. The rice soils are essentially either hydromorphic associates such as Low Humic Gley soils or alluvial derivatives of the above great soil group/sub group (Panabokke and Nagaraja, 1964).

A wide range of risk and limitations exist in rice soils in the WZ. Low fertility due to strong acidity and inherently low availability of nutrients such as Potassium, Nitrogen, Phosphorus and Zinc and low base saturation demonstrate the poor productivity of rice lands in the WZ (Kumaragamage *et al.*, 1999; Panabokke and Nagaraja, 1964). In

addition, iron toxicity in localized areas and salinity along the coastal belt due to intrusion of sea water also limit the productivity in rice lands.

There are three types of rice soils in the LCWZ and they are referred to as sandy soils, mineral soils and organic soils which includes bog and half bog soils. The similar type of classification was documented in the WZ soils by Dimantha (1977). Identification of the spread of each of the above soil types in the LCWZ is possible after careful observation in the rice landscape of a particular area. The sketch of the LCWZ rice land system and distribution of different soil types within the system are presented in Fig. 1.

Sandy Soils

Sandy soils are found in rice lands surrounded by or near uplands. In most of the instances these sandy rice lands are prone to iron toxic condition. The realized yield in these soils varies from 1-2 t/ha and 35% of the rice lands in the LCWZ belong to this category.

Mineral Soils

The mineral soils are the most potential soils in the LCWZ and these soils spread over 40% of the rice lands in this region. Under the best management conditions the rice yield of 5-6 t/ha can be obtained in these soils though the present average yield of these soils is 3.0-3.5 t/ha. A variation in yield could be seen as the iron toxicity could also affect the rice yield in these soils.

Organic Soils (Half Bog and Bog soils)

Due to poor or ill drain conditions, accumulation of organic matter results in half bog and bog soils in the region. Depending on the extent of drainage, organic matter content varies from 15% to about 30% in these soils. Most of the flood prone rice area in the LCWZ contains organic soils. Iron toxicity and salinity could also be observed in these soils which accounts for about 25% of the rice lands in the LCWZ. The average yield varies from 2½ to 3 t/ha. Poor drainage induced by mismanagement of low lands further aggravates such conditions and such lands remain uncultivated. Hence, it is advisable to implement a drainage improvement program in such lands and facilitate the farmers to re-cultivate them.

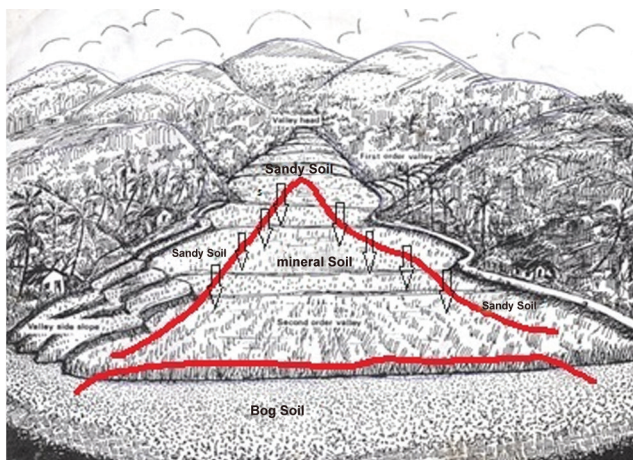


Figure 1: Sketch of the LCWZ rice land system and within system distribution of different soil types (Adopted from Panabokke and Somasiri (1980) and modified).

Socio economic constraints

More than 42% of the national population is occupying in the LCWZ so that population density in the LCWZ is very high. Hence, land fragmentation is very common. At present, average rice land holding size in the LCWZ is less than 0.2 ha. High rate of urbanization in the area increases the filling up of rice lands further aggravating the situation. Almost all the rice farmers in the area are subsistence and part time farmers. They cultivate the rice lands mostly in Maha season. Land ownership may change from time to time which also leads rice lands to become marginal and continue to yield lower than the expected. The youth contribution for rice cultivation is unusually low due to high attraction towards more convenient jobs available in the area. Hence, shortage of labor force for the rice cultivation is one of the major constraints at present.

Risk of cultivation is another socio-economic constraint in the WZ rice production. Rice farmers particularly in the LCWZ appeared to be satisfied even with very marginal yield levels as long as they are consistent over seasons with a low risk. Whenever a high risk is involved with highly inconsistent yields over seasons, farmers move out of rice cultivation (Abey Siriwardena and Sandanayake, 2000).

Other constraints

Narrow small rice lands surrounded by uplands and boggy nature of the rice soils in the WZ limits the use of machineries for land preparation and harvesting in the area. Sun drying, storage and marketing of harvested rough rice are facing many zone specific problems particularly related to unexpected rains and high relative humidity. Post-harvest losses in

harvesting, handling, threshing, drying and storage of rough rice may even go up to about 20% of the total harvest in the WZ.

Rice Varieties

Improving productivity is the key to improve rice production while reducing the risk of cultivation and increasing the farmer income. Among the many technologies to increase productivity by alleviating related constraints, development of rice varieties suitable for the LCWZ has been identified as the cheapest and the most effective technology that can be easily adopted by the farmers.

Until 1940's rice varieties grown by farmers in the country were mostly selections from large number of land races with the emphasis on grain quality. During 1940's these cultivars were subsequently improved by means of pure line selection by the researchers in the Department of Agriculture. These pure lines typically had long and droopy leaves, tall weak stems, late maturity, susceptibility to lodging when heavily fertilized, susceptibility to leaf blast and bacterial leaf blight; the most dangerous diseases in rice and low yield potential (Rajapakse, 2000). The selection aimed for comparatively a shorter stature, resistant to lodging and efficient biomass partitioning between grain and straw and breeders unintentional selection for improved canopy architecture, light penetration and other favorable agronomic characteristics resulted old improved plant type, mainly the varieties of

'H' series. However, spread of old improved varieties over traditional varieties in the WZ was comparatively slower when compared to that of LCD&IZ of the country up to 1980's.

Some of the traditional varieties grown in the WZ were Manchelalergy, Rath-al, Dahanala, Kalu Heenati, Rath Suwandal, Molligoda, Kottayar, Sudu Heenati, Beheth Heenati, Sulai, Kalubalawee, Kaharamana, Mada-el, Pachchaiperumal, Pokkali, Kuruluthuda, Kohumawi, Muhudu Kiri-el, Podiwee, Dewaraddari, Goda Heenati, Kahata Wee, Murungakayan, Suwandal, Suwanda Samba, Herath Banda, Dik Wee, Masuran, Rathu Heenati, Hondarawala, Gonabaru, Madathawalu, Harti-al and local mawee varieties. The distribution of these traditional varieties was comparatively higher in Kalutara and Galle districts followed by Rathnapura district in 1970s (Jayawardena and Peiris, 1980). The comparatively higher agro-ecological diversity exists in the WZ and addiction to the taste of the traditional varieties by the consumers may have been the reasons for difficulty in replacing traditional varieties by improved varieties. At present though the extent under traditional varieties is comparatively less, traditional rice varieties are more popular in Kalutara district than any other district (Jayawardena *et al.*, 2010) in the country.

Rice varieties developed and recommended for the Low Country Wet Zone

H4 released in 1958 was the first improved rice variety in Sri Lanka. It spreaded very fast in the WZ. The susceptibility to leaf blast and lodging in response to added nitrogen and poor grain threshability were the main drawbacks of this variety. Until late 1960s, the old improved variety H4 became more and more popular in the WZ and subsequently the semi-dwarf high yielding variety Bg 11-11 quickly replaced H4 in the high potential areas of the WZ (Jayawardena and Peiris, 1980). However, semi-dwarf varieties such as Bg 11-11, Bg 94-1, Bg 34-6 and Bg 34-8 did not spread into the problem soils and low lying areas which cover the major part of the WZ because of their susceptibility to iron toxicity, ill drained conditions, submergence, salinity and poor competitive ability with weeds. This drew the attention of authorities to initiate a separate rice improvement program for the WZ. Since then the rice improvement for the WZ was continued and paid attention.

Bw and Ld rice varieties

Regional Rice Research and Development Center (RRRDC), Bombuwela and Agriculture Research Station (ARS), Labuduwa initiated the rice improvement program for the WZ mainly to cater for the problem soils with more emphasis on iron toxicity as well as for the potential soils in the zone. As a result, many varieties have been developed and the list of

rice varieties developed at Bombuwela (Bw) and Labuduwa (Ld) and released for cultivation to date is presented in Table 3. Bw 78, Bw 100, and Ld 66 were the first set of varieties developed and released for cultivation in the WZ. Four months salinity tolerant rice varieties, Bw 400 and Bw 451 released in mid 1987 were popular up to 1990s in the LCWZ. As those varieties were in the 4 - 4½ months age group, 3 - 3½ months varieties such as Bw 272-6b, Bw 266-7, Bw 267-3, and Bw 351 having a potential yield of 5-6 t/ha became popular from 1980's up to early 2000's. Rice varieties in the 4½ months age group released in 1990's namely Bw 451, Bw 452 and Bw 453 did not become popular as their age was too long. Since the release of 3½ months Ld 356, being a red pericarped, short round (samba) grain variety, it spreaded faster than any other variety in the WZ. However, severity of susceptibility to Neck Blast and Brown Plant Hopper resulted this variety to become less and less popular in the recent past in the WZ and production of seeds in the DOA was terminated. Among all the varieties, Bw 267-3, a white pericarped variety was the most suitable iron toxicity tolerant variety for the LCWZ. However, high susceptibility to lodging and white pericarped nature of the variety were the main drawbacks as lodging led to yield losses and majority of the consumers in the region preferred red rice. In addition, appearing red grains in this white pericarped Bw 267-3 though at a very low frequency led the authorities at that time to terminate the seed production of this variety in

the DOA under the impression that maintaining genetic purity of this variety was difficult. A comparatively higher out crossing rate of Bw 267-3 was reported and this was attributed as the reason for the loss of genetic purity of the pericarp color in this

variety (Abyawickrama *et al.*, 2009). However, as the demand for seed of this variety is still high in the LCWZ, only the private sector is producing seeds of this variety at present without the problem of genetic purity of appearing red seeds.

Table 3: Improved Bw and Ld rice varieties released for cultivation and their year of release, pedigree, grain color and type and potential yield in the Wet Zone.

Variety	Year of Release	Pedigree	Released for *	Grain Color and Type	Potential Yield (t/ha)
4-4½ months age					
Bw 78	1977	H105//Podiwee A8/H5	LCWZ	White Samba	5.0
Bw 100	1979	do	LCWZ	White Samba	6.0
Bw 400	1987	Bw 259-3/Bw 242-5-5	WZ	Red Long	6.0
Bw 451	1987	Bg 400-1/ Bg11-11	LCWZ	White Samba	6.0
Bw 452	1992	Hondarawala 502/ C104	GC	Red Long	5.0
Bw 453	1992	IR 2071-586/Bg 401-1	LCWZ	White Long	7.0
Ld 66	1971	-	LCWZ	Red Long	7.0
Ld 408	2010	At 01//Ld 98-152	GC	Red Long	7.5
3½ months age					
Bw 266-7	1981	Bw 242-5-5//0b 677/ Bg 90-2-2	LCWZ	White Long	6.0
Bw 267-3	1981	Ld 125/ Bw 248-1	LCWZ	White long	6.5
Bw 351	1986	Bg 90-2/ Bg401-1	LCWZ	Red Long	6.0
Ld 355	1994	Bw451/ IR50	SP	White samba	6.0

Ld 356	1994	Bw 451/ Bw 351	LCWZ	Red samba	6.0
Bw 361	2002	IR 36/ Bw267-3-11M	I&DZ	Red long	11.0
Bw 363	2004	-do-	GC	White long	11.0
Bw 364	2006	Bw 400/Ob2552//Bg357	LCWZ	Red Long	7.0
Ld 365	2008	selection from Ld 355	LCWZ	Red Samba	7.0

3 months age

Bw 272-6b	1981	Bw 259-3/Bw 242-5-5	LCWZ	Red samba	5.0
Bw 302	1987	Bw 259-3/Bw 242-5-5	LCWZ	White Long	5.0

*LCWZ - Low Country Wet Zone, WZ - Wet Zone, SP - Southern Province, GC - General cultivation, I&DZ - Intermediate and Dry Zone, BS - Bog soils

Red pericarped Bw 361 and white pericarped Bw 363 released in year 2002 and 2004, respectively showed significant improvement in plant architecture which is similar to the “New Plant Type” proposed by IRRI (IRRI, 1989). Furthermore, these two varieties in the 3½ months age group are tolerant to iron toxicity and their potential yield is about 10-11 t/ha. However, dense plant canopy of these two varieties resulted to create a humid micro environment favorable for diseases, particularly endogenous pathogens like Sheath Blight (*Rhizoctonia solanai*) and Stem Rot (*Sclerotium oryzae*). Hence, Bw 361 was recommended for the Intermediate and Dry Zones and Bw 363, white pericarped variety was recommended to grow across the country.

White pericarped Bg 359 and red pericarped Bw 364 released for cultivation in the WZ in 1999 and 2006,

respectively showed comparatively a better plant architecture for the LCWZ. Bw 364 with comparatively a higher tolerance to iron toxicity, showed better performances in sandy soils and Bg 359 performed well in other soil types in the region. Consumer preference for cooked rice of Bw 364 is higher than that of Bg 359. However, moderate shattering of Bw 364 is a drawback. Ld 365, 3½ months red samba variety and Ld 408, 4 months (110 days) red intermediate bold grain variety developed at ARS, Labuduwa were recently released by the DOA in year 2008 and 2010, respectively to be grown across the country, particularly in the WZ.

In the WZ, long age varieties are grown in the low lying areas of coastal and inland flood plains. The only variety which is available and recommended for cultivation in the Yala season in these flood plains particularly in the

coastal flood plains is the traditional variety 'Devareddiri' which takes about 5½ months to mature during Yala season due to its moderate photoperiod sensitivity. Farmers' yield of Deveraddiri is as low as 1-1.3 t/ha (Jayawardena and Peiris, 1980) so that it is rarely cultivated at present. So far no improved variety has been developed to replace Deveraddiri so that about 6000 hectares of coastal flood plains is fallowed at present. However, several photoperiod sensitive long aged (5-6 month) improved high yielding rice varieties namely Bg 3-5 in 1973 and Bg 407, Bg 745 and Bg 38 in 1981 have been released for cultivation in the Maha season in these flood plains. As these varieties are photoperiod sensitive they can be field established well ahead of the Maha rains so that by the time of flooding they are grown enough to escape floods but still in the vegetative phase due to their photoperiod sensitive nature. Grain type of Bg 3-5 and Bg 407 is intermediate bold and of Bg 745 and Bg 38 is samba. As there are no appropriate facilities to conduct a breeding program for photoperiod sensitive rice varieties at the RRRDC, Bombuwela, the responsibility of developing long aged varieties is still with the Rice Research and Development Institute, Batalagoda.

Other rice varieties cultivated in the Wet Zone and their suitability

2½ - 3 month age varieties

Bg 250, variety in the 2½ month age group is cultivated only in few hectares. This variety is suitable for cultivation if the main crop is failed or after flood is receded in the flood prone areas without keeping the field fallow as it fits into a short growing period.

Bg 300 is the most popular variety even in the WZ under 3 months age group. Wide adaptability of Bg 300 makes it suitable for the WZ. However, the variety shows susceptibility for gall midge bio type-2 and sheath mite especially in the Yala season. Among other varieties, At 307 performs well and is expected to become popular in the WZ. At 308 a white samba variety also has a potential to spread in the WZ. Though all these varieties perform well in the potential soils in the WZ, they are not adaptable to iron toxic rice soils. Hence, there is an urgent need to have suitable varieties of this age group to cater to problem soils in the LCWZ.

3½ month age varieties

Bg 357, Bg 352, Bg 359, Bg 94-1 and At 362 (Intermediate bold grain varieties) and, Bg 358 and Bg 360 (samba grain varieties) are grown widely in the WZ. Bg 358 shows severe neck blast disease when grown under WZ conditions. However, except Bg 359 all the other varieties are not suitable for iron toxic rice soils.

4 - 4½ month age varieties

Bg 379-2 and Bg 400-1 white pericarped intermediate bold grain varieties in the 4 month age group are commonly grown in the WZ. Bg 450 a samba variety is also grown in the potential soils of the WZ.

Popular rice varieties in the Wet Zone

The most popular 10 rice varieties during the period from 2007 to 2009 in the WZ are presented in Table 4. Interestingly, out of the ten popular varieties in the WZ eight were Bg varieties while other two were one Ld and one At variety. However, no Bw variety was included in the list. One of the reasons for this may be that the limitation in seed supply of Bw varieties. At the same time farmers may not be aware of the varieties suitable for problems exist in their rice fields.

This is something that needs attention and investigation.

Nevertheless the most popular variety and the next popular variety in the WZ were Ld 356 and Bg 379-2, respectively in 2007 and 2008 but in 2009 Bg 379-2 became the most popular variety and Ld 356 became the next popular variety probably due to increased neck blast incidences with Ld 356. The varieties, At 362, Bg 359, Bg 360 and Bg 450 gained a steady increase in popularity from 2007 to 2009 probably due to either their wide adaptability to WZ conditions or their high grain quality. However, Bg 360 is highly susceptible to submerged and iron toxic conditions. Bg 400-1 is not in the seed production program in the Department of Agriculture at present and Bg 357, Bg 352 and Bg 94-1 are susceptible to sheath blight and Iron toxicity though high yielding and popular in the WZ.

Table 4: Spread of most popular 10 rice varieties as indicated by the area cultivated (ha) in 2007, 2008 and 2009 in the Wet Zone.

Rice Variety	Area Cultivated (ha)		
	2007	2008	2009
Ld 356	19913	23254	17381
At 362	5888	13072	16947
Bg 94-1	11358	14775	7720
Bg 400-1	4569	12557	9886
Bg 379-2	16579	21426	17886
Bg 450	6151	8860	10034
Bg 352	10056	9802	9743
Bg 357	5911	10801	9819

Bg 359	8787	12182	17190
Bg 360	5935	8123	11524

Increase in rice productivity as influenced by the variety in the Wet Zone

Yield potential is the maximum yield that a variety can reach under the best environment whereas potential yield is the maximum yield that can be realized by a variety $\times$ environment combination. Most of the time environment restricts the variety yield potential so that what is practically achievable is the potential yield of the variety $\times$ environment combination which changes from environment to environment (Emitiyagoda *et al.*, 2010). Potential yield of a given environment with present day varieties is a kind of an average yield and Abeyesiriwardena (2000) reported average potential rice yields for ten major rice growing

environments including the WZ with an average potential yield of 6 t/ha.

Micro catchment wise development strategies are needed to enhance the productivity of rice cultivation in the LCWZ. As mentioned earlier, identification of landscape and the soil type in the Yaya and select the best varieties and associated technologies for specific environments is advisable. When the appropriate present day varieties are used in sandy, organic and mineral soils, three specific variety $\times$ environment combinations can be identified with the present yield level, expected yield level within next five years and the potential yield with present day varieties (Table 5). Thereby present average yield of 3.3 t/ha in the LCWZ may be increased up to 4.4 t/ha within next five years.

Table 5: Present yield, potential yield and yield expected within next five years in sandy, organic and mineral soils in the LCWZ.

Variety $\times$ Environment Combination	Average Yield (t/ha)		Average Potential Yield (t/ha)
	Present	Expected within next five years	
Iron toxic soils with present day varieties	2.0	3.0	5
Organic soils with present day varieties	2.5	3.5	6

Mineral soils with present day varieties	3.5	5.0	7
*Average	2.8	4.0	6

*Averages are weighted averages by expected land area cultivated

Suitable rice varieties

By considering all the climatic and soil constrains, proposed rice varieties for the LCWZ are presented in Table 6. As the rice grown in sandy soils are always prone to iron toxicity, varieties having the high level of tolerance to iron toxicity with the ability of giving comparatively higher yields at low levels of added fertilizer are proposed for sandy soils. Varieties for the mineral soils have to be selected depending on the iron concentration of the soil. If the iron concentration is high, farmers should be advised not to use the varieties such as Ld 365, At 307 and At 308. Most of the varieties

do not show any iron toxic symptoms when grown in organic soils though the iron concentration in organic soils is as high as 300 ppm. Hence, iron toxicity susceptible variety Bw 272-6b performs well in organic soils. As Bw 272-6b belongs to 3 months age group with red, short round grains and comparatively better eating and cooking qualities the demand for the variety in the LCWZ is high. In addition, this is the only known improved variety with a very high nutritional quality as it has high protein (>10%) (Padmananda *et al.*, 2007), high iron (2.5 mg/100g) and high zinc (2.5 mg/100g) (Herath *et al.*, 2007).

Table 6: Presently available rice varieties suitable for different soil types in the Wet Zone.

Soil Type	Varieties*
Sandy	Bw 267-3 (IB,W), Bw 364 (IB,R), Bg 359 (IB,W)
Organic	Bw 272-6b (SR, R), Ld 365 (SR, R), Bg 38 (SR, W), Bg 359 (IB, W), Bw 361 (IB, R), Bg 379-2 (IB, W), Bw 363 (IB, W), Bg 300 (IB, W)
Mineral	Bw 361 (IB,R), Bw 363 (IB, W), Ld 408 (IB, W), Ld 365 (SR, R), Bg 450 (SR, W), Bg 379-2 (SR, W), Bg 359 (IB, W), Bg 358 (SR, W), Bg 300 (IB, W), At 307 (IB, W), At 362 (IB, R)

* IB – Intermediate bold, SR – Short round, R – Red, W – White

Proposed rice plant type for the Wet Zone

First major breakthrough in raising the potential yield of rice was made with the development of semi-dwarf plant type with high tillering and erect dark green leaves. This plant type was developed through the incorporation of a recessive gene called *sd1* for semi-dwarfism from a Chinese variety, Dee-geo-woo-gen into Peta a vigorous variety from Malaysia (IRRI, 1967). The germplasm of this genetic background was shared by the rice breeders in Sri Lanka and they were able to achieve the present level of potential yield of rice in Sri Lanka. Further refining of plant architecture that would be suitable for the LCWZ will be an approach to increase the potential yield of rice varieties grown in the LCWZ. Proposed plant type for the LCWZ should possess the following characteristics.

Reduced tillering and large panicles

The varieties with large number of tillers have comparatively more number of secondary and tertiary tillers (Kim and Vergara, 1990) that result in comparatively longer duration of flowering and shorter panicles with more unproductive tillers. Thus, reduction in the number of tillers could direct more nutrients to grain production and increase the number of spikelets while reducing the number of unfilled grains per panicle. It will also facilitate to achieve uniform flowering

and maturity, uniform panicle size and efficient use of horizontal space. Furthermore, it discourages the dense canopy which creates humid micro-environment favorable for fungal diseases (Mew, 1991). Stem Borer incidence is high in high tillering varieties. As the genetic resistance for stem borer is very limited, low tillering varieties will enhance the field level resistance for such pests. Inheritance of low tillering is controlled by a single dominant gene with a pleiotropic effect on the culm length and thickness and the panicle size (Ise, 1992).

High density grains

In rice, usually high density grains are found at the top of the panicle (superior spikelet positions) in the primary branches and comparatively less density grains are found in the lower portion of the panicle (inferior spikelet position) particularly on the secondary branches. Hence, having comparatively more high density grains on a panicle is desirable as 30% increase in grain yield has been estimated if all the spikelets in a variety produce high density grains (Venkateswarlu *et al.*, 1986). Moreover, high density grains also result in high milling recovery and head grain yield.

Canopy and leaf characteristics

The rice varieties with vertically oriented leaves have comparatively better carbon assimilation rate as both sides of the leaves are exposed to light in contrast to droopy or horizontally

oriented leaves. However, the droopy leaves at seedling stage up to 3-4 weeks after planting could control weeds, which is the biggest problem in the WZ rice cultivation. The best combination is to have droopy leaves at seedling stage and erect leaves thereafter. The varieties having leaves with 'V' shaped leaf blades are reported (Longpin, 1997) to have comparatively more CO₂ concentration near the leaf surface which in turn helps to increase photosynthetic efficiency. As the leaf thickness is positively correlated with leaf photosynthetic rate, the thick leaves are thought to be a desirable trait.

Crop growth duration

The varieties with growth duration of 130-140 days could give higher yields than varieties of shorter growth duration under certain environmental and cultural conditions (Vergara, 1970). The optimal growth duration for maximum yield with present day varieties in the WZ has been observed to be 3-3½ months under broadcast sown conditions. However, the constraints such as flooding and drought in the WZ demand 5-6 months varieties that can be established well ahead to escape from flooding or very short duration varieties such as 2½ months that can fit into the short cropping period after the flood recedes and escape from drought at the end of the season. In addition, water situation of some environments is such that only 4-4½ month varieties can have the best fit into those environments. Hence, development of varieties with maturity

duration ranging from 75 days to 180 days is necessary for the WZ.

Plant height and stem thickness

Short stature reduces the susceptibility for lodging which is a priority at present and also increases the harvest index (Tsunoda, 1962). Short culms require less maintenance respiration and contribute to an improvement in photosynthesis respiration balance. However, the short stature encourages the fungal disease. The varieties having culm length less than 60 cm show comparatively more susceptibility to Sheath Blight (*Rhizoctonia solanai*) and Stem Rot (*Sclerotium oryzae*) particularly in the WZ. Thus, thick sturdy culms of more than 65 cm tall would be ideal for such conditions.

Biomass production and harvest index

Though the dry matter content is negatively correlated with the harvest index, high dry matter production is required for high grain yield. However, increase in biomass production results in lodging, mutual leaf shading and increase in disease incidences and decrease in grain yield. Hence, effort should be made to increase the biomass with comparatively stronger and slightly taller culms and erect leaves while maintaining a harvest index around 0.50.

Pre-harvest sprouting

Germination of seeds while on the parent plant before harvesting is referred to as pre-harvest sprouting. Pre-harvest sprouting is a problem in some areas in some seasons in the WZ leading to considerable crop losses. During wet weather or when the crop lodges into standing water at maturity grains begin to germinate on the parent plant. Thus, tolerant to pre-harvest sprouting is important in the WZ. Abeyisiriwardena *et al.* (1995) reported variety differences for pre-harvest sprouting and an improved method to screen rice varieties for tolerance to pre-harvest sprouting.

Root system

In general, above ground plant characteristics are paid much attention with respect to grain yield of rice. However, in the WZ most of the problems that limit grain yield of rice are associated with soils so that poor root growth of the rice plant in the WZ has been observed. In addition, varieties giving high yields in the WZ generally possess a good root system. Thus, adequate attention must be paid on the improvement of the root system of the rice plant for it to be well adapted to the WZ conditions. Our observation is that the root system of the rice plant in the WZ should be large, deep and thick with a rapid renewal rate and high oxidizing power. In addition, root/shoot ratio should be kept as high as possible. However, studies are needed

to confirm this and to further identify useful root characteristics of the rice plant for the WZ.

Potential yield

The yield of the new improved high yielding varieties when grown in the best WZ environment with high management can even go up to 9.38 t/ha while the potential yield of such varieties under the same conditions in the Dry Zone is 11.73 t/ha (Emitiyagoda *et al.*, 2010). This is really encouraging for the rice improvement program in the WZ. Hence, in order to achieve sustainable increase in yield at farmer level, varieties with high potential yield and adaptable to WZ conditions have to be further developed. For this purpose general breeding objectives should be targeted to improve the potential yield of improved varieties with specific adaptability to specific environments.

Variety adaptability and stability over diverse environments

Highly adaptable and stable rice varieties are important to reduce the gap between potential and realized yields at farmer level. Identification of widely adaptable varieties across all specific environments or specific varieties with high level of general adaptability within specific environments is important to increase productivity across all environments. Use of heterosis in improving variety adaptability and

stability may be useful (IRRI, 2005). Efficient and effective biometrical methods specifically developed to test variety adaptability and stability in rice are available (Abeyesiriwardena *et al.*, 1991; Abeyesiriwardena, 2001). At the same time seed production programs for the varieties released for specific environments should be paid adequate attention to make their seed available to farmers in time with the required quantities.

Tolerance to abiotic stresses

Iron toxicity

Iron toxicity which is a physiological disorder of rice commonly known as ‘Bronzing’ is very common in the WZ. This is mainly due to high supply of iron into rice soils from upwelling of ground water and lateral seepage from adjacent uplands. The rice varieties that can tolerate high iron concentrations at cellular level though no restriction in iron absorption at root level or the varieties those resist absorption of iron from the soils at the root level can survive in such conditions. The susceptible varieties grown under such conditions show stunted growth with reddish brown leaves and produce discolored unfilled grains with poor yields.

Varying levels of iron toxicity have been reported in traditional and improved varieties (Bentota and Weerasingha, 2005). The rice variety Bw 267-3 was the best improved variety available for iron toxic rice soils until early 2000s.

Bw 361, Bw 363 and Bw 364 (Bentota *et al.*, 2010) are the recently released improved rice varieties with tolerance to iron toxicity.

Salinity

Tolerance to salinity is another important aspect to be considered in the rice improvement program for the LCWZ as intrusion of seawater is experienced in low lying rice lands along the coastal belt. Salinity becomes a problem during the dry period at the tail end of the season. Tolerance to coastal salinity should not be considered in isolation mainly in the Yala season as coastal salinity is mostly associated with submergence in Yala so that varieties tolerant to salinity should also have tolerance to submergence or should be able to escape from floods. This is also the case in some of the flood prone areas in Maha season.

Submergence

Most of the low lying areas in the coastal and inland flood plains undergo submergence during peak monsoonal rains so that rice varieties adaptable to those areas must either possess submergence tolerance or be able to escape from submergence. In addition, most of the submergence prone areas in the coastal flood plains are also prone to salinity during the dry period due to intrusion of sea water so that varieties adaptable to such areas must possess both salinity and submergence tolerance, which has not yet been successfully achieved by rice scientists anywhere in the world.

Incorporation of submergence tolerant gene (sub 1 A, a dominant QTL in chromosome 8) which allows the rice plant to undergo submergence maximally for 14 days, into rice is one way to overcome the problem of submergence (Ismail *et al.*, 2006). However, fitting ultra short age varieties (2½ month varieties) into the short growing period after the flood is receded is one of the latest approaches proposed to be adopted to escape from floods successfully with the development of the ultra short age variety, Bg 250 (Abey Siriwardena, 2005). Adaptability of Bg 250 in coastal flood plains may be tremendously improved if salinity tolerance is incorporated to Bg 250.

Tolerance to biotic stresses

Diseases

Among the diseases, fungal disease such as Leaf Blast, Neck Blast, Sheath Blight and Stem Rot are common in the LCWZ. Most of the new improved varieties have adequate resistance to Leaf Blast as it is one of the breeding objectives from the inception of rice breeding program in the country. Similarly varieties having resistance to Leaf Blast have the resistance for Neck Blast. Ld 356 is highly susceptible to Neck Blast and the variety was declared for limited cultivation in the WZ. During the selection process the breeders tend to select the varieties having field resistance to Sheath Blight and Stem Rot. Bw 363, Bw 364 and Bg 359 are such varieties recommended for the LCWZ. Bacterial Leaf Blight

(BLB) is the only bacterial disease that influences rice production in the country. This disease is reported mostly in the Dry Zone but in the recent past the disease is also frequently reported in the WZ. Hence any variety released for the LCWZ should possess some level of resistance to BLB.

Pests

Breeding for Brown Plant Hopper (BPH) and Gall Midge (Bio type 1 and 2) resistance has been highly successful and most of the varieties grown in the country have sufficient levels of tolerance to those pests. Stem Borer, Leaf Folder, Sheath Mites, Rice Bug and Thrips are the other frequently reported pests at present. Germplasm screening for Thrips resistance revealed that the traditional varieties Dahanala and Kahatawee possess sufficient resistance to thrips (Kudagama, 1977). Incorporation of thrips resistance to improved plant type is continued at RRRDC, Bombuwela. However, no germplasm resistant to Stem Borer, Leaf Folder, Rice Bug and Sheath Mites has so far been identified.

Grain quality

Vivid preference to grain size, shape, pericarp color and appearance of the milled rice by the consumers in the WZ demand a wide range of rice varieties. All the rice varieties released to date are expected to possess acceptable milling, eating, cooking and keeping qualities.

As the yield levels are generally low in the LCWZ, development of varieties with high eating quality such as Basmathi and Suduru Samba and high nutritional quality i.e. high proteins, iron, zinc, vitamins and fiber, is immense importance to fetch a high price to make the rice cultivation profitable in the LCWZ. Amylose content (AC), gelatinization temperature (GT) and gel consistency (GC) are considered to be the most important physico-chemical properties closely related to cooking and eating qualities of rice. In addition, locally developed and introduced varieties with long, slender, aromatic and soft grains are available. Sufficient variation for GT and GC are found in the traditional and improved rice varieties in Sri Lanka though the variation is low for AC content (Abeysekara *et al.*, 2008a). Almost all the tested varieties were with high amylose grains except the

traditional variety - Suduru Samba with an intermediate AC. Abeysekara *et al.* (2008b) reported that these traits did not show any difference between two prominent seasons, Yala and Maha.

The Physico-chemical properties such as protein and fat contents in rice grains were studied by Padmananda *et al.* (2007) and reported that protein and fat contents varied from 9.6 - 12.1% and 2.3 - 3.9%, respectively among varieties. They also found that the pericarp color whether red or white has no influence on nutritional quality of rice. Iron and zinc contents in the rice grain were found to vary from 1.7-3.1 mg/100g and 2.1-3.3 mg/100g respectively (Herath *et al.*, 2007). Thus genetic variability for the nutritional quality of the rice germplasm exists to be used in the WZ rice improvement program.

Conclusions

Potential for rice production in the WZ is high while the importance of rice production in the WZ cannot be under estimated with respect to national food security in the country. There are clearly identified climatic, edaphic, socio-economic and other constraints to further increase in rice production in the WZ. Rice varieties suitable for cultivation in the different environments within the WZ is available to enhance the productivity of rice to the expected level of 4.0 t/ha within next five years in the WZ. The rice plant type that may further overcome the constraints of rice production in the WZ could be defined so that the future rice improvement program should be focused on this to bridge the existing gap between potential yield and the realized yield at farmer level in the WZ.

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Potentials and Limitations of Hybrid Rice Production in Sri Lanka

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Abstract

Rice Research and Development Institute (RRDI) initiated hybrid rice research and development program in 1980 to develop high yielding (15-20% yield advantage over the best inbred) hybrid rice varieties with good grain quality characteristics and management packages for hybrid rice seed production and hybrid rice cultivation. During the last 30 years, especially in the last 10 years, RRDI has gained significant achievements in hybrid rice research and development. Four Cytoplasmic Male Sterile (CMS) lines (BgCMS1A, BgCMS2A, BgCMS3A and BgCMS4A) with desirable characteristics, five Restorer (R) lines (Bg1R, Bg2R, Bg3R, Bg4R and Bg5R) which are pre-requisites for the development of hybrid varieties, and two hybrids; Bg 407H with a yield potential of 13 t/ha (released in 2005) and BgHR8 (to be released) utilizing those CMS and R lines have been developed. Cultural practices such as nursery seed rates, spacing in transplanting, fertilizer application for cultivation of hybrids and management practices for F1 seed production have also been developed. The hybrid rice technology has been transferred to farmer level. The average seed yield F1 is around 0.5 t/ha which is comparatively low mainly due to poor out crossing rate of CMS lines that have been used to develop hybrids. In 2009, a new approach "Saruketha hybrid rice yaya program" was initiated aiming self-seed production of hybrids by farmers and achieved satisfactory progress.

Inadequacy of scientific staff, non-availability of suitable separate lands for seed production, lack of heterotic genetic materials are the major problems that need attention immediately to achieve the expected benefits from the program. Low out crossing rate of currently available CMS lines need to be addressed and good restores are also needed to be developed. Private sector involvement may also be obtained as it is expected to play a significant role in this effort and thus current policies on seed imports and other relevant matters need to be changed accordingly.

Key Words: Hybrid rice, CMS lines, Restorer lines, F1 seed production, Out crossing rate

Introduction

Among the many avenues available, the most effective and economical way available to increase rice production at present seems to be the development and use of hybrid varieties. Through

hybrid rice technology, the yield potential can be increased by 1.0-1.5 t/ha beyond the limits of semi dwarf high-yielding varieties (HYVs). This technology has been successfully developed and commercialized in China since 1976. At present, China

with more than 50% of its total rice area under hybrid rice cultivation, has successfully demonstrated the application of hybrid rice technology to increase production, income and farm employment. China, following the success of three-line hybrid rice in the 1970s successfully commercialized two-line hybrid rice technology in 1995. Two-line hybrid rice has 5-10% higher yield advantage than the three-line hybrid rice. Hybrid rice has also been successful in other countries like Viet Nam and India. With the high rate of adoption of hybrid rice cultivation, Viet Nam has become the second largest rice-exporting country in Asia. Furthermore, many other countries, such as the Philippines, Bangladesh, Indonesia, Pakistan, Ecuador, Guinea and the United States of America have also made great progress in expanding hybrid rice cultivation.

Sri Lanka needs to increase the present level of rice production of 3.65 million tones up to 4.56 million tones to meet the demand in 2020. During the last two decades, the rice sector has undergone unprecedented challenges. Diminishing profitability due to increasing cost of production, low and stagnating yields leading to low returns to investment, declining rice area because of abandoning or converting rice lands for other purposes and declining manpower for rice are some of them (Silva, 2010). Since expansion of rice lands is not possible, only possibility is to increase the productivity of rice for which hybrid rice technology has been identified as one of the possible options to overcome

the problem to increase the rice yield in the country (Abey Siriwardena and Sandanayaka, 2000). Rice Research and Development Institute initiated the hybrid rice research program in Sri Lanka in 1980. During the last 30 years RRDI has gained significant achievements in the area of hybrid rice research and development. This paper highlights the past attempts, achievements and failures and the problems to be addressed for the successful hybrid rice production program in Sri Lanka.

Stages in Hybrid Rice Research and Development Activities in Sri Lanka

Early phase: RRDI commenced its hybrid rice research program at RRDI (then Central Rice Breeding Station) in early 1980s. Unavailability of exploitable genetic material and insufficient resources hindered the progress of this program. Then hybrid rice program was reinitiated in 1994 with the objectives to develop high quality Sri Lankan rice hybrids and hybrid seed production procedures including management practices for Sri Lankan hybrids.

Chinese project phase: In 1996, RRDI and Zchuan University of P.R. China collaborated with the hybrid rice research program in Sri Lanka to find out the suitability of Chinese hybrid rice genetic materials under local conditions. After 4 years of testing, it was revealed that Chinese genetic material cannot be directly introduced to Sri Lanka.

IRRI project phase: IRRI supported the hybrid rice research program of RRDI through the IRRI-ADB project on “Development and use of hybrid rice in Asia” during 1998-2002 and “Sustaining food security in Asia through the development and dissemination of hybrid rice technology” during 2002-2005 strengthening the existing program. Several CMS lines and restores were identified or locally developed. During the second phase of the above project several experimental hybrids were developed and small scale F1 seed production program was initiated (Abeysekera *et al.*, 2005). In 2005, first Sri Lankan hybrid rice variety, Bg 407H was released and its large scale seed production was initiated.

FAO project phase: In 2007, RRDI received funds from FAO through the project on “Strengthening National Capacity for Hybrid Rice Development and Use for Food Security and Poverty Alleviation” under Technical Corporation Program (TCP) with the objectives of 1) transferring the hybrid rice technologies to farmers for rapid increase in national rice production, 2) strengthening national capacity in hybrid rice development, 3) strengthening hybrid rice seed production and 4) formulation of a medium term national program for the development and use of hybrid rice in the country.

Progress

Development of CMS lines

Three line system has been identified as the feasible system for the development and seed production of hybrid rice in Sri Lanka. In this system, CMS (A) lines, and their maintainer lines (B) and restorer lines (R) are important. Development of CMS lines with desirable characteristics and high out crossing ability is the most important step in hybrid rice development because hybrid rice industry primarily depends on the F1 yield. Studies conducted with 53 IRRI, Indian and RRDI CMS lines at RRDI during Yala 2001 and Maha 2001/2002 showed that panicle exertion rate and spikelet opening angle and duration had significant influence on out crossing rate under local conditions (Abeysekera *et al.*, 2003). Based on these results, two CMS lines (PMS 11A and Khrishna from India) and six CMS lines (IR 68281A, IR 68897A, IR 69623A, IR69616A, IR 69625A and IR 68895A from IRRI) have been identified as adaptable and stable under local conditions. Some of these good CMS lines were used to develop local CMS lines. During Maha 2001/2002, several Indian and IRRI and two locally developed CMS lines were evaluated at RRDI. The CMS line Krishna from India, two CMS lines; BgCMS1 and BgCMS2 developed in Sri Lanka and twenty eight CMS lines from IRRI have been identified as stable for both pollen and spikelet sterility and adaptable under Sri Lankan conditions. Two

CMS lines; BgCMS1A and BgCMS2A sterility with 25-27% out crossing rate developed at RRDI have shown 100% (Table 1).

Table 1: Characteristic features of two CMS lines developed at RRDI.

CMS line	Days to 50% flowering (days)	Plant height (cm)	Panicle number	Panicle length (cm)	Panicle exertion (%)	Total spikelets/ panicle	Out crossing rate	Grain type
Bg CMS 1	-	-	-	-	100.0	-	25.0	Long grain
Bg CMS 2	76	90.0	13.6	22.4	100.0	203.2	27.0	Small grain

-Data not available

Development of CMS lines with higher out crossing rates

One of the drawbacks of the CMS lines developed by RRDI and available at present is the low out crossing rate (20%). Under FAO/TCP project, CMS line with comparatively higher out crossing rates (30-40%) was developed utilizing TeA(335) and 75AYLD 12 A in 2009.

Development of maintainer lines

Initially several selected local varieties/ lines were identified as maintainer lines for some of the adaptable CMS lines and were backcrossed to their CMS sources. The most promising restores developed were Bg 99-1757, Bg 98-665, Bg 98-1684, Bg 91-2864, Bg 92-979, Bg 98-3278, Bg 99-700, At 95-2-1, (IR 68281B/Bg 1528), (IR 68275B/ At 354), (Ob 2552/Bg12-1/Bg 379-1)

and (IR 69628B/ Bg 94-4516). CMS lines are being developed in the genetic background of these maintainers.

Development of restorer lines

Development of restorer lines is one of the important steps in a hybrid rice development program. Many high yielding locally developed elite lines have been tested for restoring ability along with some restorers from IRRI and Bangladesh Rice Research Institute. The identified restorers (IR 34686- 179-1-2IR, IR 50360-121-3-3-3R, IR 29732-143-3-2-1-R, IR 60819-34-2R, IR 65575-47-2-1-9R, IR 62653-8-3-3R, IR 40750- 82-2-2-3R, BR 168-2B-23R and H4) are maintained and multiplied for producing hybrids. Recently, five restorer lines, Bg1R, Bg2R, Bg3R, Bg4R, and Bg5R have been developed. Bg1R was used to develop 1st rice hybrid variety Bg 407H and Bg4R was used to develop new early maturing hybrid variety BgHR8. Bg3R is a good “samba” type

restorer with desirable characteristics and therefore, it has been included in the National Coordinated Rice Varietal Testing program for testing its suitability throughout the country.

Development of hybrids

The following three hybrids, BgHR1 (long slender grains with red pericarp), BgHR6 and BgHR12 (long slender grains with white pericarp) were developed and yield trials were conducted in target areas to test their adaptability over diverse environments (Abeysekera *et al.*, 2005).

Two hybrids, BgHR12 and BgHR6 have recorded a significant yield advantage of 2 t/ha and 1.6 t/ha, respectively over popular inbred variety Bg 403 in Maha 2003/04 in large scale demonstration

trials at RRDI (Abeysekera and Abey Siriwardena, 2002). Growth duration of BgHR1 was comparatively longer (> 4 month) than the other two hybrids and seed production was difficult as restore (H4) was mildly photosensitive variety. Pollen fertility of CMS line (IR 250000A) used to develop BgHR12 was not stable and therefore, maintaining purity of the F1 seed was a problem. Therefore, these two hybrids were kept aside for future improvement and BgHR6 was selected as a suitable experimental hybrid. It was submitted for testing in National Coordinated Varietal Testing program. It was released as Bg 407H in 2005. This is the 1st hybrid rice variety developed and released in Sri Lanka. Further, early maturing (3½ month) hybrid variety, BgHR8, was developed and it is now in testing stage (Table 2).

Table 2: Performance of Bg HR8 during Yala 2009, Maha 2009/10 and Yala 2010 at RRDI, Batalagoda.

Variety	Yala 2009		Maha 2009/10		Yala 2010	
	Maturity duration (days)	Grain yield (t/ha)	Maturity duration (days)	Grain yield (t/ha)	Maturity duration (days)	Grain yield (t/ha)
Bg 300	104	3.6	98	2.6	103	4.4
Bg 357	117	3.9	108	3.6	115	5.3
Bg 379/2	-	-	125	3.3	124	4.5
Bg 407H	133	4.5	123	3.9	129	4.9
BgHR8	112	4.5	106	4.4	112	6.0

Source: Jayawardena and Abeysekera (2009)

Grain quality is one of the key factors that determine the consumer acceptance of hybrids. Therefore, three hybrids; BgHR1, BgHR6 (Bg 407H) and BgHR12 developed at RRDI were evaluated for milling recovery, eating quality and other grain quality characteristics during Maha 2000/01. This revealed that total milled rice percentage and other characteristics are not different from that of inbred rice. BgHR6 was the best in terms of appearance having low translucent grains and high head grain recovery (Satraj and Abeysekera, 2001).

Performance of elite Indian rice hybrids

Eight hybrids from India and three locally developed experimental hybrids (BgHR1, BgHR6, BgHR12) were tested with three standard popular inbred varieties for grain yield and other characteristics during Maha

2002/03 and Yala 2003 seasons in the Dry Zone at the Field Crop Research and Development Institute (FCRDI) and in the Intermediate Zone at RRDI. Significant grain yield differences among varieties were observed. The Indian hybrid PHB71 (DP) showed better and consistent performance than Sri Lankan hybrids. It is early maturing (93-95 days) compared to Sri Lanka hybrids (115-125 days). All other varieties were inferior in all aspects except the earliness. 30006 (Ad) matured in 85-87 days (Abeysekera *et al.*, 2004).

In Yala 2010, another set of Indian rice hybrids was tested at RRDI and results revealed that none of them was better than Sri Lankan hybrids except their earliness in some varieties. All the Indian varieties were infected by stem borer and as a result grain yields were poor (Table 3).

Table 3: Performance of Indian rice hybrids at RRDI in Yala 2010.

Variety	Maturity Duration (days)	Grain Yield (t/ha)
CRH 501	107	1.1
CRH 502	103	3.8
PAC 801	102	4.5
PAC 835	96	4.0
PAC832	113	1.6
Swarna(1)	117	4.0
Swarna(2)	110	3.6
Bg 357 (Inbred check)	106	3.2
Bg HR8 (Hybrid Check)	100	5.4
Bg 407H (Hybrid check)	128	3.5

Development of management practices for hybrids

Success in commercial hybrid rice varieties primarily depends on their high yield potential. Yan (1988) observed that growth pattern and nutrient requirement of hybrid rice differ considerably from the inbred rice varieties and therefore, to exploit this advantage of hybrid rice, it is necessary to adopt appropriate crop management practices. Most of the studies on optimizing hybrid rice yield were conducted with temperate hybrid in China and those strategies cannot be applied for tropical countries (Peng *et al.*, 1988). Therefore, development of appropriate management practices to suit local conditions is very important. Extensive studies on aspects such as nursery management, transplanting spacing, establishment methods, nutrient management etc. on the management of locally developed hybrid had been conducted. To date, a broad agronomic package for the cultivation of hybrid rice has been developed. However, there is a need to fine-tune the package for each hybrid combination.

Crop establishment

Experiments conducted using three RRDI hybrid rice varieties (BgHR1, BgHR6 and BgHR12) revealed that seedlings raised with 20-30g of seeds/m² in the nursery performed comparatively better and produced higher grain yield (Jayawardena *et al.*, 2004). Experiments conducted with BgHR6 and BgHR12 revealed

that comparatively higher grain yield could be obtained by transplanting two seedlings per hill at wider spacing (20x25cm and 25x25cm) when compared to that of narrower spacing (15x15cm and 15x20cm) recommended for inbred varieties (Jayawardena and Abeysekera, 2002).

In hybrid rice cultivation, transplanting is important to reduce the seed cost, as hybrid seeds are more expensive than that of inbred rice seeds. However, present day Sri Lankan farmers are reluctant to transplant the crop due to requirement of additional labor for transplanting. Studies conducted at RRDI and farmer fields revealed that seedling broadcasting (Parachute transplanting) could be adopted as an alternative method for manual transplanting. This would help to cut down the labor cost required for manual transplanting and seed requirement by around 50 percent (Jayawardena *et al.*, 2004). At present this method is being practiced by the farmers in many parts of the country. Jayasinghe *et al.* (2007) indicated that this method has a potential for adopting in foundation seed production programs in the government seed farms. Seedlings raised in the parachute nursery could be thrown into standing water up to 2.5 cm depth to control the weeds without affecting the yield (Hasalaka, 2006). Dharmesena *et al.* (2005) reported that insecticide used to control the thrips could be minimized using the seedling broadcasting method. Recent studies conducted by HARTI on farmer adoption of seedling broadcasting revealed that this method has a yield

advantage of more than 25% over the conventional methods (Renuka, 2009). However, high prices of seeding trays and difficulty in access to seedling trays are two problems associated with this system.

Studies conducted for developing direct sown hybrid rice cultivation technology at RRDI during 2007-2009 revealed that direct seeding at low seed rate (50-75 kg/ha) could be possible in hybrid rice cultivation compared to direct seeding of inbred rice with 100 kg/ha (Jayawardena and Abeysekera, 2009). Kahadawala (2010) also reported that the possibility of adopting low seed rate (15-30 kg/ha) for direct sown Bg 407 H in the Wet Zone in combination with straw mulch application. Another study conducted with inbred and hybrid rice on weed emergence and growth patterns found that higher weed biomass in early stages in HR plots over the inbred rice plots and suggested that the importance of control of weeds in early stages in hybrid rice cultivation (Abeysekera, *et al.*, 2001).

Nutrient management

In a nitrogen fertilizer response study conducted with an inbred and RRDI hybrids revealed that response to nitrogen of inbred and hybrid is more or less similar and maximum yield response observed up to 180 N kg/ha. The maximum grain yield obtained at 180 N kg/ha level was 6.8 and 6.4 t/ha in hybrid and inbred, respectively (Jayawardena and Abeysekera, 2002). Fertilizer recommendation made

in 2005 for Bg 407H was more or less similar to inbred rice except the application of additional dose of N at flowering (DOA, 2005). In another experiment it was revealed that application of Zinc and Sulfur is required to obtain comparatively higher grain yield from Bg 407H and application of Magnesium is not required in Low Humic Gley soils in the Low Country Intermediate Zone (Bandara *et al.*, 2009).

Seed production

Commercial viability of hybrid rice industry depends on the seed yield, economics and efficiency of seed production in CMS lines and F1 seed yield. Hybrid rice seed yield is influenced by many factors such as climatic factors, synchronization of parental lines, out crossing rate of CMS lines and special attributes such as panicle exertion rate of CMS lines etc. Seed production ability of different CMS lines in different locations was studied and found that minor season to be the best season and Intermediate Zone (Girandurukotte) as the best location for F1 seed production (Kiriwaththuduwa *et al.*, 2004). Other than those factors, row transplanting of A and R lines to a specific row ratio, application of GA3, leaf clipping and supplementary pollination are very important activities to obtain high F1 seed yield. Results of the experiments carried out at RRDI have revealed that row ratio of 3R: 4-6 A is better than the IRRI recommended row ratio of 3R:8-10A (Abeysekera *et al.*, 2009). Application of GA3 is

essential for getting high F1 yield and the rate of application depending on the type of CMS lines and the environmental condition prevailing in the seed production site. Application of GA3 at the rate of 120 g/ha has found to be the best level under Sri Lankan conditions (Kiriwaththuduwa *et al.*, 2004). However, the cost of GA3 is comparatively higher than that of other cost components in F1 seed production (Silva, 2010) and therefore it should be applied accurately to get the maximum benefit.

Large scale seed production

In a large scale study conducted by RRDI at different locations found that hybrid rice has shown around 21.6 percent (12.5 - 30.7%) yield increase over the inbred varieties. The cost of cultivation of hybrid rice estimated to be Rs. 65,900/ha (659 USD/ha) and this is little higher than that of the inbred rice cultivation (Rs. 62,962/ha or 630 USD/ha). Under transplanted condition F1 seed yield was around 0.65 t/ha and the cost of cultivation estimated as Rs. 95,230/ha (630 US\$/ha) which is 1.7 times higher than that of inbred rice cultivation. Poor synchronization of parental lines was the major factor for low seed yield. The current cost of production of F1 seed is about Rs. 147 per kilogram and this can be substantially brought down by increasing the F1 seed yield through the better synchronization of flowering of parental lines and by adopting labor saving techniques in crop establishment (Jayawardena *et al.*, 2006). In a study conducted by Silva (2010) indicated

that the cost of production has increased from Rs. 120 to Rs. 225 per kilogram at RRDI and Rs. 333 per kilogram in farmers field. Even with these prices, hybrid rice cultivation is still profitable. Small scale seed production program was started in farmer's fields in Devehuwa and Rajanganaya in Yala 2008. The results were encouraging and therefore, during Yala 2008 and 2009, a pilot scale program namely 'Hybrid Rice Saruketha Yaya program' was started in collaboration with the extension and communication center of the DOA under the FAO/TCP project. The objective of the program was to introduce self-seed production program to supply the F1 seed for the area. The program was conducted in 25 locations at Polonnaruwa, Rajanganaya, Devahuwa, System B and System C. Farmers were able to produce reasonable F1 yield (around 100-200 kg/ha), except in Systems B and C. In Maha 2009/10, a new site of F1 seed production program was started at RARDC, Makandura and a reasonable yield of around 250 kg/ha was obtained.

Other activities

Combining Ability Analysis and DNA Analysis - Studies on estimating General Combining Ability (GCA) which determines the additive genetic effects and Specific Combining Ability (SCA) for non-additive genetic effects were initiated. Restorers for additive effects for grain yield, number of spikelets, and length of panicles have been identified. DNA studies have indicated that some hybrids have a

higher heterosis level than parental lines. These methods cut down the time required to evaluate parental lines with the conventional out-grow tests.

Training - Under the FAO/TCP project, training programs for officers of the DOA and farmers who were willing to engage in hybrid rice seed production and cultivation were held at RRDI. Farmers were given seed materials and other necessary materials such as GA3 and polythene to separate the seed crop from other rice plots. RRDI officers periodically visited the farmers to give the necessary training and instructions. This experience indicated us that F1 seed production of rice hybrids is feasible; however, realizable F1 yield by farmers was comparatively poor due to poor synchronization and inadequacy of artificial flowering. Some farmers have sold their seeds at Rs. 400-500 per kilogram. Establishment of hybrid rice demonstration plot at the School of Agriculture in Sri Lanka and inclusion of hybrid rice technology as a subject matter in the curriculum of Diploma in Agriculture are also planned. IRRI published hybrid rice seed production manual was translated into Sinhala and Tamil languages. A video CD on hybrid rice seed production was prepared for the training purpose.

Problems Related to Hybrid Rice Research and Development

There are many problems related to hybrid rice research and development and the most important problems are given below:

Technical;

1. Limited grain yield heterosis is in experimental hybrids. This could be due to narrow genetic diversity among the available CMS sources and Restorers. Priority should be given to develop CMS lines with high out crossing rate.
2. Low F1 seed yield on CMS lines. This is mainly due to the low out crossing rate of available CMS lines that are used to produce hybrids and poor management practices that are adopted. Current CMS management practice to produce F1 seeds need to be refined and new practices have to be developed with new varieties.
3. Unavailability of isolated location for hybrid rice research program, especially for breeding.
4. Unavailability of rapid quality testing program for F1 seeds that are produced by research as well as by farmers.

Development;

1. Hybrid rice research program is not a priority program of the DOA. DOA must give priority for the hybrid variety development program in all crops. The least priority has been given from the inception of the program by the department. From the inception, hybrid rice program has been handled by one officer and currently another officer has been allocated. This staff availability is very much

less than the requirement. At least three qualified officers are needed, two officers on breeding and one on seed production. Both inbred and hybrid rice research programs should go parallel and then only both programs could make progress.

2. Inadequacy of qualified scientific staff for hybrid rice research program. Need additional qualified staff and resources to independently conduct its activities.
3. Lack of seed certification standards for hybrid rice.
4. Poor participation of private sector in hybrid rice seed industry. Favorable government policies and interventions are required for the establishment of a sustainable seed production program with the available physical and other resources. Hybrid seed production could be made at government seed production farms, provided that special allocation is made for the purpose.
5. Hybrid rice research program is carried out at RRDI, Batalagoda side by side with conventional breeding program where thousands

of inbred lines are handled and also surrounded by farmer's rice field. Thus, maintaining genetic purity of breeding material becomes very difficult even with adoption of remedial measures. This could be overcome by establishing a separate hybrid rice research center/institute in an isolated location.

6. F1 seed production is currently handled by the RRDI which is extremely difficult. Seed production could be undertaken by the government seed farms or some other party. However, seed farms are running on advance account. Since, HR seed production is not a profitable venture, under the current situation it is difficult to handle the HR seed production program in seed farms unless special funds and manpower and other facilities are provided. Private sector is also not interested in HR seed production because they are reluctant to invest due to inadequacy of government policies. Favorable policies such as testing of imported and local hybrids should be adopted. Seed production of selected hybrids should be done in Sri Lanka for which appropriate incentives should be given by the government.

Conclusions

During the last 16 years, RRDI has gained significant achievements in the hybrid rice research and development sector. RRDI has been able to develop local CMS lines, maintainer lines and restores that are the most important mile stones in a hybrid rice development program. Other than line development, development of two hybrid varieties Bg 407H (already recommended) and Bg HR8 (to be

recommended) are successful achievements. In addition to variety development, HR seed production has been perfected and seed production technology has been passed to farmers. The current HR research and development program need to be strengthened further. It is very important to setup a separate institute for the development of hybrid rice technology in the country. Inadequacy of scientific staff and non-availability of suitable separate land for seed production are two major problems that need to be paid due attention immediately. Low out crossing rate of currently available CMS lines need to be improved and also good restorers should be developed. Hybrid rice research and development program in Sri Lanka should be continued and expanded as a remarkable progress has been made to date.

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Identification of Candidate Genes for Salt Tolerance in Rice

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Abstract

Salinity is becoming an increasingly important abiotic stress factor constraining rice production. Several factors, including sea level rise and increased evapotranspiration rates due to global warming, contribute to expand both coastal and inland salinity in rice lands. Rice is one of the most sensitive cereal crops to salinity, with growth and yield being reduced at ECe levels above 1.9 dS m⁻¹, which is much lower than the threshold of 4 dS m⁻¹ for most agricultural crops. Development of salt tolerant varieties is one of the main options to maintain yields in salinity-prone rice lands. Breeding of salt-tolerant rice varieties requires identification of candidate genes which contribute to salt tolerance, which is a complex trait determined by the interplay between a large number of genes, variation in the growing environment and timing, intensity and duration of salt stress. The two-phase growth response to salt stress forms the basis for identification of mechanisms of salt-tolerance, which in turn provides the basis for exploration of candidate genes. Phase I of salt stress is an osmotic stress brought about by the reduced uptake of water by plant roots due to excess salts in the external soil solution. Phase II is due to salt ion toxicity in photosynthesizing leaves due to gradual accumulation of salts. Hence, traits of salt tolerance can be broadly categorized into three major groups as those contributing to, (a) osmotic stress tolerance, (b) prevention of salt ion build-up in leaves by salt-exclusion and (c) protection of tissues from excess concentrations of salt ions. While it has been possible to develop salt-tolerant rice varieties using conventional breeding techniques, molecular biological methods that have emerged during the last decade have the potential to identify candidate genes and incorporate them more rapidly and effectively into the existing germplasm having other desirable traits. This paper describes the early results of a study carried out at the University of Peradeniya to identify candidate genes conferring salt-tolerance in the salt-tolerant Sri Lankan rice variety At 354. The paper first describes the two-phase growth response to salinity and reviews the effects of salinity on growth, physiology and yield formation of rice. It then provides an overview of the molecular biological techniques available for studying gene expression leading to identification of up- and down-regulated genes under salt stress. Finally, the putative candidate genes identified from At 354 during Phase I of salt stress are presented and their possible role in conferring salt tolerance to At 354 is discussed.

Introduction

Rice is one of the most sensitive crops to salinity (Maas and Hoffman, 1977; Munns and Tester, 2008), which is an abiotic stress factor of

increasing importance during the last two decades (Flowers and Yeo, 1995). With the probable increase of terrestrial evapotranspiration rates and sea level rise due to global warming, the arable land area affected by salinity

is likely to increase during this century (Wassmann *et al.*, 2004; IPCC, 2007). Salinity or salt stress, in relation to agricultural crops, is defined as the presence of an excessive amount of soluble salts to an extent that affects the physiological functions responsible for normal crop growth and yield formation. As growth of most plant species are affected at a threshold E_{Ce} (i.e. electrical conductivity of the saturated paste extract) of 4 dS m⁻¹, saline soils are defined as those with an E_{Ce} of more than 4 dS m⁻¹ (USSL, 2005). As sodium chloride is the most abundant and soluble salt that is released from weathering of rocks (Szabolcs, 1989), salinity is quantified in terms of the concentration of NaCl in the soil or nutrient solution. The threshold E_{Ce} of 4 dS m⁻¹ is equivalent to 40 mM NaCl.

In Sri Lanka, as in many other countries, salinity is a major environmental problem, causing crop yields to drop and land to become unfarmable. Two types of salinity are found in Sri Lanka. Inland salinity occurs in seasonally dry areas due to high evaporation rates, which bring salts in deeper layers of the soil profile to the surface, and due to irrigation, using water with dissolved salts. Coastal salinity occurs in areas near the sea mainly through intrusion of underground sea water and deposition of salts carried inland by the sea spray. Apart from these processes, in Sri Lanka, the Tsunami disaster in 2004 resulted in a temporary build-up of salinity in about 4500 ha of agricultural lands.

Reclamation of salinity-affected lands by drainage of excess salts is not possible on an extensive scale. Therefore, improving the salt tolerance in crops that are cultivated in salinity-affected or salinity-prone lands is the only viable strategy to face the challenge of spreading salinity. While development of salt-tolerant crop varieties is possible through conventional plant breeding strategies (Yeo and Flowers, 1986; Shannon and Noble, 1990; Shannon, 1997; Abeyasiriwardena, 2000; Gregorio *et al.*, 2002; Schubert *et al.*, 2009), molecular-based plant breeding strategies offer a more focused, efficient and effective approach to develop salt-tolerant varieties (Flowers and Yeo, 1995; Flowers *et al.*, 2000; Borsani *et al.*, 2003; Rabbani *et al.*, 2003; Yamaguchi and Blumwald, 2005; Blumwald and Grover, 2006; Ashraf, 2009; Ashraf and Akram, 2009; Cuartero *et al.*, 2009). Molecular-based breeding strategies could be especially useful in rice, where the entire genome has been sequenced. Such strategies require identification of physiological mechanisms that confer salt tolerance and the putative genes that are responsible for such mechanisms of tolerance (Yeo *et al.*, 1990; Hasegawa *et al.*, 2000). Determination of gene expression patterns under salt stress in identified salt-tolerant varieties is one approach to identify putative genes of salt tolerance (Cushman and Bohnert, 2000; Kawasaki *et al.*, 2001). Altering the expression of identified putative genes of salt tolerance has been shown to improve tolerance (Apse *et al.*, 1999; Fukuda *et al.*, 1999; Zhang *et*

al., 1999). The present paper discusses the molecular and genomic approaches to increasing salt tolerance of rice and describes the early results of the application of the above approach to identify candidate genes for salt tolerance in a Sri Lankan rice variety, At 354, which is known as salt-tolerant.

Effects of Salinity on Growth and Yield Formation of Rice

Being a glycophyte, rice is highly sensitive to salinity (Yeo *et al.*, 1990; Aslam *et al.*, 1993; Munns and Tester, 2008). Although the general ECe threshold for defining salinity for agricultural crops is 4 dS m⁻¹ (USSL, 2005), it has been shown that growth and yield formation of rice are adversely affected at much lower levels of salinity. Mass *et al.* (1999) and Hanson *et al.* (1999) had shown that for each 1 dS m⁻¹ increase above a threshold of 3 dS m⁻¹ in salinity, rice yields decrease by 12%. However, Grattan *et al.* (2002) showed through a series of field and plant house experiments that rice yields begin to decline from a salinity threshold as low as 1.9 dS m⁻¹, with the rate of decline being 9.1% per increase of salinity by 1 dS m⁻¹.

The effects of salinity on rice may vary depending on the stage of its life cycle at which it is exposed to salinity. Several studies indicate that rice is tolerant to salinity during germination but young seedlings are sensitive (Moormann and Breeman, 1978; Flowers and Yeo., 1981; Aisha *et al.*, 2005). Damage to the plant during transplanting increases

its sensitivity to salinity (Moormann and Breeman, 1978). Salt tolerance increases during tillering, but the plant again becomes sensitive during flowering (Moormann and Breeman, 1978). Thus, susceptibility to salt stress is most severe at the seedling and reproductive phases, eventually leading to significant reductions of grain yield and yield components. Zeng *et al.* (2001) identified the stage from the initiation of the third leaf to panicle initiation as the most sensitive to salinity in terms of seed yield. However, the spikelet number per panicle was most sensitive to salinity between panicle initiation and booting (Zeng *et al.*, 2001). The stage/s at which rice is sensitive to salinity has shown varietal variation as well. Some varieties were highly sensitive to salinity at the germination and early vegetative stages but were tolerant at the reproductive stage while others showed tolerance at germination but sensitivity at the subsequent stages (Heenan *et al.*, 1988). Moradi and Ismail (2007) concluded that tolerance of rice to salinity at the seedling stage is only weakly correlated with tolerance at the reproductive stage.

Increasing salinity significantly reduces the number of surviving rice plants (Grattan *et al.*, 2002; Zeng and Shannon, 2002), the number of tillers per hill (Grattan *et al.*, 2002) and leaf area index (Zeng *et al.*, 2003) indicating the high sensitivity of the vegetative stage of rice to salinity. Significant varietal variation has been shown in the magnitude of reductions of tiller and spikelet numbers (Zeng

et al., 2001 & 2002). Salinity reduces both shoot (Zeng *et al.*, 2001) and root dry weights, with the shoot growth being more sensitive than root growth (Grattan *et al.*, 2002).

Salinity decreases the number of fertile tillers per hill, fertile florets per panicle and pollen viability (Khatun and Flower, 1995). Yield reductions of plants grown under salt stress but treated with fertile pollen from non-stressed plants have shown that salinity affects the stigma as well (Khatun and Flower, 1995), probably reducing stigma receptivity (Khatun *et al.*, 1995). Although salinity reduces the seed weight per panicle (Zeng *et al.*, 2001), the individual seed weight may remain unaffected (Grattan *et al.*, 2002). Salinity also decreased the panicle length and the number of primary branches in the panicle and yield reductions due to salinity were substantially greater than reductions in growth (Khatun *et al.*, 1995). They further concluded that overall yield reductions due to salinity are dominated by the effects on panicle development, stigma and grain-filling than by the effects on pollen.

Several physiological parameters of rice are also influenced by salinity. It reduces the net photosynthetic rate, stomatal conductance and transpiration rate with relatively tolerant cultivars showing lower reductions and partial recovery after a period of acclimation while sensitive cultivars showing continuous decreases (Dionisio-Sese and Tobita, 2000; Moradi and Ismail, 2007). Chlorophyll fluorescence,

measured in terms of quantum yield of photosystem II (Fv/Fm), does not seem to be affected by salinity in rice (Dionisio-Sese and Tobita, 2000; Moradi and Ismail, 2007) although it has been shown to decrease significantly with salinity in other cereals such as sorghum (Netondo *et al.*, 2004). However, Moradi and Ismail (2007) observed that non-photochemical quenching increased (i.e. greater dissipation of absorbed radiation energy as chlorophyll fluorescence and heat) and electron transport rate decreased under salt stress.

Two-Phase Growth Response to Salinity

Development of salt-tolerant rice varieties requires a clear understanding of the physiological responses of different rice varieties to salt stress. The two-phase growth response of plants to salinity as developed by Munns (1993 & 2002), forms the physiological basis of plants' response to salt stress. The two-phase model postulates that plants respond in two sequential phases to increasing salinity. The first phase of salt stress development (Phase I) occurs immediately upon exposure to a salt ion concentration above a threshold ECE of 4 dS m⁻¹ (1.9 dS m⁻¹ for rice), which is equivalent to 40 mM NaCl (19 mM NaCl for rice), for most plant species (Munns and Tester, 2008). During Phase I, excess salt ions in the external soil solution decreases its osmotic potential thereby decreasing the ability of plant roots to absorb water. In this phase of 'osmotic stress',

plants show the typical responses to drought stress by decreasing shoot and root growth rates, with the shoot growth being more sensitive than root growth (Munns and Tester, 2008). Reduced expansion growth occurs primarily due to hormonal signals, mainly abscisic acid (ABA) (Munns and Cramer, 1996), originating in roots and translocating to the shoot (Munns, 2002; De Costa *et al.*, 2007). Usually, salt ion concentrations (principally Na⁺ and Cl⁻) in meristematic tissues of the shoot do not increase to levels which would inhibit their growth during Phase I (Fricke, 2004; Munns, 2005). This is primarily because the rapidly-expanding meristematic cells can compartmentalize the excess salts that reach them into the expanding vacuole (Intra-cellular compartmentalization). However, with increasing salt concentrations in the external soil solution and/or with increasing duration spent in a saline soil, salt ion concentrations in the meristematic tissues increase beyond the capacity of cells to compartmentalize the excess in to the vacuole. As a result, the plant enters the second phase of salt stress development (Phase II) in which salt concentration increases in the cytoplasm, where all the important physiological processes take place, to toxic levels to cause gradual injury and death of tissue. During this second phase of salt-specific ion toxicity (i.e. 'ionic stress'), older tissues (e.g. old leaves), which are no longer expanding their cells and therefore are not able to dilute the salts that are coming in, are affected first.

Osmotic stress during Phase I develops rapidly after a plant is exposed to salinity. In contrast, development of ionic stress in Phase II is a slow process. The duration of Phase I can vary depending on the evaporative demand of the atmosphere, which determines the rate at which salt ions build-up in the plant tissue via the transpiration stream and the degree of sensitivity of the plant to salt ion toxicity. In reality, the two phases of salt stress development may not be mutually-exclusive with ion toxicity taking effect during Phase I (Sümer *et al.*, 2004) and osmotic effects being in operation during Phase II. Inter-variatal variation in salt tolerance has been shown in both Phase I (De Costa *et al.*, 2007) and Phase II (Yeo *et al.*, 1990; Zeng *et al.*, 2002 & 2003).

Mechanisms of Salt Tolerance

The two-phase model forms the basis for identification of mechanisms that are responsible for salt tolerance. Accordingly, three major mechanisms can be identified. Salt tolerance during Phase I can be achieved primarily by tolerance to osmotic stress (Mechanism No. 1), which primarily involves maintenance of shoot and root growth even under salt stress. Hence, a tolerant genotype may show a lower growth reduction than a sensitive genotype. Tolerance during Phase II may be achieved by either salt exclusion from photosynthesizing leaves (Mechanism No. 2) and/or by increasing the tissue tolerance to excess salt (Mechanism No. 3). As sodium chloride is the

most abundant and soluble salt that is released from weathering of rocks (Szabolcs, 1989), salt-stressed plants have to develop tolerance mechanisms primarily against Na^+ and Cl^- ions. Out of these two ions also, Na^+ toxicity often develops earlier than Cl^- toxicity (Munns and Tester, 2008). Consequently, the majority of research on tolerance to salt ion toxicity has focused on mechanisms of exclusion of and tissue tolerance to Na^+ .

Osmotic tolerance involves mechanisms such as osmotic adjustment, modification of long-distance signalling involving ABA (Munns and Cramer, 1996; De Costa *et al.*, 2007) and possibly other growth regulators (Dodd, 2005) and decreased inhibition of cell expansion and tillering (Bartels and Sunkar, 2005; Munns, 2002; Zhu, 2002; Munns and Tester, 2008). Sodium exclusion involves reducing the absorption (i.e. exclusion at the root cortex) and long-distance transport of Na^+ (i.e. exclusion at the root xylem parenchyma) to the photosynthesizing leaves and its key organelles such as the chloroplast (Schubert and Läuchli, 1990; Munns and Tester, 2008). Increasing selective uptake of K^+ over Na^+ also contributes to Na -exclusion (Koyama *et al.*, 2001). Tissue tolerance to excess Na^+ involves several mechanisms such as compartmentalization of excess Na^+ in the leaf vacuoles, thereby preventing toxic salt concentrations in the cytoplasm, delaying the development of ion toxicity in chloroplasts, delaying of premature senescence of old leaves and accumulation of high

concentrations of compatible solutes in the cytoplasm.

Importance of Elucidating Salt Responsive Genes which Confer Salt Tolerance

The expression of numerous genes is regulated by salt stress. By studying which genes are regulated by salt stress and how they are regulated, the function of various gene products in conferring salt tolerance can be elucidated. A general assumption has been that stress up-regulated genes may be important for salt tolerance. Genes that are up-regulated by salt stress belong to several groups based on their possible function. These specific genes could encode for proteins implicated in Na^+ sequestration (H^+ -ATPase, NHX -type transporters), in synthesis of compatible osmolytes (proline, glycine betaine, polyols), in the detoxification of toxic compounds (ROS scavenging enzymes), in signal perception and regulation factors and other unknown functions (Munns, 2005; Munns and Tester, 2008).

Salt tolerance in rice is known to be a complex character which is controlled by a large number of genes (Gregorio *et al.*, 2002). The above mechanisms of salt tolerance are brought about by the expression of genes in different tissues, cells, organelles and membranes. Therefore, identification of candidate genes that are responsible for various salt tolerance mechanisms and quantifying their expression would be extremely useful for development of salt tolerant rice varieties. Molecular

techniques that can be used in gene expression studies are described below.

Molecular Techniques Used in Gene Expression Studies

With advances of Molecular Biology, especially during the last two decades, techniques have been developed to monitor expression of a single gene at a time or monitor the expression of an array of genes simultaneously. Broadly, the methods available to study the expression of genes can be divided as hybridization-based and PCR-based techniques. Expression of a gene requires the transcription of a given DNA segment into mRNA which is then used to synthesize a protein. Hence, gene expression is quantified by measuring the amount of mRNA produced. Hybridization techniques, based on either radioactively- or non-radioactively-labelled methods, can be used to detect the timing of gene expression and the tissue in which a given gene is expressed. Hybridization also enables quantification of the amount of mRNA produced and thereby quantify the degree of expression of a given gene. Northern-, slot- and dot blotting and in situ hybridization for RNA detection are some examples of frequently used hybridization techniques to study the expression patterns of individual genes.

In addition to hybridization techniques, reporter gene constructs have also been used to monitor the level, timing and cell/tissue specificity of expression of a gene. Semi-quantitative and quantitative reverse transcription and

PCR and nuclease protection assay are some other molecular biological methods used for the analysis of expression of single, specific genes. To overcome practical limitations of the methods by which a limited number of genes are studied at a time, subtractive hybridization, sequencing of expressed sequence tags, serial analysis of gene expression, hybridization to microarrays, cDNA-AFLP technique, differential display reverse transcription PCR (DDRT-PCR) and differential screening of cDNA libraries can be used to investigate the expression of thousands of genes at a time, on a genome-wide scale (Moody, 2001). Brief descriptions of these techniques are given below.

cDNA Microarray hybridization analysis

A complex mixture of labelled polynucleotide samples (such as cDNA derived from mRNA) are tested against sequences from thousands of different genes (e.g. unique products of the polymerase chain reaction (PCR) generated from cDNA libraries or genomic DNA), fixed on small solid supports (usually glass microscope slides). Depending on their sequences, the samples will hybridize with different spots in the array, which is analyzed by specialized image software (Schena *et al.*, 1995). cDNA microarrays were first developed for *Arabidopsis thaliana* (Schena *et al.*, 1995). However, microarrays have already been used to study gene expression of many plant physiological

processes such as circadian rhythms, plant defence mechanisms, oxidative stress responses, fruit ripening, phytochrome signalling, seed development and nitrate assimilation (Aharoni and Vorst, 2002). Moreover, Wulschleger and Difazio (2003) have highlighted the use of this technique in profiling the expression of genes of barely under water deficit, Chinese cabbage under temperature stress and aluminium-tolerant genes in wheat. This technique is widely used to identify salt-stress responsive genes in many plants including rice. For example Kawasaki *et al.* (2001) examined transcript regulation in response to high salinity in the salt-tolerant rice cultivar Pokkali with a microarray employing 1,728 cDNAs from different libraries of salt stressed tissues. They have found that about 10% of the transcripts in Pokkali were significantly up or down-regulated within 1 hour of salt stress. Also, Chao *et al.* (2005) used the cDNA-array method to compare the salt-tolerant line Nona Bokra and the salt-sensitive line IR29. They showed earlier and stronger expression of genes related to transcription factors and signal transduction under salt treatment. Walia *et al.* (2005) used the Affymetrix rice genome array containing 55,515 probe sets to explore the transcriptome of a salt tolerant and a salt-sensitive rice genotype under salt-stressed and unstressed conditions during the vegetative stage.

cDNA-AFLP technique

Screening for differentially expressed genes is one of the most straightforward approaches to reveal the molecular basis of a biological system. Wang *et al.* (2009) has highlighted that, as a differential screening method cDNA-amplified fragment length polymorphism (cDNA-AFLP) is more stringent and reproducible than many other methods because it can amplify low-abundance transcripts. Moreover, the cDNA-AFLP technique is a robust, high-throughput, genome-wide expression tool for gene discovery, where prior knowledge of sequences is not required. In this method, RNA isolated from different organisms or the organism subjected to different treatment conditions is converted into double stranded cDNA and then digested with two restriction enzymes, a frequent-cutter and a rare-cutter. Synthetic adapters are ligated to the cDNA ends and primers complementary to the adapter sequences (plus small extensions of 1, 2, or 3 nucleotides) are used to amplify fragments with asymmetric ends. These fragments are displayed on sequencing gels and a comparison will be done either between the different organisms or the organism exposed to different conditions. Specific fragments can be eluted from gels and sequenced to identify genes with differential expression (Bachem *et al.*, 1996). cDNA-AFLP have also been applied to identify differentially expressed genes in cold-tolerant and cold-sensitive alfalfa genotypes and rice (Ivashuta *et al.*, 1999).

Differential Display Reverse Transcription PCR

This is a fingerprinting technique to identify and compare mRNAs expressed during different cell processes. The requirements of this protocol are: (i) reproducibility; (ii) comparison of all mRNA species in the cell populations of interest, and (iii) the ability to isolate the corresponding cDNA (Sturtevant, 2000). The principle of the DDRT-PCR technique is to amplify specific sub-populations of mRNA, using reverse transcriptase and PCR to produce a population of PCR fragments of different lengths. To do this, mRNA is reverse transcribed in subsets using 12 anchored oligo(dT) primers that recognize different fractions of the total mRNA population. The resulting cDNA is then amplified with the same anchored oligo (dT) primer and a short arbitrary primer. Labelled dATP is introduced in the reaction and the labelled products are separated on a DNA sequencing gel and visualized by autoradiography. The DDRT-PCR technique has also been used to isolate several cDNAs induced in plants under various stress conditions such as salinity (Muramoto *et al.*, 1999) and heat shock (Visioli *et al.*, 1997). As detailed by Sturtevant (2000), several methods such as RNA fingerprinting by AP-PCR (RAP-PCR), DDRT-PCR with selected primers (SPR), targeted display and Ordered Differential Display (READS), which have been derived from DDRT-PCR are also available to assess the differential expression of genes.

Differential screening of cDNA libraries

Differential screening of cDNA libraries allows the determination of transcript profiles and patterns or identification of differentially-regulated genes in organisms under two contrasting conditions (e.g. an organism subjected to salt stress and not subjected to the stress, an organism inoculated with a pathogen and not inoculated with the pathogen etc.). By differential screening of plasmid or phage cDNA libraries, differentially expressed cDNAs under two contrasting conditions can be isolated. For this technique to be effective, two samples of mRNAs derived from two tissues or two plants (i.e. subjected to the two different treatments) should differ and the mRNA which is not common in both samples will be identified through differential screening using the following steps:

- i. mRNA isolated from tissues treated with a particular treatment (i.e. salt stress, temperature stress) is used for synthesis of cDNA using reverse transcriptase and a cDNA library will be constructed;
- ii. cDNA obtained from both samples of mRNA (i.e. treated and non-treated tissues) are labelled and used for sequential probing of the cDNA library;
- iii. Clones that are more strongly hybridized with the cDNA probe from one mRNA sample than the other contain genes that are

differentially expressed (i.e. up- or down-regulated) in response to the given treatment in the sample.

This technique has been used by Xiong *et al.* (2001) for identification of defence-related genes in rice. Iuchi *et al.* (2000) also used this technique to identify dehydration responsive genes from cowpea. In the present study, we used this technique to identify differentially expressed (up and down-regulated) genes responsible for salt tolerance in the rice variety At 354, which has been identified as being salt tolerant.

Case Study: Detection of Candidate Genes for Salt Tolerance from the Sri Lankan Rice Variety At 354

Most of the previous work to identify candidate genes for salt tolerance has been based on plants which had been subjected to a sudden salt shock, which would drastically alter plant water relations and reduce cell volume causing sudden disruption of a large number of physiological processes (Munns, 2005). In such a situation, it would be difficult to identify gene transcripts which are specifically induced by salt stress. Therefore, as proposed by Munns (2005), the present study was conducted on rice plants which were subjected to gradual salt stress so that putative candidate genes for salt tolerance during Phase I and Phase II of stress development could be identified separately.

Pokkali and Nona Bokra, which had originated in the coastal zones of Southern India, are the rice varieties which have been used most in physiological and molecular studies to understand mechanisms of salt tolerance (Moons *et al.*, 1995; Kawasaki *et al.* 2001; Munns *et al.*, 2006). However, both these varieties have a tall plant type and are low-yielding. The relatively salt-tolerant Sri Lankan rice variety At 354 has been bred using Pokkali and the semi-dwarf new-improved Sri Lankan variety Bg 94-1 as parents. Accordingly, At 354 was used in the present study to identify candidate genes for salt tolerance in the Sri Lankan rice germplasm. A parallel study has been conducted to screen about 100 traditional, old - and new-improved varieties from the Sri Lankan rice germplasm to identify further sources of salt tolerance.

Identification of Salt Stress Responsive Genes in Shoot Tissues of At 354 at Phase I of Salt Stress

This work was carried out at the University of Peradeniya, Sri Lanka. A cDNA library was constructed using mRNA extracted from shoots of the rice variety At 354, grown hydroponically in Yoshida nutrient solution (Yoshida, 1972) under a salt stress of 100 mM NaCl (imposed in four incremental steps of 25 mM each at two-day intervals) and harvested at Phase I (i.e. 24 hours after increasing salinity up to 100 mM). In total, 2892 cDNA clones from the above-constructed cDNA library were blotted on to nylon

membranes, fixed by baking at 80°C for 2 hours and subjected to differential hybridization screening using total cDNA probes prepared from the cDNA of salt stressed (100 mM NaCl salt stress) and non-stressed (1 mM NaCl) plants. cDNA clones carrying up-or down-regulated genes were detected by the chemiluminescence method based on relative signal intensity (Plates 1 and 2). A clone which is up-regulated in response to salt stress showed a higher signal intensity when hybridized with the probe prepared from salt-stressed tissues (i.e. salt probe) in comparison to the signal intensity with the 'control probe' prepared from non-stressed plants (Plate 1). Clones which were down-regulated showed high signal intensity with control probe relative to the salt probe (Plate 2).

Differential hybridization results of the present study identified fifteen cDNA clones as salt-responsive, out of which ten were up-regulated

and the rest were down-regulated. Thereafter, the plasmid DNA harboured in the selected 15 cDNA clones were isolated by the alkaline lysis method. The cDNA fragments of these differentially expressed clones were amplified by using M13 universal primers. The amplified cDNA regions were subjected to DNA sequencing at GeneTech, Colombo, Sri Lanka in order to detect putative genes encoded by the cDNA fragments contained in the selected cDNA clones. To this end, homology search was done for the DNA sequences of nine cDNA fragments which could be sequenced successfully, using available DNA and protein databases by BLAST and FASTA analysis programmes.

Table 1 summarizes the putative genes and their possible functions of ten cDNA clones which were differentially-expressed in the Sri Lankan rice variety At 354 in response to salt stress during Phase I.

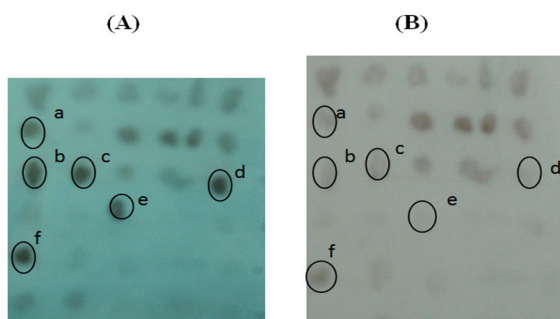


Plate 1: Identified up-regulated cDNA clones from salt-stressed At 354. Signals developed by salt-treated cDNA probe (A) and control cDNA probe (B) when selected cDNA clones were subjected to differential hybridization. Six differentially hybridized clones corresponding to Clone Nos. (a) 30 (b) 32 (c) 147 (d) 194 (e) 928 (f) 1271 are indicated by circles.

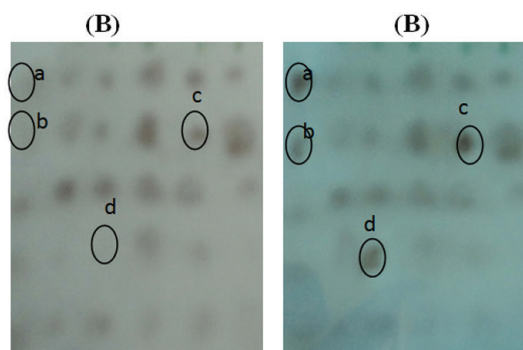


Plate 2: Identified down-regulated cDNA clones from salt-stressed At 354. Signals developed by salt treated cDNA probe (A) and control cDNA probe (B) when selected cDNA clones were subjected to differential hybridization. Four differentially hybridized clones corresponding to Clone Nos. (a) 230 (b) 2792 (c) 2805 (d) 2855 are indicated by circles.

Table 1: Putative genes and probable proteins produced by the differentially-expressed cDNA clones of variety At 354 at Phase I of salt stress development.

Clone code	Status of differential expression due to salt stress [†]	Homology information	Identity [‡]	Probable function under salt stress
30	UR	Putatively expressed salt-tolerant gene of rice identified previously from <i>Oryza sativa</i> Japonica (variety Nipponbare)	D 75	No information available on physiological function. May be a novel unreported gene
32	UR	Homologous to eukaryotic translation initiation factor eIF5A	D 96	Protection of older leaves from salt ion toxicity; Exclusion of Na ⁺ from younger leaves
194	UR	Homologous to a gene similar to LSTK-1-like kinase of <i>Oryza sativa</i> Japonica (variety Nipponbare)	D 99	Osmotic stress/Drought tolerance
147	UR	Homologous to a gene of <i>Oryza sativa</i> which produces a putative Serine Decarboxylase	D 97	Increased biosynthesis of Glycinebetaine, an osmoprotectant

928	UR	Homologous to a putative gene of <i>Oryza sativa</i> that produces AP domain DRE binding factor	D 100	Osmotic stress/Drought tolerance
1271	UR	Homologous to a putative gene which produces a chromosomal replication initiator protein	D 96	No information available on physiological function. May be a novel unreported gene
230	DR	Homologous to amine oxidases	D 89	Reduced degradation of Polyamines and ensure tolerance of osmotic stress and ion toxicity; Reduced production of H ₂ O ₂ ; Regulation of stomatal movements
2792	DR	Homologous to amine oxidases	P 70	
2855	DR	Homology to agmatine deiminase protein product	P 100	Reduced production of Putrescine to increase osmotic tolerance
2805	DR	Homology to protein product of rap-GTPase-activating protein	P 100	Stomatal closure through down-regulation of Cyclic Nucleotide pathway; Cytoplasmic ion homeostasis

†UR - Up-regulated; ‡DR- Down-regulated.

*D - Percent identity in nucleotide sequence; P - percent identity in amino-acid sequence.

Probable Involvement of the Up - and Down - Regulated Genes in the Salt Tolerance of At 354

Genes up-regulated in At 354 in response to salt stress

Results of homology search on the cDNA Clone No. 30 from salt-stressed At 354 showed 75% identity with a nucleotide sequence (Accession No. AK068953), which has been identified from the Japonica rice variety Nipponbare and is expressed under salt stress. At present, no further information is available in databases on this gene to deduce its possible physiological role in conferring salt-tolerance.

The cDNA fragment of Clone No. 32 from salt-stressed At 354 is homologous to an eukaryotic translation initiation factor eIF5A. This factor, formerly called eIF-4D, is the only cellular protein known to contain the unusual amino acid hypusine (Chou *et al.*, 2004). As reported by Bartig *et al.* (1992), this is a small (18 kDa) abundant protein that is highly conserved in eukaryotes. In plants, eIF5A has been so far isolated from *Arabidopsis thaliana*, alfalfa, tobacco, maize, tomato and rice. Chou *et al.* (2004) found that abiotic stresses such as salt and heavy metal induce the accumulation of OseIF5A in rice cells. While being expressed in rice leaves and panicles, high relative amounts of OseIF5A were observed in old

leaves (Chou *et al.*, 2004). During salt stress development, salt ion toxicity is initially experienced by the older leaves which are unable to dilute incoming salts by expanding. Hence, it can be speculated that OseIF5A expressed in salt-stressed At 354 may play a role in protecting older leaves from salt ion toxicity. Although in the present study, up-regulation of OseIF5A has been observed during Phase I of salinity, where salt toxicity is not expected to be significant, the high salt-sensitivity of rice (Grattan *et al.*, 2002; Munns and Tester, 2008) may mean that salt toxicity has already begun in Phase I. This was confirmed by measurements of tissue Na⁺ concentrations in old and young leaf tissue in our experiment (data not shown). Chou *et al.* (2004) further observed that OseIF5A was spatially regulated during rice leaf development. Therefore, being more abundant in older leaves, OseIF5A may be contributing to Na⁺ exclusion from younger leaves by helping in the transport of excess Na⁺ to the vacuole of older leaves. Alternatively, through its expression in panicles and older leaves, OseIF5A may be contributing to a protective function from salinity. Chou *et al.* (2004) also observed that transcripts of OseIF5A in suspension-cultured cells were substantially reduced upon sugar starvation. Our observation of increased expression of OseIF5A during Phase I of salinity in At 354 agrees with this observation because sugar starvation is unlikely during Phase I of salinity where photosynthetic rate is not substantially reduced (W.A.J.M. De Costa, Personal communication). Thompson *et al.*

(2004), while noting that eIF5A is not a conventional translation initiation factor, states that eIF5A plays an important role in the determination of cell proliferation and death in eukaryotic cells through its multiple isoforms. As expression of eIF5A in rice has been shown to be high in older leaves (Chou *et al.*, 2004), it may also contribute to maintenance of older leaves alive under salt stress by increasing their tissue tolerance to salt ion toxicity.

DNA and protein homology search of the present study showed that cDNA Clone No. 194 from salt-stressed At 354 is homologous to a gene similar to LSTK-1-like kinase of *Oryza sativa* Japonica Nipponbare rice variety. Li *et al.* (2009) found that LSTK-1-like kinase enzyme is expressed in maize plants exposed to drought stress. As drought, induced by osmotic stress, is the major stress that operates during Phase I of salt stress development, up-regulation of clone 194 in At 354 provides evidence that it may be coding for a gene product that is responsible for osmotic stress tolerance.

One effective mechanism to reduce damage from salt stress is the accumulation of high intracellular levels of osmoprotectant compounds (Hasegawa *et al.*, 2000; Rhodes *et al.*, 2002; Munns, 2005). According to our findings, the cDNA Clone No. 147 obtained from salt-stressed At 354 was homologous to a gene of *Oryza sativa* which produces a putative serine decarboxylase. As reported by Blumwald *et al.* (2004), this enzyme

is needed to produce ethanolamine, a precursor of choline (McNeil *et al.*, 2001), which is needed to produce glycinebetaine (Huang *et al.*, 2000), which acts as an osmoprotectant conferring salt tolerance in plants (Blumwald *et al.*, 2004). Hence, the gene products of this cDNA clone could be providing a protective function, primarily from osmotic stress that occurs during Phase I.

The cDNA Clone No. 928 obtained from salt-stressed At 354 gave homology to a putative gene of *Oryza sativa* that produces the AP domain DRE binding factor. The DRE (dehydration responsive elements) related motifs are at promoter regions of many cold and drought inducible genes (Liu *et al.*, 1998; Jaglo *et al.*, 2001; Dubouzet *et al.*, 2003). The over-expression of these regulatory elements, which are transcription factors, has been reported to increase drought (Agarwal *et al.*, 2006; Chen *et al.*, 2008; Kumashiro and Yamaguchi-Shinozaki, 2008), salt and low temperature tolerance (Wang *et al.*, 2008). In a comparison of expression profiles of 36,926 genes or gene models from the shoot, flag leaf and panicle of rice under drought and salt stress, Zhou *et al.* (2007) showed that promoter regions of most of the genes induced by both drought and salinity were enriched with the two transcription factors, DRE (ABA-independent Drought-responsive element) and ABRE (ABA-responsive element). Similarly, Rabbani *et al.* (2003) found these transcription factors in several genes induced in rice

by drought, salinity and cold stresses and ABA treatment. Comparison of expression profiles for several stresses showed that there was considerable overlap between profiles induced by drought, salinity and ABA treatment. Therefore, it is logical that the gene product of this particular cDNA clone is up-regulated in the salt-tolerant rice genotype At 354 during Phase I of salt stress, which is dominated by osmotic stress-induced drought.

The cDNA Clone No. 1271, obtained from salt-stressed At 354, which was up-regulated gave homology to a putative gene which produces a chromosomal replication initiator protein. There have been no reports of the possible involvement of this gene product in stress tolerance in plants.

Genes down-regulated in At 354 in response to salt stress

The protein homology search predicted that cDNA fragments harboured by the Clone Nos. 230 and 2792, which have been identified as down-regulating genes in our study, encode for amine oxidases. Amine oxidases catalyze the oxidative de-amination of polyamines (Cona *et al.*, 2006), which are an important group of compounds regulating plant responses to various stresses such as salinity, drought, low temperature and ozone (Liu *et al.*, 2007). Liu *et al.* (2007) provide evidence that overexpression of polyamine biosynthesis could lead to abiotic stress tolerance via stabilization of negatively-charged

molecules such as proteins and membrane lipids, reduction of free radicals by anti-oxidant properties and improvement of ionic balance. By down-regulating genes encoding for polyamine degradation, gene products of cDNA Clone Nos. 230 and 2792 obtained from salt-stressed At 354 are likely to reduce polyamine degradation during salt stress, thereby allowing the stress tolerance functions of polyamines to operate. This could contribute to the salt tolerance of At 354 in Phase I of salinity. Furthermore, polyamine degradation produces H_2O_2 , which mediates cell death and thereby increase stress sensitivity. Down-regulation of genes that trigger the above process would also contribute to the stress tolerance of At 354.

The cDNA Clone No. 2855 obtained from salt-stressed At 354 was homologous to agmatine deiminase protein product, which is an enzyme in the biosynthesis pathway of Putrescine, an important member of the polyamine group (Gill and Tuteja, 2010). It has already been noted that polyamines can play an important role in stress tolerance as stress messengers in plant responses to different stress signals (Liu *et al.*, 2000 & 2007; Alcázar *et al.*, 2006 & 2010). Interestingly, Zapata *et al.* (2003 & 2004) observed on a range of vegetable crops that salt stress decreased tissue Putrescine (Put) levels while increasing the levels of the other two polyamines, Spermidine (Spd) and Spermine (Spm) so that the ratio (Spd + Spm)/Put was increased. They further observed that salt-tolerance increased with the above concentration changes in

different polyamines. In a comparative study with six crop species differing in salt-tolerance, Zapata *et al.* (2008) showed that Put concentration in shoot was negatively correlated with salt-tolerance of the species, with the most tolerant species (*Beta vulgaris*) having the lowest shoot Put concentration. Similarly, in a screening of nine rice cultivars differing in salt-tolerance, Krishnamurthy and Bhagwat (1989) observed significant increases in Spd and Spm in salt-tolerant cultivars, but no significant alteration in Put when subjected to salt stress. In contrast, the salt-sensitive cultivars showed an excessive accumulation of Put under salt stress but Spd and Spm concentrations remained low. Therefore, the observed down-regulation of the gene product responsible for biosynthesis of Put in the salt-tolerant At 354 under salinity in the present study probably contributed to its salt tolerance. This finding is in agreement with the observations of Zapata *et al.* (2004) and Krishnamurthy and Bhagwat (1989).

Polyamines have been shown to influence stomatal movements through their influence on voltage dependent K^+ channels in the plasma membrane of guard cells. Liu *et al.* (2000) showed that all natural polyamines including Put, Spd, Spm and Cadaverine strongly inhibited stomatal opening. However, interestingly, at a concentration of 1 mM, while Spd and Spm achieved complete inhibition of light-induced stomatal opening, Put achieved only 63% inhibition. Hence, it is clear that out of the different polyamines, Spd and Spm have a greater influence on

stomatal closure than Put. Therefore, the observed down-regulation of the biosynthesis pathway of Put in salt-stressed At 354 is not likely to inhibit stomatal closure, which is one of the earliest responses to salt stress in Phase I, initially and transiently due to reduced turgor (Termaat *et al.*, 1985; Munns, 2002) and subsequently due to salt-stress induced ABA synthesis (Fricke *et al.*, 2004; De Costa *et al.*, 2007; Munns and Tester, 2008).

The cDNA fragment harboured by Clone No. 2805 obtained from salt-stressed At 354 was homologous to a protein product of rap-GTPase-activating protein. This protein is needed to activate the cyclic nucleotide pathway in mammals. Although the presence of cyclic nucleotides (i.e. 3',5'-cyclic adenylyl monophosphate, cAMP, and 3',5'-cyclic guanylyl monophosphate, cGMP) and their functional roles have been well-established in animal cells, their presence in plant cells have been confirmed only recently (Newton and Smith, 2004; Martinez-Atienza *et al.*, 2007). Cyclic nucleotides have been implicated in the signalling and regulation of many important plant processes such as stomatal functioning, cation fluxes, chloroplast development, gibberellic acid signalling, pathogen response and gene transcription (Martinez-Atienza *et al.*, 2007). There are several reports of evidence indicating a role for cAMP and/or cGMP in stomatal opening (Newton and Smith, 2004). Results of the present study have shown a down-regulation of a gene product (i.e. rap-

GTPase-activating protein), which is responsible for activation of the cyclic nucleotide pathway in At 354 during Phase I of salt stress. As osmotic stress-induced water deficits are the main effect of salinity during Phase I, partial stomatal closure by down-regulating the cyclic nucleotide pathway may help alleviate adverse effects of water stress and thereby contribute to salt-tolerance of At 354 in Phase I.

On the other hand, Maathuis and Sanders (2001) and Essah *et al.* (2003) observed that cGMP and cAMP reduced Na⁺ uptake in salt-stressed Arabidopsis, probably by down-regulating the non-selective cation channel proteins through direct binding. Donaldson *et al.* (2004) observed that salt and osmotic stress caused a rapid increase of cGMP levels in Arabidopsis. These observations indicate that down-regulation of the cyclic nucleotide pathway that was observed in the present study may increase Na⁺ uptake in salt-stressed At 354 during Phase I.

In view of the multiple functions of cyclic nucleotides in plants, it is not possible to identify the precise contribution of down-regulated cyclic nucleotide pathway to salt-tolerance of At 354 in Phase I. It is possible that different processes regulated by cyclic nucleotides are differentially adapted to salt stress. For example, salt-tolerance requires partial stomatal closure (which needs down-regulation of the cyclic nucleotide pathway) while reduced uptake of Na⁺ (which requires up-regulation of the cyclic

neucleotide pathway). In the case of salt-stressed At 354 during Phase I, the requirement for partial stomatal closure may be stronger than reduced Na^+ uptake. This is highly likely because tolerance to osmotic stress would take precedence over tolerance to salt ion toxicity during Phase I. Tuteja and Sopory (2008) provide evidence that cGMP and cAMP protects the cell from ion toxicity. As protection from ion toxicity is of lesser importance during Phase I, down-regulation of the biosynthesis pathways of these two cyclic nucleotides is not likely to affect the salt-tolerance of At 354 during Phase I. It would be interesting to see whether the cyclic nucleotide pathway is up- or down-regulated in salt-stressed At 354 during Phase II. This work is in progress.

There is evidence of direct effects of cyclic nucleotides on intra-cellular cation fluxes raising the possibility of a role in regulating the homeostasis of K^+ , Ca^{2+} and Na^+ ions (Maathuis, 2006). Ion homeostasis involves maintaining cytoplasmic ratios of $\text{K}^+:\text{Na}^+$ and $\text{Ca}^{2+}:\text{Na}^+$ within optimum ranges. This may require both increases and decreases of the uptake of these cations in plants experiencing salt stress. Therefore, it is possible that down-regulation of the cyclic nucleotide pathway in salt-stressed At 354 during Phase I may contribute to maintenance of cytoplasmic ion homeostasis. Zeng (2005) observed that the $\text{K}^+:\text{Na}^+$ selectivity increased while the $\text{Ca}^{2+}:\text{Na}^+$ selectivity decreased with increasing

salt stress in rice and that $\text{K}^+:\text{Na}^+$ selectivity was negatively correlated with shoot Na^+ content. It is possible that the cyclic nucleotide pathway may be involved in determining the selectivities of $\text{K}^+:\text{Na}^+$ and $\text{Ca}^{2+}:\text{Na}^+$ in a complex manner.

Concluding Remarks and Future Prospects

In the present study, differential hybridization clearly demonstrated that salt stress during Phase I (i.e. osmotic stress) caused changes in the expression of several genes in the salt-tolerant Sri Lankan rice variety At 354. Several genes whose expressions were either up- or down-regulated due to salt stress were linked to important physiological mechanisms responsible for tolerance of osmotic stress, which is the dominant stress factor operating during Phase I, and/or tolerance of salt ion toxicity, which may also be operating even during Phase I because of the high sensitivity of rice to salinity. These genes may enable exploration of newer avenues for engineering salt tolerance in the Sri Lankan rice varieties. Such genes may also be used in marker-assisted selection to screen the large germplasm of rice available in Sri Lanka for new sources of salt-tolerance.

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Weedy Rice: A Threat to Direct Seeded Rice Culture in Sri Lanka

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Abstract

Emergence of off-types of rice which are morphologically and physiologically different from the inbred varieties, their parental material, land races and traditional varieties in farmer fields has been a significant phenomenon ever since broadcast sowing became popular a few decades back. However, this phenomenon became a major issue when it started causing significant yield losses which is now referred as weedy rice. Weedy rice, reported for the first time in the Eastern Provinces of Sri Lanka in 1990s, has now spread into many rice growing areas irrespective of the agro-ecological zones in the country. Efforts were made over the last decade to understand morphological and physiological properties of weedy rice and its epidemiology to help design control methods and to disseminate such information to the farmers for immediate action to prevent further spread. Some of the distinct features of weedy rice are; diverse grain color, grain shape, presence of awn, leaf color and angle, and panicle characters. In addition, weedy rice collected from different locations showed some features specific to the location. Of more than 2,000 weedy rice types collected so far, agro-morphological features of 1,300 weedy rice types were evaluated. Weedy rice has well developed root system compared to cultivated types indicating their high competitive nature. The seeds of weedy rice have a high dormancy period and a lengthy viability (> 80 weeks) under submerged conditions in paddy soils as compared to rice cultivars. This feature facilitates weedy rice to develop a seed bank in paddy soil. Weedy rice has various biological properties, for instance more than 30% weedy rice types tested found to possess resistance to brown plant hopper.

Weedy rice could cross pollinate with cultivated rice when grown together; and hence the harvested seeds of rice from weedy rice infested fields may contain cross pollinated seeds. Off-type plants emerging in fields where the seeds of improved varieties originated from weedy rice infested fields have been observed. Weedy rice can not be differentiated from cultivated rice at early stage of the crop. There are no weedy rice specific post-plant herbicides to control emerging seedlings. Chemical control is only possible at heading stage through wiping of weedy rice panicles that flower before the rice crop using glufosinate ammonium. Farmers who opt to up-root weedy rice plants at heading sometimes discard the seed bearing plants to water ways and unknowingly help spread the seeds to adjacent fields. The package of practices developed for weedy rice control has been found effective in controlling weedy rice. The package include; use of uncontaminated seed paddy, burning of straw soon after harvesting to encourage dormant weedy rice to germinate, proper land preparation, use of pre-plant herbicides to control weedy rice seedlings that emerge before crop establishment, use of row seeding, water seeding or seedling broadcasting

or transplanting instead of direct seeding, roguing of weedy rice plants, application of glufosinate ammonium for wiping of weedy rice panicles at heading and crop rotation.

Introduction

Weedy rice is defined as undesirable biotypes or morpho-types of rice growing as a weed in and around arable lands showing high shattering of seeds at early stage of grain development affecting quantitative and qualitative losses to rice production (Tang and Morishima, 1996); Sometimes the annual types of wild rice (*Oryza nivara* or *O. spontanea*) are also referred to as weedy rice (Vaughan *et al.*, 1999). Hence, weedy rice includes all unwanted *Oryza* species including wild rice growing in farmer's rice field. Weedy rice has now become a major issue in more than 50 countries in Asia, Africa and Latin America (Baki *et al.*, 2002).

The rice weed flora in the new millennium is dominated by weed species like *Echinocloa crusgalli* complex, *Ischaemum rugosum*, *Isachne globosa*, *Leptochloa chinensis* and weedy rice spontaneous (*O. sativa* complex) (Abeysekara *et al.*, 2006). Weedy rice has emerged as a serious problem only where direct sowing is practiced extensively as compared to transplanted crops. Direct sowing expanded rapidly in countries like Sri Lanka from 1990s as a result of labor shortage and heavy wages incurred making rice cultivation unprofitable. In order to maintain a good crop stand under direct sowing, farmers were compelled to rely heavily on post-emergent herbicides and ignore

the most standard land preparation methods, fertilizer management and manual weeding (Abeysekara, 1999; Marambe and Amarasinghe, 2000). Surveys in weedy rice affected area revealed that weedy rice appeared harmless at the initial invasion stage with no trace of the grain shattering trait. This compels farmers to assume that these off types add to increase their yield. However, off types could only be differentiated as true weeds once these plants start showing weedy characters in 2-3 year period. This is considered as one of the reasons for rapid spreading of weedy rice within the different agro-ecosystems.

Other contributory factor that helped the emergence of weedy rice is the unpredictable drought spells experienced during the recent past that made farmers to forgo investments on land preparations and weed management. Some paddy kept uncultivated for a longer period for climatic, economic and social factors also helped to enhance weedy rice populations. Those fields have acted as propagation area and major contributor to the emergence of weedy rice. The movement of seed paddy contaminated with weedy rice can not also be over ruled. In addition movement of heavy machinery that may have helped spread of seeds can not be ignored (Abeysekera, 2004).

Origin of weedy rice

Weedy and cultivated rice assumed to be evolved from wild *Oryza species* (Vaughan *et al.*, 2005). The genus *Oryza* has 21 wild rice species, of which nine are tetraploid (BBCC, CCDD) and the rest are diploid (Khush, 1997). Several hypotheses have been put forward by scientists to explain the origin of weedy rice. Some consider weedy rice as natural hybridization among different cultivars as well as between wild and cultivated rice. The hybridization among different cultivated rice could be the main reasons for the development of many morphological variations of weedy rice. Some scientists consider them to be a result of natural mutations (Azmi *et al.*, 1994; Caling, 1992; Abdallah *et al.*, 1996). Weedy rice could occur due to several factors like gene flow between wild and cultivated japonica types and old rice varieties becoming feral as well as crosses between cultivated rice and wild rice such as *O. nivara* and *O. rufipogon* (Marambe, 2005). Cultivated rice and most of weedy rice have the same genome; AA and the same number of chromosome $2n = 24$ (Azmi *et al.*, 1994). Species with same genome group are of most concern as they can easily hybridize and produce weedy rice (Oka, 1990; Gressel, 1999; Lu, 2004). Marambe (2005) analyzed two weedy rice sample collected from Puttlam and Galle districts using morphological as well as molecular techniques and found that the weedy rice to possess intermediate characteristics to those of cultivated rice of Bg 300 and annul wild rice; *O. nivara*.

Distribution of weedy rice

Weedy rice has become a major threat to rice cultivation in many rice growing countries in Asia, Africa, Europe and USA (Labarada 1999; Watanable *et al.*, 2000). In Sri Lanka weedy rice emerged as the most distinct field problem in some major rice growing areas i.e. Ampara, Matara and Puttalam districts since late 1990s. At present it is spreading at an alarming rate in almost all rice growing areas in the country. This problematic weedy rice is known by different names in different rice growing areas such as Uru wee, Walgoyam, Irratta wee, Hulan wee, Pandisamba, Pachcha wee, Pachchaperumal.

In 1992, two perennials and 4 annual types of weedy rice were collected in Addalachenai, Akkaraipattu and Ninthavur in Ampara district (Abeysekera and Iqbal, unpublished). By 1995, the area infested to weedy rice in the district rose to 1,000 ha and most of the affected lands were in the coastal belt, where some farmers abandoned rice cultivation for want of low cost method for weedy rice control (ARS/Samanthurei, 1997); Marambe and Amarasinghe, 2000). Continuous production of seed paddy by farmers by themselves and dry sowing with very high seed rate (250 kg/ha) may have helped for such situation. Farmers in that area are reluctant to cultivate rice due to heavy infestation of weedy rice that has increased the cost of production and reduced the quality and quantity of the crop yield. Furthermore, it was found that weedy

rice infestation was higher in Maha season compared to Yala season and infestation increased during post Tsunami period in all over the costal paddy fields (ARS/Samanthurei, 2004).

Surveys conducted up to Yala 2006 season revealed that the weedy rice infested area in Ampara district alone has reached 2,000 - 3,000 ha (Table 1).

Table 1: Weedy rice affected area (ha) in the coastal area of Ampara district from 2003 to 2006.

Affected Area (AI Range)	Yala 2003	Maha 2003/04	Yala 2004	Maha 2004/05	Yala 2005	Maha 2005/06	Yala 2006
1. Addalaichenai	1200	1780	1125	2135	1340	2120	830
2. Akkaraipattu	1110	1867	875	1355	823	1455	785
3. Ninthavur	320	457	235	327	115	235	113
4. Sammanthurai	684	845	436	368	786	543	214
5. Thirukovil	187	624	321	155	143	230	115
6. Pottuvil	845	410	213	150	120	415	133
7. Palamunai	116	224	130	187	88	229	115
Total	4462	6209	3336	4677	3415	5227	2305

In Puttlum district, weedy rice was identified in 2000 at Madampe Mahawelayaya (360 ha), Ponnakanniya (30 ha) Mukunuwatawana (16 ha), Helabawatawana (16 ha) and Elakkattuwa (8 ha) (Personal communication, Mr. Justin Perera,

SMO Paddy, 2002). The weedy rice samples collected from 2000 to 2002 had 12 different variants according to the grain characters and some morphological similarities to *O. nivara* (Table 2) (RRDI, 2002).

Table 2: Weedy rice morhpo-types classified according to grain characters.

District	Hull colour			Pericarp colour		Grain shape	
	Black	Brown	Straw	Red	White	Short <20mm	Long >20mm
Ampara	108	212	202	408	114	312	210
Matara	98	165	222	298	187	265	220
Puttalam	170	70	5	180	65	172	73
Baticoloa	150	24	36	196	14	185	25
Polonnaruwa	132	46	194	210	162	169	203
Kandy	98	24	100	108	114	115	107
Galle	135	41	102	96	182	116	162

Kegalle	78	46	1	104	21	78	47
Anuradhapura	165	44	116	210	115	142	183
Hambantota	210	28	42	186	94	164	116
Rathnapura	162	34	164	132	228	146	214
Kurunegala	180	74	331	301	284	352	233
Paranthan	128	58	266	231	221	236	216
Kalutara	165	76	147	210	178	165	223
Total (%)	1,979 40.8%	942 19.4%	1,928 39.8%	2,870 59.9%	1,979 40.8%	2,617 54.0%	2,232 46.0%

In 2004 a severe weedy rice infestation was observed in Matara district. Studies carried out during 2004 to 2006 showed a very high morphological diversity and eleven distinct types were identified based on the grain morphology. The yield losses caused by weedy rice was apparent and in highly infested fields, the density of weedy rice plants found to be three times higher than the density of cultivated rice plants (Darshana *et al.*, 2007). During 2006 to 2008, 64 different morphological

variants on the basis of grain characters were collected from Kaburipitiya, Thihagoda, Hakmana farmer fields in Matara district (Abeysekera, unpublished). In addition, a survey conducted in Matara district in 2009 revealed that more than 40% of rice lands in Pasgoda, Kamburupitiya, Kirinda-Puhulwelwella, Thihagoda, Akuressa and Mulatiyana DS areas had higher contaminations of weedy rice compared to area under other DS regions (Table 3).

Table 3: Area surveyed and percentage of area found infested to weedy rice classified according to Divisional Secretariats in Matara district.

Divisional Secretariat	Area Surveyed (ha)	Area Infested (%)
Akuressa	1,207	58
Pitabeddara	393	47
Kotapola	255	9
Pasgoda	786	86
Athuraliya	243	19
Mulatiyana	790	8
Kamburupitiya	1,302	62
Hakmana	755	17
Dickwella	256	14
Kirinda-puhulwella	605	68

Thihagoda	1,090	41
Malimbada	748	18
Dondra	93	23
Weligama	27	24
Welipitiya	621	9

Source: Jeevani, 2010

Abeyssekara *et al.* (2010) identified 517 morphologically different weedy rice types from 1500 samples collected from different rice growing areas in Kurunegala, Polonnaruwa, Anuradhapura and Kandy districts during 2002 to 2004. As at present 2,800 weedy rice accessions have been collected from different rice growing areas throughout the country and 1,300 distinct and abundant morphological variants of weedy rice have been identified.

hull color, presence or absence of awn, and its length, grain size and shape, panicle type, panicle length and plant characters such as plant height, leaf color, leaf angle, number of tillers per plant etc. (Table 4). As reported elsewhere by Chin (1997), Azmi *et al.* (2000) and Liuqing *et al.* (2000), the local weedy rice collected also found to be taller, produce more tillers and panicles per hill, had greater panicle length and more grain per panicle, longer flag leaf and flowered earlier than cultivated rice.

Characteristics of weedy rice

Analysis of more than 1,300 weedy rice types collected island wide showed high degree of variation in grain color,

Table 4: Morphological variation of weedy rice collected from different parts of Sri Lanka.

Morphological Characters		Range
1.	Plant Height	90 -165cm
2.	Tiller per plant	6 – 21
3.	Flag leaf length	24 - 68cm
4.	Flag leaf angle	Erect (40%)’ Droopy (60%)
5.	Panicle type	Compact (80%); Open (20%)
6.	Seed shattering	High 57% (coastal districts) Moderate (43%) (Inland districts)

7	Hull color	Black 68%; Brown 9% Black streak with straw 18% Straw-5%
8	Awn color	Black; Brown; Dark brown Straw
9	Awn length	0.1 - 9cm
10	Pericarp color	Red (64%); White (36%)

Source: Abeysekara *et al.* (2010)

The morphological analysis indicated that weedy rice types have intermediate characters to those of cultivated and wild rice (*O. nivara*, and *O. rufipogon*). It varies from location to location and depends on the types and age of the growing rice cultivar. Most of the weedy rice identified near to the coastal areas in the country viz Ampara, Mannar, Putlam, Matara are more similar to wild rice. They have high degree of seed shattering (>90%), long seed dormancy, red pericarp, long awn, light seeds tillers /plant and panicles /plant though they look like cultivated rice during the vegetative stage. Samples collected from inland areas of the country viz. Kurunegala, Polonnaruwa, Anuradhapura, Mahiyangana showed less degree of seed shattering (40-70%), different grain shapes (long, intermediate and bold), red and white pericarp, and black, golden or brown hull. They were with and without awn, always taller than the cultivated rice, early maturing and having less tillers/plant and less panicles /plant and looked very similar to traditional rice cultivars. These weedy rice types are also difficult to be distinguished from cultivated rice during vegetative period (A.S.K. Abeysekara, unpublished). It may be due to the natural hybridization between different cultivars as well as

between perennial or annual wild rice and cultivated rice (Moody, 1994).

Growth pattern of cultivated rice and weedy rice were different. It is difficult to be distinguished from cultivated rice in early vegetative stage but is easy to be distinguished while/after tillering stage and early flowering stage due to many morphological variances in comparison with the cultivated rice such as longer and slender tillers, droopy and pubescent leaves, rooting in stem nodes, tall plants, early flowering and easy shattering before the maturity of cultivated rice.

Reaction to brown plant hopper

Studies conducted on the level of resistance to brown plant hopper in 52 weedy rice types collected from 5 districts showed that 15 had high level of resistance while 22 were moderately resistant to the pest (Wickremasinghe *et al.*, 2010). The underline reasons for these observations are not very clear. However, it is worth noting that most of the improved rice grown in these areas carries Ptb 33 as a parent, a land race from southern India as the donor source for brown plant hopper resistance.

Effect on yield

Competitiveness is the ability to deplete rapidly consumable resources, such as water and nutrients or to grow better than associated species when these resources are limiting. Higher competitiveness was observed in weedy rice plants compared with the cultivated plants which results in a significant yield loss of cultivated rice. A pot experiment with different ratios of weedy rice and cultivated rice

indicated that the yield loss could be 100% when the ratio of weedy rice to cultivated rice exceeds 6:9 (Table 5). Similarly a field observation in Matara revealed a linear relationship ($R^2 = 0.71$) between weedy rice density and yield of cultivated rice (Darshana *et al.*, 2007).

Table 5: Yield loss in improved rice varieties due to competition from weedy rice.

Ratio of weedy to cultivated Rice	Yield loss (%)
9:9	100
8:9	100
7:9	100
6:9	100
5:9	82
4:9	68
3:9	43
2:9	32
1:9	20

Viability and longevity of weedy rice

The high seed shattering and prolonged seed dormancy have helped enrichment of soil seed-bank of weedy rice in infested fields. The soil seed bank of weedy rice consists of viable and dormant seeds. The non dormant seeds in the soil germinate with the

availability of moisture and favorable temperature. Hence, weedy rice seed bank plays an important role in determining the severity of infestation in rice fields. Therefore, several studies have been conducted to determine seed dormancy, seed viability and longevity and germination ability of weedy rice biotypes in an effort to develop management strategies to control weedy rice.

An experiment conducted with seeds of 3 common weedy rice types from Ampara (eastern coastal area), Puttlum (western coastal area) and Matara (southern coastal area), and 3 popular improved varieties (Bg 300, Bg 94-1 and Bg 379-2) revealed that seeds of improved varieties lost their viability in 20 weeks when buried 15cm deep in paddy soil as compared to a period of 2

years for the weedy rice seeds (Fig. 1). The germination percentage of weedy rice remained 40% even after 80 weeks in paddy soil. These results suggest the reasons for increasing trends of the weedy rice seed-bank in soil in infested fields and the importance of long-term strategies to minimize the soil seed-bank of weedy rice.

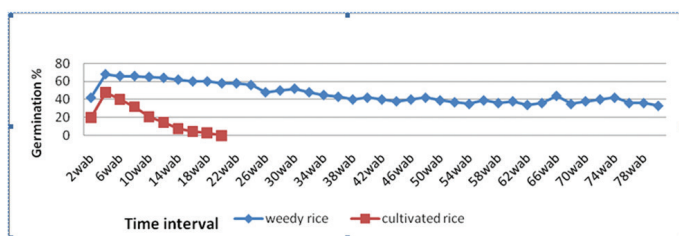


Figure 1: Mean germination percentage of improved varieties and weedy rice seeds from 2 to 80 weeks after burying in paddy soil (RRDI, Batalagoda, 2006).

Sustainable management options

The Department of Agriculture and the Provincial Agriculture Departments have taken steps to disseminate weedy rice control methods to farming community through training programs, demonstrations and other mass media. The control methods included use of uncontaminated seed paddy (harvested from weedy rice free areas), burning of straw soon after harvesting to encourage germination of dormant weedy rice seeds before crop establishment, proper land preparation, use of pre-plant herbicides to control weedy rice seedlings that emerge before crop establishment, use of row seeding or water seeding or seedling transplanting or transplanting instead

of direct seeding, roguing of weedy rice plants, application of glufosinate ammonium to weedy rice panicles at heading and crop rotation.

A field experiment conducted in Matara district revealed a total success of the weedy rice management through application of some of these methods. A significantly low number of weedy rice panicles were recorded in plots subjected for these management methods (Jeevani, 2010) (Table 6). Plots treated with the transplanted crop have significant by lowest weedy rice than the broadcasted. Results indicated the effectiveness of transplanting and other control methods for weedy rice management. The weed control efficacy was at the highest in control package which was 90.7%.

Table 6: Average number of weedy rice panicles/m² (n=3) in plots managed with different weedy rice control measures.

Treatment		Average no. of Weedy Rice (panicles/m ²)	*Control Efficacy
Crop transplanted + Control package		3.1d	90.7
Crop transplanted + Control package except glufosinate ammonium		16.4c	76.7
Crop broadcasted + Control package except glufosinate ammonium		32.1b	60.0
Crop broadcasted + No control package		94.1a	
CV%	19.42		

Means with same letter are not significantly differed at $p = 0.05$

Control efficacy = (Average no. of panicles/m² in control - Average no. of panicles/m² in treatment) x 100/ Average no. of panicles/m² in control

Source: Jeevani, 2010

Weed management has become more challenging in rice fields because of the occurrence of weedy rice. Herbicides are effective against many rice weeds, if they are applied at correct rate and time. Hence herbicides have become the most commonly used weed control practice. However, management of weedy rice through herbicides is difficult as there are no selective herbicides that can be used against weedy rice during plant growth. In addition, it would be extremely difficult or impossible to define a single technique for effective control of weedy rice due to the specific attributes the weedy rice possesses. The attributes of weedy rice that pose challenges to weed management specialists include their higher competitive ability, early seed shattering, prolonged dormancy and morphological similarity to cultivated rice, ability to disperse

through irrigation water, contaminated seed paddy and farm machinery etc. These transmission mechanisms are more difficult to regulate. Farmer field studies conducted at RRDI during 2000-2008 identified cohesive management practices are the only possible way to control weedy rice in many rice growing areas. One of the focal issues that need to be highlighted is the use of uncontaminated seed paddy. However, it is not successful in fields where weedy rice infestation is very heavy due to prolonged viability of weedy rice seeds in soil. Hence, weedy rice management at pre-plant stage should be vital. Irrigation of the fields enhances the early germination of seeds in the soil and helps control weedy rice mechanically or chemically through a pre-plant herbicide. This is considered as one of the effective methods. If infestation level is high

the above process can be repeated 2 or 3 times before crop establishment. However, this method cannot be adopted in all rice growing areas as it depends on soil and climatic factors.

In some areas especially in Ampara and Puttalam, burning of straw after harvesting to enhance emergence of weedy rice that can be controlled easily is practiced. Although this method is environmentally destructive, more than 80% success in controlling weedy rice can be achieved though this method but it is not promoted for large scale application by the Department of Agriculture. Many demonstrations conducted by the Department of Agriculture have shown the proper tillage practices could reduce weedy rice infestation by 60%. Furthermore, farmer participatory research have shown that broad holistic approaches are the best alternative for weedy rice management in all rice growing areas in Sri Lanka than the single control method adopted by many farmers. Preventive and cultural management are the major means of controlling weedy rice in rice in Sri Lanka. Probably the most important is the use of clean seed - seeds free from weedy rice. It is, therefore, critical that farmers be made aware of the importance of using clean, uncontaminated seed as planting material.

Summary of recommended control methods

In wet seeded rice in Sri Lanka, management programs to control weedy rice include (1) Identification of

weedy rice; It is proposed that the off-types clearly visible in the rice fields should first be positively identified as weedy rice based on the specific characters - high shattering of seeds at early stage of panicle maturity. In addition, weedy rice could be detected at early stage of plant growth on the basis of root and leaf morphology. (2) Use of quality seed paddy, irrigation water and equipment. (3) Proper land preparation; It is proposed extra efforts to be taken at the land preparation, in weedy rice infested areas, emphasizing on the control of seed bank of the weedy rice in the field. This includes burning of stubbles, if possible, soon after harvesting to kill the weedy rice seedlings on the surface of the soil and to stimulate the submerged seeds to germinate once the field irrigated. (4) Use of effective stand establishment method; Use of transplanting/ seedling broadcasting/ row seeding/ water seeding establishment methods instead of broadcasting. Weedy rice seedlings poorly adapt for transplanting. The transplanting shock kills many seedlings generated through weedy rice. Row seeding found to be the best direct seeding method for weedy rice control. (5) Use of herbicides; Application of pre-plant total killer (glyphosate, paraquate) before and after initial land preparation to kill weedy rice seedlings and minimize the seed bank of weedy rice, application of pre-emergent herbicide pretilachlor (sofit) to kill newly emerged weedy rice seedlings and application of glufosinate-ammonium 15% SL (Busta) wiping method; The panicle wiping technique recommended by the

DOA as the final attempt for weedy rice control. This technique works well when applied just at heading of the weedy rice and when the weedy rice density is high (> 10 weedy rice plants/ m^2). (6) Manual weeding. (7) Roguing of panicles and (8) Crop rotation. Therefore, integrated approach is the effective way to manage weedy rice.

Recommendation

An analysis of the present situation of weedy rice in Sri Lanka indicates that weedy rice will become one of the major national issues in the rice production in the country, demanding comprehensive plans for its management. Proposed work plan for weedy rice management should include:

1. Awareness programs to extension staff and farmers on the initial phase of weedy rice infestation to epidemic level out breaks.
2. Development of detection methods for early stage of weedy rice infested areas.
3. Mapping of weedy rice infested areas with the infestation levels by the methods adopted in Matara district by Jeevani (2010).
4. Basic studies on the mechanisms of the emergence and spread of weedy rice.
5. Development of management methods based on the infestation level, type of weedy rice and rice ecosystem.
6. Development of a system to increase pure seed paddy production in areas where weedy rice not recorded especially through community based seed paddy production programs.
7. Encouragement of control of weedy rice seedlings that emerge during the early stage of land preparation through repeated ploughing or use of pre-plant herbicide
8. Encouragement of transplanting methods or row seeding methods.
9. Encouragement of burning of straw in highly infested fields and panicle wiping to burn weedy rice seeds.
10. Development of a system to monitor areas affected to weedy rice and to evaluate the effectiveness of weedy rice control program.

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A Review of Rice Pest Management Research in Sri Lanka

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Abstract

Progress of research and development activities undertaken in Sri Lanka on rice pests during the last decade is reviewed with future directions towards the development of economical and eco-friendly pest management technology options. Damage by insect and mite pests is one of the major constraint in increasing rice production in Sri Lanka. It is proposed that estimates of 10-20% yield loss for pest infestations should be revised through a planned surveillance system to help design new tools for their management. During the period of review, efforts have been made to develop varieties resistant to pests, to identify emerging pests and new bio types of existing pests and important natural enemies associated with rice eco system and to find out efficient eco friendly insecticides. In view of the emerging demands for sustainable management methods for rice pests, emphasis should be given to develop strategies, techniques, options and policies to enhance naturally occurring biological functions such as predation, parasitization and pathogenesis for the management of pests. Actions need to be taken to improve knowledge and competence among the farmers and spray operators of the proper use of pesticides including selection of correct pesticide, decision of correct dose and dilution, selection of appropriate nozzles and spraying procedure. The advanced information technological tools should be used to maximize the utilization of available resource to achieve high level of success in pest management operations in the country.

Introduction

A multitude of societal and environmental events and transformations have motivated intensive research and innovations on sustainable systems for the management of pests of rice that affect greatly on the quality and quantity of the produce. One such driving force has been the escalating demand for quality-enriched rice for local consumption. It also would be

unjustifiable to ignore the climate and human influenced changes taking place in the rice ecosystems that affect on the intricate natural balances evolved over million years and now obviously visible through the emergence of once cryptic pests into prominence and for farmers' to become over dependant on external inputs for pest management. Concerns on environment and human health and farmers' occupational safety have also created pest management a crucial issue to the farming community

(Heong *et al.*, 1994; Norton, 1996; Lechman, 2010). These accumulations of facts during the last decade put the farmers to incline more towards synthetic chemical inputs for managing their pest problems ignoring other available options. This is observed through a gradual withdrawal of enthusiasm among farmers on the time tested, farmer and environment friendly pest management methods disseminated during the early post-green revolution era in 1970s, 1980s and 1990s. The underline reasons for these unwholesome changes are intricately uncomplex where a detailed analysis is warranty (Lechmann, 2010). Lack of strong knowledge-based information-rich extension programs to transform attitudes and improve competence of the farming community on novel pest management options, and weakness of the regulatory mechanisms and policies have collectively undermined the farmers' right to access the best management options for their pest issues.

It is anticipated that initiatives would be taken to re-establish environment based management strategies that would help enhance the ecosystem services for the purpose of pest management. Establishment of area-wide surveillance systems to assist farmers on decision making on pest management issues need to be highlighted as a priority (Hidaka and Nugaliyadde, 2000). Much debated standardization of spraying technology has to be addressed along with the training of pesticide applicators on safe and effective application of pesticides.

It is anticipated that these changes would help improve the health and safety of the occupation of pesticide applicators and farmers and bring another professional dimension into rice farming.

This paper attempts to review the research and development activities undertaken on rice pests during the last decade (from 2000 to 2010) with a view of redirecting and revitalizing continued efforts to improve crop protection technologies that are economical, environmentally safer and acceptable to farmers and fulfill the desires for a prosperous Sri Lanka.

Pest Profile

Based on the reported incidences of pest epidemics, rice thrips (*Stenchaetothrips biformis*), gall midge (*Orseolia oryzae*), leaf folder (*Cnaphalocrocis medinalis* and *Marasmia* spp.), yellow stem-borer (*Scirpophaga incertulas*), brown plant hopper (*Nilaparvata lugens*) and paddy bug (*Leptocorisa oratorius*) are considered as major insect pests of rice in the country (Wickramasinghe, 1980; Kudagamage and Nugaliyadde, 1995; Nugaliyadde *et al.*, 2000a; Nugaliyadde *et al.*, 2003a). With the revelation of the damage symptoms of sheath mite (*Steneotrasionemus spinkii*) and leaf mite (*Olygomychus* spp. and *Hemitarsonemus* spp.) in rice in Sri Lanka in 2000, incidences of severe infestations of these pests accumulated from several locations of the country led to make these pests a significant position in the pest profile (Caberera *et*

al., 2002; Wijesekera, 2002; Amarasiri, 2008; Sandamali, 2009; Pushpakumari *et al.*, 2010).

Reports of sporadic out breaks of white-backed plant hopper (*Sogatella furcifera*), ear cutting caterpillar (*Mythimna separata*), cut-worm (*Spodoptera* spp.), black bug (*Scotinophara lurida*) (Hemachandra and Nugaliyadde, 2007) and nematode (*Meloidogyne graminicola*) (Nugaliyadde *et al.*, 2001a) from some areas of the country required concerted attention from research and development for concerted solutions. The epidemic of leptospirosis, reported from some rice growing areas, has prompted extra attention for the control of rice field rats (*Bandicola bengalensis*) under the purview of agriculture and health authorities (Plant Protection Service, Department of Agriculture). An out break of a whitefly (the species yet to identify) in Ampara and Batticaloa districts during 2009/2010 major season has created a new threat to rice farming requiring attention for immediate solutions (Mr. Y.B. Iqbal, personal communication).

With increasing awareness of the importance of managing pests of stored paddy and consumption rice, mainly, the rice moth (*Sitotroga cerealella*), rice weevil (*Sitophilus oryzae*) and *Tribolium castaneum* prompted attention towards differentiating the responses of these pests on commercial varieties and to determine safer alternatives for their management.

In order to design, develop and disseminate sustainable innovative

solutions for the management of rice pests, continued efforts were made to breed varieties with inherent resistance to pests, to field evaluate improved integrated pest management methods and to search for efficient, eco-friendly high-technology-base insecticides.

Incidences of Pest Out Breaks

Lack of a well established monitoring and reporting system prevent compilation of reports on the incidences of pest infestations, incurred yield losses, and investments made on interventions including pesticides (Hidaka and Nugaliyadde, 2000). Hence, it should be reiterated the need for a pest surveillance system for information collection and sharing to enable designing tools for their management, approaches to educate farmers for preventive options and early interventions and much required policy directives. The extrapolated estimates of 10 - 20% yield loss due to pest infestations in rice in Sri Lanka (Kudagamage and Nugaliyadde, 1995) need to be revised in the context of present increasing trends in cultivated extents and cropping intensities and cropping methods.

The mechanisms operating within the Department of Agriculture including the provincial agriculture system were found insufficient in providing quantitative information on pest incidences of rice. However, it is admiring the efforts taken by the Extension and Training Center of the Department of Agriculture to collect

pest incidence records in the inter-provincial areas of the country from the 2010 minor season (Table 1). The agenda items in the Provincial Technical Working Group (PTWG) meetings, news paper reports on

localized pest out breaks and the public out cry against heavy loading of pesticides to rice systems are some of the justifications that can be put forth for investments on rice pest management research.

Table 1: Rice area damaged to different rice pests during 2010 minor season (inter provincial area in Sri Lanka).

Pest	Damaged area (ha)						
	Ampara	Anuradhapura	Kandy	Polonnaruwa	Monaragala	Hambantota	Total
Thrips	-		10				10
Leaf folder	3,555		47	71	12	52	31
Gall midge	0		0	0	0	0	0
BPH	1,300		2	5	-	-	-
Leaf mites	3,750		120	10	105		55
Sheath mites	2,750		-	-	-	-	200
Stem borer	-		-	1.5	-	-	-
Cutworm	-		-	-	-	-	-
Black bugs	-		-	-	-	-	-

- = not reported

Source: Mr. Jagath Sudasinghe, Assistant Director, Extension and Training Center, Department of Agriculture, Peradeniya

Brown plant hopper (BPH)

Among the rice pest complex, brown plant hopper (BPH) continued to be the main focal point for discussions and interventions. However, during the last 10 year period, only a few districts have collected information on areas affected

to BPH (Table 2). This is despite of the information received by the Plant Protection Service of the Department of Agriculture of several out breaks of BPH in major rice growing areas during major seasons 2004/05, 2006/07, 2007/08 and 2009/10 (Plant Protection Service, Department of Agriculture).

Table 2: Rice area affected (ha) to BPH in different districts from 2000 to 2010 (years with no records are not included).

District	Year					
	2000	2005	2007	2008	2009	2010
Anuradhapura	-	100	-	-	669	-
Gampaha	106	-	-	12	03	-
Galle	-	-	21	-	-	-
Hambantota	-	-	466	1,510	-	-
Kandy	-	-	-	-	40	1,140
Kurugedala	-	-	167	-	-	-
Matara	-	-	347	-	-	-
Mahaweli C	-	-	-	-	5,700	-
Puttalam	-	-	-	-	-	-
Polonnaruwa	-	-	-	-	-	5,710
Ratnapura	-	-	55	-	-	-
Total	106	100	1,056	1,522	6,412	6,850

- = not reported:

Source: Provincial Departments of Agriculture and Extension and Training Division, Department of Agriculture

Biotypes of BPH

Heavy infestations of BPH on Bg 379-2, a variety introduced for commercial cultivation in 1980 possessing resistance to BPH, were reported from Kegalle since 2005/06 (Mr. Hemmathagama, personal communication). Investigations conducted revealed that BPH populations in Kegalle produce significantly higher amounts of honeydew on Bg 379-2 than laboratory grown culture from Batalagoda, indicating that the population has adapted to the inherent resistance in Bg 379-2. Molecular analysis done at the PGRC revealed distinct variations

between these virulent and avirulent BPH populations suggesting presence of a new BPH biotype (Fernando *et al.*, 2007). Other resistant varieties (Bg 300, Bg 352, Bg 358 and Bg 360) derived from the same resistant source for BPH, Ptb33, however found to show same level of field performance. This is attributed to the differences in the genetic composition and associated mode of resistance of the Ptb33 derived commercial varieties (Nugaliyadde, 1994; Nugaliyadde and Wilkins, 2000). BPH resistance in Bg 379-2 found to be associated with its amino acid composition in the phloem sap where as the resistance in other

varieties (Bg 300) found to derive from the epicuticular wax composition. These information signals needs for comprehensive approaches in the breeding programs utilizing wider genetic base for BPH resistance.

Rice leaf folder

In spite of innumerable reports and queries on the issues related to the control of rice leaf folder, only Nuwara Eliya district reported the damage in 145 ha, in total, from 2006 to 2010 (Plant Protection Service, Department of Agriculture; Central Provincial Department of Agriculture). This, we consider a large gap of information flow unhealthy for the development of the rice sector.

Rice sheath (panicle) mite

Rice sheath mite or panicle mite (*Steneotarsonemus spinki*) was recorded for the first time in Sri Lanka in 2000 (Caberera *et al.*, 2002; Nugaliyadde *et al.*, 2003b; Pushpakumari *et al.*, 2008; Pushpakumari *et al.*, 2010). Presently, it has become a major pest in most of the rice growing areas in the country and only Hambantota and Ampara districts so far have areas affected to this damage (Table 1). This cryptic mite causes substantial losses in grain yield and quality, demanding an urgent attention for its management (Amarasiri, 2008; Sandamali, 2009). Prolong periods of drought found to influence greatly on population growth of this mite and intermittent

heavy showers influence negatively on this mite requiring more attention for management in the dry seasons (Wijesekera, 2002).

At field level, at least a 20x magnifying lens is required to observe the mites on the adaxial surface of the leaf-sheath. The fungal pathogen, *Sarocladium oryzae*, causal agent of sheath rot was isolated from field collected sheath mites indicating that this mite could act as a vector for this disease (Sandamali, 2009). It has been detected that the most susceptible stage for the mite damage is the booting stage. Moderate to severe infestations lead to purple to brown patches on the leaf sheath, grain sterility deformed and discolored grains and considerable yield losses. This mite can be transmitted by wind, water movement, field workers, machinery and infested seeds (mites are available inside the seed hull). After harvesting they remain in the rice stubble and ratoon. Cultural practices such as burning of rice stubble, ploughing of rice stubble, rotation with an alternate crop fallowing infested fields for 3 months may reduce the infestation. Drying process may eliminate mites from seeds. Tentative recommendation of profenofos 500g/l EC was given in view of the urgency of a control method for rice sheath mite until a safer product could be made available to the farmers. It is important that this recommendation is only for the fields where severe infestation of mites, sheath rot and sheath blight were observed in the previous season.

Field rats

The outbreak of leptospirosis, a notifiable disease in Sri Lanka, over the last decade in several areas prompted health and agriculture authorities to design and implement eradication programs for the rice-field rat at national level (Plant Protection Service, Department of Agriculture). The incidence reached the highest value of 7,099 in 2008 with more than 60% confirmed cases had a history of exposure risk from paddy fields (MOHCN, 2008). Coincidentally, 374 cases has been recorded from Gampaha district during 2003 and 2004. The tools available for rodent control, such as cleaning the breeding grounds, application of rodenticides and prophylactic treatment of antibiotics for those with exposure risk helped prevent this disease to a greater extent. A need exists for detail research to develop effective rodent control methods, among which microbial control as extensively used in other Asian countries may have high potential.

Whitefly

Sudden outbreak in minor season 2010 of a whitefly species in Ampara and Batticaloa in the eastern rice belt in Sri Lanka prompted attention of both extension and research to identify the problem and to develop methods for its management. Several species of whiteflies (*Bemisia tabaci*; *Aleurocyboticus indicus*) are known to infest rice in the drier, irrigated areas in the African continent (Heinrichs, 2004).

Whiteflies are minute (2-4 mm), have a wide alternate host range including weeds and could multiply extensively on rice under high temperature and low humid conditions leading the crop to wilt and burn. This situation prompted farmers to apply heavy doses of both systemic and contact insecticides, available and the out come of these interventions need to be assessed for new directions.

Pests of stored rice (*Sitotroga cerealella*, *Sitophilus oryza* and *Tribolium castaneum*)

It is estimated that nearly 15% of the harvested paddy is lost at various stages of post-harvest operations and the 6% loss is accounted for rice grain moth (*Sitotroga cerealella*) and rice weevil (*Sitophilus oryzae*). While many farmers follow traditional methods for the management of these pests and at industry level, phosphene fumigation and OP are used extensively to protect seed paddy and stocks of consumption paddy. An intensive survey conducted in a rice-farming community in Mapalana to update the quantitative estimates of yield losses on grain moth (Table 3) revealed that 93% of farmers considered grain moth as the most damaging pest of paddy and samples taken revealed 3.2% of the stored grains were damaged by the moth at any given time (Kanthi *et al.*, 2006).

Table 3: Damage assessment of stored paddy in Mapalana (n = 30) in 2006.

Criteria	Mean and range (%)	
Average number of grains / 50 g sample	2,017	
Grains damaged by grain moth	3.24	(1.2 – 35.4)
Grains damaged due to other reasons	1.35	(0.2 - 13.9)
Total damage	4.69	(0.8 - 27.5)

Source: Kanthi *et al.*, 2006

Varietal Resistance/ Tolerance

Rice thrips

Efforts were continued to elaborate the mechanisms involved in thrips-resistant local traditional varieties and to transfer this resistance into improved varieties. Thrips resistance found to be poly-genic, and many complexities encountered in breeding, like incompatibility in the form of grain sterility hindered the progress of the breeding program (Javier *et al.*, 1989). With the several drawbacks in the past, Rice Research and Development Institute (RRDI), Batalagoda has been successful in developing advanced generation materials with high level of resistance to thrips through a back-cross program - the progenies developed are: Heenati (03998) (resistant) / Bg 94-1-BC5F7 (maturity 90 d) and Heenati (03264) (resistant) / Bg 360-BC5F6 (maturity 90 d). Similarly crosses between resistant varieties with improved varieties developed at Regional Rice Research and Development Center (RRRDC), Bombuwela yielded

compatible progenies with resistance to thrips making the opportunities to come out with breeding lines having thrips resistance. In addition, mutation programs were initiated to develop a Dahanala mutant with similar resistance.

Density of trichome (long curved trichome) on the adaxial leaf surface was related to the thrips resistance in some traditional rice varieties like Dahanala (trichome density > 60/ mm²) and it was used as an important criterion to select segregating populations for thrips resistance (Nugaliyadde and Heinrichs, 1984a and b; Nugaliyadde *et al.*, 2003a; Wickramasinghe *et al.*, 2007). However, further evidences revealed that some traditional varieties showing moderate level of resistance to thrips have low density of trichomes (<20 /mm²); indicating thrips resistance is a character with multiple modes of resistance; both morphological as well as chemical (Lewis, 2000). With the available rich information and experience, one should go forth in encouraging the utilization of this rare and valuable resource for the advancement of rice production in the country.

Rice gall midge and brown plant hopper

RRDI and RRRDC continued to invest resources for routine evaluation of breeding lines for resistance to BPH in greenhouse and GM at field (Table 4) and helped releasing 14 varieties with multiple resistances to both BPH and GM (Table 5) (Nugaliyadde *et al.*, 2001b).

In efforts to initiate pyramid resistance during last decade, 14 varieties with multiple resistance to BPH and GM were recommended. Molecular markers have been identified for BPH (BPH10 & BPH2) and GM (GM 4) genes (Fernando *et al.*, 2007).

Table 4: Number of progenies tested for BPH (green house) and GM (field) resistance from 2000 to 2010 (RRDI, Batalagoda).

Season	Brown plant hopper		Gall midge		Total
	R*	R/MR*	R*	R/MR*	
Maha 1999/2000	432	116	17	547	933
Yala 2000	193	287	128	185	1,360
Maha 2000/2001	686	197	372	159	1,303
Yala 2001	344	342	177	169	1,074
Maha 2001/2002	510	100	165	470	1,159
Maha 2002/2003	129	159	144	199	1,160
Yala 2003	312	234	687	586	1,989
Maha 2003/2004	667	750	341	132	2,224
Yala 2004	574	730	1,027	228	2,192
Maha 2004/2005	284	367	579	174	1,227
Yala 2005	553	434	278	444	1,856
Maha 2005/2006	633	414	394	147	1,834
Yala2006	133	328	189	194	1,101
Maha 2006/2007	520	388	394	147	1,470
Yala 2007	424	478	105	1256	1,344
Maha 2007/2008	493	373	216	1025	1,633
Yala 2008	374	368	477	319	1,293
Maha 2008/2009	313	252	217	160	1,157
Yala 2009	615	383	595	1140	2,198
Maha 2009/2010	687	235	71	209	1,453

*R-resistant MR-moderately resistant

Table 5: Levels of resistance to brown plant hopper (BPH) and gall midge (GM) of the rice varieties released for commercial cultivation in Sri Lanka from 2000-2010.

Variety	Maturity (days)	Year released	Reaction to *	
			BPH	GM
Bg 250	80	2005	R/MR	MR
At 306	90	2004	R	MR
At 307	90	2005	R/MR	R/MR
At 308	90	2008	MS	R/MR
Bw 361	105	2002	MR	MR
At 362	105	2002	MR	MR
Bw 363	105	2004	MR	MR
Bw 364	105	2006	MR	MR
Ld 365	105	2008	R	MR
Bg 366	105	2009	R	R/MR
Bg 406	120	2005	R	MR
Ld 408	120	2010	MR	MR
Bg 407H	120	2005	R	R
Bg 454	135	2005	MR	MR

*R-resistant MR- moderately resistant

Rice sheath mite

Sheath mite damage found to be significantly higher on 135d varieties (Bg 379-2 and Bg 450) as compared to 120d variety, Bg 403 and 105d variety, Bg 358. This could be partly attributed to the long vegetative period in 135d varieties that enhances population growth (Wijesekera, 2002)

Pests of stored rice (*Sitotroga cerealella*, *Sitophilus oryza* & *Tribolium castaneum*)

A study on the tolerance level of recommended varieties to grain pests, revealed that the weight loss in Bg 403 (4 month) and Bg 379-2 (4½ month) within a period of 8 months was significantly low 8 - 12% as compared to the weight loss in At 405 and Bw 450. The information collected from this study could be directly applied in the sanitation of stored grains against invading pests (Dharmasena and Abeyasiriwardena, 2003).

Interventions for Pest Management

In spite of the plethora of preventive options available at hand for managing pests, farmers found to place highest level of trust and confidence on interventions through pesticide for their pest problems (Nugaliyadde *et al.*, 1997). One such option is the use of natural enemies of rice pests that occur in abundance in the ecosystems (Hidaka *et al.*, 2001). A study on the insecticide usage by rice farmers has revealed that they use a least amount of insecticides (1.2 insecticide applications/farmer/season) in Sri Lanka as compared with similar tropical Asian countries where the number may range from 3 to 6 applications). Inquiry into the recent data on insecticide usage in rice would provide a closer look at the rice

farmers' perception on insecticides in the country and develop avenues for minimizing and regularizing pesticide usage in rice in the country. In addition, very few studies have been undertaken to evaluate the non target effects of insecticides on pests which is considered as essential component in recommending pesticides for pest management (Dissanayake and Nugaliyadde, 2002).

Botanicals for rice pest management

An water based emulsion containing 2% Neem-oil (v/v), 2% garlic extraction (w/w) and 0.5% soap powder (w/w) applied 2 times in 10 days apart at the flowering stage found effective in reducing BPH and paddy bug populations (Table 6 and 7).

Table 6: BPH populations in fields treated with neem based products at RRDI, Batalagoda.

Treatment	BPH population (No. / hill)*			
	Before treatment	3 days after 1 st treatment	10 days after 1 st treatment	10 days after 2 nd treatment
Neem oil + Garlic	20.7 a	03.7 b	04.0 b	0.7 b
Neem oil	23.0 a	06.9 b	02.8 b	0.9 b
Control	30.9 a	24.1 a	11.5 a	3.9 a
CV%	23.5	22.8	32.8	20.7

*Means in a cloum followed by the same letter are not signifiant at 5% probability level(Source: H.L.K. Liyanage – unpublished)

Table 7: Paddy bug populations in fields treated with neem based products at RRDI, Batalagoda.

Treatment	Paddy Bug Population (no./m ²)*	
	Before Treatment	After Treatment
Neem oil + Garlic	3.0 a	0.1 c
Neem oil	2.8 a	1.1 b
Control	3.4 a	2.9 a
CV%	19.7	51

*Means in a cloum followed by the same letter are not signifiant at 5% probability level(Source: H.L.K. Liyanage – unpublished)

New recommendations

In order to provide the best option for farmers for their pest problems, RRDI and RRRDC continued testing of new molecules / formulations against the major pests using the guidelines available. Low dose technology, environment and applicator safety target specificity and novel mode of action were the main concern in selecting the products for testing at laboratory and filed levels. BPH and leaf folder found to be the main target of these new molecules indicating, indirectly, of the importance of

these two pests in rice cultivation. During past 10 year period, RRDI has tested more than 17 compounds and recommended 16 insecticides for commercial use (Table 8). It is expected that the large profile of insecticides would help farmers to select the best of their choice and indirectly in managing resistance development in insects to insecticides. Framers lack of understanding of the different modes of action of these insecticides and the importance of correct handling and use of these products have hindered realization of the concepts underline these recommendations (DOA, 2003).

Table 8: Insecticides tested against rice pests.

Common Name	Pest	Recommended dose (ml or g FP /ha)	WHO Class
Novaluron 100g/L EC	BPH, Leaf folder	320 -400	III
Carbosulfan 200g/L SC	BPH	800- 1,000	II
Thiamethoxam 700g/kg WS	Thrips (seed dressing)	20g/100 kg	III
Neemgrow 20g/kgWP	Leaf folder		IV

Azadirachtin 10 g/L EC	Leaf folder	1,600 - 2,000	IV
Chlorfluazuron 50g/L EC	Leaf folder	256 – 320	III
Chromafenozide 50g/L	Leaf folder	320 – 400	III
Thiamethoxam 25% WG	BPH	96 – 120	III
Imidacloprid 200g/L OD	Thrips, BPH	160 – 200	III
Imidacloprid 700g/kg WG	Thrips	48 – 60	III
Ethiprole 100g/L SC	Thrips, BPH	480 – 600	III
Thiocyclam 4G	BPH	25kg	III
Thiocyclam 500g/kg Sp	BPH, Paddy bug	800-1,000	III
Flubendiamide 240 g/kg WG	Leaf folder	48	III
Flubendiamide 480g/L SC	Leaf folder	24	III
Profenofos 500 g/L EC	Sheath mites	1,000	II
Acephate 97 DF	Leaf folder	Not recommended	II

Revalidation of recommended insecticides for the management of rice leaf folder

It is well documented that rice leaf folder is the main target pest against which farmers initiate application of insecticides and that the crops treated with broad spectrum neuro-toxic insecticides for the control of this pest become more vulnerable to BPH damage (Nugaliyadde *et al.*, 2001c; Nugaliyadde *et al.*, 2001d; Nugaliyadde *et al.*, 2003a; Chien *et al.*, 2010). RRDI has conducted several investigations to demonstrate negative impacts of neuro-toxic compounds on natural enemies of rice pests.

One such investigation has revealed that an organophosphate compound (acephate) and insect growth regulator (tebufenozide) are equally effective against leaf folder. Contrary, the

organophosphate compound performed negatively on the natural enemies of rice pests as compared to the insect growth regulators (Table 9). Therefore, RRDI would initiate steps to change the insecticide recommendations in view of this situation to enable obtain maximum services of the natural enemies and to avoid secondary pest out breaks. Furthermore, efforts will be made to encourage farmers to avoid unnecessary applications of insecticides at the vegetative stage of rice, especially against the leaf eating pests such as leaf folder and cutworms and to promote safer and target specific insect growth regulators.

Table 9: Leaf folder and common natural enemy populations* in plots, before and after treating with an organophosphate (OP) and insect growth regulator (IGR) separately at RRDI, Batalagoda in 2009.

Pest/ Natural Enemy	Time of Sampling**	Untreated Control	Tebufoenozide (IGR)	Acephate (OP)	LSD
Leaf folder	BT	81.6	82.8	67.8	4.5
	14DAT	0	0	31.2	
Othoptera	BT	7.3	7.4	8.9	1.4
	14DAT	8.4	5.4	0.3	
Odonata	BT	0	0.3	0.5	0.2
	14DAT	1.3	2.6	1.5	
Hymenoptera	BT	10.9	5.8	7.3	0.5
	14DAT	12.7	4.0	1.5	
Spiders	BT	2.6	1.0	3.4	1.3
	14DAT	6.0	4.5	0	

* Leaf folder – No. of larvae / 100 folded leaves, Other fauna – No. per 10 sweep net sample

**BT-Before treatment, 14DAT- 14 days after treatment

More supportive information in available from the toxicity studies conducted at RRDI on leaf folder larvae. It has been revealed that thiodicarb (Carbamate) and chlorpyrifos (Organophspate) had

higher LD50 values than acephate (Organophosphate) and nuvaluron (an Insect Growth Regulator) (Table 10). These results showed the high efficacy of the latter for leaf folder control (Weerasiri, 2000).

Table 10: LD50 values (ug/g body weight) of some insecticides on 3rd instar leaf folder larvae at RRDI, Batalagoda in 2002.

Insecticide	Ld 50 ug/g body weight	Fiducial limits	
		Max.	Min.
Chlorpyrifos 400g/L EC	30.0	45	19
Acephate 75SP	12.5	24	6
Thiodicarb 375F	188.5	230	153
Nuvaluron 10EC	7.6	13	4

Resistance development of pests to insecticides

Studies conducted elsewhere have shown that BPH have high potential for the development of resistance to insecticides when the product is used extensively (Ling *et al.*, 2009). Similarly BPH population at Batalagoda has been found to have developed resistance to two neo-nicotinoid insecticides (imidacloprid and acetamiprid) from 2000 to 2009 (Nugaliyadde *et al.*, 2000a; Senadhipathi, 2009; Ling *et al.*, 2009). These observations, if available over a longer period of time could provide some guidelines for revision of the recommendations.

Laboratory and field experiments conducted at RRD I have shown that chloropyrifos, which is extensively used by farmers against rice leaf folder, is becoming less effective. Similarly, fenobucarb was also found to be less effective against BPH as compared to other recommended insecticides (DOA, 2003). Extensive studies on the toxicity of commonly used insecticides to pest populations from Batalagoda and Angunakolapelessa revealed that most of the tested rice insects have developed resistance to pyrethroids, chlorpyrifos and fenobucarb, as compared to pests of vegetables (Weerakoon *et al.*, 2001; Karunaratna *et al.*, 2007). This information demands immediate attention from the regulatory body for further action.

Pesticide residues

Non availability of local information on the fate of pesticides in the environment after their release has been a matter of debate over decades. Lack of a regularized mechanism for such activity has resulted scattered interest in collecting information on residues of insecticides in ecosystems. As a result, suggestions are made to follow “Precautionary Principles” in dealing with issues like this and to help redress pesticides recommendations, particularly the OPs and Carbamates that have been in use over 3 decades in rice. A study conducted at RRD I on the fate of carbofuran 3G revealed that its residue level fall below detectable levels in 2 days in water and 10 days in plant (Ranwala *et al.*, 2008). Similar studies conducted elsewhere on the carbofuran 3G in ground and surface water have revealed residue levels at low alarming rates (Prof. J. Liyanage, Personal communication). It is suggested that concerted efforts are timely needed for the advancement of food and environment and human health quality in the country.

Effect of botanicals on storage pests

At present many farmers use a number of locally available plants such as citrus, cinnamon, neem to protect stored paddy against the common pests. A test conducted at RRRDC revealed, papaw leaves at the rate of 50 g/5kg was effective in controlling *Sitotroga cerealella* in stored paddy followed by lime and Nika leaves. Similarly

a comprehensive study conducted to determine the effect of essential oils of some medicinal plants, *Cymbopogon citratus*, *C. nardus* and *Cinnamomum zeylanicum* found that these oils act on grain weevil (*Sitophilus oryzae*) and rice moth (*Sitotroga cerealella*) to reduce their survival and reproductive rates and hence could be used for the control of stored pest and inhibit growth of *Aspergillus flavus* (Link) and aflatoxin formation (Paranagama *et al.*, 2002a and 2002b; Jayaratne *et al.*, 2002; Nugaliyadde *et al.*, 2003a; Paranagama *et al.*, 2003).

Use of pheromones for rice moth management

In order to develop a user friendly approach for household and large scale paddy moth control, a pheromone synthesized at the Phero Bank, Netherlands was tested to determine the efficacy in attracting paddy moth. Rubber septa loaded with 10 mg of the pheromone attracted an average of 5.9 moths/day during the testing period of 2 weeks (Kanthi *et al.*, 2006). The attraction to pheromone found to be dose dependant within the tested range of 1mg - 10 mg/septa. It is expected these pheromones would reduce the in house use of toxic insecticides and minimize the human health hazards.

Natural Enemies of Rice Pests

Rice ecosystems in the tropics are enriched with complex guilds of fauna with a predominant feature of

abundant natural enemies (Hidaka and Nugaliyadde, 2000). As such, the government declared Integrated Pest Management (IPM) as the Sri Lanka national policy for the management of agricultural pests where natural enemies of pests play a major role. This is exemplified by the numerous activities on IPM and Integrated Pest and Vector Management (IPVM) programs launched by the Plant Protection Service in collaboration with other line institutes of the Department of Agriculture since early 1980s through assistance from FAO and other foreign donor organizations. However, available literature on the natural enemies and their role in managing pests to prevent them emerging into epidemic levels is noticeably scanty. This low profile given on this significant sector in rice lead to misunderstanding of the services one can obtain from the ecosystem for their pests problems. Hence a need is extremely topical for an uplift of awareness programs on natural enemies of rice pests.

Egg parasitoids of paddy bug

Being the main key pest of rice, emphasis has been given to study the natural control mechanisms of paddy bug since early 1920s (Austin, 1922; Kobayashi and Nugaliyadde, 1988). Studies on natural enemies of paddy bug have focused mainly on the egg parasitoids, *Gryon nixoni*, as it is the common and significant parasitoid known in many parts of the rice growing countries. Field studies conducted at RRDI, RRRDC and in farmer's

fields revealed that the egg parasitism may vary from 8% to more than 70% depending on the parasitoid load. A need to use ecosystem engineering tools in rice ecosystems to enhance the parasitoid abundance is highlighted as the paddy bug eggs occur in the field only during the flowering stage of the crop. Growing of pollen and honey bearing weeds in the rice systems for the parasitoid adult to survive during the 'off season' and making alternate egg source for the sustenance of this parasitoid have been discussed (Ms. S. Ratnapala, unpublished).

Spiders

Spiders are considered as the predominant predators of rice pests in rice ecosystem in Sri Lanka. In house trials revealed that wolf spider (*Paradosa pseudoannulata*) could prey an average of 14 BPH per day and can tolerate a temperature range of 10-40°C while their peak predatory activity was observed between the temperature range of 25-35°C (Abeyratna, 2002). *Argiope aurantia* is a late colonizing orb-web spider common in rice ecosystems in Sri Lanka. In rice fields received no insecticides, the web density estimated to increase from 4/ha to 80/ha from 9th to 14th weeks old crop. This is accounted to a high reproductive rate of the spider (450 spiderlings /egg case) that may help population explosions in a short period in insecticide free rice fields (Shyamalee, 2009). Furthermore, *Argiope* preferred to prey on homopterans (8-10 homoptera/day) than coleopterans (0.1 beetles/day),

indicating the valuable ecological service provided by this spider in managing the homoptera pests of rice.

Future Research and Development Needs

Enhancing biodiversity in rice ecosystem

In view of the emerging demands for sustainable management methods for rice pests, emphasis should be given to develop strategies, techniques, options and policies to enhance naturally occurring biological functions such as predation, parasitization and pathogenesis for the management of pests. For this purpose, an in depth knowledge on the natural ecosystem is imperative, to enable manipulate habitat and enhance the floral and faunal biodiversity. Through improved floral diversity, ample food and refuge resources could be provided for the parasite and predator populations and enhance their survival and services for pest management (Kalubowila *et al.*, 2007; Catindig *et al.*, 2006)

In this respect, research programs should be initiated to catalogue nectar-producing weedy plants that can be grown in rice bunds and adjacent lands; and to determine the abundance of natural enemies in fields planted to such weeds. These activities should help design methods to educate farmers and enhance their competence on non-chemical pest management.

Improving pesticide spraying technology

The present understanding on the proper use of pesticides including selection of correct pesticide, decision of correct dose and dilution, and selection of appropriate nozzles and spraying procedures are scanty, faulty and misleading among the farmers and spray operators (Nugaliyadde *et al.*, 2003b). This is largely attributed to the lack of training provided to the farming community and non-availability of rich information for extension programs. Hence, it is suggested to initiate activities to develop locally suitable spraying technologies and educate farmers on the correct method of spraying, nozzle selection etc. This program would help obtain maximum benefits of the products, minimize misuses of pesticides and provide opportunities to improve occupational health and safety standards of the farming community (DOA, 2009).

Revision of pesticide recommendation

Information received from the extension service of the DOA and farmers on the low effectiveness of some recommended insecticides against rice pests need to be investigated to determine whether this is a faulty judgment or otherwise. The methods adopted in other countries on similar issues are exemplary, such as, regular monitoring of the toxicity levels of insecticides that provide a greater insight into this issue need to be noted. Actions to prevent resistance development in pests to pesticides

should be given a priori. Policy directions to make relevant revisions to the management recommendations should be assured.

Pest surveillance

Surveillance is one of the principle tools in integrated management of pests. As an island country, Sri Lanka has ample opportunities to implement a successful surveillance system on administrative district basis or agro-ecological zone basis to help research and extension services to implement relevant management programs and for farmers to practice best options available to them (Hidaka and Nugaliyadde, 2000; Hidaka *et al.*, 2001). The pest surveillance program implemented by the Plant Protection Service of the DOA a few decades ago has provided useful information to the extension and research arms of the DOA on the incidences and intensities of pests out breaks on district basis. This system, however, ceased to continue with the restructuring of the extension activities of the DOA and for the lack of required initiatives and policy backing. The initiatives taken recently by the DOA to collect data on pest incidences on monthly intervals from inter provincial systems should be highlighted.

Institutional linkage

The advanced information technological tools would provide opportunities for the rice sector to maximize the utilization of available resource to achieve high level of success in pest management operations

in the country. Out sourcing of responsibilities would also be a possible answer in situations where human resources and other valuable facilities are limiting. It would provide avenues to improve productivity and efficiency. University System, private sector and non-governmental organizations have all potentials in complementing the government sector responsible in

rice industry provided avenues and mechanisms are provided for efficient integration. What is articulated here yet to be realized as the mainstream of pest management; we need to understanding that overcoming decades of believes is an emotionally taxing exercise.

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Management of Rice Diseases in Sri Lanka During the Last Decade

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Abstract

Rice diseases are one of the major biological constraints that cause yield losses worldwide. Among the diseases, Bacterial leaf blight, Rice blast, Sheath blight and Brown spot are the economically most important diseases in Sri Lanka. Recent research findings in respect of these rice diseases are discussed and strategies to overcome them are suggested. Breeding for disease resistance has to be continued a priority. Plant pathologist should pay attention to study the nature of host resistance and interaction between the plant and pathogen. This will enable to find suitable donor varieties to develop the varieties with multiple resistances. Use of new technologies such as marker-aided selection is also important during the diseases screening to incorporate multiple resistance into new varieties. The agronomist may be involved in developing an appropriate package of cultural and management practices to prolong the durability of resistance and to minimize the occurrence. Use of fungicides should be the last effort for disease management.

Key words: Rice diseases, Bacterial leaf blight, Blast, Sheath blight, Brown spot

Introduction

Rice diseases are one of the major biological constraints to increase rice yield in Sri Lanka and also in all the rice growing countries around the world. Among the rice diseases, Bacterial leaf blight, Rice blast, Sheath blight and Brown spot are the economically most important diseases.

Rice blast

Rice blast disease (caused by *Pyricularia grisea*) incidence is seen mainly at two crop growth stages. The leaf blast is at the seedling stage and panicle blast is prevalent

at the flowering stage. Panicle blast is reported specially in the Maha seasons where the flowering period coincides with conditions favorable for the development of disease from December to January.

Among the control measures, the use of resistant varieties is more practical than use of fungicides, which are more expensive. Advance progenies obtained from the varietal improvement programmes at Batalagoda, Bombuwela, Labuduwa and Ambalantota were screened at Rice Research & Development Institute (RRDI) at Batalagoda and Regional Rice Research and Development Centre (RRRDC) at Bombuwela.

While many advance progenies were identified as resistant and among the varieties recommended during the last decade, Bg 454, Bg 406, Bg 250, At 362, Bw 361, Bw 363, Bw 364, At 306, At 307 and At 308 were identified as resistant to the leaf blast disease.

The analysis of genetic ancestry of varieties recommended in Sri Lanka has clearly shown that the overall gene base is very narrow especially in respect of rice blast (Dissanayake 1998). Therefore, in the rice varietal improvement programme, identification of new sources of resistance from introduced materials is important. Nineteen entries were

identified as resistant for leaf blast from the 30th International Rice Blast Nursery (IRBN) of International Network for Genetic Evaluation of Rice (INGER) programme which was conducted at RRRDC, Bombuwela (Personal communication). The selected blast resistant entries are given in Table 1. These entries are being used by breeders in their breeding programmes and also conserved at Plant Genetic Resource Centre (PGRC), Gannoruwa for future use. Further, traditional varieties Kottayar, Sudu Heenati, Beheth Heenati and Kattamanjal could also be used as blast resistant sources (Vidanapathirana *et al.*, 2006).

Table 1: Characters of the Rice blast resistant entries of the 30th International Rice Blast Nursery.

Entry No.	Designation	Origin	Days to 50% Flowering	Culm length (cm)	Grain Type
57	IR 73971-87-1-1-1-1	IRRI	79	64.3	White, Long
80	WAR115-1-2-4-2-9-B-B-2	Sierra Leone	77	69.4	Red, Long
40	HTAFR81051-1-5-17	Thailand	77	64.1	White, Long
33	ARN CH 3-2	China	82	85.7	White, Long
75	IR 77701-129-1-1-2-2	IRRI	79	80.7	White, Long
69	IR 66043-2B-R-3-2-3-3-3	IRRI	78	57.8	White, Long
76	RD 19	Thailand	79	63.2	White, Long
79	VANDANA	India	68	63.9	White, Long
62	IR 77186-123-3-3-3	IRRI	79	72.8	Red, Long
85	WAR 72-1-1-1-4	Sierra Leone	82	59.8	White, Long
77	REJNINI(MO.12)	India	89	53.5	White, Long
81	WAR 115-1-2-4-2-9-B-B-6	Sierra Leone	80	79.3	White, Long
52	IR 72911-191-1-1-2	IRRI	92	64.5	White, Long
89	WAR90-6-1-1-B-B_3	Sierra Leone	79	87.5	White, Long
72	IR 71146-97-1-2-1-1-3	IRRI	83	81.5	White, Long
82	WAR 120-1-5-1-1-B-B-1	Sierra Leone	78	54.0	White, Long
60	IR 76240-B-R-R-R-29-3	IRRI	74	66.3	White, Long

The location differences of blast resistance among rice varieties and breeding lines at the blast screening sites at Batalagoda, Makandura and Peradeniya have been observed (Dissanayaka *et al.*, 1998). Therefore, the genetic variation of rice blast pathogen isolates in different agro ecological zones in Sri Lanka is now being studied.

Even though fungicide application is expensive, when there are blast outbreaks under favorable weather conditions, recommended application of fungicides is important. Fungicides Tebuconazole Ew, Isoprothiolate Ec and Tricyclazole 75% WP have been recommended during past decade.

Bacterial leaf blight disease (BLB)

Bacterial leaf blight disease (caused by *Xanthomonas campestris* pv. *oryzae*) is the only bacterial disease found in the rice growing regions in the country. The disease incidence depends on cultivation practices, soil fertility, weather conditions and the season. This disease is known to cause seedling blight or 'Kressek' during the early growth stage. During Maha 2009/10, Kressek stage was recorded in variety Bg 358 at Kalutara, Gampaha, and Galle districts (Personal communication). The problem was overcome by draining of water and keeping the fields dry. Fertilization of the crop improved the growth of the plants and the yield loss recorded was minimum. Pathogenicity of the pathogen varies depending on the location (Noda *et*

al., 1990) and as such the genetic variability and pathogenicity of seven isolates of BLB causal organism were studied by Randomly Amplified Polymorphic DNA based - Polymerase Chain Reaction (RAPD-PCR) using three primers namely OPD7, OPD20 and OPK16 (Fernando *et al.*, 2005). It was found that there was high variation of the BLB causal organism.

Incorporation of resistance to BLB is a major objective of the rice improvement programme of Sri Lanka in order to overcome this problem. Variety Bg 358 which is BLB susceptible was crossed with IRBB 60 - a pyramid line carrying resistant gene Xa-21 for BLB. The breeding program was carried out until F5 generation. Marker Assisted Selection was carried out with the specific primer pTA 248 for Xa-21 gene. Plants of filial generation of IRBB x Bg 358 were resistant to most virulent isolate of BLB in the field (Fernando *et al.*, 2007).

The elite breeding lines developed at Batalagoda, Bombuwela, Labuduwa and Ambalantota are screened for disease resistance under natural condition at Batalagoda. Out of the varieties recommended during the last decade, Bg 454, At 362, Bw 363, Bw 364, At 306, At 307 and At 308 were identified as resistant to BLB. As the known sources of resistance are limited acquisition of new sources of resistance from introduced materials is important. Eight BLB resistant entries were selected to BLB from the 24th International Rice Bacterial Leaf Blight Nursery (IRBBN 2009)

of INGER programme which was conducted at Rice Research Station, Labuduwa (Personal communication). The resistant entries and their

characters are given in Table 2. These entries are being used in rice breeding programme and also conserved at PGRC, Gannoruwa.

Table 2: Characters of the Rice Bacterial Leaf Blight resistant entries identified from the 24th International Rice Bacterial Leaf Blight Nursery (IRBBN 2009).

Entry No.	Designation	Origin	Days to 50% Flowering	Culm Length (cm)	Pericarp Colour	Grain Type
IRBB 21	IR-BB59 (IR72920-1-2-40)	IRRI	88	85	White	Long slender
IRBB 52	GZ 5830-63-1-2	IRRI	91	83	White	Long slender
IRBB 53	UPR 84-21	IRRI	87	82	White	Long slender
IRBB 58	IR 728890-81-3-2-2	IRRI	93	85	White	Long slender
IRBB 59	IR 72894-35-2-2-2	IRRI	93	86	White	Long slender
IRBB 60	IR 72903-121-2-1-2	IRRI	89	87	White	Long slender
IRBB 65	IR 79584-38-2-1-4	IRRI	87	86	White	Long slender

Sheath blight disease

In early 1980, Sheath blight disease (caused by *Rhizoctonia solani* Kuhn) was a major problem in the Low Country Wet Zone. After studying the epidemiology of the disease a package was developed to minimize the disease incidence (Mithrasena *et al.*, 1989). With the introduction of the new improved varieties with high tillering ability and droopy leaves, the disease has also spread into the Dry Zone areas too where the high seed rates ranging from 200 - 400 kg/ha are used.

As there are no disease resistant varieties for sheath blight disease, fungicide application is the only available method for disease management. Hexaconazol Ec, Thiophonate 70% WP and Propiconazole EC were recommended recently by the Department of Agriculture (DOA, 2010) in addition to the previously recommended fungicide Pencycuron 25% WP (Mithrasena *et al.*, 1989).

There are several reports on the richness of the rice plant and its ecosystem in terms of beneficial microorganisms having antagonistic effects against

rice pathogens (De Costa *et al.*, 2006a; De Costa *et al.*, 2006b). The rice phyllosphere harbours antagonistic bacteria and fungus that can inhibit the growth of the sheath blight pathogen of rice.

The epiphytic microbial antagonist against rice sheath blight disease has been studied (De Costa *et al.*, 2008b). Among the total of 196 bacterial and 91 fungal isolates, 12 bacterial and 2 fungal isolates have shown more than 50% growth inhibition of fungus. It is the general belief in biological control that most of the antagonistic organisms that are effective in vitro are not so successful in vivo or vice versa. But this study has shown in-vitro and in-vivo antagonistic efficiency of phyllosphere epiphytic microorganism of rice in controlling sheath blight. Among the microbial antagonists tested under field condition were *Bacillus megaterium*, *Bacillus subtilis* and *Aspergillus niger*.

A formulated product of the microbial antagonists is more suitable and would be long lasting in comparison to cell suspensions of antagonist that have been used in the pot and field experiments. A talc-based formulation of the microbial antagonists (i.e. *A. niger*, *B. megaterium* and *B. subtilis*) has shown to have the potential to reduce sheath blight incidence and severity under field conditions (De Costa *et al.*, 2008a).

Moreover, field experiments were conducted at the RRRDC, Bombuwela to study the relationship between soil factors and pre-plant density of *R. solani*

sclerotia in the soil (Nanayakkara, 2005). The effect of polythene mulching, straw incorporation and their combined effect on the viability of *sclerotia* were studied. According to the results, use of polythene mulching five weeks prior to rice seed sowing is recommended to minimise sheath blight diseases in rice cultivation.

The early detection of the pathogen *R. solani* in the soil is also important in effective control of the disease. Traditional diagnosis methods that involve isolation and microscopical examination of the mycelium are time consuming and laborious. Therefore, DNA probe and PCR based diagnostic assay have been developed for specific detection and identification of *R. solani* from the soil (Weerasena *et al.*, 2004).

Leaf spotting and grain discoloration

This is caused by a number of fungi, bacteria and environmental conditions. Fungal pathogens *Cuvularia*, *Fusarium*, *Colletotrichum*, *Helminthosporium* were isolated from infected leaves and grains (Vidanapathirana *et al.*, 2006). Studies carried out at RRDC, Bombuwela found that this pathogen population was higher inside the leaf canopy during day time and it was higher above the canopy during the night (Personal Communication). This grain discolouration is a serious problem in the wet condition during the milking stage up to ripening. With changing climatic conditions, application of fungicides is required during the period of flowering to

maturity to control this disease. The fungicide testing experiments are being conducted at Batalagoda, Bombuwela and Aralaganwila to recommend effective fungicide to overcome this problem.

Recently, rice sheath mite (*Stenoeotarsonemus spinki*) infestation was also identified in fields subjected to grain discoloration. Initial mite infestation at the tillering to booting stage of the crop could be induced by the pathogen as a secondary infection of grain discoloration. Further investigations are being carried out at Bombuwela and Batalagoda.

In addition to the major disease problems mentioned above, Sheath rot (*Sarocladium oryzae*) and Leaf scald (*Rhynchosporium* sp.) diseases have also been reported at low intensities. Virus diseases have not been recorded in rice fields during the last two decade. Leaf yellowing was observed at various crop growth stages from different parts of the island. However, plants recovered when fertilizer was applied suggesting the cause to be nutrient deficiency.

Future strategies

Future strategies for the control of most of these diseases should focus on integrated effort of the plant breeder, agronomist and plant pathologist. Breeding for disease resistance has to be considered a priority. The major task of the plant pathologist should be to study the nature of host resistance and interaction between the plant and pathogen. This will enable to find suitable donor varieties and develop the varieties with multiple resistances. Use of new technology such as marker-aided selection is important in diseases screening to incorporate this multiple resistance into new varieties. The agronomist should develop an appropriate package of cultural and management practices to prolong the durability of resistance while keeping the occurrence and frequently to a minimum. Use of fungicides should be the last effort for disease control.

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Management of Rice Sheath Blight with Emphasis on Biological Control

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Abstract

*Rice sheath blight, a significant biotic constraint of rice cultivation, can be managed effectively and sustainably using biological control measure particularly the formulated microorganisms isolated from naturally-inhabiting microflora on different tissues of the rice plant in combination with other available control measures. Being indigenous to the rice ecosystem, these fungal and bacterial antagonists possess desirable features which would enable them to survive better and effectively target the pathogen, to control the disease successfully. In addition to the direct antimicrobial action against the pathogen, *R. solani*, many of the microbial antagonists exhibit additional desirable attributes such as growth promoting ability and induction of systemic resistance through production of defence-related proteins. This paper describes the economic significance of rice sheath blight and limitations of its control measures presently available. Then it goes on to describe a case study from Sri Lanka to highlight the prospects of controlling rice sheath blight using indigenous microflora, the modes of antagonism of these microbial antagonists and biopesticides, especially against *R. solani*. A series of studies conducted using the indigenous microbial antagonists aimed at the development of biopesticides to control rice sheath blight in Sri Lanka is described in detail with intended future research.*

Introduction

Sheath blight of rice caused by *Rhizoctonia solani* AG1, is an emerging threat to rice cultivation worldwide including South and Southeast Asia, with Sri Lanka being no exception. A combination of environmental and crop management factors have contributed to the upsurge of sheath blight in rice ecosystems. Crop intensification and changes in cultivation practices which are indispensable to achieve comparatively higher yield per unit

area to meet the growing food demand of ever-increasing human population have created favourable conditions for the sheath blight pathogen in rice ecosystems. Intensified production systems of rice rely heavily on the use of high-yielding varieties, high crop densities and high rates of application of fertilizer, especially nitrogen. Moreover, environmental conditions, generally experienced by many of the rice growing regions of the world, particularly the ambient temperature ranging from 23-35°C, frequent

rainfalls and high relative humidity (>95 %) accelerate the sheath blight development (Mew *et al.*, 2004a; Savary *et al.*, 1995). For example, the estimated yield reduction from sheath blight in plots receiving higher N rates than the recommended rate ranged from 20 to 42% in artificially inoculated plots (Cu *et al.*, 1996). Not only the Asian countries, but also the rice growing areas of the Southern United States are experiencing 4-50% rice grain losses due to sheath blight under conditions favourable for disease development (Groth and Bond, 2007).

Among the rice-growing regions of Sri Lanka, the Low Country Wet Zone faces the strongest threat from sheath blight infestation (Wickramasingha and Mithrasena, 1989) because the environmental conditions in this agro-ecological zone are ideal for sheath blight development and spread. Apart from the Low Country Wet Zone, rice ecosystems in the Amparai district of Sri Lanka also have a micro-environment conducive for development of sheath blight because of the high seed rates and consequently high plant densities used by farmers in this region. Losses due to sheath blight have been attributed to reductions in grain yield, grain quality and milling yield and increased lodging of the crop (Groth and Bond, 2007). As highlighted by Cu *et al.* (1996), under tropical intensive rice production systems, yield reductions due to sheath blight occurred primarily due to reduction in mean seed weight and comparatively a lower percentage of filled spikelets. Under Sri Lankan conditions, empty ear heads have been

observed in sheath blight infested fields and yield losses as high as 25% could be experienced, depending on the growth stage of the crop infested and the level of infection.

Management of rice sheath blight

No currently available method is adequately effective in controlling rice sheath blight because of several reasons involved with its host-pathogen interactions (Mew *et al.*, 2004a). One reason is that the pathogen has a wide host range including wild rice. Other than a very recent report in 2010 on identification of genetic resistance to sheath blight in several wild rice species and mapping of a genetic region in the rice genome which has a major effect on control of sheath blight (Yao, 2010), no significant resistance has been identified among cultivated rice varieties, despite testing of more than 30,000 accessions of rice germplasm (Mew *et al.*, 2004b). Production of transgenic rice by introduction of defence-related genes (e.g. Chitinase and Pathogenesis-related proteins) has been attempted with limited success because of the lack of consistent expression of the transgene (Datta *et al.*, 1999). Strict use of the recommended dosage of N fertilizer, low tillering varieties, low planting densities, application of silicon and removal of crop debris infected with the sheath blight pathogen are the agronomic practices that could minimize the development of sheath blight (Rodrigues *et al.*, 2003). Chemical control is practiced using

a few types of effective fungicides mostly in temperate or sub-tropical environments when the infestation levels are severe (Mew *et al.*, 2004a). Fungicides, Pencycuron (Monceren) and Tri phenyl tin hydroxide have been reported to control sheath blight effectively under Sri Lankan conditions (Mithrasena *et al.*, 1987). However, as highlighted by Mew *et al.* (2004a), use of chemicals by small-scale rice farmers is not sustainable and has many adverse impacts on the environment. Furthermore, declining efficiency of routinely-used fungicides has been observed. A striking example is the decreasing efficiency of Jinganmycin, which has been used continuously in China for 30 years to control sheath blight.

Biological control

Biological control using naturally-inhabiting microbial antagonists has been suggested as a promising option for management of rice sheath blight, especially for resource-poor farmers (Mew *et al.*, 2004a; Kanjanamaneesathian *et al.*, 2007; Someya *et al.*, 2005). The Rice ecosystem including its rhizosphere and phyllosphere is a rich source of microbial antagonists against many rice pathogens such as *R. solani*, *Magnaporthe grisea*, *Fusarium moniliforme*, *Sarocladium oryzae* (Mew and Rosales, 1992; Cottyn *et al.*, 2001; De Costa *et al.*, 2006). Among a series of fungal and bacterial antagonists isolated from the rice ecosystem, bacterial antagonists such as *Bacillus subtilis*, *B. pumilus*,

B. laterosporus, *Pseudomonas fluorescens*, *Burkholderia cepacia* and *Pseudomonas putida* (Mew *et al.*, 2004a; Rosales *et al.*, 1993; De Costa *et al.*, 2006) and fungal antagonists such as *Aspergillus niger*, *Trichoderma* spp. have been recorded as potential candidates to control rice pathogens (De Costa *et al.*, 2006; Krishnamurthy *et al.*, 1999; Kandhari *et al.*, 2000).

Production of fungal-inhibitory substances by *Bacillus* spp. (Rosales *et al.*, 1995), secondary metabolites such as 2, 4- diacetylchloroglucinol and pyrrolnitrin by *Pseudomonas putida* and *Burkholderia cepacia*, respectively (Rosales *et al.*, 1995) and phytohormones by *Bacillus* spp. (Utkhede and Rahe, 1983) have been reported. *B. coagulans* has shown the ability to inhibit the growth of *R. solani* by inhibition of ergosterol biosynthesis (Czarczyk *et al.*, 2002). In addition to their fungicidal/fungistatic nature, these microbial antagonists perform growth promoting activities such as increased seed germination and plant vigour and induce systemic resistance of the host plant (Mew *et al.*, 2004a). Application of *Pseudomonas fluorescens* strains has increased the systemic resistance of rice tissues against *R. solani* by increasing the chitinase and peroxidase activity of tissues treated with the bacterial antagonist (Nandakumar *et al.*, 2001; Radjacomare *et al.*, 2004).

Production of mycolytic enzymes such as chitinases and 1-3 β glucanases by fungi and the detrimental effect of these enzymes on lysis of fungal cell

walls and subsequent degradation is well documented (Elad *et al.*, 1983). Ability to produce an endochitinase, chitobiosidase and N-acetyl β -glucosaminidase by *Trichoderma harzianum* and its ability to control soil borne fungal pathogens such as *R. solani*, *Sclerotium rolfsii* and *Pythium ultimum* has been reported (Cook and Baker, 1983; Pe'er and Chet, 1990). Ability to control *R. solani* by several strains of *Trichoderma viride* and *T. harzianum* has been demonstrated (Krishnamurthy *et al.*, 1999). Production of chitinase in culture filtrates of *A. niger* has been reported. Colony growth inhibition of several plant pathogenic fungi has been observed by the culture filtrate of *A. niger* (Mohamed, 2001) and production of citric acid and enzymes such as glucose oxidase and lysozyme has been reported (Bennett, 2010). In addition, potential microbicidal/microstatic effects of these chemicals can be anticipated.

Potential of developing commercial formulations of biopesticides

According to the definition by Schisler *et al.* (2004), a formulated microbial product is composed of biomass of a biocontrol agent and ingredients to improve the survival and effectiveness of the product. Spores, vegetative cells or combination of the two types of the propagules can be used as the biomass of the microorganism, although use of spores is more preferable due to the durable nature of the spores. A broad array of amendments or carriers have been used to formulate biocontrol

agents that improve the chemical, physical and nutritional properties of the formulated product, to name a few stabilizers, binders, stickers, thickeners, surfactants etc. (Schisler *et al.*, 2004). Following are some examples of microbial formulations available in the market containing different strains of *Bacillus spp.*; Serenade (*Bacillus subtilis* QST 713), Kodiak (*B. subtilis* GB03), Subtilex (*B. subtilis* MBI600), Yield Shield (*B. pumilus* GB34) and EcoGuard (*B. licheniformis* SB3086) for the control of fungal pathogens, while Serenade has shown effectiveness against several bacterial pathogens. In addition, *Pseudomonas fluorescens* as a talc-based formulation (Radjacomare *et al.*, 2004), *B. megaterium* and *B. pumilus* as pellet formulations (Pengnoo *et al.*, 2000), *B. megaterium* as an effervescent fast-disintegrating bacterial formulation (Wiwattanapatapee *et al.*, 2007) and *B. megaterium* as a floating pellet formulation (Wiwattanapatapee *et al.*, 2004) have been tried successfully under laboratory, greenhouse and field conditions. *A. niger* (commercially available as Kalisena) is in the Indian market to control rice sheath blight and several other soil-borne fungal pathogens (Kandhari *et al.*, 2000).

Potential of indigenous microbial antagonists for the control of rice sheath blight - a case study in Sri Lanka

Considering the beneficial attributes of biocontrol agents in controlling rice sheath blight, potential microbial antagonists were isolated from the phyllosphere of the rice plant,

especially from the tissues of rice sheath. Initiation of infection of *R. solani* and subsequent disease development takes place in the rice sheath. Hence, the microflora dwelling on the same site and having common space and nutritional requirements as the target pathogen would be more effective in controlling the pathogen. Therefore, microflora were isolated from the rice sheath of traditional and new-improved varieties grown in six locations belonging to the agroecological zones IL1, IL2 and WM2. All the fungal and bacterial isolates obtained from the isolations

were tested for antagonism against *R. solani*, in vitro by dual culture plate method as described by Adebola and Amadi (2010). Fungal and bacterial isolates which gave more than 50% antagonism under in vitro conditions were identified by morphological, biochemical and rapid identification systems such as BIOLOG (Biolog Inc. USA). The identified candidate antagonists, the rice varieties from which they were isolated and the possible modes of antagonism, based on in vitro antagonism, are presented in Table 1.

Table 1: Selected microbial antagonists, their identity, rice variety from which the antagonists were isolated and modes of antagonism shown on dual culture plates.

Identification Code	Microorganism	Rice variety	Mode of antagonism
HMWB4	<i>Dermaococcus nishinomiyaensis</i>	Horana Ma Wee	Chemical inhibition
Bg450B11	<i>Brevibacterium mcbrellneri</i>	Bg450	Chemical inhibition
Bg300B1	<i>Bacillus amyloliquefaciens A</i>	Bg300	Chemical inhibition
Bg352B1	<i>Bacillus amyloliquefaciens A/ Bacillus subtilis</i>	Bg352	Chemical inhibition
Bg379-2B15	<i>Bacillus licheniformis</i>	Bg379-2	Chemical inhibition
Bg379-2B16	<i>Bacillus megaterium A</i>	Bg379-2	Chemical inhibition
Bg94-1B5	<i>Bacillus licheniformis</i>	Bg94-1	Chemical inhibition
GbB5	<i>Bacillus amyloliquefaciens A</i>	Gonabaru	Chemical inhibition
B4	<i>Bacillus subtilis B</i>	Bg450, Bg350, At354	Chemical inhibition
Bg357B3	<i>Chryseobacterium indologenes</i> (syn. <i>Flavobacterium indologenes</i>)	BG357	Chemical inhibition
MB40	<i>Pseudomonas aeruginosa</i>	Murungakayan	Chemical inhibition
MB4	<i>Aquaspirillum dispar</i>	Murungakayan	Chemical inhibition
Bg379-2F1	<i>Aspergillus niger</i>	Bg379-2	Competition
Bg379-2F2	<i>Trichoderma spp.</i>	Bg379-2	Competition

The potential fungal and bacterial antagonists were tested for their ability to control sheath blight, in vivo in an open-field pot experiment using two different rice varieties namely, Bg 94-1 and IR8. The rice plants grown in 5l plastic pots were treated with the suspensions of antagonists as a soil drench 45 days after transplanting. Plants were inoculated with *R. solani* at

47 days after transplanting by applying 4 sclerotia/hill. The inoculated plants were treated with suspensions of antagonists, separately, 2 days after inoculation of the pathogen. Percentage disease incidence and disease severity were recorded ten days after inoculation and the number of days taken to cover 50% of the rice sheath by lesions was also recorded.

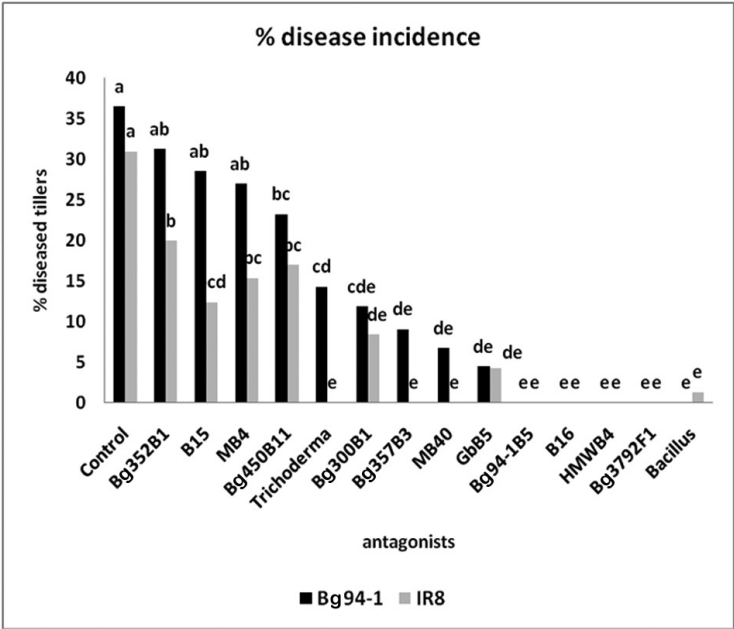


Figure 1: Percentage disease incidence of rice varieties Bg94-1 and IR8, treated with 15 different potential antagonists under open-field pot experimental conditions. Common letters across the treatments of a given variety are not significantly different at $p=0.05$ (Adapted from De Costa *et al.*, 2008).

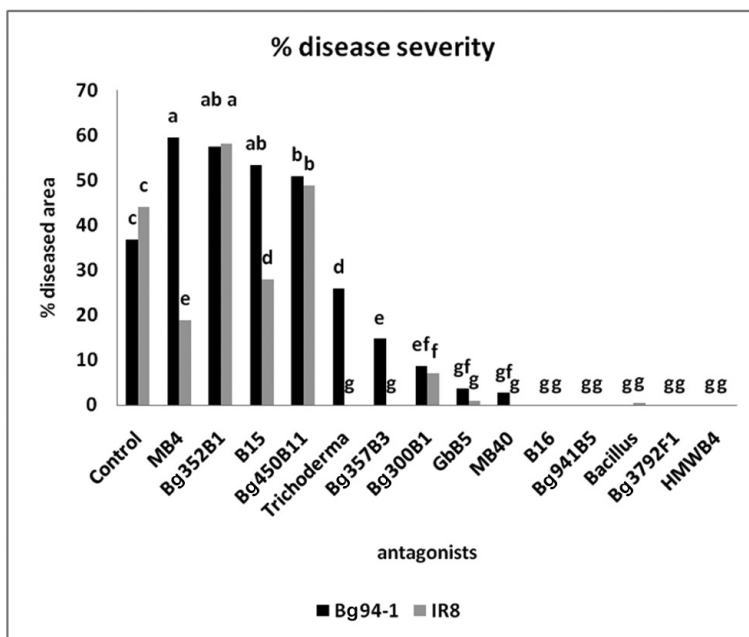


Figure 2: Percentage disease severity of rice varieties Bg94-1 and IR8, treated with 15 different potential antagonists under open-field pot experimental conditions. Common letters along the treatments of a given variety are not significantly different at $p=0.05$ (Adapted from De Costa *et al.*, 2008).

Based on the findings of % disease incidence (Fig. 1) and disease severity (Fig. 2) six antagonists namely, GbB5 (*Bacillus amyloliquefaciens* A), Bg 94-1B5 (*Bacillus licheniformis*), Bg 379-2B16 (*Bacillus megaterium* A), B4 (*Bacillus subtilis* B), HMWB4 (*Dermaococcus nishinomiyaensis*) and Bg 379-2F1 (*Aspergillus niger*) were identified to be the most effective antagonists to be used in further studies under field conditions.

Variety Bg 94-1 was used in the field experiment conducted for two consecutive growing seasons (i.e. May to September 2005 and November 2005

to March 2006) at the Rice Research and Development Institute, Batalagoda. Broth cultures of the above selected five antagonists (4 bacterial antagonists and a fungal isolate) at cell concentrations of 108 cfu/ml were sprayed on to the rice sheath (De Costa *et al.*, 2008). Fig. 3 shows the disease severity of the field grown plants in season 1 (i.e. May to September 2005) with reference to % diseased tillers/hill and % disease area on sheath. As highlighted by De Costa *et al.* (2008), severity of sheath blight was significantly reduced by all the tested antagonists when sprayed as a broth culture with a cell concentration of 108 cfu/ml, starting three days after

the development of symptoms and continuing for three applications at 10-day intervals. In both seasons, the antagonists performed in a consistent manner with *B. megaterium* A (B16)

and *A. niger* (Bg379-2F1) being the most effective candidates resulting in a reduction of disease severity by 50% and 61%, respectively.

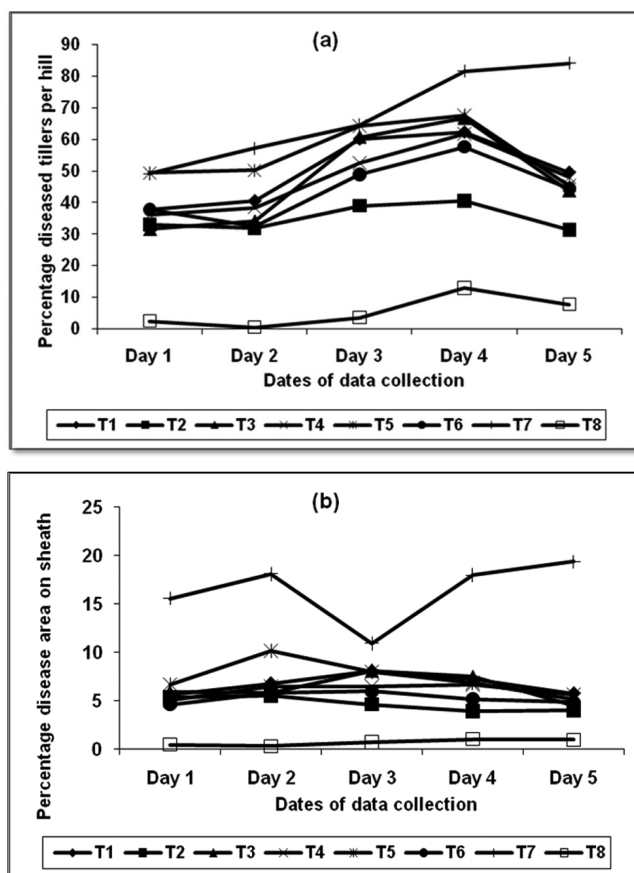


Figure 3: Percentage disease severity with reference to diseased tillers/ hill (a) and diseased area on sheath (b) of field-grown rice plants, variety Bg94-1, when treated with five different antagonists in growth season 1. T1= *B. licheniformis* + *R. solani* T2= *Bacillus megaterium* A + *R. solani*, T3= *B. amyloliquefaciens* + *R. solani*, T4= *Bacillus subtilis* B, T5= *Dermacoccus nishinomiyaensis* + *R. solani*, T6= *Aspergillus niger* + *R. solani*, T7= *R. solani* only, T8= no antagonist and no *R. solani* (Adapted from De Costa *et al.*, 2008).

Future studies towards the development of a commercial product

Efficiency of the above antagonists was confirmed under laboratory, pot and field experimental conditions. In all these experiments, suspensions of fresh (vegetative cells) cells were used for treating the rice plants. Even though, these antagonists are successful under laboratory and controlled-environmental conditions, like most of the potential biocontrol agents identified so far, it is essential to produce a final product out of these biocontrol agents which is able to maintain its disease-controlling efficiency together with durability, user-friendliness and cost-effectiveness. Towards the formulation of a long lasting product from these antagonists, the sporulation ability of *Bacillus spp.*, and the heat- and desiccation-resistant nature of the spores of *Bacillus spp.* can be considered as desirable attributes (Emmert and Handlesman, 1999).

Moreover, the presence of *Bacillus spp.* and *A. niger* ubiquitously in natural habitats and the growth promoting action of most of these antagonists can also be considered as favourable additional advantages when selecting a prospective antagonist for further development towards a commercial product. Growth promoting activities of *Bacillus spp.* and *A. niger* have been documented in previous studies (Compant *et al.*, 2005; Kandhari *et al.*, 2000). Furthermore, the potential to induce systemic resistance through defence chemicals by the biological control agents is worth investigating.

Considering the potential of the already identified antagonists, investigations are underway at present to develop a spore-based product of *B. megaterium*, *B. subtilis* and *A. niger* using a suitable carrier material which is readily-available, additionally-useful and cost-effective to rice growers.

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Meeting Challenges to Realize Potential Yield in Rice Cultivation in Sri Lanka

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Abstract

With the implementation of Best Management Practices through various farmer centered programs, paddy production in Sri Lanka increased from 2.85 million tons in 1999 to 3.65 million tons in 2009. The rate of increase was significantly greater during the second half of the decade. The national average grain yield increased by 17.4% while the total cultivated extent increased by 13.0% from 1999 to 2009. In spite of increase in productivity, a significant yield gap between farmer realizable potential and actual farmer yield still exists. Depleting soil fertility and thereby reduction in tissue nutrient concentrations in the high density broadcast sown environment, suitability of different stand establishment methods including seeding broadcasting, depleting organic matter in the soil, increasing seedling density to overcome deficiencies in weed management, increasing subsurface soil salinity, deficiencies in management of abiotic stresses including plant and soil N management using leaf color chart, organic rice farming and management of Wet Zone rice cultivation including staggered cultivation with less emphasis on rain water were addressed by means of research during the last decade in order to realize potential yields in different rice growing systems.

Introduction

Though, Asia contributes about 90% of the world rice production, the contribution to global rice production from Sri Lanka is just 0.63%. Still, rice is the livelihood and the staple food of Sri Lankans with a per capita consumption of about 115 kg of rice/year. Rice crop occupies 17.6% (0.7 million ha) of the total agricultural land area in the island which is cultivated by about 1.8 million farm families and 11.6% of the total population or 32%

of the total labor force is engaged in rice. In 2009, Sri Lanka produced 96% of the rice requirement amounting to 3.65 million tons of rough rice with a national average yield of 4.34 t/ha (Department of Census and Statistics, 2009). Rice contributed 14.2% to the total agricultural GDP of the island (Central Bank, 2008). Sri Lanka will have to achieve its future requirement of about 5.03 million tons of rough rice by year 2030 with less land, less labor and less water. Furthermore, serious challenges with biotic pressure

have to be faced with limited use of agro-chemicals while sociological changes continue to limit productivity and profitability in rice farming. To avoid imports and meet the demand of the nation's increasing population, there is an urgent need to improve rice productivity and production by introducing novel technologies. This paper reviews the favorable changes made in rice cultivation and agronomic technologies developed and disseminated during the last decade to overcome challenges faced by the rice cultivation in Sri Lanka.

Area cultivated and production trends

The area under rice cultivation fluctuated between seasons (Fig. 1). The cultivated extent stagnated during the first half of the indicated period

and from 2004 there was an increasing trend in major and minor irrigated rice ecosystems during both Maha and Yala seasons. There was no change in the extent under rain-fed system during Yala but a small increase could be observed in Maha season (Fig. 1).

Along with the increase in cultivated extent, there was also an increase in productivity. National average rice grain yield increased by 17.4% from 3.67 t/ha in 1999 to 4.3 t/ha in 2009. The increase was highest with 16.8% in the major irrigated rice growing ecosystem while it was 11.9% and 12.0% in minor irrigation and rain-fed ecosystems, respectively (Fig. 2). Similar to the cultivated extent, the productivity also showed a clear increase but from 2003.

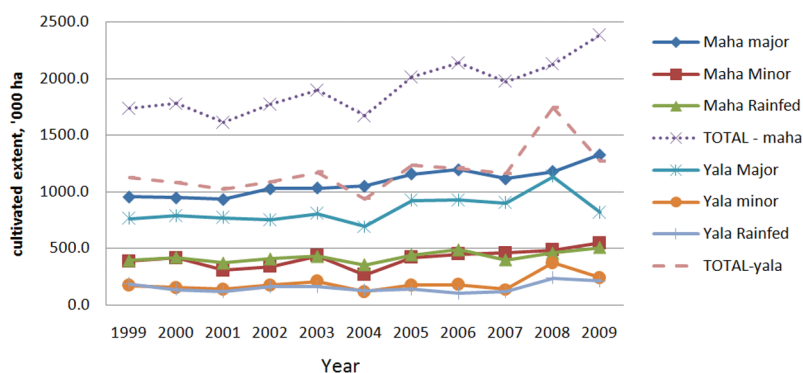


Figure 1: Cultivated extent of rice in different ecosystems during Maha and Yala seasons from 1999 to 2009.

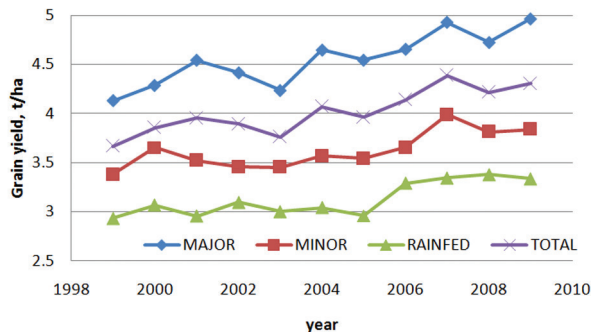


Figure 2: National average grain yield of rice in major rice growing ecosystems in Sri Lanka from 1999 to 2009.

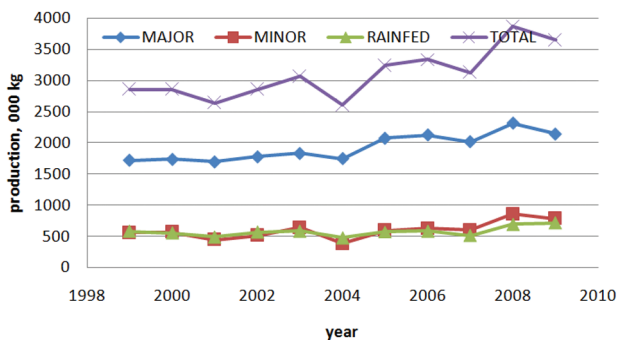


Figure 3: Total rice production in different rice growing ecosystems in Sri Lanka from 1999 to 2009.

Contribution from different ecosystems to total rice production in the island remained same over the last decade. About 60% of the total rice production was received from the major irrigated ecosystem while rain-fed and minor irrigated ecosystems had an equal share of each about 20%. About 60-65% of the total rice production came from Maha season. In general the extent cultivated, productivity and production increased significantly from 2004. Favorable government policies for rice cultivation made a

significant impact on the revival of rice production in the island. These developments in the rice sector during the last decade helped to overcome the adverse impact of increased global scarcity of rice in 2008. Further, to overcome the crisis situation in the rice market, government of Sri Lanka had to introduce a price ceiling for rice while encouraging enhancing rice production through the “Api Wawamu Rata Nagamu” program supported with the fertilizer subsidy scheme.

Production constraints in different agro-ecologies and yield gap

Expression of potential grain yield of a rice cultivar is determined by its' interaction with climatic, edaphic and biotic components of the environment. The highest potential in rice productivity is in the Low Country Dry and Intermediate Zones where solar radiation and other climatic parameters are conducive and the edaphic environment is relatively more favorable for rice. Alluvial soils and Non Calcic Brown soils have recorded grain yields of 11-12 tons/ha which is the highest yield ever recorded in Sri Lanka (Emitiyagoda *et al.*, 2010). Even though the realizable potential yield in favorable rice growing ecosystems exceeds 11 and 9 t/ha in the Dry and Wet Zones, respectively (Emitiyagoda *et al.*, 2010), average grain yields obtained in those ecosystems are comparatively much lower (Table 1).

The obvious reasons cited for this yield gap in many instances are crop management deficiencies resulting from socio-economic and/or institutional constraints. The yield gap was due to confounded effects of many factors depending on the farmer. In many instances the predominant factors for the yield gap were deficiencies in nutrient supplying capacity of the soil, unfavorable climate, weeds, insect pests and inefficient management. Practices such as non-seasonal, untimely staggered cultivation neglecting the optimum use of climatic parameters hinder the increase in

productivity. These practices resulted in buildup of pest and disease pressure due to overlapping cropping cycles. Weeds are the most disastrous constraint to rice production across all agro-ecologies followed by either excess and/or deficit of water. Problem of weeds is severe in the Dry Zone and could be related to the insufficient availability of water as most farmers use standing water in controlling weeds. Due to land fragmentation over generations, the undulating terrain in the Dry Zone and the lack of resources to maintain large blocks, average block size of rice paddies in both Dry and Wet Zones are small. This would increase cost of management while decreasing net block area. Farmers were using seed rates ranging from 87 to 220 kg/ha for intermediate bold type grain and 71 to 176 kg/ha for short round type grain. They considered yield potential, maturity duration and eating quality as criteria in selecting a variety for cultivation. Among the farmers surveyed, only 21% in the Dry Zone, 13% in the Intermediate Zone, and 29% in the Wet Zone adhered to the recommended method of basal fertilizer application. These management deficiencies resulted in a large variation in the grain yield realized by farmers in different eco systems. To overcome these management deficiencies and to increase resource use efficiency and grain yield, a package of practices for the direct seeded rice culture was introduced by the Department of Agriculture in 2004. This package of practices included timely cultivation within the season, improving quality

of land preparation, growth duration and appropriateness of the cultivar for a given ecology, reduction in pre- and post-harvest losses, methods that enhance soil chemical, physical and biological properties to improve and sustain soil fertility, the major determinant of crop productivity and management of biotic factors.

Based on the rice requirement for the next two decades, sustainability of the ecosystem and availability of natural resources, productivity and expected production targets were calculated. Accordingly, most important ecosystems will continue to be the Dry Zone major and minor irrigated ecosystems followed by the rain-fed system in the Dry and Intermediate Zones (Table 1).



Table 1: Area cultivated and average yield in 2009, realizable yield, targeted average grain yield, targeted rice production in 2015, 2020 and 2030 during Maha and Yala seasons from different rice ecosystems in Sri Lanka.

Ecosystem	Season	Area cultivated (ha)	Average yield (t/ha)	Realizable yield (t/ha)	Targeted average yield (t/ha)			Targeted rice production (t)		
					2015	2020	2030	2015	2020	2030
Dry & Inter. Zone major irrigation	Maha	253,027	5.14	8.0	6.00	6.50	7.00	1,518,162	1,644,676	1,771,189
	Yala	163,256	4.85	7.0	5.50	6.00	6.50	897,908	979,536	1,061,164
Dry & Inter. Zone minor irrigation	Maha	112,766	4.19	8.0	5.20	5.50	6.50	586,383	620,213	732,979
	Yala	47,441	3.77	7.0	5.00	5.50	6.00	237,205	260,926	284,646
Dry Zone rainfed	Maha	47,214	3.55	6.5	3.80	4.00	4.50	179,413	188,856	212,463
	Yala	1,462	3.2	5.5	3.80	4.00	4.50	5,556	5,848	6,579
Intermediate Zone rainfed	Maha	44,190	3.73	6.0	4.00	4.50	5.00	176,760	198,855	220,950
	Yala	21,801	3.26	6.0	4.00	4.50	5.00	87,204	98,105	109,005
Wet Zone	Maha	82,075	3.39	5.0	4.00	4.00	4.50	328,300	328,300	369,338
	Yala	68,462	3.22	5.0	4.00	4.00	4.50	273,848	273,848	308,079

Table 2: Maturity duration, grain yield, total biomass, Leaf Area Index (LAI) and total above ground N, K and Zn of different rice varieties grown under broadcast sown (BS) and transplanted (TP) conditions during Yala 2006 at RRDI, Batalagoda.

Variety	Maturity duration (days)		Yield (t/ha)		Total biomass at flowering (g/m ²)		LAI at flowering		Total above ground N (g/m ²)		Total above ground P (g/m ²)		Total above ground Zn (g/m ²)	
	BS	TP	BS	TP	BS	TP	BS	TP	BS	TP	BS	TP	BS	TP
Bg 379-2	115	121	5.63	7.34	1064	1137	4.60	4.20	972	988	1842	2748	4.83	6.42
Bg 407H	124	130	6.19	7.69	1280	1206	4.87	4.26	841	1117	2797	2145	8.83	6.18
Bg 94-1	104	111	5.69	5.84	1125	895	5.46	3.77	1313	961	1501	1402	7.63	5.00
Bg 357	104	106	6.80	5.57	1063	928	5.42	3.69	1117	1016	2505	1199	5.71	4.64
Bg 307	96	109	5.50	6.13	899	852	3.56	4.22	915	807	1826	1073	7.03	5.14
Bg 250	81	89	5.60	5.81	790	734	4.16	3.55	915	897	1395	1326	4.62	3.43

Methods of stand establishment

About 96% of the total rice extent in Sri Lanka is broadcast sown. Therefore, any variety or technology for the broadcast sown environment must be based on the studies done under such conditions. Studies conducted at RRDI, Batalagoda proved that, days to maturity, crop growth, nutrient concentration in plant tissues, total nutrient uptake, grain yield and other parameters differ depending on method of stand establishment and maturity duration (Table 2). Under the Dry and Intermediate Zone conditions due to transplanting shock, transplanted crop using 2 weeks old seedlings matured about 6 days later than that of direct seeded crop. The 4-4½ month varieties performed well when they were transplanted. However, when direct seeded, grain yield of 4-4½ month varieties were comparable to 3½ month varieties. This suggests that under direct seeded environment where inter plant competition is high, there is no yield advantage with presently available 4-4½ month varieties. In the 4-4½ month maturity duration class when same plant density is used, increased vegetative duration increased biomass in the direct seeded environment but due to overcrowding, leaf senescence occur leading to less green tissues at flowering which leads to decrease in canopy C assimilation capacity.

The wider spacing and maximum tillering is the basis of System of Rice Intensification (SRI) where eight day old single seedling per hill is

transplanted at 30cm x 30cm spacing and soil is maintained below saturation level. It was revealed that only new practices in SRI are wider spacing and planting very young seedlings in a special manner but adding organic matter, no impounding of water and manual weeding are already recommended practices as part of the best management practices (BMPs) recommended for rice cultivation in Sri Lanka. Although SRI is reported (Uphoff, 2001; CIIFAD, 1999) as a system of rice cultivation by which extraordinarily high grain yields could be easily obtained even in the tropics, SRI has been identified as one of the two extreme ends, the end with the lowest seed rate in the range of stand establishment methods in rice (Abeyasiriwardana *et al.*, 2009)

Different methods of stand establishment in rice namely broadcasting (BC), transplanting (TP), System of Rice Intensification (SRI) and transplanting 12 day old seedlings and BMPs recommended by the DOA including integrated nutrient management (BMP) were compared for rice varieties of 3 month (Bg 300) and 4 month (Bg 403) maturity durations. As expected the BC crop due to its greater initial density covered the ground very quickly suppressing weeds in contrast to SRI treatment which covered the ground fully at late booting stage requiring manual weeding several times. However, excess biomass at early stages and mild N deficiency symptoms with BC were not observed with TP, SRI or BMP. With SRI, leaf N concentration was greater than other

treatments and was higher than the optimum but the LAI was lower due to wider planting, which could reduce canopy C assimilation. However, wider spacing induced late season tillering which were mostly unproductive. Maintaining soil moisture below saturation is unavoidable for SRI as flooding suppresses tillering which is essential for the success of SRI. However, profuse tillering with wider spacing and no inundation increased unproductive tillers that would have helped in absorbing nutrients as new roots are formed with new tillers which increase nutrient absorbing capacity (Weerakoon *et al.*, 2000). Even with Bg 403 of 4 month maturity group, the

canopy closure was poor suggesting that SRI is not geared to utilize the sunlight effectively in the tropics due to its short growing period. The grain yield of Bg 300 at harvest was significantly greater with BMP than SRI, TP and BC suggesting that SRI is comparable with conventional systems of rice cultivation (Table 3). With Bg 403, the decrease in yield with SRI and BMP are not significantly different to other treatments. However, if the availability of seed paddy is very costly or limited, SRI has an advantage over other systems but at the expense of labor.

Table 3: Influence of method of stand establishment and best management practices on grain yield of Bg 300 and Bg 403 during Yala 2001.

Variety	BC	SRI	TP	BMP
Bg 300	6.95 a	6.90 a	6.82 a	7.14 b
Bg 403	6.50 a	5.98 a	6.54 a	5.68 a

Above findings were further confirmed with different experiments in Yala 2001 and Maha 2001/2002 seasons. Both methods of management of rice, the DOA recommended method and SRI, decreased grain yield when organic matter and/or NPK fertilizer were not applied (Table 4). In the first season the impact of OM and NPK was small but during the subsequent season when soil supplying capacity decreased, there was almost 50% reduction in yield in both methods. The SRI entails application of OM and no chemical fertilizer.

These findings suggest that grain yield of SRI only with OM is not superior to the method of rice cultivation using recommended practices of the DOA. Further, the impact of organic matter was lower during the first season than with the second season suggesting that with the addition of chemical fertilizer, addition of organic matter is essential to maintain high grain yields (Table 4).

Table 4: Grain yield (t/ha) of Bg 357 under SRI and DOA recommended method with and without organic matter (OM) and NPK fertilizer.

Method of establishment	Grain yield (t/ha)			
	Yala 2001		Maha 2001/2002	
	With OM	Without OM	With OM	Without OM
SRI with NPK	8.86 a*	8.10 a	7.56 b	6.35 a
DOA with NPK	8.43 a	8.04 a	8.82 a	6.61 a
SRI without NPK	7.21 b	6.89 b	6.18 c	4.68 b
DOA without NPK	8.34a	7.61ab	6.52 c	4.98 b

* means followed by the same letter are not significantly different at $p=0.05$.

Increased spacing with the SRI system resulted in increased labor cost for weeding using the rotary weeder and subsequent manual weeding. In addition, transplanting very small seedlings was practically difficult and frequent irrigation with small quantities of water was needed to keep the soil in moist condition. When the soil was dry there were practical problems of applying prilled urea. Even though, application of pesticides is not recommended for SRI we were compelled to apply insecticides against leaf folder and paddy bug to save the crop. However, the attack of leaf folder was lower than that of other treatments but not with paddy bug. With all the difficulties in managing SRI, the grain yield was not different between SRI and DOA method of management. Thus SRI cannot be recommended for farmers in lowland rice culture. This is in agreement with Abey Siriwardena and Dissanayake (2009) who reported that SRI is practically the least feasible and technically the least adoptable

method of stand establishment of rice, which saves seed at the expenses of labor without any advantage in grain yield expressed in terms of t/ha.

Broadcasting, the most common method of direct seeding in Sri Lanka is an age old practice. About 96% of the rice extent is broadcast sown (Fig. 4) and transplanting is not popular in Sri Lanka due to increased labor cost and unavailability of labor during the planting season. Further, as a method of saving seed and also to overcome labor scarcity, seedling broadcasting technique was recently introduced in rice cultivation in Sri Lanka (Jayawardena *et al.*, 2004). This is the best method of stand establishment for Hybrid rice as Hybrid rice seed is comparatively more expensive. Grain yield of seedling broadcasted hybrid rice variety Bg 407H was comparable to broadcast sown or transplanted crop. Furthermore, this method could reduce 50% of the labor requirement and seed requirement for crop establishment

(Jayawardena *et al.*, 2006). Studies on the influence of depth of water at seedling broadcasting on the establishment of crop and weed growth revealed that seedling broadcasting could be practiced even with standing water with a depth of up to 2.5 cm without affecting crop establishment and grain yield. Maintaining standing water depth up to 2.5 cm significantly decreased the weed density which could reduce cost for weed control. This technique was also superior for direct seeding where water cannot be drained. Seedling broadcasting technique also preserved water in situ, suppressed weeds and prevented soil and nutrient loss as there is no need for draining of water.

Seedling broadcasting is now gaining popularity among farmers and it is being used even for inbred varieties as well. Broadcasting of rice under adverse rainy weather conditions is difficult due to stagnation of water. It was revealed that, except for small grain types, rice seeds soaked in water for 24 hrs and incubated for 48 hrs could be broadcast sown directly to leveled rice fields with 2-5 cm standing water without affecting seed germination, growth and grain yield of the crop. This water seeding method minimized the damage by birds and reduced weed incidence due to standing water while it helps to minimize nutrient losses and conserve water as no draining of water at sowing.

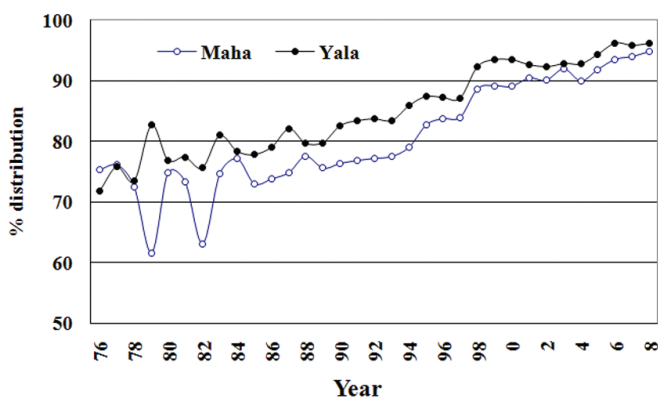


Figure 4: Percentage distribution of rice area under direct-seeding from 1975 to 2008 during Yala and Maha seasons.

In an attempt to test the influence of different methods of crop establishment on weed population and grain yield of rice revealed that even with greater rice-weed competition, row seeding gave a significantly higher grain yield than that of broadcasting, seedling

broadcasting and water seeding. Weed population in unweeded row seeded control plots was lower than that of water seeding, seedling broadcasting and broadcasting.

Screening rice lines for broadcast sown environment

Sri Lanka, unlike many other rice growing countries in Asia cultivates rice predominantly as a broadcast sown crop and has a shorter growing period. Hence, cultivated varieties also should suit this growing environment where competition for space and nutrients is high. Hence, rice varieties to be recommended for cultivation must be tested under the broadcast sown condition. Therefore, since 2003, the method of stand establishment of rice in all trials in the National Coordinated Rice Variety Testing (NCRVT), Variety Adaptability Testing (VAT) and Large Scale Variety Adaptability Testing (LSVAT) programs were changed from transplanting to direct seeding allowing new rice lines to be tested in a greater competitive environment for space and resources. During the last decade from 1999 to 2009, more than 110 new rice lines of 2½, 3, 3½ and 4 month maturity durations were tested in all rice growing agro ecological conditions. After testing under VAT and thereafter in LSVAT, 19 rice varieties were recommended for cultivation during the period from 1999 to 2010 (Table 5). Even though these varieties covers about 34% of the total cultivated extent of rice, about 25% of the total cultivated extent is still being cultivated with four varieties released in 1999. This suggests that it takes more than 10 years for a variety to be fully established in farmers' fields. A significant percentage, nearly 50% of rice lands are cultivated with varieties released in 1975, 1980, 1987 and 1992

and some of them are considered inferior to new varieties with regard to pest and disease resistance (Jayawardena *et al.*, 2010). This suggests that most of the new varieties released recently may not be comparatively more acceptable to farmers. The decentralization of the extension system and the reduced attention by the authorities to rice variety development have reflected in the poor acceptance of new rice varieties released in later years by farmers. This negative approach towards rice variety development has to be corrected immediately to avoid further deterioration of farmers' faith on rice varieties developed by the Department of Agriculture.

Screening rice varieties for water limited situation

Rice cultivation in Sri Lanka is faced with many abiotic stresses particularly water stress. A significant extent, about 38% of the land cultivated during Maha is not cultivated during Yala season. Further, about 12% of the cultivated extent is not harvested mainly due to terminal water stress. Therefore, a program to screen rice varieties for water limited situation was initiated in 2004. The variety Bg 304 was tolerant to water stress than other varieties tested (RRDI, 2005). There was a need to identify a variety which matures earlier than Bg 304, to escape drought in a very short growing season. In 2004, Bg 250, an ultra-short duration high yielding rice variety with maturity duration around 80 days in the Dry Zone with improved plant type was recommended for cultivation.

In a water limited rice growing environment, to compete with heavy weed pressure, a medium height rice variety is essential. Further, higher the weed suppression ability, better the variety. Therefore, a systematic program to screen varieties for drought and aerobic rice growing environment was initiated with the support

from IRRI's International Genetic Evaluation of Rice program (INGER) and subsequently Green Super Rice (GSR) program. Two lines from the GSR program and four lines from the INGER program were ear marked for further testing in farmers' fields in 2010.

Table 5: Year of release, cultivated extent, no. of districts reported and percentage of the total cultivated extent of different rice varieties grown in Sri Lanka in 2009.

Variety	Year of release	Cultivated extent, (ha)	% of total extent	No. of districts reported	Variety	Year of release	Cultivated extent (ha)	% of total extent	No. of districts reported
Traditional	1940	298	0.033		At 402	1992	4297	0.469	8
H4	1958	1573	0.172	6	At 354	1992	4045	0.442	10
Bg 11-11	1970	4	0.000	1	At 401	1992	787	0.086	2
Bg 34/8	1971	760	0.083	2	Bg 403	1993	12234	1.336	17
Bg 34-6	1971	11	0.001	1	Bg 304	1993	4449	0.486	24
Bg 3-5	1973	87	0.010	5	Ld 356	1994	23203	2.534	18
Bg 94-1	1975	82187	8.976	15	Ld 355	1994	4253	0.465	10
Bg 90-2	1975	3	0.000	1	Ld 356	1994	299	0.033	2
Bg 94-2	1978	187	0.020	2	Bg 357	1997	26769	2.924	18
Bg 379-2	1980	22941	2.506	20	Bg 358	1999	142977	15.61	23
Bg 400-1	1980	1942	0.212	23	Bg 359	1999	51246	5.597	22
Bw276-6B	1981	5325	0.582	4	Bg 360	1999	25686	2.805	24
Bw 267-3	1981	1684	0.184	3	Bg 305	1999	13815	1.509	13
Bg 38	1981	556	0.061	7	At 362	2002	52209	5.702	14
Bg 745	1981	234	0.026	6	Bw 361	2002	7694	0.840	13
Bg 407	1981	61	0.007	4	Bw 363	2004	3993	0.436	7
Bg 380	1982	50	0.005	3	At 306	2004	710	0.078	6
Bg 450	1985	6055	0.661	18	At 307	2005	9899	1.081	14
Bg 350	1986	12599	1.376	17	Bg 454	2005	2566	0.280	5
Bw 351	1986	5073	0.554	14	Bg 250	2005	2564	0.280	11
Bg 300	1987	173194	18.916	23	Bg 406	2005	1320	0.144	4
Bg 301	1987	515	0.056	4	Bg 406	2005	261	0.029	4
Bw 302	1987	59	0.006	3	Bg407H	2005	85	0.009	3
Bw 400	1987	22	0.002	1	Bg 454	2005	8	0.001	5

Bw 451	1987	4	0.000	1	Bw 364	2006	2964	0.324	6
At 303	1990	5900	0.644	17	At 308	2008	1861	0.203	4
Bg 352	1992	162547	17.753	23	Bg 366	2009	-	-	-
At 353	1992	31542	3.445	20	Ld 408	2010	-	-	-

Source: Jayawardena *et al.*, 2010 - Data not available

Nutrient management of rice varieties using SPAD meter and Leaf Color Chart (LCC)

Direct seeded rice crop absorbs greater quantity of nutrients than that of transplanted rice crop (Table 2). The uptake of N, K and Zn with 2½, 3 and 3½ month rice varieties were significantly greater in the broadcast sown rice crop compared to transplanted crop. However, with 4 month varieties, the uptake was greater with transplanted rice. Further, the LAI of broadcast sown rice of 4 month crop was also lower than that of 3½ month crop. With the same initial density, varieties with longer maturity duration tended to produce more biomass with time than varieties with shorter maturity duration which showed increased senescence. Therefore, with the senescence of tillers and leaves of 4 month age class, even though the total nutrient uptake over time was greater with DS, actual amount of nutrients available with the plant at flowering was comparatively lower.

In broadcasting, the panicle number was very much greater than transplanting, which decreased green leaf area per panicle in all varieties of broadcast sown crop (Fig. 6). This suggests that either greater quantity of N as fertilizer should be applied or competition for

photosynthates among rice panicles should be reduced by manipulating initial density, to avert leaf N dilution in broadcast sown rice crop which would maximize canopy assimilation capacity at flowering and thereby increase grain yield. The leaf K and Zn concentration did not show any clear pattern of change with broadcasting compared to transplanting. To increase the N uptake N should be applied in required quantities based on crop demand and soil supplying capacity at required intervals.

Under lowland conditions at RRDI, five rice varieties of 3 month maturity group; Bg 300, Bg 304, Bg 305, At 303 and six varieties of 3½ month maturity group; Bg 352, Bg 357, Bg 94-1, Bg 358, Bg 359 and Bw 328-1 (Bw 354) were direct seeded in rows of 15cm apart at four different leaf N levels using SPAD meter; 33, 35, 36, 38 and SPAD values at no added N. Bg 300 was used as the standard variety to check the SPAD value. N in the form of urea was applied to all varieties based on the Bg 300 SPAD values. Therefore, other varieties may not have had the optimum conditions and probably they were with excess or deficit in N.

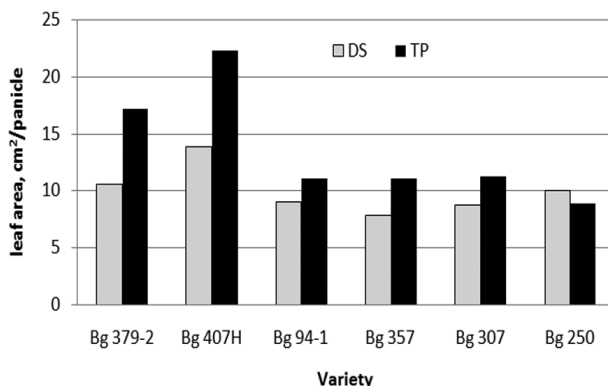


Figure 6: Green leaf area per panicle at flowering with direct seeding and transplanting.

In the 3-3½ month maturity group, to maintain leaf SPAD values of Bg 300 and Bg 94-1 over the season, recorded averages were closer to the set target but the other varieties always had a higher value across the season. This suggests that the N requirement vary among the varieties. Bg 300, At 303 and Bg 94-1 showed some similarities while Bg 304 and Bg 305 were in need of comparatively less N to maintain the same leaf N status. Thus, a fixed rate of N cannot be recommended for all the varieties when N is applied. This is indeed proved in many occasions when N response was studied for elite breeding lines. Lines differ in their N uptake pattern and in partitioning of N to biomass produced, thus the N concentration of component plant parts vary with the same amount of N applied. To maintain SPAD levels at 33, 35, 36 and 38 about 50, 75, 95, 100

kg of N/ha, respectively were needed.

Average SPAD value recorded over the season with the grain yield obtained for each variety showed a non-linear relationship (Fig. 7 & 8). The pattern was clear with Bg 300, Bg 94-1 but At 303, Bg 304 and Bg 305 behaved differently. A similar trend was observed with 3½ month varieties of Bg 352, Bg 358, Bg 359 and Bg 357. The trend showed that SPAD value between 34 and 36, recorded highest yield in all varieties (Fig. 7, 8 & 9). Grain yield did not change or rather decreased when average SPAD value exceeded 36 in all varieties suggesting that irrespective of the variety the optimum SPAD value was around 35 for direct seeded 3 - 3½ month rice varieties under lowland direct seeded conditions.

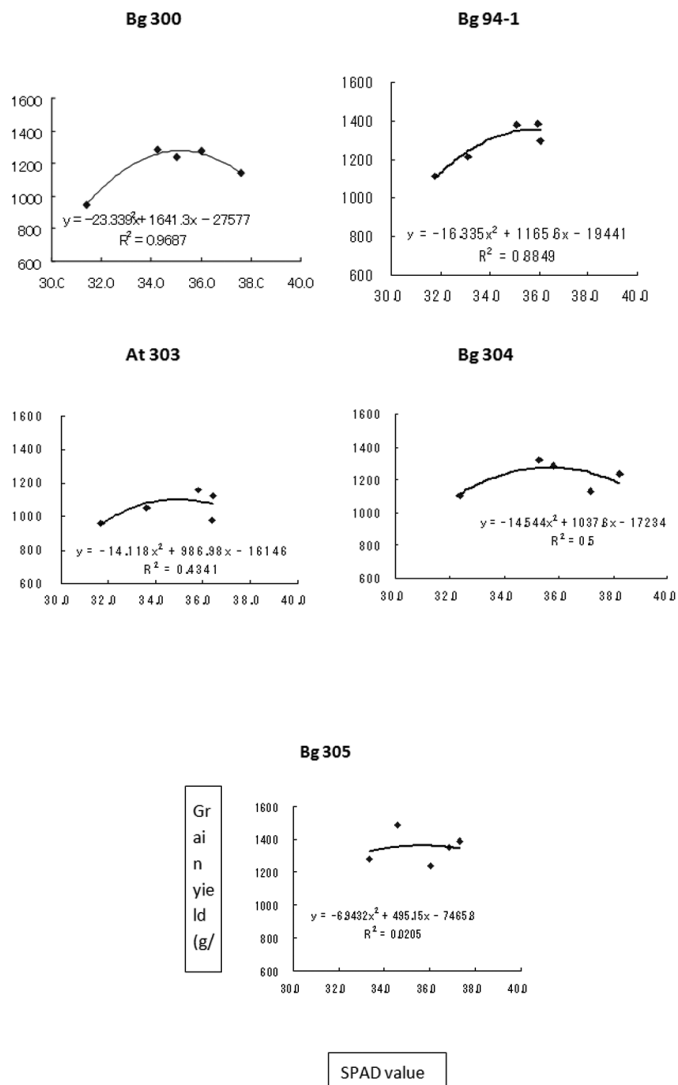


Figure 7: Relationship between grain yield per plot vs average SPAD value recorded for different varieties of 3 month maturity class.

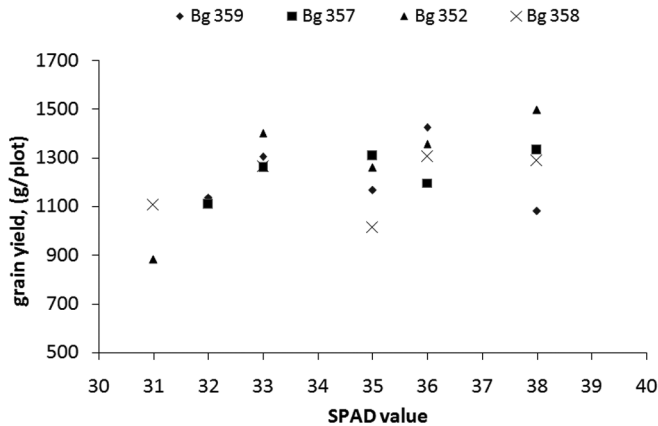


Figure 8: Change in grain yield per plot with different SPAD values (average over season) of different rice varieties of 3½ month maturity class.

SPAD cannot be recommended to the farmers as the equipment is too costly, so that a Leaf Color Chart (LCC) developed by IRRI was calibrated with the SPAD values for all varieties at the vegetative stage. A remarkable linear relationship was observed when the SPAD value was plotted against the LCC of the same leaf, irrespective

of the variety (Figs. 9a and 9b). Therefore, for both 3 month (Bg 300, Bg 304, Bg 305, At 303) and 3½ month rice varieties (Bg 94-1, Bg 352, Bg 357, Bg 358, Bg 359), to realize the leaf SPAD value around 35, the LCC value throughout the vegetative period until late booting should be maintained around 4.

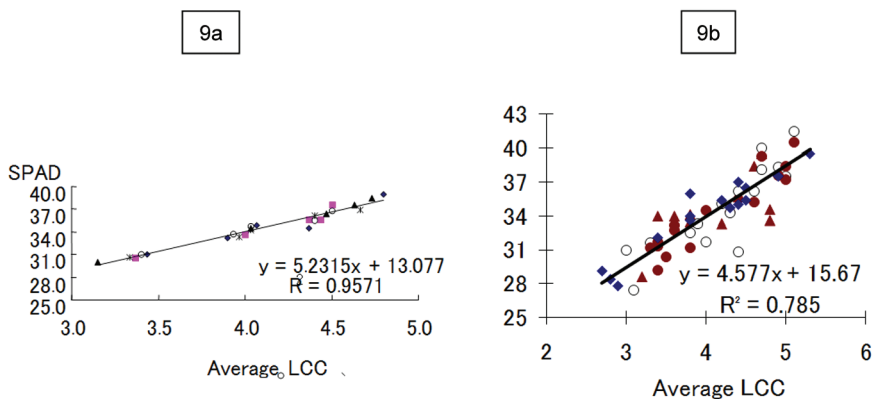


Figure 9: Relationship between LCC and SPAD value (average over season) of different rice varieties of 3 month and 3½ month maturity groups.

Timely cultivation and maximizing the use of rain water

Apart from the biotic and edaphic factors the climatic factors are also responsible for the yield gap. In the early days, farmers used to cultivate the rice crop with the onset of the monsoon. Timely cultivation and collective participation were primary considerations in the rice culture. Togetherness and correct timing in planting and harvesting within the season reduce the risk of terminal drought, and pest and disease incidences. Practice of collective cultivation of rice cultivars of same growth duration is rare today. Further, early crop establishment helped the farmers to cultivate medium duration rice varieties (3½ month) with comparatively a higher potential grain yield by exploiting the growing environment to a maximum through harnessing the solar energy for a longer period. However, a survey revealed that timely cultivation with the onset of monsoon is a seldom practice in the irrigated rice eco-system which comprised of over 75% of the cultivated extent in Sri Lanka (Weerakoon *et al.*, 2006).

Change in the amount of rainfall in the river catchments (Premalal, 2009) and increased demand for water extracted for domestic and industrial needs decreased the quantity of water flow by about 27 million m³/year in the Mahaweli river which could have otherwise provided supplementary

irrigation for cultivation of over 100,000 ha in the Mahaweli system (Weerakoon and Costa, 2009). Even though the water diverted for agriculture from Mahaweli river remains the same over the years, the water available for hydropower generation and irrigation for the downstream cultivation decreased. Further, increase in population and changes in life styles in these agricultural districts would further increase pressure to enhance the value of irrigated water used for rice cultivation. The vulnerability of the North East monsoon to climate change has increased along with an increase in consecutive dry days in the Dry Zone over the last 20 years (Premalal, 2009). Increase in number of dry days and less reliability on the monsoon would increase demand for irrigation water, especially for rice cultivation in the Dry and Intermediate Zones. These gradual changes of water availability would not only expose rice crop to terminal drought but also force the farmer to cultivate comparatively low yielding ultra-short duration rice varieties in place of 3½ month high yielding varieties if water saving irrigation systems were not practiced. It was revealed that impounding water in rice paddies is not essential for maximizing rice yields (Weerakoon *et al.*, 2010). Numbers of irrigation rounds in low land rice fields in the Intermediate Zone for maintaining saturated soil culture was the same as impounding water but the total water required to maintain saturated condition was about 24% and 72% lower than that is required for continuous impounding during Yala

and Maha, respectively (Weerakoon *et al.*, 2010). Maintaining rice plants at saturated condition through the growing period significantly increased total biomass, root dry weight and leaf greenness than the impounded or impounded to dry system of rice cultivation (Weerakoon *et al.*, 2010). Increased root dry weight and increased greenness under saturated condition suggests that the ability of the rice plant to take up nutrients would have also increased with saturated soil culture. Grain yield in both seasons remained the same in both saturated and impounded treatments but significantly decreased with drying of soil below saturation (Weerakoon *et al.*, 2010). In general, maintaining saturated soil culture in lowland rice fields could save a considerable amount of irrigation water while maintaining the same level or even greater grain yield. The saving of water by maintaining saturated soil culture could be very significant especially during Maha season which could be used for the cultivation of additional area under rice or other crops during Yala in both Dry and Intermediate Zones of Sri Lanka.

Soil salinity and mid- season yellowing

Subsurface salinity is becoming a serious problem in Yala season especially in the Dry Zone irrigated low lands. Investigations on mid-season yellowing particularly in the Polonnaruwa district during Yala 2004 revealed that it was mainly due to decrease in nutrient uptake with the buildup of excess minerals in the root

zone. With the shifting of cropping season, plant water requirement increased due to increased evaporative demand with heavy dry winds during latter part of Yala season which coincided with the peak rate of growth of the rice crop. This peak nutrient absorption period which is about 6-8 weeks after sowing for a 3-3½ month rice crop, coincided with the period of increased salt concentration in the root zone due to upward movement of salt from deeper layers with upward movement of water. Building up of salinity in the root zone increased competition among minerals for absorption, especially K, thus showing yellowing symptoms. Increased Mg^{2+} concentration in the rhizosphere and relative decrease in K^+ ion with the upward movement of water resulting from high evaporation due to high winds and moisture stress caused yellowing in rice. Salinity tolerant At 354 exhibited significantly lower level of yellowing than that of Bg 352.

Impact of Cl^- ions

Coastal salinity is characterized by increased NaCl concentration in the rhizosphere, which affects rice plant growth and yield. We studied the influence of Na^+ and Cl^- ions on rice plant. The impact of Na^+ in different forms such as NaCl and Na_2SO_4 and concentrations placed at different depths of soil on yield and other attributes of different varieties of rice were studied. All varieties tested were susceptible to both surface and sub-surface soil salinity but the degree of susceptibility varied with

the variety. Depending on the severity of soil stress, there was a difference in yellowing of leaves in all tested varieties. Yield of Bg 300 when grown in soils with 4 ds/m level of salinity reduced grain yield by 43% while the same salinity level but placed at sub-surface of soil reduced yield only by 34%. The reduction in yield of saline resistant At 354 was 15% and 6% when the same concentration of NaCl was placed in whole soil and sub-surface soil, respectively.

The influence of Na_2SO_4 was much lower than that of NaCl suggesting that Cl^- is much damaging to rice than Na^+ . The reason for the reduction in growth and yield with increased salinity was the reduction in leaf area and individual leaf C assimilation. Planting salinity tolerant varieties at the correct time helped in overcoming this problem. Further, if sufficient water is not available for flooding of fields which are having salt in deeper soil layers, shallow ploughing was recommended. Screening rice varieties for reaction to salinity to identify salinity tolerant varieties is difficult under field condition due to uneven nature of occurrence of salinity in the field and the high cost involved for transportation, labor and materials. Therefore, a screening method was developed for screening large number of rice varieties at seed germination stage. The ability of rice seeds to germinate after soaking in saline water with an EC between 40 - 50 ms/cm for 9 days is a good indicator of salinity tolerance (Abey Siriwardena, 2004).

Organic rice farming

Until 1950's almost all the rice lands in Sri Lanka were under organic farming with traditional rice varieties. However, since then traditional rice varieties were gradually replaced by fertilizer responsive new improved rice varieties. These improved varieties increased the national rice production level to self-sufficiency, but with added cost on inorganic fertilizer, chemicals for the control of weeds, pests and diseases. However, it is much cheaper than importing rice to feed the nation. Nevertheless increasing concerns on human health & environment have led to research on chemical free rice cultivation methods. Though the rice grain yield under organic condition is lower than that of high input rice farming, use of appropriate technological package for organic farming will provide chemical free food. Consumers opt to pay a higher price for such products. RRDI initiated a research program in Yala 2006 to identify an appropriate agronomic package including suitable varieties for organic rice farming systems.

Potential grain yield of 4-5 t/ha could be obtained under organic farming conditions with the incorporation of 10 - 15 t of green manure, 4 t of rice straw and 350 kg Eppawala rock phosphate per ha and with foliar application of green leaf and neem extracts for controlling of pests from improved rice varieties such as Bg 360, Bg 300, At 306 and Bg 358. Yield of traditional rice varieties such as Herathbanda,

Gonabaru, Suduheenati and Kahatwee under the same conditions was around 3 t/ha.

Under the Dry and Intermediate Zone conditions, the following traditional varieties were promising for organic farming; ‘Gonabaru’, ‘Herath Banda’, ‘Godawee’ and ‘Dikwee’ (4 month varieties) and ‘Moddikaruppan’, ‘Kakimahathaya’, ‘Kahatawee’ and ‘Kahatasamba’ (3-3½ month varieties) (Bandara *et al.*, 2007). It has also been recommended that either 16 t/ha of *Sesbania rostrata* or 10 t/ha of *Gliricidia* should be applied to the field for organic rice cultivation (Bandara *et al.*, 2007).

Management of Wet Zone rice

The area under rice in the Wet Zone (WZ) (121,000 ha) accounted for about 25% of total extent and 20% of the total national rice production in 1980 (Jayawardena and Pieris, 1980). With an average holding size of 0.2 ha, low attention had been given to the improvement of the rice crop in the WZ, although the farmer families / ha had been greater in the WZ than in the Dry and Intermediate Zones (Jayawardena and Pieris, 1980). The cultivated extent of rice in the WZ in mid 1980s decreased by 10% during Maha and by 20% during Yala by mid 1990s (Abeyasiriwardena and Sandanayake, 2000). Athukorala and Hettiarachchi (2005) showed that the percentage of rain fed paddy fields in the WZ, which was as high as 45% of the total cultivated area in the early 1960s dropped to about 28% by 2000.

Even though insignificant, the tsunami in 2004 destroyed a total of 3520 ha of paddy lands located in the coastal belt out of which 275 ha were in the WZ (Emitiyagoda *et al.*, 2005).

Since 2005 there had been an increase in the extent of paddy in 4 main districts in the LCWZ viz. Colombo, Gampaha, Kalutara and Galle. The extents had increased from 75728 ha in 2000 to 79113 ha in 2008 (Census and Statistics Department, 2009). Majority of the rain fed rice area in Sri Lanka lies in the LCWZ. Therefore, it could be safely surmised that there is a re-awakening of farmers in the entire country towards cultivating rice which is also reflected in the LCWZ and that the LCWZ with green water cannot be overlooked as an area of insignificance.

In the LCWZ the average yield had increased from 2.7 t/ha (mid 1980's) to 3.39 and 3.22 t/ha during Maha and Yala, respectively in 2009 (Table 1). Farmers were content with a low input - low output scenario. However, a system - ‘moderate input and moderate output’ was suggested (Panabokke, 1980). For the LCWZ rice ‘high’, ‘medium’ and ‘low’ potential areas need varieties with specific characteristics to suit those land types and not merely high yielding varieties (Jayawardena and Pieris, 1980). Many region specific varieties have come out in the recent past.

In the LCWZ, there are lands with different potential yields which have to be identified and effort should be made to make them profitable using

adaptable high yielding varieties and associated technologies while in particular medium potential and low potential lands should be dealt with the production of rice with grain quality. The inherent grain qualities and those that are induced by way of production, leading to environmentally and health friendly rice, which would invariably add a high market value, may bring about an economic gain to the already low yield levels of such lands, making the venture profitable. There is already an awakening in the private sector in regard to healthy food, thus creating an expanded market locally. There is already an expanding market globally for organic food, which has to be properly harnessed.

To increase yield levels in the LCWZ up to 6 t/ha of rough rice, husk could be incorporated to the soil just before or after the 2nd ploughing. Rice husk at 4 t/ha with DOA recommended fertilizer and other practices produced a significantly higher grain yield of 3.6 t/ha over the DOA recommendation without rice husk (2.3 t/ha) (Unpublished data, RARDC, Bombuwela). Farmer field testing of the same package in two locations in comparatively wetter region of the Ratnapura district revealed an increase in yield up to 85% (6 t/ha) and 46% (4.7 t/ha) over the farmer practice (Unpublished data, RARDC, Bombuwela). In another experiment, the beneficial effects of different organic materials such as straw, cattle manure, broiler and layer litter and *Salvinia* compost applied at 3 t/ha along with the DOA recommendation

included foliage of darker green color and consistently higher grain yields (>3.00 t/ha) than that of the control (1.69 t/ha) (Unpublished data, RARDC, Bombuwela).

A research had been conducted to test the efficacy of root clipping before transplanting on inducing root growth in inbreds. Root clipping was found to be very successful giving significantly higher yields (4.23 t/ha) over the farmer practice, the non-clipped transplanting (3.69 t/ha) (Unpublished data, RARDC, Bombuwela). In another experiment conducted to evaluate the storability of seed paddy, it was found that white pericarp varieties could be stored under the conditions of the LCWZ for more than 6 months (80% germinability) when packed in polysack bags at any given seed moisture level at packing irrespective of the seed size (bold or long type). However, red pericarp seeds could not be stored under similar conditions for more than 4½ months. There was a rapid decline in seed germination after the above indicated periods particularly in bold type seeds. The situation was aggravated if the seeds were stored at low seed moisture level of 12% (Unpublished data, RARDC, Bombuwela). Abey Siriwardena (1985) also reported that polysack was the best packing material for packing of seed paddy with high moisture under the LCWZ conditions.

In an attempt to boost the yield levels in the LCWZ, the performance of hybrid rice was tested using a series of hybrid rice combinations at the

RRRDC, Bombuwela. Two hybrid rice combinations along with the control viz., Bg HR 06 (Bg 407H), Bg HR12 and Bg HR 1 showed good adaptability and an average level of tolerance to Fe toxicity.

To overcome the intricacies of crop establishment, experiments were initiated to see if broadcast sowing method practiced with inbreds could also be used with hybrids. It was revealed that at a lower seed rate of 30 kg/ha, a successful crop establishment could be achieved by broadcast sowing which produced an average yield of 4.3 t/ha compared to the 'Parachute method' (3.94 t/ha) and the row transplanting method (4.17 t/ha). To overcome adverse effects of high rainfall, a thin layer of straw (2 t/ha) may be spread over the sown crop (Kahandawela, 2010).

To cater to the needs of organic farming, following indigenous varieties, OIVs and NIVs have been selected: 'Molligod', 'Murungakayan', 'Behethheenati', 'Suduheenati', 'Herath Banda', 'Kahata wee', 'Sulai', 'Madathawalu', Bw 272-6b, Bw 364 and Bg 250 (Unpublished data, RARDC, Bombuwela).

An agronomic package has been developed for the LCWZ conditions where different methods of stand establishment, different types of organic manure and different ways of their application and plant protection methods have been included. It has been observed that a rice crop grown organically could yield up

to 3 t/ha when green manure or compost was added at the rate of 20 t/ha (Unpublished data, RARDC, Bombuwela). Organic manure can be given either as a basal application or in 2 splits at planting and at 4 weeks after sowing. Studies have shown that half the compost requirement can be substituted with an equal amount of green manure (10 t/ha). The standard green manure crops of the Dry Zone- *Gliricidia spp.*, *Tithonia diversifolia*, *Sesbania rostrata* and *Crotalaria juncea* are not available. The existing weed mulch has been analyzed and found to be comparable to the standard green manure crops.

Studies conducted on compost preparation have generated knowledge on reducing the time taken for process of composting. Using tender parts of plants and a slurry of cow dung and cow urine of low C:N value, composting process was accomplished in 3 weeks as compared to the 3 months taken in the conventional system. Compost made using this rapid method repeatedly showed better results in rice grain yields (3.3 t/ha) compared to 2.97 t/ha of compost conventionally made (Unpublished data, RARDC, Bombuwela). Nutrient contents of different compost types analyzed at RARDC, Bombuwela are presented in Table 6.

Table 6: Nutrient contents of different compost types analyzed at RARDC, Bombuwela.

Compost Type	N%	P%	K%
Rice husk compost	1.33	0.107	0.058
Vegetable refuse compost	0.77	0.321	0.133
Salvinia compost	1.33	0.321	0.133
Adhoc compost	2.66	0.385	0.205
Saw dust compost	1.19	0.043	0.046
Animal based compost	2.17	0.535	0.080

An organic package developed for organic rice production in Fe-toxic soils was found to be efficient if compost at the rate of 10 t/ha (basal) and green manure at the rate of 15 t/ha (at 4 weeks after establishment) were included in the package (Unpublished data, RARDC, Bombuwela).

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Land Resource Planning for Achieving Higher Productivity in Rice Lands

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Abstract

During the past decade, national average rice yields and the production have been increasing and this trend will continue. Therefore, rice land utilization strategies for managing excess production should be given a high priority. Dry Zone exhibits the highest rice production potential followed by the Intermediate Zone and the Wet Zone exhibiting the lowest. Thus policies and implementation strategies should be formulated to maximize the production potentials especially in the high potential areas where as marginal lands should be scientifically used for other food crops production. Rice farming in Sri Lanka is a well established system with a long history. Hence, no radical interventions are necessary, but efforts should be geared to eliminate the present problems in rice farming which are mainly provoked due to its mismanagement. This paper examines key areas to be addressed to increase the productivity in terms of land resource planning.

A significant spatial variability exists within landscape in a rice tract with respect to yield. It is mainly due to the differences in crop management at block level, land drainage and their interactions within yaya on soil fertility and land productivity. Holistic approach is needed to resolve these issues and use of Geographic Information System & Remote Sensing technologies with limited field activities have shown effectiveness of making yaya based recommendations on land and crop management. At present, estimated extent and the production figures of rice are approximate values and reliable rice production forecast depending on the weather conditions during the season is not possible. As a measure to get accurate estimates of area, production at regional and national levels and thereby to allocate resources judiciously, use of RS and GIS is a prime necessity. Further diversification of rice lands should be done using indicator mapping approach. To reduce cost of production, land related constraints should be minimized with land consolidation must be done where possible. Unauthorized conversion of rice lands for other uses mainly due to rapid urbanization is on the rise, particularly in urban and its suburbs. This should be evaluated in terms of both technical and social point of view. As a long term measure, preparation of proper zoning plans considering all aspects and strict implementation is essential.

Introduction

Rice farming is a tradition in Sri Lanka and rice is the staple food of Sri Lankan for many centuries. At present, rice sector contributes around 30% to the agricultural GDP which accounts for 12% of the national GDP (Central Bank, 2009) and more than that rice farming is still the livelihood of approximately 32% of the total labour force in the country. The maximum estimated asweddumized land for rice cultivation at present is 730,000 ha. and the present cropping intensities for Yala and Maha seasons are 47.2% and 88.5%, respectively making annual cropping intensity around 136% (Department of Statistics, 2009). During the past decade, national average yield of rice has been rising from 3.86 to 4.58 t/ha.

According to forecasts, rice production in year 2010 would be about 4.24 million tonnes (SEPC/DoA, 2009) which would be the highest recorded production in the recent history. This production is equivalent to 2.75 million tonnes of milled rice. As per to the estimates, the total rice requirement for year 2010 is 2.25 million tonnes, thus it appears that the rice production exceeds the total consumption (SEPC/DoA, 2010). This excess production trend will be continued with intensified adoption of improved varieties and technologies and addition of rice lands in the North and East which have been neglected over the past three decades. Thus, while improving the yields through effective research and extension, rice land utilization strategies for managing

excess production should be given a high priority in the near future.

It must be noted that the rice farming in Sri Lanka is a system with long history and it is a well established farming system. Therefore, emphasis should be given to scientific management of the existing system through eliminating problems arisen due to mismanagement of lands and issues such as inadequate or shallow ploughing, poor maintenance of irrigation and drainage channels, land fragmentation, haphazard land filling etc. should be dealt with. The objective of the present paper is to identify land related issues and constraints in rice farming, which are limiting land productivity and suggesting remedial measures that can be incorporated at different planning levels ranging from national to farm level.

Emphasizing more on areas with higher production potentials

Data on average yields in each district gives a reasonable clue for production potential under present resources, technical and socio-economic context. Careful observation of such data (Department of Statistics, 2009) indicates that average yield in areas such as Ampare, Polonnoruwa, Anuradhapura, Hambantota, Vavunia, Mullathive, Mannar and Udawalawe is above 4,500 kg/ha while in Gampaha, Colombo, Kandy, Kalutara, Ratnapura, Galle, Kilinochchi and Matara is less than 2,500 kg/ha. However, districts such as Kurunegala, Puttalam, Matale,

Nuwareliya, Mahaweli H, Badulla and Monaragala are exhibiting moderate potentials and produce 3,500 - 4,500 kg/ha (Fig. 1). The situation clearly indicates that Dry Zone exhibits the highest rice production potential, Intermediate Zone shows moderate potentials and the Wet Zone has the lowest potential. However, notable exception is Mahaweli H area which should have produced high yields, but fallen under moderate category. Thus policies and implementation strategies should be formulated to reap the maximum production potential in the high potential areas while effectively managing natural resources of these

areas to cater the needs of the systems. Hitherto, two practical approaches i.e. a.) Intensify the rice production in high potential areas to provide the local requirement and use of low potential areas to cultivate high quality rice for export market and b.) Identify best areas for production of high quality rice for the export market under intensified management, to manage the excess production in the country while paving way to earn foreign exchange can be given.

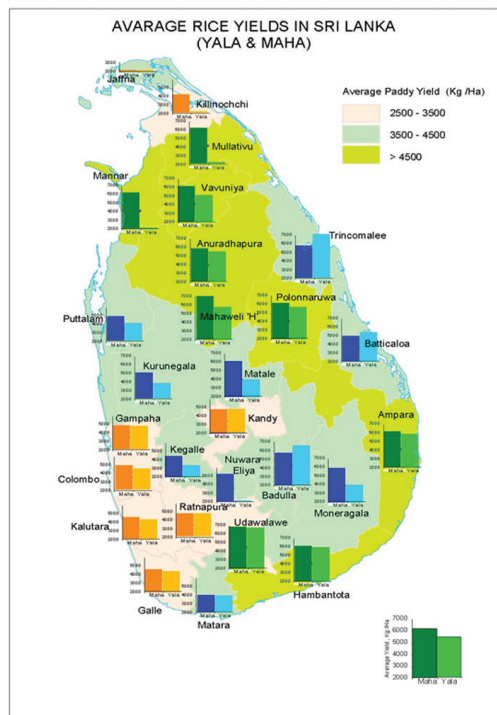


Figure 1: Average rice yields in Sri Lanka on district basis.
Source: Department of Statistics

Holistic approach on solving land related issues

In a rice tract, significant spatial yield variability exists within landscape and even this variability can be visualized within a block. Reasons for this variability are differences in crop management at block level, land drainage and their interactions within a yaya on soil fertility and land productivity. Productivity of such lands can be further improved by addressing the above issues, but holistic approach is needed rather than individual and location specific approach as the problems are compounded. This situation demands the need of developing yaya based recommendations on land and fertilizer management practices. However, the existing approaches in this regard are individual, labour intensive, time consuming and require lot of financial and other human resources. Therefore, lot of issues related to above remain unresolved, thus necessitates an efficient and cost effective method. In this context, use of information technology in resolving these problems is very useful and is timely needed. A novel method has been developed using Geographical Information System (GIS) of which data inputs were from Remote Sensing (RS) and limited field activities. This approach produces various thematic maps and highlighted land related limitations such as development of salinity, soil nutrient loss, waste of fertilizer through overland flow etc. causing the deterioration of resource base in a spatial context due to defective,

leaking or unsystematic designing of irrigation channels, drainage blocking, under capacity culverts, improper placement of farm roads, raising of tank bunds in neighbourhoods etc. which could be easily identified. In this approach, sampling and data collection is done on a grid basis at reasonable resolution ignoring individual farmer plot boundaries, but recommendations could draw for each and every farmer plot using GIS generated thematic maps. The approach reduces the cost involved in the present process of soil testing and fertilizer recommendation and also addresses the topography specific land related issues and thereby identifies land rehabilitation needs on yaya basis. As the approach highlights the causative factors in a spatial context, yaya based land management techniques including soil test based fertilizer recommendations for individual farmers can be formulated. Investigations conducted in the Anuradhapura and Kurunegala districts in the Dry Zone revealed the effectiveness and practical feasibility of the approach. In one such location namely, Kadurugasdamana in Anuradhapura, the spatial relationship of a leaking irrigation channel with available Zn content in soil is shown in Fig. 2. This indicates the impact of a leakage of irrigation water which causes continuous flooded condition in a certain section of the land had low Zn availability. The approach is already field tested and have shown the effectiveness, thus there is a necessity of implementing this programme at national level.

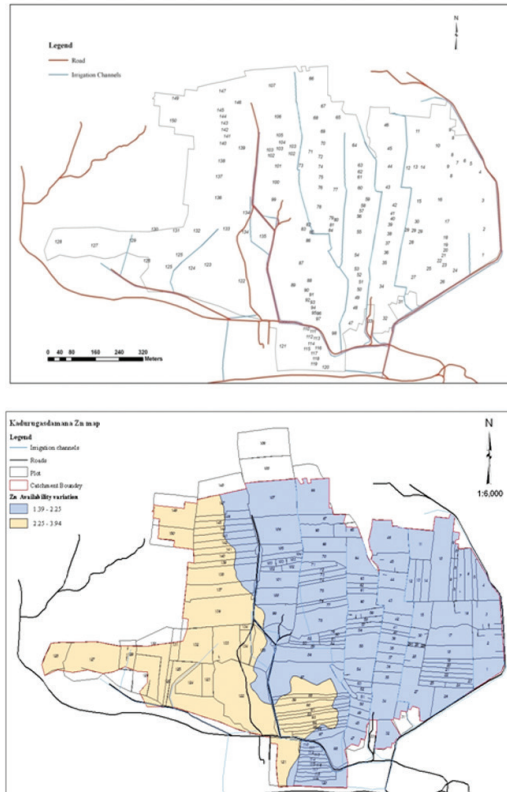


Figure 2: Availability of Zn in relation to drainage condition in Kadurugasdamana yaya (Source: Kendaragama *et al.*, 2009).

Need of a proper monitoring, planning and implementation of rice cultivation

Approximate rice extent in Sri Lanka is about 73,000 ha, but accurate information on total rice extent and cultivated extent in each season is not available hitherto. Having more accurate figures are impotent at present context for efficient distribution and usage of natural resources and inputs. Moreover, there is no reliable rice production forecasting system at

present depending on the weather condition during the season. The situation leads to dilemma of cluelessness of the production, thus affecting import export policies, preparation of production targets and precise allocation of resources. Therefore, need of a proper area estimation of rice cultivation extents and forecasting of production in regional and national levels depending on the weather situation before the ending of the season is a prerequisite. Related studies are limited in local context, but are practically implemented for yield

forecasting, nutrient management, biomass production and soil moisture assessments etc. in various countries, especially countries in the South Asian Region (Maas, 1988; Mass, 1993; Paul *et al.*, 2003). Madana and Kadupitiya (2003) have shown on a pilot scale that area estimation of rice is possible through remote sensing under Sri Lankan condition, but further investigations have never attempted afterwards. The technique is successfully used in many other countries at present (Paul *et al.*, 2003) and it is timely necessity to adopt such methodologies and approaches to generate more accurate information with less cumbersome. Other than the area estimation and yield forecasting, this technique can be used in managing natural resources. Diminishing of natural resources particularly land and water in terms of quality and quantity have been observed and the resources allocation should be done to maximize the productivity and the total production.

Conversion of rice lands for other uses

With the increasing population, the pressure on land particularly for housing and urbanization and the issue has been reviewed by Kendaragama and Bandara (2000). There is no systematic method or land use zonation to cater the need of urbanization and therefore unauthorized conversion of rice lands for other uses; mainly to cater the needs of rapid urbanization is in the rise. This is mainly due to lack of suitable lands at present for residential /commercial

ventures, particularly in urban and its suburbs. In the next decades, a rapid growth rate in the economy can be expected, and more and more people will move to urban areas and therefore rice lands in the flat terrain located within the urban neighbourhoods will have more pressure; thus some lands will wean out of rice cultivation. In addition, more rice lands will have to give away for the other development activities such as new highways and industrial zones and cities. Therefore, issue of conversion of rice lands should be evaluated not only in terms of technical point of view, but social perspective as well. Economic potential of rice lands and the environmental sensitivity of the locality of rice lands should be studied and preparation of proper zoning plans and strict implementation will be the long term solution for such issues. In such zoning planes, residential / commercial expansion should be considered but the lateral expansion beyond the identified boundaries should not be entertained at any instance.

Diversification of rice lands for crops other than paddy

Rice is mainly cultivated in azweddumized lands but other crops are also grown in a considerable extent of these lands. This has been done over decades (Kendaragama, 2001; Kendaragama, 2006). Diversification of rice lands was given a boost in seventies and eighties with the policies of then governments. However, situation was not sustained and thereafter faded away due to technical,

socio-economical and policy related reasons. The time has come again to think about diversification of rice lands for other crops, as the country is facing excess rice production and increasing production trend. At past the diversification was promoted basically as a water saving option which is still a valid factor and even in future agriculture as well. However, diversification of rice lands should be promoted not only as a measure of water saving, but as an improvement of land productivity. Diversifiable potentials /suitability for rice fields should be evaluated and mapped based on an indicator mapping approach, of which should be essentially based on the proportion of well drain soil member of each Yaya. In this context, a policy decision should be taken that only Other Field Crops (OFC) should be cultivated in such lands during the Yala season and allowing to cultivate rice in the Maha season. In such a way much needed other field crops requirement of the country can be met systematically while improving the productivity of the said rice lands. Moreover, attempts for diversifications should be implemented in low rice yielding areas, thus the diversification impacts on rice production will be marginal. However, diversification of rice lands on permanent basis should not be promoted in large scale, as it will not only cause rice lands completely being weaned out of rice culture and also it can cause hydrological issues through restricting storage for overland flow.

Consolidation of blocks (Liyaddes)

Cost of production of rice cultivation in Sri Lanka is high compared to that of other countries in the region. Availability of labour for rice cultivation is withdrawing gradually, thus cost of labour becomes significantly high. One of the solutions is the promotion of mechanization in paddy culture. The present momentum for mechanization is substantial, but land related constraints such as smaller block size remarkably reduces the efficiency and the applicability of the mechanization. Although, block size data in different regions are not available, the situation is severe particularly in Mid Country due the terrain conditions. Furthermore, the mechanization is further reduced in situations where soil profile is deep and such situations are prevalent particularly in the Low Country Wet Zone. However, most of the rice lands are comparatively small in many of the high potential major rice growing areas except some rice lands in the eastern region. Out of the above two land related limitations, the consolidation of blocks (Liyaddes) of smaller size into larger blocks is a feasible solution to eliminate difficulties for handling of mechanical equipment. Other advantages are the increase of the land availability for cultivation by reducing the area occupied by bunds and less labour requirement for cleaning and preparation of bunds.

As per to the present trends, mechanization of agriculture, to the maximum possible extent is essential to make the rice culture profitable and policies and strategies should be geared to increase the farmer to land extent ratio. Thereby, not only the farmer occupied in farming gets a better income, but also those who are in excess will find some other non-land dependent income generation activities, and reduces the socially lead land fragmentation issues as well.

However, practical implementation of consolidation seems not moving, the main reason being the cost of land development /reshaping, thus an incentive for those who are for consolidation would be a feasible solution.

Landuse rights for commercial rice/other crop cultivation

Abandonment of rice lands, particularly in the Low Country Wet Zone is a frequent happening basically due to unavailability of labour and also due to low rate of returns. However, at present the tendency of abandonment

has been low, due to many reasons. Although extensive surveys to asses abandonment have not been conducted recently, expert judgments through cross-sectional observations reveals that the percent abandonment is within the range of 10 - 20% as opposed to the abandonment rate in early years of the present decade, which was around 30 - 40%. At present the remaining uncultivated balance needs considerable investment for land development and individual farmers are either not willing or not capable of doing so. However, unsystematic infrastructural development and its resultant drainage impedances also have contributed to the issue by making the lands uncultivable. The situation suggests that whenever development activities are taking place, its impacts should be thoroughly studied and they should be planned in such a way that impacts are mitigated as much as possible. In this effort provincial or local level planning needs government intervention. However, in order to utilize abandoned rice lands, a possible alternative is, entrusting the land use rights for private organizations and working out of a practically feasible profit sharing system among the land owners.

Conclusions

Rice farming in Sri Lanka is a system with long history and it is a well established farming system within the Island. However, it must be emphasized that the existing problems in rice farming are repercussions of its mismanagement. Thus, importance should be given to scientific management of the existing system. In the process, extravagant introductions or innovations in terms of land management would not be required. Identification of high production regions, holistic management

approach coupled with systematic monitoring, policy implementation based on zonation plans, consolidation of rice lands for promotion of mechanization, policies for minimization of rice land abandonment etc are the key areas which needs the attention of rice land managers and policy makers.

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Recent Advancement in Mechanization of Rice Production in Sri Lanka

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Abstract

Farm Mechanization plays a major role in increasing productivity in rice farming in the country. Use of high efficient machinery in rice farming is significantly increasing as a precaution to labour scarcity problem in peak demanding period. Among the machinery the combine harvester is relatively new and impressive equipment which reduces the cost of rice production, minimizes the post harvest losses and reduces the farm drudgery. This machine has been effectively utilized in paddy harvesting in the Eastern part of the country. The Government support significantly contributed to the advancement of farm mechanization in the countries like Japan, Korea, Vietnam and the Philippines. The common factor leading to successful farm mechanization programs has been identified as the strong political will. Country's effort on mechanization must be anchored on a coherent strategy based on the actual needs and priorities of the farmers.

Introduction

Use of tools, implements and powered machinery in agricultural land development, crop production, harvesting, processing and storage is termed as "Agricultural Mechanization". It includes three main power sources: human, animal, and mechanical. Manufacture, distribution, repair, maintenance, management and utilization of agricultural tools, implements and machines is covered under this discipline with respect to supply of mechanization inputs to the farmer in an efficient and effective manner.

The level, appropriate choice and subsequent proper use of mechanized inputs in agriculture has a direct and significant effect on achievable levels of agricultural production, the profitability of farming and the environment. In general, in a situation where the expansion of agricultural land is limited, the application of advanced tools and machines does not, by itself, lead to increase unit yields. However, the full benefit achieved through the use of many advanced crop husbandry inputs such as improved seed, fertilizer, and pesticides, cannot be realized without the use of improved tools. In situations where land is not a

constraint, increased farm power can lead to direct increases in production by simply increasing the land area or animal numbers that one man can handle. Among many advantages of agricultural mechanization, main advantages include, increase quality and quantity, timeliness, reduce drudgery, and reduce production cost and low labour requirement. Above all, in the attraction of present generation back to agriculture, machinery plays a major role.

Sri Lanka has a written history more than 2500 years and being an agricultural country since civilization. Hand tools and animal draught implements were used at the cultivation of the staple crop rice in the known history and moved to mechanization from oxanization (using water buffalo draught implements). Reasons behind moving to mechanization are; to meet the increasing demand of food and as an alternative for labour intensive work because cost of labour is continuously increasing.

As labour migrates away from agriculture, the productivity of those who remain on the land needs to be increased significantly. In actuality, the percentage share of labour employed in agriculture decreased from 47% of total employment in 1990 to 36% in 1999. Such a low level of performance puts rural Sri Lanka on a collision course with demographic projections that foresee the need for agriculture to absorb over a quarter million new

workers by 2010. This shortfall can be only filled by mechanization, the only alternative for labour.

In many parts of the country, the most pressing need is to feed a growing human population. This requires sustaining food production, which can be realized by increasing land and labour efficiency in agriculture through farm mechanization.

Farm mechanization plays a significant role in every nation's economy. However, it is often misconstrued to mean modernization, beneficial only to industrialized countries with highly mechanized agriculture. Countries like Sri Lanka have to rely on a variety of imported farm machines, which are seldom appropriate for small farms.

The need of appropriate farm machinery for small scale farming systems would be a greater benefit of small-scale farmers. The activity provided a venue for the sharing of practical application of existing systems and technologies, as well as of strategies that will be most effective in tackling the very complex socioeconomic and environmental factors influencing the adoption and utilization of small farm machineries. Topics included: status and directions of small farm mechanization in the Asian and Pacific region; factors influencing the successful development, adoption, and utilization of small farm mechanization systems and technologies; problems, issues, and constraints in developing and

introducing small farm mechanization systems and technologies to the end users; and recommendations to enhance small farm mechanization development and adoption, particularly in less-developed countries. Lessons from the comparatively more developed countries in the region were shared, particularly in terms of their experiences in corporate farming and land consolidation, which make the joint use of land physically easier, including the communal use of large farm machineries and other modern farm facilities.

Status of farm mechanization in the country

The world agricultural scenario indicates that food security is the paramount concern of every nation. All technological advances in both developed and developing countries must gear towards increasing food production. Both the large-scale, specialized commercial agriculture and small-scale mixed semi-subsistence types of agriculture play vital roles to attain this objective.

The average operational farm size in Sri Lanka is 1 hectare where as in Asia it is ranging from 1 to 3 ha. Research funding for agriculture in the country is very low compared to the middle income countries, which is about 2 percent. The share of the Asian and Pacific region in the global agriculture machinery consumption is only 10 percent compared with Europe, which

has the highest share of 80 percent. As of 2002, Japan had the highest number (1,042,000 units) of harvesters/threshers in use, followed by China (197,000 units), while Sri Lanka had the lowest number of only 10 units of harvester/threshers.

Details of farm machinery in the country in operational wise

Land preparation

Cattle and buffaloes were used mainly before tractorization (before 1960s) for farm operations in Sri Lanka. One pair of male cattle can develop 1 hp and can work six hours a day. The economic life is 6-10 years. A pair of buffaloes can develop 1-1.5 hp but work slowly. Their economic life is 10-12 years. Today the total buffalo population in Sri Lanka is estimated to be about 0.8 million and few of them are used for draft. The total cattle population is about 1.3 million and about 12% of them are used for draft. However, the use of cattle and buffaloes for farm work is drastically diminishing specially in the major rice cultivating areas. A simple mould-board plough and the wooden country ploughs are used as tillage implements. Wooden boards are used for leveling lowland paddy fields. In overall, less than 10% of the paddy fields in the country, where the land holding size is very low, are ploughed using animal power (actual figures are difficult to obtain).

During past four decades power tillers or two wheel tractors and the rotary plough attached to it became popular in all kinds of cultivation (Fig. 1). The rotary plough is used for primary and

secondary tillage operations. The two wheel tractor population in Sri Lanka is approximately around 250,000 in 2009.



Figure 1: Most popular version of riding type power tiller.

Four-wheel tractors are used for land preparation and transportation. The tine tiller attached to the four wheel tractor is widely used for tillage operations in paddy cultivation especially in the Eastern province. The four wheel tractor population in the country is approximately 50,000 and is gradually increasing after the ending of conflict in the North.

Plant establishment

The most common practice is broadcasting. However, parachute seedling broadcasting system and direct row seeding (Fig. 2) by paddy

seeders are slowly catching up. Use of manual transplanter is still popular among very few farmers but the number is not significant. Though the row transplanting with mechanical plant establishment methods is a proven technology to increase yield, a suitable transplanter is yet to be selected and introduced.



Figure 2: Drum type direct row seeder.

Pest and weed control

In paddy cultivation weed control is done mainly using chemicals. But occasionally, mechanical weed control systems are practiced when crop is established in rows. Knapsack type hand sprayer is commonly available at almost every house hold for insect pest control purposes. Locally produced hand sprayers are very competitive with the imported brands in the market.

Harvesting and threshing

Harvesting in paddy cultivation is done mostly by manually but mechanical harvesting is gaining popularity due to scarcity of labour. Threshing and cleaning are now done mainly by machines such as threshers (Fig. 3)

and fans which could be coupled to two or four wheel tractors. Recently four wheel tractor coupled with high capacity threshers became more popular due to its attractive capacity and adaptability over different climatic conditions. Combine harvester (Fig. 4) was introduced in Maha 2002/2003 season. It is relatively new and impressive equipment, which reduces the cost of rice production, minimizes the post harvest losses and reduces the farm drudgery. A Study revealed that the adoption of combine harvester was rapid (Fig. 5) and it is significantly determined by the age, education, income and experience of the farmer. Almost all the respondents in this study were having positive attitude towards combine harvester. Further, this study

revealed that the low cost, low labour requirement and low time consumption were the main factors for high diffusion of combine harvester. The results indicated that when switching

from manual harvesting to combine harvester a farmer can save Rs. 3500 per acre and obtain extra yield of 290 kg/acre.



Figure 3: Two wheel tractor coupled to high capacity thresher.



Figure 4: Conventional type combine harvester.

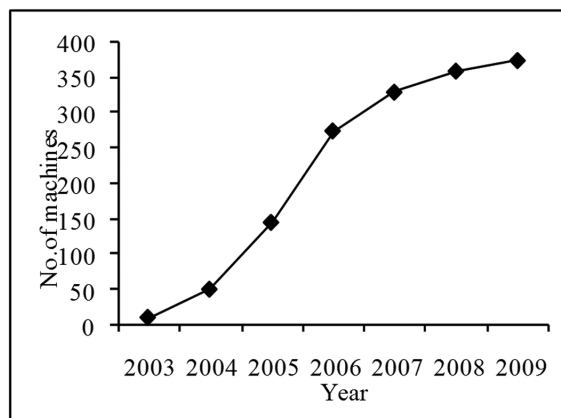


Figure 5: Diffusion of combine harvester in Ampara district.

Productivity of Sri Lankan rice cultivation is rapidly increasing with the rise in farm mechanization support. Most of the major cultivation areas are now in transition from labour intensive to control intensive agriculture. Precision agriculture and automation is the current trend in agricultural mechanization in the world. Irrigation system machines, planting machines, power sprayers, combine harvesters, dryers using biomass fuel, silo and storage handling and advanced and high quality rice mill machines are likely to be adopted by Sri Lankan farmers in the near future.

Case studies on systematic farm mechanization: strategies and approaches in Asia

Land consolidation is a successful approach in Japan to mechanize the previously small farm landholdings. The government's strong political will and commitment coupled with

the farmers' cooperation contributed to the success of land consolidation. The consolidated fields allowed for the entry and travel of tractors and other implements, provision of efficient field irrigation and drainage, road access systems, and the establishment of on-farm post harvest facilities and other infrastructures.

The experience in Korea proves the appropriateness of mechanizing the production and post-harvest operations of horticultural crops planted mostly in the uplands with 7 percent slope and above. These sloping lands comprise 61.6 percent of the country's total cultivated area. The country's integrated mechanization system employs the use of appropriate farm machinery and implements, from the seeding nursery to production to post-harvest. Focus is on competitive crops like ginseng, carrot, garlic, chinese cabbage, chinese leek, red pepper and onions.

The key to the successful implementation of the village-based cashew nut processing in the Philippines lies on several factors. They are appropriateness of small-scale machine, skill-building and market-linkaging activities, networking with other government agencies, full support of the local government unit, initial funding assistance from outside source, and positive and active collaboration of the cooperator.

Rice mechanization in Vietnam has achieved significant developments in terms of acquisition and utilization. Most machinery are imported and utilized in operations like tillage, sowing, transplanting, and harvesting. Also, the country has developed and improved axial flow threshers, rice combines and dryers. Research and development activities on rice mechanization have gained continuous progress through the endless efforts of administrators, researchers, machine designers and farmers.

Technology trends and directions

Japan, Taiwan, and Korea are among the countries with highly mechanized paddy cultivation farming operations. Common among these countries is the prevalence of strong political support and farmers' cooperation, paving the way for advancements in agricultural mechanization technologies and systems.

Taiwan's agriculture is 98 percent mechanized. Manufacturers of dryers

in this country are able to produce competitive products. Products using biomass as fuel are also becoming popular. Mini-power tillers have the highest market share in both domestic and international markets.

Government support significantly contributed to the advancement of farm mechanization in Korea. This involves institution of policies on financing for farm machinery and projects as well as subsidies by supplying farmers with machines at half the price. Machines and equipment have been developed for specific farm operations from land preparation to planting, to harvesting, to post harvest and transport.

Indonesia, Thailand, Malaysia, Vietnam, and the Philippines have been receiving similar support from the government for its special projects and programs on farm mechanization. However, level of mechanization is medium to low due to such factors as: lack of resources, infrastructure, and institutional arrangements; prevalence of manual labour/operations; and lack of policies that support the general economic welfare of the different stakeholders in the agricultural machinery industry.

Problems, issues and constraints

The barriers that impede the growth and sustainability of farm mechanization industry and programs in the country can be classified into technological constraints, socio-cultural and behavioral barriers,

financial and economic problems, and environmental issues. In Sri Lanka, local manufacturers are still in the “cut and weld” level of manufacturing technology. Low or substandard machines continue to be peddled in the market, while some machines are not suitable to the farming conditions of the users. Farm machines are likewise beyond the reach of most farmers owing to high acquisition and maintenance costs.

Small-size farm is a big issue when it comes to mechanization because it is against the “economies of scale”. Mechanizing small, non-contiguous parcels of lands may prove to be inefficient especially in operations like land preparation and harvesting. Poor rural infrastructures such as roads, bridges, canals, and power network also pose as a major obstacle to farm mechanization. However, the income of farmers has remained very low and then the purchasing power is also very low.

In terms of environmental issues, high energy-consuming large machines with emissions pollute the environment and have become an important concern. Human being, as a component of the environment, is also now being considered in machine development. “Machines with a human face” is the byword in some developed countries, which put great consideration on the plight of ageing and women farmers and on safety issues.

Recommendations and directions

Due to the nature of small landholdings, the following criteria for designing and adopting machines have been identified by Farm Mechanization Research Centre, Maha Illuppallama - the main agricultural machinery research institute of the Department of Agriculture.

Technology and systems development

- Machines should be compact, light, low-powered and multi-purpose. Locally available materials must be incorporated in fabricating machines to reduce the manufacturing costs. Manufacturing and designing parts must also be precise.
- Small-size tractors, mini-power tillers and small farm equipment must meet the needs of small farmers. Operator’s safety and comfort must also be considered.
- The high cost of fossil fuel and also unavailability of fossil fuel emphasize the need to develop energy-efficient machines by harnessing non-conventional sources of energy.

Adoption and utilization

- Information dissemination activities through multi-media, fairs and exhibits, and similar modes should be actively pursued in strategic locations where machines are actually needed.

- Training local craftsmen in manufacturing technology, operation, repair and maintenance would promote local manufacture of agricultural machinery.
- Farmers' organizations or cooperatives can also be tapped particularly in setting up joint use of farm machineries and other modern farm facilities.
- Government-private sector complementation is vital in promoting farm mechanization. Service centers could be established in rural and remote areas; and financial assistance, loans, and subsidies must continue to provide machinery owners and users of continued farm production operations.

Conclusion

The level and appropriate choice of agricultural mechanization has direct effects on land and labor productivity, farm income, environment and the quality of life of farmers in the country. Hence, basic farm mechanization requirements to cater to needs of the different levels of farmers in different agro- ecological regions must be met, such as: suitability to small farms; suitability to large farms, simple design and technology; versatility for use in different farm operations; affordability in terms of cost to farmers; and most importantly, the provision of support services from the government and the private sector/manufacturers.

Cases of highly mechanized countries point to a common factor leading to successful farm mechanization programs, that is, strong political will. Hence, each country's effort on mechanization must be anchored on a coherent strategy based on the actual needs and priorities of the farmers.

Past Achievements and Future Directions of the Seed Paddy Sector in Sri Lanka

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Abstract

Rice is the most important food crop of Sri Lanka's domestic agriculture sector. Usage of quality seed is one of the most important aspects in raising rice yield. Both formal and informal seed supply systems play important roles in meeting annual seed requirement of about 96000 mt. Basic seeds of twenty eight paddy varieties developed by the Rice Research and Development Institute, Batalagoda and its satellite stations namely, Bombuwala, Ambalantota and Labuduwa and recommended by Department of Agriculture are produced in seven government seed production farms. Certain amount of certified seed is produced by Seed and Planting Material Development Center. In addition, private sector is also involved in production of certified/ quality assured seeds. On average, 2,012 mt of basic seed (2.2% of national seed paddy requirement) per year and 14,391 mt of certified/quality assured seed (15.7% of national seed paddy requirement) per year were produced during the period from 2000 to 2009. Seed certification process is handled by 19 Seed Certification regional units and four Seed Certification laboratories across the country. Basic seed distribution is implemented through the 12 regional offices of the Assistant Directors of Agriculture (SPMDC) and both private and public sector participate in certified seed distribution. It is estimated that minimum of 25% of national seed requirement should be supplied as certified/ quality assured seed in order to obtain a significant increase in quality and productivity of rice. Hence, enhancement of private sector participation, coordination of activities involved in seed industry, strengthening of infrastructural facilities of government seed production farms and capacity building of the public and private sector staff are essential.

Introduction

Rice cultivation dates back to ancient times and rice is the staple food of Sri Lankans and is the most important food crop in the domestic agricultural sector occupying a central position

in the rural economy, society and culture. Rice has contributed to 14% of agricultural GDP in 2009 (Central Bank, 2009). Successive governments since independence have aimed at achieving self sufficiency in rice in order to protect food security of the nation.

The average annual extent cultivated during the last 5 year period (2005 to 2009) is around 0.94 million ha and the national rice production revolves

around 3.4 million tons while the average yield is around 4.2 t/ha (Table 1).

Table 1: Annual extent cultivated (ha), production (t) and average yield (t/ha) of rice in Sri Lanka during 2005-2009.

Year	Cultivated extent ('000 ha)			Production ('000 t)			Average yield (t/ha)		
	Maha	Yala	Annual	Maha	Yala	Annual	Maha	Yala	Annual
2005	581	357	938	2012	1233	3245	3.955	3.976	3.963
2006	591	319	910	2136	1206	3342	4.069	4.263	4.137
2007	525	291	816	1973	1158	3131	4.299	4.543	4.389
2008	582	471	1053	2125	1750	3875	4.181	4.195	4.187
2009	632	345	977	2384	1268	3652	4.421	4.186	4.336
Average	582	357	939	2126	1323	3449	4.185	4.233	4.202

Source: Department of Census and Statistics

If the factors affecting rice cultivation and production remain favorable, country is in a position to meet her entire demand of rice. However, with current population growth rate of 1.2%, assumed annual per capita rice consumption of 113 kg, annual seed requirement of 0.1 million tones and post harvest losses of 5%, the projected annual rough rice requirement of the country would be 4.53 million tons in 2020. The forthcoming decade would be comparatively more challenging for rice production as it may not be possible to further expand cultivated extent significantly. Therefore, increasing productivity of the paddy

lands by enhancing the use of inputs, adoption of appropriate technology and providing efficient extension services to satisfy the future demand and ensure food security of the country become imperative. Among the inputs which are used in paddy cultivation, seed paddy plays a unique position as it is the carrier of genetic potential for high yield.

The present average annual national seed paddy requirement is around 100,000 tons (Table 2). Both formal as well as informal seed supply systems in Sri Lanka play important roles in meeting this requirement.

Table 2: Seed paddy requirement in Sri Lanka during 2000-2009.

Year	Seed paddy requirement					
	Maha		Yala		Annual	
	Bu ('000)	000 ton	Bu ('000)	000 ton	Bu ('000)	000 ton
2000	2745	56.27	1645	33.72	4390	90.00
2001	2395	49.10	1595	32.70	3990	81.80
2002	2550	52.28	1710	35.06	4260	87.33
2003	3010	61.71	1905	39.05	4915	100.76
2004	2605	53.40	1290	26.45	3895	79.85
2005	2905	59.55	1785	36.59	4690	96.15
2006	2955	60.58	1595	32.70	4550	93.28
2007	2625	53.81	1455	29.83	4080	83.64
2008	2910	59.66	2355	48.28	5265	107.93
2009	3160	64.78	1725	35.36	4885	100.14
Average	2786	57.11	1706	34.97	4492	92.09

The majority of farmers in Sri Lanka depend on informal system to meet their seed requirements in which seeds saved from their previous crop or seeds borrowed from neighboring farmers are involved. Yet, the formal seed supply system is more important in assuring the quality of the seeds used. The formal seed paddy production system includes a wide range of activities i.e. variety development, seed production, certification and distribution. This paper examines the important attributes of the formal seed paddy supply system in order to provide a comprehensive picture of the current status and past achievements of seed paddy sector and also to make suggestions on where improvements can and should be made in the seed paddy industry.

Variety development

The development of new varieties is an essential element of the formal seed production programme. Varietal development is conducted by the Rice Research and Development Institute (RRDI) Batalagoda, and Regional Centers at Bombuwala, Ambalantota and Labuduwa.

Seed production

There are two steps in the seed production program in Sri Lanka namely production of Basic seed (Foundation and Registered) and Certified/Quality assured seed production. At present, there are 28 recommended varieties (Table 3) in seed production program.

Table 3: Number of rice varieties included in the seed production programme in year 2009.

Age Group (Months)	Varieties					Total
	White Nadu	White Samba	White Basmati	Red Nadu	Red Samba	
2½ months	Bg 250					1
3 months	Bg 300, Bg 304, Bg 305, At 307	At 308	At 306	Bw 272-6B, At 303		8
3½ months	Bg 94-1, Bg 352, Bg 357, Bg 359, Bw 363	Bg 358, Bg 360		Bw 351, At 353, Bw 361, At 362, Bw 364	Ld 356	13
4 -4½ months	Bg 379/2, Bg 403, Bg 454	Bg 450		At 402, Bg 406		6

Basic seed (Foundation and Registered) production

Basic seed production starts with a small quantity of breeder seeds obtained from the research centers and two multiplications are done at the government seed farms. The first multiplication is referred to as foundation seed and second is referred to as registered seed. Both productions of foundation and registered seed

are managed by the Seed and Planting Material Development Center (SPMDC) of the Department of Agriculture. Currently, seven government seed production farms of SPMDC are involved in basic seed production (Table 4). In addition to that, a few selected farms of Provincial Departments of Agriculture, private seed producing companies and contract growers are involved in registered seed production.

Table 4: Basic and certified seed paddy production in government seed farms in 2009.

Seed Farm	Maha 2008/09 (t)			Yala 2009 (t)			Total (t)		
	FSP	RSP	CSP	FSP	RSP	CSP	FSP	RSP	CSP
Aluttarama	10.2	198.3	22.7	17.4	53.6	58.8	27.6	251.9	81.5
Ambalantota	1.6	31.1	6.4	2.67	51.4	13.8	4.2	82.5	20.2

Bataata	1.5	34.9	0.74	1.17	11.39	3.3	2.6	46.2	4.04
Kantale	6.8	214.2	-	16.2	238.8	7.1	23.0	453	7.1
Mahailluppallama	14.8	161.9	-	0.57	67.9	-	15.3	229.8	0
Malwatta	5.9	168.8	-	7.4	165.8	11.4	13.3	334.6	11.4
Polonnaruwa	11.5	325.9	7.7	13	303.2	-	24.5	629.1	7.7
Total (t)	52.3	1135.1	37.5	58.4	892.1	94.4	110.7	2027.2	131.9
Total (bu)	2551	55377	1836	2851	43516	4593	5402	98893	6429

FSP – Foundation seed paddy, RSP-Registered seed paddy, CSP-Certified seed paddy

(Source: Annual Performance Report 2009, Department of Agriculture)

The highest basic seed production by SPMDC during last 9 years was recorded as 2558.75 t (124820 Bu) in 2006. It was equal to 2.7% of the total seed paddy requirement of the country. The average of basic seed production during the last 9 years was 2012 t (Table 5) and it was equal to 2.2% of the total seed paddy requirement of the country. The expected multiplication of seed paddy under favorable

conditions is 1:40. Accordingly, basic seed production equaling 2.5% of the national seed paddy requirement is logically sufficient to meet the entire seed paddy requirement in Sri Lanka. As 100% use of quality seed is not realistic, the current basic seed production level is satisfactory. However, efficiency should be further improved to enable smooth functioning of seed multiplication process.

Table 5: Seed paddy requirement and availability (2001-2009).

Year	Requirement (t)	Seed Availability (t)			Percent Seed Availability		
		Basic	Certified / Quality assured	Total	Basic	Certified / Quality assured	Total
2001	81822	1890.0	11219.0	13109.0	2.3	13.7	16.0
2002	87384	1665.0	13062.0	14727.0	1.9	14.9	16.9
2003	100718	1807.0	15144.0	16951.0	1.8	15.0	16.8
2004	79774	2058.4	14715.8	16774.2	2.6	18.4	21.0
2005	96061	2577.0	16286.0	18863.0	2.7	17.0	19.6
2006	93324	2558.7	14648.7	17207.5	2.7	15.7	18.4
2007	83713	1827.2	16581.9	18409.1	2.2	19.8	22.0
2008	107931	1632.1	11123.6	12755.7	1.5	10.3	11.8
2009	100200	2137.8	16740.5	18878.3	2.1	16.7	18.8
Average	92325	2012.0	14391.3	16408.3	2.2	15.7	17.9

Certified/ Quality assured seed production

Collaborative certified seed paddy production programme is conducted with the involvement of the public and private sector as shown in Table 6. Under this programme registered

seed paddy is issued by the SPMDC to the relevant agencies. In addition to this collaborative certified seed paddy production programme, SPMDC itself produces certified seed paddy through contract growing programme and utilize the lowlands of government seed farms that are not used for basic seed production.

Table 6: Details on collaborative certified seed paddy production program in 2009.

Agency	Maha 2008/09		Yala 2009		Year 2009		Production	
	Quantity Issued by SPMDC (t)	Extent Certified (Ac)	Quantity Issued by SPMDC (t)	Extent Certified (Ac)	Quantity Issued by SPMDC (t)	Extent Certified (Ac)	t	Bu
Provincial Agriculture Department	108.6	994.5	44.9	768	153.5	1762.5	3613.1	176248
Inter Province	53.5	630.5	30.2	415.5	83.7	1046.0	2144.3	104600
Agrarian Service Department	14.7	139.8	29.4	41.5	44.1	181.3	371.7	18132
Mahawelli Authority	42.7	344.1	29.5	241.5	72.2	585.6	1200.5	58541
Cooperative Society	13.4	190.0	8.3	123.5	21.7	313.5	642.7	31351
Farmer Organization	77.7	842.2	72.0	699.5	149.7	1541.7	3160.5	154171
Individual Farmers	88.5	864.3	55.4	416.0	143.9	1280.3	2624.6	128029
Private Company	41.3	516.0	34.5	482.5	75.8	998.5	2046.9	99849
Others	1.8	17.4	1.3		3.1	17.4	35.7	1741
Total	442.2	4538.8	305.5	3188	747.7	7726.8	15840	772662

Fig. 1 provides an overview of production of certified seed by SPMDC contract growing system. There was a drastic reduction in the production of certified seed paddy in 2003 and 2004 when compared to 2001. This was mainly due to government decision to hand over certified seed production starting from Yala 2002 onwards to private sector with a view to increase the private sector participation in accordance with the seed policy of 1996. However, decision was

reversed later with the election of a new government and SPMDC was requested to initiate production of certified seed at the latter part of 2005 under contract growing system.

Unfavorable weather conditions prevailed during the harvesting periods, high farm gate prices, inconsistent government policy and inadequate funds badly affected the quantities purchased under the contract growing programme.

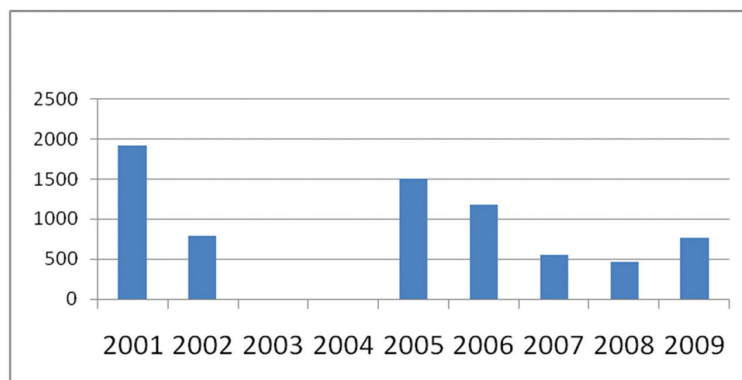


Figure 1: Certified seed production (t) of SPMDC under contract growing system during 2001-2009.

It is clear that SPMDC cannot shoulder the entire responsibility of supplying certified seed requirement of the country with limited resources at its command. Therefore, involvement of private sector in certified seed production is a prerequisite to keep favorable seed renewal rate of the country. About 20-25% of the total seed requirement should be supplied as certified/quality assured seed in order to obtain a marked increase in rice productivity (Senevirathna *et al.*, 2008). Therefore, farmers need to renew the seed paddy

with certified seed/quality assured seed preferably in every season or at least once in 4 seasons.

Fig. 2 indicates the quantities of certified seed produced under collaborative certified seed production programme. It reached the highest amount of 15,889 t followed by 15,840 t in 2007 and 2009, respectively. These production levels are equal to 18.9% and 15.8%, respectively of the total seed paddy requirement of the country.

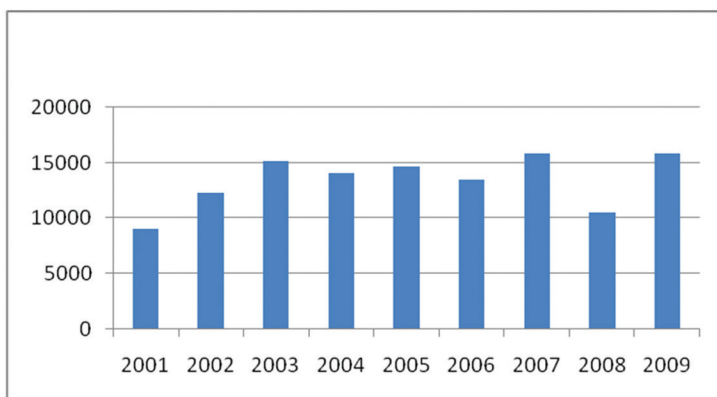


Figure 2: Collaborative seed paddy production (t) during 2001-2009.

Average quantity of certified seed produced from 2001 to 2009 under both SPMDC certified seed production programme and collaborative certified seed production programme was around 14,391 t (Table 5) and it was equivalent to 15.7% of the total seed paddy requirement of the country. This level and achievement can be considered to be a satisfactory situation. The highest quantity of certified seed paddy amounting 16,741 t was produced in 2009 while the highest percentage availability of 19.8% was recorded in 2007. Both the lowest amount of production (12,756 t) and availability percentage (10.3%) of certified seed paddy were recorded in 2008 (Table 5). It was mainly due to unfavorable weather conditions prevailed in 2008.

Still there is no proper data base to confirm actual quantities produced by the private sector. However, actual amount should be higher than the

amount at the records of SPMDC as some companies who have their own quality control system do not participate in the government seed certification programme. Thus qualities of these products are questionable. Therefore, a proper system should be developed under the Seed Act to regulate the certified seed paddy production programme. It will be helpful for future planning of the seed paddy industry in Sri Lanka.

Seed certification

Seed certification is an important activity in any formal seed production programme. Main objective of the seed certification is to ensure the quality of seed in any multiplication stage without dilution of the genetic identity (Madawanarachchi and Weerasena, 1990). Therefore, it assures the quality of seed available to the farmer. To obtain the seed certification, the seed should satisfy the field and laboratory

level standard as shown in Table 7. Field level and laboratory level standards are monitored by the nineteen Seed

Certification Regional Units and four seed testing laboratories, respectively.

Table 7: Field level and laboratory level standards for seed paddy certification in Sri Lanka.

Seed Class	Field Standard						Laboratory standard						
	ODV (Earheads/100sq m ²)	Weedy Rice	Isolation (m)	Moisture(%)	Germination (%min)	Viability (%min)	Pure seed (%max)	Inert matter (%max)	ODV (max/500g)	Weed seed (max/500g)	Insect & Mechanical damage seed (max/500g)	Smell	Appearance
Foundation	9	0	1	13	85	90	99	1	20	5	200	Good	Good
Registered	18	0	1	13	85	90	99	1	40	10	200	Good	Good
Certified	18	0	1	13	85	90	98	2	100	10	200	Good	Good

Source : Seed Certification Service, Department of Agriculture.

Seed distribution

Seed should be made readily available on time to farmers at affordable prices. Therefore, proper seed distribution is a very important activity in any seed production programme. The basic seed is distributed through 12 regional Assistant Directors of Agriculture (Seed and Planting Materials). Regional seed committees and head office of the SPMDC coordinate the distribution of the basic seed among the seed producers.

Most crucial issue in seed distribution is mismatch of the time of availability with the time of demand for seed. It is mainly due to poor infrastructure facilities in implementing, harvesting,

processing, and storage at the government seed producing farms. Furthermore, seed production farms do not receive priority in national irrigation schedules. It is essential to take every effort to increase efficiency of the government seed production farms to streamline the seed production industry in Sri Lanka.

Certified seed paddy produced by SPMDC and excess basic seed are sold via extension division (provincial and inter provincial), Agrarian Service Centers, farmer organizations, individual farmers, etc. However, distribution channel and system of SPMDC do not function effectively and efficiently. Therefore, the dealer net work should be further developed.

Even though private sector is relatively efficient and effective in seed paddy distribution, sometime their market produce has not undergone the complete certification process and/or is below the accepted quality standards. Therefore, it is important to regulate the seed paddy marketing system under the present Seed Act.

Future direction for the seed paddy production program

01. Enhancement of private sector participation

An organized seed production programme was seen with the advent of high yielding varieties as far back as 1950s. The Department of Agriculture was the sole supplier of seed in the country up to late 1980s. However, as the demand for quality seed have shown an increasing trend which cannot be met by the Department of Agriculture alone, the government took the initiative to involve the private sector in seed production and marketing with the dawn of the decade 1990.

As the demand for quality seed keeps on increasing, private sector involvement in local seed production also has increased. This has led the government to declare the National Seed Policy (NSP) in year 1996. The main aim of the NSP with respect to seed paddy was to facilitate establishment of viable seed enterprises and to enable the farmers to have easy access to quality seed from the local supply.

During the last two decades private sector has significantly contributed

towards increasing the quality seed paddy production of the country. At present, four leading agribusiness companies, few small scale seed enterprises, farmer organizations, multipurpose Cooperative Societies are mainly involved in seed paddy production and marketing. Private sector involved in the local seed production programme has shown in increasing trend over the years. In 2009, it was estimated that 8.5% of the total seed requirement is supplied by the private sector, which was a quite significant achievement.

In order to keep seed entrepreneurs to remain in the industry as well as getting more participation in the industry, due consideration should be given to the following aspects.

i. Supply of basic seed (Registered seed paddy)

Department of Agriculture (DOA) is the sole supplier of basic seed. The basic seed of the varieties requested on time and in correct quantities should be available to enable incorporation of basic seed timely into out grower programs. Most often, the basic seeds are not available timely leading to frequent changes in the out grower programs.

ii. Training programs for seed producers

It is important to have well trained seed producers so that the quantity of seed produced should be in conformity with the accepted certification standards.

This has become important as the regulations stated in the Seed Act impose a label being exhibited on the seed bag with the quality parameters belonging to the particular seed class. Most of the seed producers in the industry are not adequately trained and do not possess correct technical know-how on quality seed production. Therefore, it is important to train this category of people before embarking any new programs. At present, there is hardly any organized training programme on seed production for these enterprises. The DOA as a government agency responsible for ensuring high quality seed production of the country should assist the private sector by providing them the necessary training to upgrade and update the quality of the human resources of the private sector. This will enable the private sector to produce seeds of desired quality stranded. These training programs should cover all aspects of seed production from field inspection, processing and even seed testing. The training programs should not be confined to technical aspects, but also should cover the entrepreneurship aspects of the seed industry.

iii. Supply of seed processing machines

One of the major constraints to promote quality rice seed production was the shortage of seed processing capacity in the country. Large numbers of seed processing machines were owned by the DOA numbering to about 26. Seven Multipurpose Co-operative Societies involved in rice seed production also

have one machine each. Four large scale agri-business companies also possess some processing facilitation, and other enterprises are without adequate facilitates for seed processing. Seed processing machines are not available for sale in the country and have to be imported. To boost the seed processing capacity of the DOA, 10 small seed processors were imported on duty-free basis in the latter part of 1990 s, and sold at the imported cost to selected seed enterprises. Seven co operative societies involved in seed production at that time were benefited from this offer. As more and more entrepreneurs enter the trade, it becomes opportune to formulate a similar scheme to import small scale seed processing machines and to sell them to private sector seed enterprises on concessionary terms. This benefit should be awarded on the basis of past performance and contribution made to seed industry.

02. Co-ordination of activities involved in seed industry

As a result of the private sector involvement, the seed supply system today has become a multi-institutional entity involving public sector, co operatives and private sector. Due to its multifunctional nature, co-ordination of activities through National Seed and Planting Materials Committee (NSPMC) becomes an important function. Although, such a mechanism existed in the past it is not functioned at present. Therefore, it is suggested to review the NSPMC and institutionalize it under the Ministry of Agriculture. The secretary of the Ministry of

Agriculture should chair the committee and the Director (SPMDC) of DOA should function as the secretary. Representatives from the private sector organizations, Co-operatives and all committed public institutions should be members of this committee. This committee should meet at least twice a year before each cultivation season and whenever it becomes necessary by the industry. Proposed Terms of Reference (TOR) for the committee are briefly indicated below;

- i. Review the progress of the seed industry.
- ii. Review the present policy and advice the government on any new changes to be made.
- iii. Review the present data base of the seed industry and develop a mechanism to obtain reliable data pertaining to the industry.
- iv. Development of a forecasting system for future seed paddy demand.
- v. Training need identification for both production and processing of seed and liaise it with relevant government agency for obtaining successful results.
- vi. Identification and facilitation of technical assistance required for seed industry.

03. Strengthening of the government seed production infrastructural facilities

Inadequate attention was paid during this decade for strengthening of infrastructure facilities in seven government seed paddy production farms of the

SPMDC. This has resulted in lowering the efficiency of seed production leading to delays in supply of registered seed paddy on time. Therefore, seed drying, processing, storage, and irrigation facilities for seed production fields need to be rehabilitated.

The seed processing machines in most of the government farms are over 25-30 years old and can be considered obsolete at the present context. Frequent shut downs and repairs are reported during peak processing times. Even though, adequate storage facilities are available in government farms, they need to be rehabilitated in order to improve the quality of basic seed produced. Similarly, drying facilities too needed to be rehabilitated to enable drying of seed be performed effectively.

Availability of quality seed is one of the key factors that contribute the national rice production. As such, the facilities are needed for improvement in the production process should be adequately funded from the national budget. About Rs. 50-80 million will be sufficient for the improvement of those facilities, and improvement should be completed soon. This investment will lead to improve the quality of basic seed as well as timeliness in seed supply.

04. Capacity building of the public and private sector staff

Very little attention has been paid to this aspect in the recent past. In both public and private sector organizations, there

is a need to upgrade the knowledge and skills of staff involved in seed production. This is especially needed for the private sector as most of the staff involved in seed production has not been adequately trained. The DOA which has the technical capabilities on quality seed paddy production could help in improving knowledge and skills on seed production in the private sector. Training should cover both theoretical and practical aspects. Field identification of ODVs, characteristics of varieties, weeds and their removal at correct times and seed processing are important aspects that should be covered on these training programmes. Therefore, providing all staff engaged in seed production with this type of training at least once in every season will no doubt contribute to raise quality of seed produced.

The staff of the DOA Seed Division has not been adequately exposed to in-country or out side training opportunities during the past decade. Only a very few officers, have got the opportunity of attending foreign training. Exposure to foreign training

is valuable in updating the knowledge and skills on recent development on seed technologies.

It was very strongly felt that none of the staff members of the division got any opportunity to pursue post graduate studies on Seed Technology or any other related field during the last decade. An organization like SPMDC of DOA should possess well qualified competent staff, so that it would by no means second to any other institution of this nature. However, when compared to other institution of the DOA, very scant attention has been paid to improvement of technical skills of its staff. Therefore, this issue should be addressed and a comprehensive programme should be drawn up to rectify the situation. It is quite justifiable as this division contributes significantly to national rice production by providing basic seed material for the production of quality seed paddy. The measures that will be taken to improve the capacity building of the staff will be reflected in the quality of the seed produced.

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Climate Change and Rice Cultivation in Sri Lanka: Impacts, Adaptation and Mitigation Strategies

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Abstract

Rice is grown in 31 of the 46 agro-ecological regions of Sri Lanka and its productivity and cultivated extent are highly sensitive to climate change. Global climate change initiated by the increased atmospheric concentrations of greenhouse gases including carbon dioxide and methane has already brought about many changes in the climate of rice-growing regions of Sri Lanka. Increased maximum and minimum temperatures beyond the threshold for rice, reduced quantity and quality of irrigation water, increased frequency of drought periods, expanding soil salinity and altered pest and disease populations are some of already experienced changes influenced by the climate change that are relevant to rice production in Sri Lanka. Increased atmospheric CO₂ concentration has largely positive impacts on the growth and yield of rice, with the positive impacts being greater for longer duration varieties provided the crops are grown with adequate nitrogen and other inputs. Analysis of long-term changes of national average rice yields and atmospheric CO₂ concentrations since 1960 has shown that while the varietal improvement programs in Sri Lanka have effectively captured the beneficial effects of increasing CO₂ under irrigated rice ecosystems, it has not been so in the rainfed ecosystem. Increasing temperatures, especially when coupled with high relative humidity levels could increase spikelet temperatures above the threshold level resulting pollen sterility which may encourage cross fertilization. Although this would not cause a yield reduction initially, there are important long-term implications such as reduced genetic purity of seed paddy resulting increased number of off-types in the nuclear seed pool. Decreasing trend of water availability for irrigated rice production in the Dry and Intermediate Zones necessitate increased water use efficiency and water productivity. Saturated rice culture has the potential to increase the water use efficiency substantially. Climate change-induced sea level rise could increase salinity in the coastal saline floodplains while increasing the incidence of submergence in the inland floodplains. Salt-tolerant and submergence-tolerant varieties and land management systems such as the 'sorjhan system' in submergence-prone rice lands are some available adaptation options.

Rice cultivation is identified as a major source of greenhouse gas (GHG) such as methane and nitrous oxides emission to the atmosphere. Methane emission from rice paddies can be minimized by maintaining the plough layer in an oxidative condition. Even though addition of organic matter improves soil fertility and grain yield, such additions could increase methane emissions especially under submerged conditions. Rice cultivation could also be

a source of nitrous oxide emissions because of the high usage of nitrogen fertilizers and the low nitrogen use efficiency (NUE). Therefore, any agronomic management measure to increase NUE which can reduce fertilizer usage without sacrificing yield may reduce emission of nitrous oxide.

Introduction

Rice is the staple food of over 20 million Sri Lankans and is the livelihood of more than 1.5 million farm families. Sri Lanka produced about 3.65 million tons of rough rice in 2009 with a national average rough rice yield of 4.3 t/ha. With the present population

growth rate of 1.1%, increasing per capita consumption, requirements for seed, and for wastage in handling and storage, Sri Lanka needs about 4.56 million tons of rice in the year 2020 (Fig 1). In order to meet the national rice requirement in the future, the productivity in the Dry, Intermediate and Wet Zones should be increased.

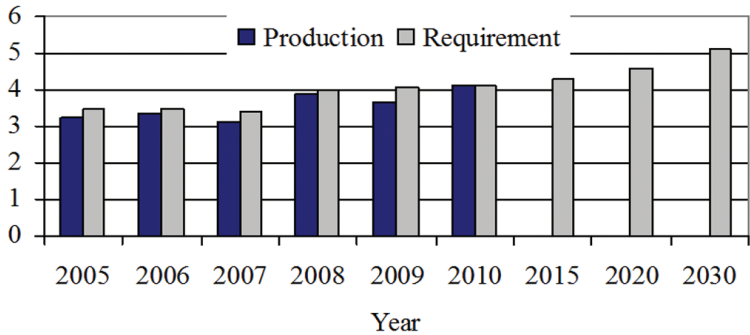


Figure 1: National rice production and requirement since 2005.

Rice in Sri Lanka is grown under a wide range of agro ecological conditions varied with elevations, soils and hydrological conditions. It is grown in 31 of 46 agro ecological regions. The rainfall, temperature and soil conditions vary widely in these different rice ecosystems in the island. Owing to these differences and management deficiencies, wide yield gaps between realizable potential and actual productivity exists in different rice ecosystems.

National rice production could be increased by increasing the cultivated extent, productivity or both. Cultivated extent of paddy lands in 2008 increased by 22% to a record level of 1,049,093 ha from 816,714 in 2007. However, in 2009 cultivated extent of paddy again decreased by 6.8% to 977,560 ha (Fig. 2). Therefore, total rice production in 2009 compared to 2008 decreased by 5.8%. Furthermore, on average about 20% of the rice lands cultivated during Maha in the Dry and Intermediate Zones major irrigation schemes and 40%

of rice lands cultivated during Maha in the Wet Zone were not cultivated during the Yala seasons (Fig. 2). Also, a considerable extent of cultivated rice land has not been harvested during

both Maha and Yala due to various reasons (Fig. 2). This indicates that the cultivated and harvested extents fluctuate significantly between seasons.

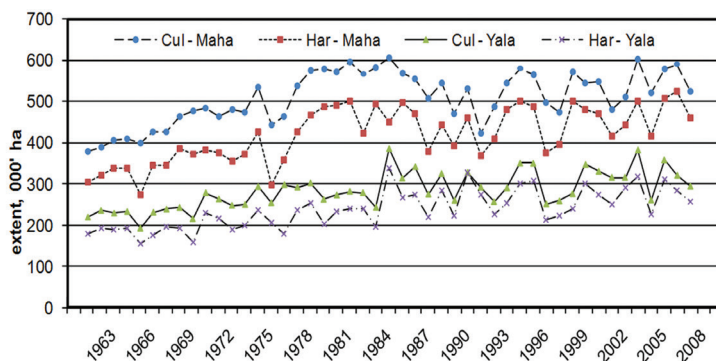


Figure 2: Cultivated and harvested rice extents during Maha and Yala seasons from 1961 to 2007.

Productivity of rice lands also fluctuated between seasons and ecosystems (Fig. 3). Since the introduction of new improved varieties in the 1970s along with their associated management technologies and the opening up of new lands for cultivation after the inception of the accelerated Mahaweli development program, there was a gradual increase in productivity particularly in the Dry Zone which has helped Sri Lanka to achieve near self-sufficiency in rice production. However, technology generated so far has been mainly to increase productivity in favorable irrigated ecosystems. In general, there was no scarcity of water for irrigated rice cultivation. Field problems such as the

build-up of salinity were confined to a small extent and high temperature was not a serious problem. Incorporation of certain pest and disease resistant genes to rice in the past has allowed effective control of major biotic stresses. However, at present, increasing productivity to meet the future demand has to be met with many abiotic and biotic constraints such as inadequate water, nutrient deficiencies, soil toxicities, comparatively higher day and night temperatures and changes in weed, insect and disease composition and pressure. Challenges due to increased biotic and abiotic factors are expected to be significantly greater for rice cultivation in the future.

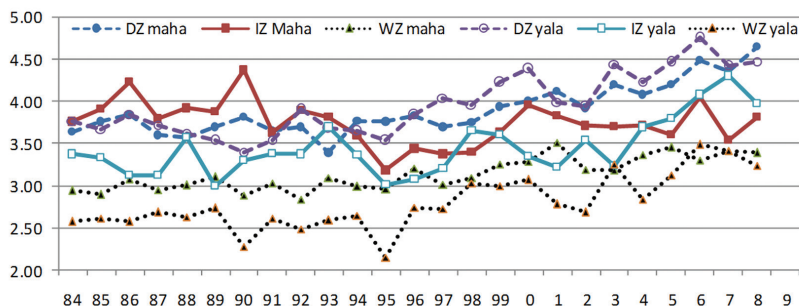


Figure 3: Change in productivity of different rice eco systems during Yala and Maha seasons from 1985-2009.

Climate change

Increased concentrations of atmospheric greenhouse gases have led to a continuous change in the climate. Atmospheric CO_2 concentration ($[\text{CO}_2]$) has increased gradually and is predicted to increase up to about 550 $\mu\text{mol/mol}$ by the end of this century (IPCC, 2007). A major portion of atmospheric methane is said to be generated from rice paddies with standing water. Increased atmospheric $[\text{CO}_2]$, methane, nitrous oxide and other greenhouse gases have led to an increase in average global atmospheric temperature and under different scenarios it is predicted to increase it between 1.1°C to 6.4°C towards the end of this century (IPCC, 2007). Eleven out of the last 12 years from 1995 to 2006 have been ranked as the warmest years of global surface temperature (IPPC, 2007). There could be decreased global precipitation and increased sea level (IPCC, 2007).

Sri Lanka is no exception to these changes in global environment. In fact, the rate of change in average temperature in Sri Lanka has been significantly greater than the global average (Premalal, 2009). Continuous warming has occurred since 1930s until 2007 in several locations representing the major agroecological zones in Sri Lanka (De Costa, 2008). In a major rice growing location, Anuradhapura, the rate of increase in air temperature has been around 0.0078°C per year (De Costa, 2008). There has been a significant increase in the number of warm days and reduction in the number of cold nights (Premalal, 2009). There was a significant increase in both maximum and minimum temperatures in both Yala and Maha seasons in the major rice growing environments in the Dry Zone.

The variability of North-East monsoon which brings rain to the Dry Zone has

increased and there has been a slight change in annual total rainfall over the island (Premalal, 2009). Furthermore, there has been a continuous reduction in rainfall received in the catchments of Mahaweli river that would affect the water flow in the river. This would in fact decrease the quantity of irrigation water and decrease of hydropower generation.

Therefore, in future with climate change, the rice crop in Sri Lanka will be exposed to greater atmospheric $[\text{CO}_2]$ and many abiotic constraints such as increased maximum and minimum air temperatures above the threshold during the cropping season, reduction in the quality and quantity of irrigation water resulting in an increase in the frequency of water stress periods and the deteriorating soil environment such as increased build-up of salinity and biotic constraints.

Impacts of increased atmospheric $[\text{CO}_2]$

The benefits of increased atmospheric $[\text{CO}_2]$ are greater for C_3 plant species, such as rice, than for C_4 plant species. Productivity of modern rice cultivars depends strongly on N availability (Cassman *et al.*, 1994). If N availability does not limit plant growth, increased atmospheric $[\text{CO}_2]$ increases photosynthesis and growth of terrestrial plants (Weerakoon *et al.*, 2005). Increased atmospheric $[\text{CO}_2]$ increases leaf photosynthesis

of all varieties of rice (Baker, 1991; Weerakoon *et al.*, 1999; De Costa *et al.*, 2003a). However, increases of leaf and canopy photosynthesis and radiation use efficiency (RUE) with increased atmospheric $[\text{CO}_2]$ were dependent on leaf N concentrations (Weerakoon *et al.*, 2000). Rice leaves acclimate to long-term exposure to high $[\text{CO}_2]$ and does not respond to increased $[\text{CO}_2]$ beyond 550 $\mu\text{mol/mol}$ (Weerakoon *et al.*, 1999). Even though, the leaf stomatal conductance and transpiration per unit leaf area were lower with increased atmospheric $[\text{CO}_2]$, the daily total canopy transpiration of elevated treatment was approximately the same or sometimes greater than that of canopies grown in ambient environments suggesting no reduction in plant water requirement in rice (De Costa *et al.*, 2003a).

Increased atmospheric $[\text{CO}_2]$ resulted in higher root density due to enhanced tillering resulting in an increase in N uptake and fertilizer N recovery however the partitioning towards leaves had decreased (Weerakoon *et al.*, 2005). Ultimately, the impact of elevated atmospheric $[\text{CO}_2]$ was an increase in dry matter accumulation and final grain yield of rice (Weerakoon *et al.*, 1999). Response to increased atmospheric $[\text{CO}_2]$ was always positive for all Sri Lankan rice varieties tested (De Costa *et al.*, 2007) and the increase in grain yield could be as high as 30% depending on the variety (Fig. 4).

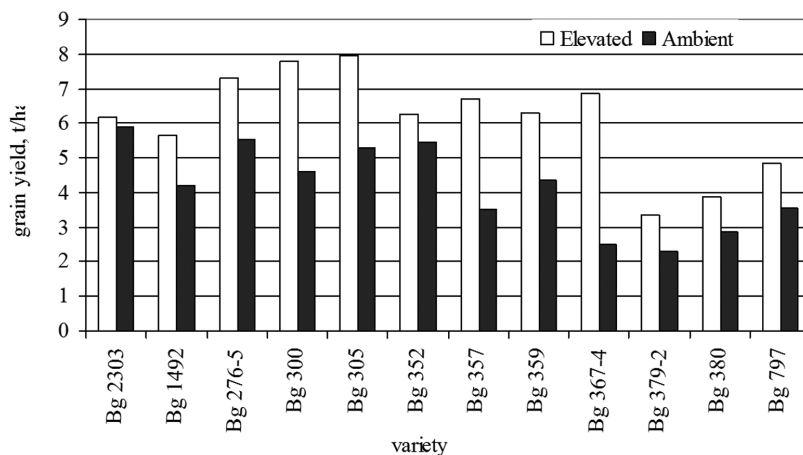


Figure 4: Change in grain yield of different rice lines with enhanced atmospheric [CO₂].

Moreover, longer duration varieties have a greater positive response to increased atmospheric [CO₂] (Fig. 5).

Therefore, increasing atmospheric [CO₂] will have a definite positive effect on rice production in Sri Lanka.

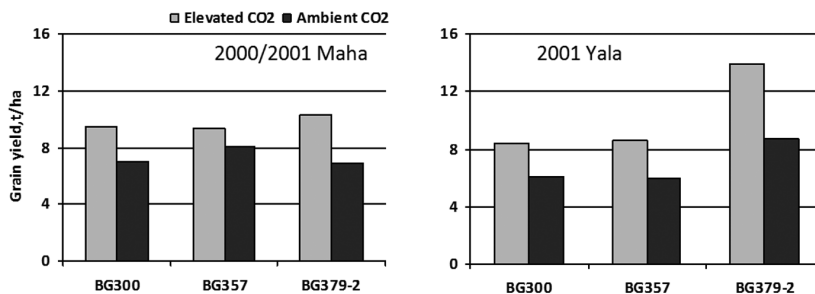


Figure 5: Grain yield response of rice varieties of different maturity groups to enhanced atmospheric [CO₂] during Maha 2000/2001 and Yala 2001.

We investigated the benefit gained by cultivated rice varieties in terms of productivity with the increase in atmospheric [CO₂] from 1960 to 2007. The change in atmospheric [CO₂] since 1960 was obtained from the records of the Maunaolao observatory. In the irrigated system, the increase

in grain yield per unit increase in atmospheric [CO₂] over the years was 0.042 t/μmol of CO₂ while in the rainfed system it was 0.019 t/μmol of CO₂ (Fig. 6). This suggests that varieties and technologies developed in Sri Lanka since 1960 were able to effectively capture the beneficial effect

of increased atmospheric $[CO_2]$ in the irrigated rice ecosystem effectively. However, the beneficial effect of increased atmospheric $[CO_2]$ was not effectively utilized in increasing grain yield of rice in the rainfed ecosystem. This could be mainly because of the high heterogeneity in the abiotic factors in the rainfed ecosystem and the rice varieties developed so far were mainly targeted for the non-

stressed environments. However, the rainfed system will be more vulnerable to climate change, thus the future productivity from these rice lands could be severely affected. Therefore, there is an urgent need to pay equal or more attention to develop varieties and technologies for rainfed ecosystems in order to maximize the potential yield gains with future increases in atmospheric $[CO_2]$.

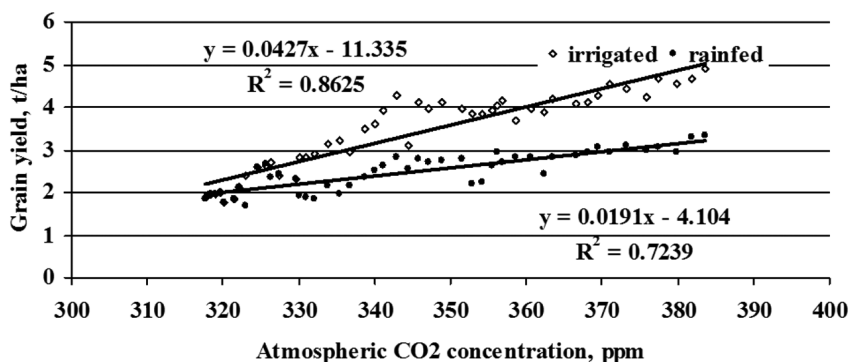


Figure 6: Change in average grain yield in irrigated and rain fed rice ecosystems with the increase in atmospheric $[CO_2]$ since 1960.

It is understood that the impact of increase in atmospheric $[CO_2]$ on rice cultivation is always positive. Furthermore, all rice variety and cultivation technology improvement attempts would automatically capture the beneficial effects of increased atmospheric $[CO_2]$ as all these improvements are made under such natural conditions. Therefore, there is no necessity to conduct field experiments specifically for increased atmospheric $[CO_2]$ in the future.

Impact of increased air temperature

A change in few degrees in temperature often leads to a change in growth and development rate of rice. The critical temperature of a rice variety varies with its phenological stage (Yoshida, 1980). In all major rice growing ecosystems in Sri Lanka, both maximum and minimum air temperature have increased over the years. In certain ecosystems, during certain periods of the year, air temperature has reached its maximum threshold levels for rice. This would have a profound negative

impact on the productivity of the rice crop in Sri Lanka.

Increasing temperatures hasten crop development resulting in a reduction in absorption of solar radiation by the canopy. Furthermore, it increases respiration and may affect C assimilation depending on the current level of air temperature and its magnitude of increase in relation to the optimum temperature. Increased temperatures will also change partitioning, resulting a change in harvest index. However, the most sensitive stage where rice plant is affected with increased air temperature is the reproductive phase specifically at the time of anthesis.

It is a common experience in the farmer fields that rice varieties released in the past mature at relatively earlier than they were at the time of release. It could also be due to genetic drift over the years but the major reason may be the impact of increased both maximum and minimum air temperatures, which would have accelerated the crop growth developmental processes. A special breeding effort is not necessary to correct this shortening of the growth cycle over time as the continuous identification of new rice varieties would eventually select for a specific duration in a particular environment. However, it could be very important in changing standard rice varieties for specific maturity duration as breeders

tend to detect maturity duration of a rice line by comparing it with its respective standard.

Minimum air temperature, which is the lowest air temperature during the night, has a significant negative impact on grain yield of rice (Peng *et al.*, 2004). This suggests that the rice productivity would decrease with the possible increase in night temperatures, so that adjusting the cropping calendar to avoid periods of high minimum air temperatures becomes important.

Increased maximum air temperature above threshold value at flowering has a significant negative impact on the productivity of the rice crop in Sri Lanka caused by spikelet sterility (Weerakoon *et al.*, 2008). Under field open top chambers, there was a significant reduction in pollen fertility at air temperatures of $32.1 \pm 1.9^{\circ}\text{C}$ (Fig. 7) (Weerakoon *et al.*, 2005). Increased temperature inside the spikelet is a function of air temperature and relative humidity (Abeyasiriwardena *et al.*, 2002). Increased spikelet temperature above threshold causes pollen grain death, disturbs dehiscence of anther, disturbs pollen shedding resulting in reduction of fertilization and ultimately increasing empty grains. Extremely high temperatures and lower humidity may also desiccate pollen grains.

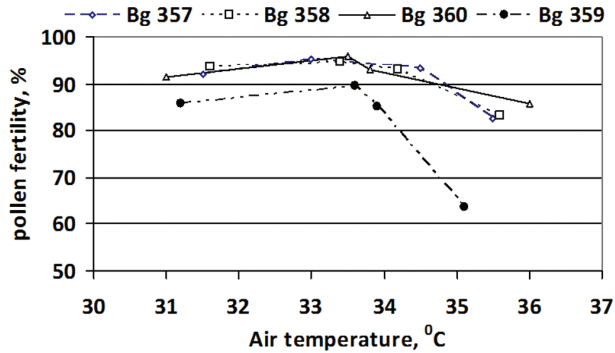


Figure 7: Change in pollen fertility of different rice varieties with increased air temperature.

The relationship between air and spikelet temperature was linear but the difference between air and spikelet temperature increased with decreased humidity and the magnitude of change varied with variety (Weerakoon *et al.*, 2008). Therefore, the effect of humidity on spikelet fertility is an “indirect effect” - A way of avoidance. Spikelet fertility is nearly related to spikelet temperature with an average critical spikelet temperature of about 31°C

and varieties differ in their response (Fig. 8). Differences in tolerance to high temperature is a combination of varietal “true tolerance” and its ability to modify spikelet temperature (avoidance) with changing Rh by way of transpiration. Variability in “avoidance” of high temperature could be used as a tool for selection of varieties to high temperature tolerance (Weerakoon *et al.*, 2008).

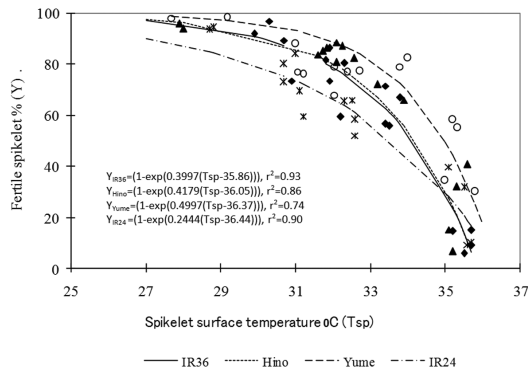


Figure 8: Change in spikelet fertility of two Indica and two Japonica rice varieties with the change in spikelet surface temperature. Lines indicate the non-linear relationship between spikelet surface temperature (T_{sp}) and spikelet fertility (Y.) for different varieties. Spikelet surface temperature was measured just before the opening of spikelet using an infrared thermometer.

These findings suggest that even at present, rice spikelets in certain locations in Sri Lanka are subjected to air temperatures above the threshold which could cause sterile pollen grains. Therefore, if a rice variety in the Dry Zone flowers in late March or August (high maximum temperature period) and if the humidity is very high, there is a great chance of increasing the percentage of sterile or empty grains. With increased air temperatures, areas in Sri Lanka which are not suitable at present for rice cultivation due to lower temperature could become favorable for rice cultivation. Therefore, in the extreme situation of climate change, rice growing area in Sri Lanka may shift from its traditional ecosystems to higher elevations where rice is not grown at present due to low temperature stress.

Grain sterility observed under continuously high air temperature inside growth chamber conditions suggests that even under day time moderately high temperature situations, spikelet sterility should occur inside open top chambers and in the open field. Such sterility has indeed been observed frequently in farmers' fields especially in the humid Wet Zone and certain parts of the Dry Zone of Sri Lanka during month of August which is locally named as "Ehala Pussa". Morita and Dhanapala (1990) also observed reduction in spikelet fertility under field conditions in Sri Lanka but attributed it mainly to damages caused by the paddy bug. The day time air temperature in most rice growing areas, even though not continuous, reaches

its maximum threshold level during anthesis suggesting that widespread grain sterility should occur in farmers' fields. However, apart from relatively greater empty grain percentages during certain seasons, no formal widespread evidence exists indicating that grain sterility is still not a serious issue in the rice cultivation of Sri Lanka.

Rice is considered as a predominantly self pollinated crop. Rice varieties released in Sri Lanka even in early 1960's are still cultivated suggesting that they are very stable. However, in the recent past, there has been a significant increase in the weedy rice population in many rice-growing districts in the island (Abeysekara *et al.*, 2006). It was first reported in late 1980's in the North-Western and Eastern Dry Zone of Sri Lanka where air temperature during flowering was considered higher than optimum. If the crop is strictly a self-pollinated crop, the chances of existence of weedy rice and such variation within cultivated rice species must be minimal. Only spontaneous mutation or expressing of an environmentally sensitive gene could trigger such variation. Recent surveys conducted by the Rice Research and Development Institute (Abeysekara *et al.*, 2006) showed that there is a mixture of off-types comprising of lines related to wild rice and lines related to the cultivated varieties. This suggests that there could be an increased out-crossing within cultivated rice in farmer's fields or between cultivated rice and wild rice resulting in a segregating population of "weedy rice".

Further evidence for such out-crossing in rice was revealed from the nuclear seed production program in Sri Lanka. Nuclear seed production of cultivated rice varieties is carried out by the breeder himself using progenies of the same cultivar. They do it with extreme care without allowing physical mixtures. To make sure that there is no genetic drift, any suspected progeny is completely removed even if a single plant showed a slight difference. There has been an increase in the number of rejected progenies in the breeder seed program indicating an increased number of off-types in the nuclear seed stock (Fig. 9). There was a clear

increasing pattern of rejection of progenies over different seasons (Fig. 9). Even though the magnitudes are different, in almost all seasons within a given age class, rejection remains consistent among varieties. Varieties of the same age class flowered simultaneously exposing spikelets to similar air temperatures at anthesis. If these differences in rejection of progenies were purely due to physical mixing, such patterns could not be observed. Therefore, it is evident that out-crossing due to high temperature induced pollen sterility is probably occurring in rice fields in Sri Lanka.

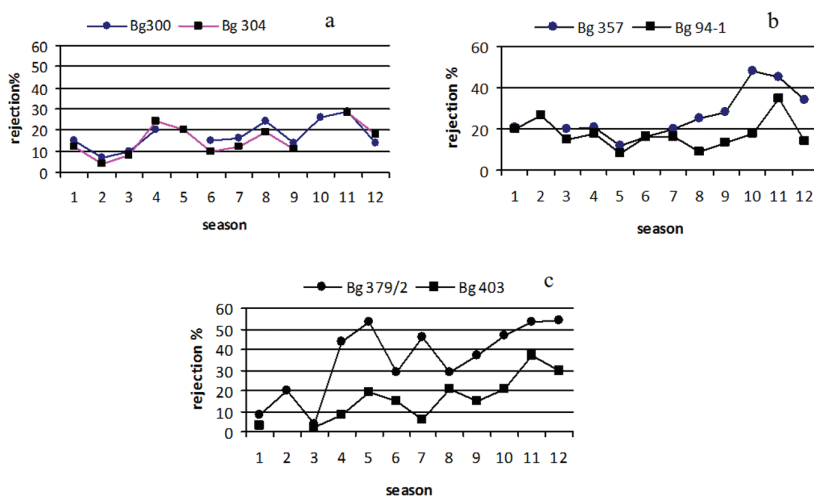


Figure 9: Change in rejection of breeder seed progenies of 3 month (a), 3½ month (b) and 4-4 ½ month (c) maturity over different seasons from year 2001.

Occurrence of morphologically different rice types within a homogeneous rice population could occur due to mutation or an out-crossing by different male parents. Spontaneous mutation is very rare

and continuous occurrence of mutated lines is extremely rare. However, there is a possibility of out-crossing as the out-crossing percentage of rice has increased (Abeywickrama *et al.*, 2009). Therefore, it is evident that out-

crossing within the same population has produced slightly different genotypes but with morphologically very similar. This is reflected in minute differences in plants of F1 generation within the progeny.

Kim *et al.* (2001) showed that when heat shock was applied to *Arabidopsis thaliana*, floral organs developed normally except pollen inside anthers suggesting that stigma is unaffected and out-crossing could occur. Observations show that anthesis of a rice panicle starts from the top and continues towards the base and the duration depends on the variety. The duration of anthesis of a panicle could vary from 4 to 8 days. Thus, it is hypothesized that under field conditions when air temperature and humidity is increased to the critical level, pollen sterility occurs. Therefore, even if the whole panicle is exposed to temperature stress only certain spikelets which are at the susceptible stage of growth are affected. Pollen of such spikelets would become sterile and the self-fertilization is hampered but depending on the duration and timing of high temperature stress, pollen from other spikelets of the same panicle or from the adjacent panicles could fertilize the spikelet inducing out-crossing. This is indeed happening in the field. When anthers were removed from some spikelets and allowed exposure to foreign pollen, it was observed that a significant number of emasculated spikelets became fertile suggesting that when self-fertilization is hampered, out-crossing from the same panicle or other panicles occurs making the spikelets fertile. If the high

temperature persists for more than 7-8 days, the whole panicle may become sterile as no viable pollen could exist in such an environment.

Under farmer field conditions when spontaneous pollen sterility occurs due to increased air temperature and humidity above the threshold levels, pollen from adjacent rice plants or from the same panicle could deposit on the stigma making the spikelet fertile. However, if these F1 seeds are used for cultivation, the F1 generation plants could give rise to F2 seeds and show a segregating population in the field. If cross-fertilization occurs with pollen of the same panicle, no segregating population is visible even after two generations. However, if the pollen for fertilization is from a different plant, a segregating population is observed. This phenomenon could be happening in the tropical sub-humid rice ecosystems as there has been an increased frequency of the number of days with high temperature when compared to past.

These observations suggest that the impact of climate change on pollen sterility in rice is a common occurrence in humid ecosystems with high temperature. However, as the temperature increase is not enough to affect the stigma, cross-pollination occurs making the impact of climate change not visible. Thus, there cannot be any decrease in productivity and thereby the national production. However, the segregating population would cause a significant impact on the

quality of rough rice produced for both consumption and as seed paddy for next season's cultivation. Therefore, the impact of climate change is expected to cause a serious impact on the quality of paddy produced in the sub-humid tropics.

Furthermore, the out-crossing would have a profound impact on seed paddy production especially on the self seed paddy program where roguing is usually not practiced. New seed stocks would have to be supplied more frequently than in the past.

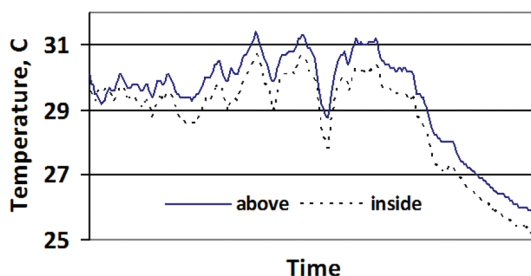


Figure 10: Change in air temperature above and inside the canopy (panicle height) of rice within a day.

Impact of an increase in air temperature on spikelet temperature could be minimized if the panicle is not exposed to direct radiation. It was observed that the temperature inside the canopy was much lower than that at the top of the canopy (Fig. 10). Therefore, if the panicle is hidden inside the canopy, spikelet temperature could be effectively reduced while exposing leaves to direct radiation. This could be used as a selection criterion for varieties to be grown in high temperature environments. Further there is an additional advantage as the panicle is not shading the upper leaves leading to increased canopy C assimilation.

With the increase in air temperature we could expect an increase in insect and pathogen pressure which would ultimately end up in reducing rice productivity is also expected.

Furthermore, a shift in weed flora towards temperature tolerant C_4 weeds in rice paddies would create an additional burden on weed management.

Deteriorating quality and quantity of irrigation water

Over the past, variability of the North East monsoon has increased along with an increase in the movement of consecutive dry days in the Dry Zone (Premalal, 2009). Even though the reduction in annual total rainfall over the island is negligible, increased movements of dry days and less reliability on monsoon and inter monsoon rains would increase demand for irrigation water, especially in the Dry and Intermediate Zones. However, due to changes in climate and other reasons, the demand for water from

sectors other than agriculture also would increase, resulting in a further reduction in the availability of water for irrigation.

Changes in the rainfall pattern in the catchments of major rivers in Sri Lanka have been observed (Premalal, 2009) and the change in catchment characteristics have also changed the water retention by the soil which had affected the water flow of rivers. Furthermore, significant increase in extraction of water from rivers for domestic and industrial needs has been observed (Weerakoon and Costa, 2009). As a consequence, water availability in rivers has changed. For example there has been a significant

decrease in the water flow in the Mahaweli river over the last 25 years (Fig. 11). In 1985, the average total river flow of the Mahaweli river at Polgolla was about 2,500 million cubic meters, and this has decreased to 1,400 million cubic meters in 2009 (Fig. 11). Even with the reduction in water flow, there was no change in the amount of water diverted towards the Dry Zone while there was a reduction in the water discharge towards Victoria depriving water for hydropower generation which would increase emission of CO₂ as the additional national electricity requirement had to be generated through thermal power.

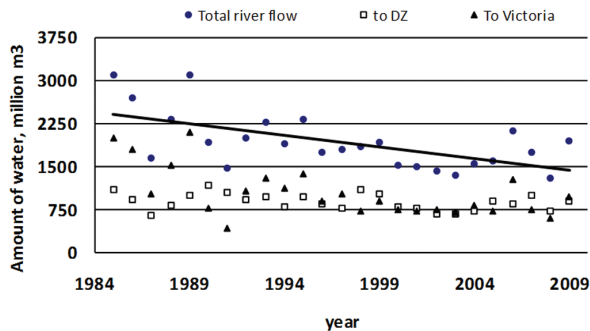


Figure 11: Annual water flow, amount diverted and downstream release of the Mahaweli river from 1985 to 2009.

Even though, the amount of water diverted was essentially the same, increased population, by about 29% from 1981 to 2001 in Anuradhapura and Polonnaruwa districts and increased industrial demand have effectively reduced the water available for agriculture, especially for rice cultivation. Moreover, there is also an

increase in the discharge of polluted water towards rivers from major cities which ultimately end up in the Dry Zone resulting in a buildup of salinity. Therefore, the adverse impacts of climate change on rice will be further aggravated with the reduction in quality and quantity of irrigation water. Hence, there is an urgent

need to increase quality of water in major rivers and irrigation water use efficiency. Therefore, by maximizing the use of rain water and using water saving cultivation practices, a significant amount of irrigation water used in lowland rice culture could be reduced. Timely cultivation could reduce the risk of exposure of the rice crop to terminal drought while increase in the extent cultivated with 3½ month high yielding rice varieties could save irrigation water.

The amount of irrigation water required to maintain saturated condition in direct-seeded lowland rice culture was 24% and 72% lower than that required for flood irrigation in Yala and Maha seasons, respectively while maintaining the same yield level (Weerakoon *et al.*, 2010). Irrigation water productivity was the highest with saturated soil culture and saturate to dry soil culture but, drying of soil below saturation until cracks are formed would reduce yield (Weerakoon *et al.*, 2010). If cultivated at the correct time, maintaining saturated culture in lowlands during Maha season in the Intermediate Zone could be done mostly without irrigation water. The saving of water could be used to increase the cultivated rice land extent while maintaining the same or obtaining a greater grain yield with medium duration varieties.

Impact of sea level rise

It is expected that towards the end of this century, the projected mean sea level rise under different climate scenarios would be between 20 to 59

cm (IPCC, 2007). The rice land extent under inland flood plain and costal saline flood plain ecosystems in Sri Lanka are approximately 16,000 ha and 6,000 ha, respectively. There could be a direct impact of sea level rise through submergence and increased salinity levels to a certain percentage on these coastal saline rice lands. Eventhough the direct impact is low, there could be a serious indirect impact due to difficulties in draining water in the inland flood plain. Therefore, there is a need to develop salt and submerge tolerant rice varieties and associated technologies for these ecosystems. A rice line, 96-74, has been identified with submergence-tolerant. Further, management options such as “sorjhan system”, a ridge and dyke system would enable cultivation of these lands integrated with inland fishery.

Greenhouse gas (GHG) emission from paddy farming

The main components of agricultural emissions of GHG are CO₂ from land use change, N₂O from the application of nitrogenous fertilizer, CH₄ from livestock enteric fermentation, CH₄ and N₂O from manure management, CH₄ from cultivation of rice, and CH₄ and N₂O from burning of savannah, forest, and agricultural residues. Of the wide variety of sources, rice paddy fields are one of the major anthropogenic sources of atmospheric CH₄ (IPCC, 2007; Nue, 1993). Preliminary studies indicated that about 68% of the total methane emission of Sri Lanka is generated from rice fields (Weerakkody, 1996). The average daily emission from rice

paddies in Intermediate Zone was observed to be 0.79 g/m²/d (Sirisena *et al.*, 2004). Recent information collected in the Second National Communication by the Ministry of Environment and Natural Resources to be submitted to the IPCC, revealed that the major sources of GHG emission are the puddled paddy fields, crop residue burning and the emission of N₂O from nitrogen fertilizers. It was reported that methane emission from the total paddy cultivated area of about 0.8 million ha in the year 2000 was 121.8 Gg. From the total crop residue generated (3170.2 Gg) in the same year, emission of CH₄, CO, N₂O and NO_x from burning were 1.06, 22.3, 0.05 and 1.71 Gg, respectively. Direct and indirect emission of N₂O from N inputs used for paddy farming (about 60% from the total usage) was 0.974 Gg (303 GgCO₂e). CH₄ emissions from rice fields are controlled by various factors under field conditions which include the types and amount of fertilizer application (Cai *et al.*, 2007), application of organic matter, water regime, soil characteristics (Yan *et al.*, 2005), rice cultivar (Watanabe and Kimura, 1999) and weather conditions. However, the accompanied changes in water regime, applied fertilizers, organic matter amendments make it complex to predict the CH₄ emissions from rice fields (Cai *et al.*, 2007). Research evidence showed that peak emissions from rice fields are during land preparation (puddling) and at the maximum vegetative growth stage such as end of the tillering stage when secretion of root exudates and products of root autolysis are highest

(Holzapfel-Pschorn *et al.*, 1986), during reproductive stage (Minoda and Kimura, 2007), and heading (Chidthaisong and Watanabe, 2007). These exudates are carbon sources such as sugars and organic acids which are used by methanogenic organisms to form CH₄.

GHG emission mitigatory options

Methane emission mitigation through genetic improvements

The improvements in rice yields reduced the need to convert additional land for rice farming which generates carbon dioxide and other greenhouse gases. Extent of rice cultivation, total production and productivity has changed from 0.59 million ha, 0.902 million tons and 1.86 t/ha in 1961 to 1.062 million ha, 4.30 million t, and 4.56 t/ha, in 2010, respectively (Department of Census and Statistics, 2010). These data clearly indicate more prominent intensification rather than increasing the extent which avoided a massive amount of GHG emission to the atmosphere. The very prominent genetic improvement programs and improved management options may also have varying impacts on CH₄ generation. Prioritizing research and development on genetic improvement programs for greater yields together with efficient resource use would be the ideal mitigatory option for rice farming.

Methane emission mitigation through crop management

Water management

Possible strategies for mitigating CH₄ emissions from rice cultivation include the control of production, oxidation or transport processes. Water management by increasing water percolation influences chemical conditions in flooded soil, adds oxygen-rich water to the reduced soil layer, minimizes decreases in soil Eh during the cultivation periods and decreases methane production. CH₄ emission rates significantly decreases with increase in the percolation rates. Since methanogens require highly reducing conditions for their activity arresting the development of soil reduction by intermittent drying is an effective way of decreasing CH₄ generation. However, such mitigation practice is only feasible where complete control of water supply and drainage is possible. However, intermittent aeration may increase gaseous losses of nitrogen as nitrous oxide (nitrification-dinitrification).

Organic residue management

In rice cultivation, apart from the biomass of previous crops and weeds remaining in the soil, organic matter is often applied as crop residues (straw), green manure in fresh forms, and sometimes animal waste before or during land preparation. Such organic matter undergoes anaerobic decomposition under flooded

conditions and generates CH₄. Therefore, organic matter management, while being important for maintaining soil fertility, has a significant influence on CH₄ production. Many research work have demonstrated that incorporation of rice straw and green manure into rice paddy soils dramatically increases CH₄ emission. A study done at RRDI, Batalagoda has shown that there was no significant increase in CH₄ emission when compost is incorporated during land preparation as the inherent soil organic C content is low (Sirisena *et al.*, 2004).

Recent trends in the agriculture sector in Sri Lanka has been the conversion of traditional systems into high-tech advanced systems which rely heavily on modern technologies and mechanization which demand more energy. This would definitely increases baseline emissions of GHGs in the agriculture sector. Therefore, energy inefficient technologies and machines used in the agriculture sector must be replaced with energy efficient technologies and systems as an effective mitigation mechanism.

Overall results indicate that the rice sector could be considered as a source of GHG emission to the atmosphere due to inherent crop characteristics and manmade management practices introduced to enhance the production and productivity. However, there are many ways that GHG emissions could be reduced without sacrificing yields, by implementing appropriate, planned, and sustainable management measures.

For that, baseline information (GHG footprints) need to be prepared by collecting available information and formulating new research for gap filling to find out appropriate mitigatory options to address GHG emission issues in the rice sector in the most viable and effective way.

Conclusions and future research directions

Climate change is a slow process and its impacts are gradual. The positive impact of climate change, which is the increase in atmospheric [CO₂] will be automatically captured through the continuous rice crop improvement programs. However, the adverse impacts of high temperature, water stress, submergence, salinity and depleting resources cannot be captured through natural selection as occurrences of these are sporadic. Thus, a well-focused program is needed to develop high yielding, pest and disease tolerant/resistant, short to medium duration varieties which are water and fertilizer use efficient, and drought, high-temperature, salinity and submergence tolerant.

Suitable agronomic and cultural management options are needed: (a) cultivate short to medium duration cultivars to conserve natural resources while maintaining same yield level; (b) avoid flowering during periods of high air temperature by adjusting the time of planting if necessary, (c) determine a new cropping calendar to maximize yield while conserving natural resources and (d) conserve water and nutrients by maintaining saturated soil moisture conditions in the field and thereby increasing cultivable land extent and hydro-power generating capacity of the country.

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Geo-informatics for Efficient Use of Resources in Rice Production

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Abstract

Potential rice yield can only be obtained by providing optimum levels of resources in suitable environments. Geo-informatics can be effectively used to identify and assess the spatial and temporal availability of resources from land preparation to paddy marketing or even up to the consumption stage. Combination of the capabilities of Remote Sensing (RS), Geographical Information Systems (GIS) and Global Positioning Systems (GPS) can be used to gather spatial data and to derive information for decision support systems in various stages of rice production. A basic understanding of rice physiology and crop management is essential for the success of applying Geo-informatics in rice-based agricultural systems. The wide span of application potentials of Geo-informatics is immensely helpful in assessing demand for rice in different spatial locations, agricultural resource management, rice crop identification and area estimations, assessment of land suitability for rice, identification of cultivation calendar, assessment of inputs and inventory of resources, efficient water management, water use and water availability mapping, efficient use of fertilizer, assessment of crop damage and identification of control measures, and yield forecasting and marketing. Pilot scale work has been conducted with the use of Geo-informatics at the Rice Research and Development Institute at Batalagoda and suitability maps of rice lands with respect to soil fertility and climatic conditions have been developed. In addition, few studies related to rice cultivation also have been carried out in Sri Lanka by using Geo-information technology. Non availability of spatial data, financial constraints, poor data sharing mechanisms, limited mapping capabilities, lack of expertise and computer facilities are described as the limitations in using Geo-informatics with respect to rice production in the country. Exploration of possible pathways to improve the use of Geo-informatics, maintenance of GIS databases on different spatial parameters such as soils, land use, hydrology, population density, agricultural land distribution etc., formulation of a proper data sharing policy are some of the strategies to improve the future prospects of using this technology in rice production. If the Geo-informatics technology can be appropriately utilized to suit the local requirements, it is possible to achieve a greater success in production, marketing and utilization of rice in an agricultural country like Sri Lanka.

Key words: GIS, GPS, Remote Sensing, Rice

Introduction

Rice can be cultivated in a wide range of environmental conditions. However, the potential yield can only be obtained by providing optimum levels of input in suitable environments. Therefore, areas having appropriate environmental conditions for rice cultivation should primarily be identified to achieve a profitable rice production and to diversify the rest of the lands which are not suitable for rice cultivation. This would ensure high land productivity in its best use. There is a spatial and temporal variability of inputs and resources for rice cultivation in Sri Lanka. Hence, spatial and temporal variability of inputs can be used with the support of Geo-informatics for enhancing resource use efficiency in rice production in the country.

Geo-informatics can be used in rice production from the stage of land preparation to paddy marketing or even to the consumption stage. The wide span of application potentials of Geo-informatics will be immensely helpful in assessing demand for rice in different spatial locations, resource availability for rice cultivation and crop management, status of crop growth, potential yield, market availability, etc. This involves a considerable amount of spatial information that can be derived using existing maps, field surveys and also through the use of remote sensing technology. Combination of the capabilities of Remote Sensing (RS), Geographical Information Systems (GIS) and Global Positioning Systems (GPS) in Geo-informatics

can be used to gather spatial data and to derive information for decision support systems in various stages of rice production.

Use of Geo-informatics in agricultural resource management

Geo-informatics encompasses GIS, RS, GPS, and other spatial disciplines including cartography, photogrammetry and digital multimedia. This technology has a vast potential in assessment, monitoring, planning and management of agricultural activities. Increasing crop production is the prime objective in agriculture. This can be achieved by bringing more lands under cultivation, using improved high yielding, stress tolerant, input responsive crop varieties, through effective pest and disease control, water management, etc. In addition, the knowledge on the extent of area under different varieties of crops, surface and groundwater availability for irrigation purposes, nature of soils and their strengths and weaknesses, management of natural calamities are important to achieve a sustainable crop production (Rai *et al.*, undated). The potentials of Geo-information technology in identifying new resources for agriculture, improving existing resources and assessing suitability of the lands have been accepted worldwide. The ability of remote sensing in capturing information from a large area at once, repetitive coverage, potential of acquiring information in a wide range of spatial and spectral resolutions together with the data handling,

manipulation and analysis potential of GIS provide a wide array of applications in the field of agriculture. The most common and important applications include basic mapping of cultivated areas, disease identification, assessment of water stressed areas, crop inventory, irrigation command area mapping, yield estimations and forecasting, crop damage assessment, soil nutrient assessment, land suitability assessment for crops, etc. Although, Geo-information technology has gained a high popularity in Sri Lanka at present, only very few studies have been carried out in the rice sector adopting this technology. Land use/ land cover mapping is the most commonly practiced application of remote sensing in agriculture with the objective of identifying crop area and crop types. Both aerial photographs and satellite remote sensing can be used for this purpose. This paper attempts to identify and discuss the potentials of using RS, GIS, GPS and related technologies in rice production and related farming practices in the Sri Lankan context.

Use of Geo-informatics in rice crop identification and area estimations

Van Niel and McVicar (2001) have commented that a basic understanding of rice physiology is essential for the success of remote sensing applications in rice-based agricultural systems. They have mentioned that this knowledge can play a critical role in the planning stages in which

identifying optimum acquisition dates to purchase imagery and also in the final stage of analysis. There are many examples around the world to highlight the usefulness of microwave remote sensing data (RADAR) in identifying and estimating cultivated area of rice. Ogawa and Saito (1999) have used RADARSAT satellite data together with Landsat TM data to determine the rice cultivated areas in Kantoh plain in Tochigi, Japan. RADARSAT is a microwave satellite sensor which is suitable in collecting data under all weather conditions. The researchers have managed to identify the rice cultivated area with a high precision using RADARSAT data combined with the Landsat TM; a multispectral sensor with seven spectral bands. Nguyen *et al.* (2009) have identified the usefulness of the ENVISAT Advanced Synthetic Aperture Radar in rice crop inventorying in Vietnam. Remotely sensed data have been used for administration of rice in Coleambally irrigation area in Australia (Van Niel and McVicar, 2004). This process relies upon identification of individual rice paddocks and summarization of each land holder's rice area. They have mentioned that rice administration was strictly based on high resolution aerial photographs in the past due to low spatial accuracies in satellite data for management of rice at paddock level. However, in recent times, they have developed simplified technologies using satellite remote sensing coupled with vector data layers in GIS. According to Aplin *et al.* (1998), higher crop classification accuracies have been achieved by the combination

of GIS, GPS and remotely sensed data. This method of classification depends on accurate GIS data and is sensitive to missing boundary lines. By using GIS data, Aplin (1998) achieved greater classification accuracies in 'constrained' areas having well-defined boundaries (e.g., agricultural fields and urban areas). GPS surveys also can be implemented in area measurements in small scale projects.

GIS and RS for assessment of land suitability for rice

Recognizing the spatial diversity of climate, soil and other resources, rice cultivation can be made profitable by identifying suitable areas for cultivation. GIS has the potential of integrating a number of spatial data layers together to assess the land suitability for various purposes using a set of criteria. The factors affecting rice production could be combined in a simple overlay model in GIS for the suitability analysis of paddy lands. Available field data, maps and satellite images can be used as information sources to prepare basic thematic maps required to identify different suitability characteristics.

Rathnayake *et al.* (2010) have carried out a suitability analysis for rice lands in Kurunegala district considering temperature and relative humidity as the factors which are the most effective in combination at the reproductive stage of rice. Different areas have been

identified as highly suitable, suitable, moderately suitable and not suitable for rice cultivation in Yala and Maha seasons based on temperature and relative humidity. Land suitability was also categorized in relation to soil fertility parameters in Kurunegala district (Rathnayake *et al.*, 2008) with the support of GIS. When climatic, soil, management, and the other resources are incorporated into the spatial database for analysis, the most suitable areas for rice cultivation can be identified.

Geo-informatics to identify the cultivation calendar

Identification of the best time to start cultivation and the length of growing season mainly relate with the spatial and temporal distribution of rainfall and the water availability. Though rainfall data is collected at point locations, spatial distribution pattern of rainfall can be derived with the interpolation capabilities of GIS. Spatial distribution and temporal availability of water can be assessed with the help of remotely sensed data in both major and minor irrigation systems. Multi temporal satellite imagery is useful in identifying and mapping water bodies and calculating capacities using relationships derived with the water surface area. By incorporating this information together with the ground realities, best time to start cropping season can be decided.

Assessment of inputs and inventorying resources in rice cultivation using Geo-informatics

Most of the farmers use equipment and machinery at various stages of rice cultivation such as land preparation, harvesting and processing. However, the availability of these machinery and other resources is not uniform across rice growing areas. GPS can be utilized to survey and map the availability of farm equipment and these maps can be used to make arrangements to share them efficiently. When seed paddy becomes scarce for the farmers, they have no information on the possible sources of supply. If there is a process to provide real time information on the availability of inputs to farmers, it avoids the stress to the farmers in searching for the required inputs. Accordingly, Geo-informatics can be used to prepare an updated spatial database for the available varieties of seed paddy, the places where the seed paddy and other inputs are available and the amount of inputs needed for the next season etc. Not only for seed paddy, even for fertilizer subsidy scheme, it is imperative to maintain a database for the efficient management of available resources. Such a database would ensure that the amounts of subsidies provided are proportionate to the lands cultivated.

Geo-informatics for water use and water availability mapping

Surface water availability mapping is a very straight forward use of remotely sensed data. This can be easily carried out with the Near Infrared imagery during wet season where water levels of the reservoirs are at the maximum. Water provides zero or minimum reflectance in the Near Infrared region hence, the boundaries of the water surfaces can be easily identified. Ground water availability and water table fluctuations can be assessed with the ground based Ground Penetration RADAR systems. In addition, water use of the crops including rice can be estimated by identifying the area cultivated under each crop and assessing their water use under different climatic conditions.

Geo-informatics for water management in rice farming

Assessing and improving Water Use Efficiency (WUE) in agricultural systems are extremely important as the demand for food production is continuously increasing. Water savings from these systems, in particular, could affect the regional and global water balances as the area of land placed under agriculture is expected to increase considerably (Lund and Iremonger, 2000).

Three main approaches are available to assess WUE regionally using Geo-informatics. Remote sensing, which can estimate both evapo-transpiration and CO₂ exchange of large areas at specific times of the day, can be used to present regional WUE estimates at specific times (Schuepp *et al.*, 1987). This requires access to much ancillary ground based meteorological data. Limitations have been identified such as difficulties in extending this from the field sites to regions and also temporally extending the data from specific times to entire growing seasons.

Remote sensing based estimates of yield and crop water requirements will provide an insight into the production status in advance. Regional databases of yield, precipitation, irrigation and initial soil water availability can be developed allowing an 'input-output' (Zoebl, 2000) definition of regional WUE. 'Input' is the water available over the crop growing season and 'output' is the yield. This approach is well suited to spatial assessment of regional WUE.

Satellite remote sensing will be helpful in assessing area coverage under each crop to calculate crop water requirement. Capabilities of Geo-informatics could be utilized to determine seepage, percolation and evapo-transpiration in water balance studies.

Layout for the distribution of canal water from the source to the field can be easily mapped with GIS for the planning and management of water

conveyance systems. When the crop water requirement is estimated and the extent of cultivated area is known the required amount of water can be delivered according to the crop growth stage efficiently and effectively. Remote Sensing and GIS can be combined in providing strategies for such an application.

Geo-informatics for efficient use of fertilizer

Identification of the requirements of the crops at micro scale is essential to formulate strategies in precision farming. The site specific fertilizer management is an important component in precision agriculture. This needs the identification of areas under various fertility levels. In large farms or farming areas, a field survey will not capture this diversity very effectively. However, remotely sensed data with high spatial resolution can provide information on the spatial variability of crop growth at different locations with same input levels to identify location specific problems. Once these troubled locations are identified with the remotely sensed data, ground sampling can be carried out to understand the conditions prevail in the location to provide site specific management options. GIS is also helpful in mapping fertility levels in soils with sampled data to assess the spatial variability of soil fertility. Sirisena *et al.* (2009) and Rathnayake *et al.* (2009) have used geo-information data to improve site specific fertilizer application in Ambanagga and Pulasthigama, respectively. This

has helped to improve rice yields by utilizing resources effectively. According to them, baseline soil survey had been carried out to estimate the fertility parameters in specific locations which are marked by GPS. Thematic maps were prepared for each parameter to identify the areas needed to apply fertilizer.

Assessment of crop damage and identification of control measures

Rice crop damage and subsequent yield reduction can occur due to many reasons including, infestations due to pathogens, insects and weeds. Cropping practices such as differences in water depth and fertilizer applications can also affect crop production. In addition, irrigation with waste water can cause heavy metal contamination in soils, also resulting in crop damage and yield reduction. Natural variables, such as climate, can also cause damage from either seasonal temperature or precipitation (drought and flood) variations.

Crop damage research using remote sensing can be simplified into two different categories as the source of damage (direct measurement) and the effect of the damage (indirect measurement). In certain situations, the source of crop damage can be measured directly from remote sensing. These studies include assessment of salinisation, weed infestations and water logging. In most cases, however, it is impossible to measure the source of the damage directly using remote

sensing. It is well established that physiological stress in crops is related to incident reflectance from crop canopies and that the altered reflectance can be detected with remote sensing. Hence, the disease infestations and the impacts of stress caused due to excess or inadequate water, heavy metals, etc. can be detected by the canopy reflectance. Distribution pattern of weed species can be mapped spatially and temporally and management packages can be introduced accordingly. Pest and disease distributions also can be identified using remote sensing and the spreading patterns can be modeled in GIS to monitor them.

Geo-informatics for forecasting yield and marketing

Standard yield estimation procedure include the analysis of crop cuttings at randomly sampled ground plots during harvest (Murthy *et al.*, 1996) or meteorological regression models using rainfall and past yield data (Karimi and Siddique, 1992). These methods often produce results that are either not timely or spatially explicit. Though still used, these methods are being replaced by estimation of crop yields using remote sensing because of its ability to produce results quickly and spatially and yield forecasting can be made as early as one month prior to harvest. The most popular methods of forecasting yield using remote sensing is through the use of LAI and using vegetation indices. Positive correlations exist between measurements of Leaf Area Index (LAI)

and Vegetative Indices. Normalized Difference Vegetation Index (NDVI) can be correlated with the LAI and also with the yield to derive empirical relationships to forecast yield. In addition, accurate estimation of area under rice cultivation is also helpful in estimating overall production of rice in the country with the use of average yield statistics. Rice yield forecasts can greatly influence the management decisions such as fertilizer applications and water delivery, farm income assessment as well as in arranging marketing facilities. The appropriate places for the construction of storages or markets can be identified using GIS capabilities. Consequently, individual farmers and decision makers show great interest in producing rapid and accurate estimates of crop yield.

Limitations in the use of Geoinformatics for rice production in Sri Lanka

Field survey with the help of GPS and Remote Sensing are valuable sources of data in rice-based agriculture, where GIS act as a decision support system with combining all spatial and non spatial data sources together. The following sections describe the limitations of using these technologies with respect to rice production in Sri Lanka.

Data availability

Non-availability of remotely sensed data due to many reasons including financial constraints, poor data sharing mechanisms and high cloud coverage

are serious constraints. It is important to monitor rice crop continuously using multi temporal imagery in order to make accurate decisions with respect to management at different crop growth stages. It may not be possible to acquire data at a high frequency mainly due to financial constraints. Though, several agricultural research outputs have been accomplished with freely available satellite imagery like NOAA AVHRR data, they are not appropriate for many applications especially with respect to the small land holding in Sri Lanka.

Land fragmentation in the country demands the use of satellite imagery with high spatial resolutions to monitor individual rice paddocks in most of the areas. High cost associated with this imagery limits their use in this sector. Not only remote sensing data, spatially distributed climatic data, soil data and other information are also very restricted for mapping purposes in GIS. It is required to develop mechanisms for data sharing for research work as most of these data have been acquired using public funds.

Limited mapping capabilities

Remote sensing is limited to mapping single crops to slightly higher than 90% accuracy when multi-date, multi-sensor data coupled with GIS data are used. Remote sensing alone, however, is more limited at discrimination of multiple crops to this level of accuracy. Yield predictions are also limited from remotely sensed data. Remote sensing is also often unable to detect direct sources of crop damage. However,

the resolutions needed for this type of detection will probably not be available for some time for some areas. Although remote sensing from satellites has high potential for providing spatially variable data needed for precision agriculture, the limitations of fixed spectral bands, coarse spatial resolutions, inadequate repeat cycles and long delivery times make the remotely sensed data difficult to use under some occasions.

In addition, the heterogeneity of crop cultivations, fragmented land holdings, varietal differences, different methods of crop establishment (broadcasting, row seeding, transplanting manually or with machines), standing water in the rice fields can make the remote sensing estimations less accurate due to spectral confusion.

Lack of expertise and computer facilities

Agricultural research with geo-informatics requires a moderate level of expertise and computing support. The processing of remotely sensed data, GPS data and GIS analysis require an investment in training of personnel as well as adequate computers and data storage. Computer hardware and software are important, but perhaps more important is mind ware to ensure the correct use of remote sensing and GIS to assist in the decision making process. These facilities are lacking in Sri Lanka.

Future prospects of using Geo-informatics technology in rice production in Sri Lanka

Though there are several limitations in using remotely sensed technology in the Sri Lankan context, the possible pathways should be explored to improve the use of this technology overcoming the constraints for the betterment of the agriculture sector in the country. Immediate requirement in the country with respect to rice production is the estimation of area under paddy cultivation during main agricultural seasons. Accurate statistics on area under paddy is important to forecast yield, assessment of seed paddy requirement, regulate the rice imports and also to develop marketing strategies for the forthcoming seasons. It is very important to assess the new areas which came under rice production from the North and East provinces after the civil war. Due to large area coverage, it will become a costly and time consuming task to implement field surveys for this purpose. Hence, the only solution is to utilize remotely sensed data with suitable spatial resolutions at different rice producing areas to identify the areas under cultivation. In addition, it is imperative to assess the land suitability for rice cultivation in different areas to improve the land productivity. GIS databases on different spatial parameters such as soils, land use, hydrology and the statistics on population density, agricultural land distribution, farming communities are readily available in the country at various government

departments. If a proper data sharing policy can be formulated, it is possible to gather all these data into a centralized location to get the maximum utilization. In addition, the country should explore the possibilities of identifying the sources which provide free remotely sensed data or data at a low cost. The technology should be promoted at the government departments and the research programmes should be designed collaboratively with the universities and non governmental organizations working in agricultural sector. Training opportunities in this

sector at local, regional and global level should be explored to train the individuals to provide theoretical and practical knowledge in the field of Geo-informatics. Collaborative research programmes can be organized with the rice based research institutions in the world to share there experiences in utilizing Geo-informatics technology in rice production. If this technology can be appropriately utilized to suit the local requirements, it is possible to achieve a greater success in production, marketing and utilization of rice in an agricultural country like Sri Lanka.

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Determination of Onset Time of Rains During Yala and Maha Seasons in Major Rice Growing Regions of Sri Lanka

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Abstract

As both Yala and Maha rainy seasons are short in an absolute scale, it is important to establish the rice crop in the field as early as possible to avoid wasting valuable time within the growing period. The reliable determination of the onset of the rains in each growing season is therefore of crucial importance for sustainable rice production. Otherwise, misinterpretation or misidentification of onset of seasonal rains always leads to crop failures and high economic losses in both Dry and Intermediate Zones where country's major productive rice tracts are located. Thus, the present study was undertaken to determine, from a retrospective point of view, the spatial distribution of onset time of rains in Yala and Maha seasons in major rice growing regions of Sri Lanka, namely, Dry and Intermediate Zones with a simple procedure using only weekly rainfall data.

There is a high spatial variability of onset time of rains in both Yala and Maha seasons. However, it is also clear that conventional onset time of the major rice growing areas of the country remained unchanged despite changes of rainfall regime attributed to climate change. As the depicted probable onset times of both Yala and Maha rains have been based on the long series of quality controlled historical rainfall data, they can be used in decision making on the cropping calendar and water issues from the tanks in respective regions without wide margin of errors.

Key words: Dry and Intermediate Zones, Maha, Onset, Rains, Rice, Yala

Introduction

Even though Sri Lanka exemplifies an average annual rainfall of about 1,860 mm with a range of 900 mm (Maha Lewaya, Hambantota) to over 5,500 mm (Kenilworth Estate, Ginigathena), its distribution over the months in a year shows very high spatial variability resulting relatively

large number of growing environments across the country compared to countries with a similar geographical extent and exposure. Identification of 46 Agro Ecological Regions (AERs) based on climatic and other edaphic features within an area of about 65,000 km² have clearly justified the presence of such a high spatial variability of rainfall across the island (Punyawardena, 2007).

Rice is grown under more diverse environmental conditions than any other food crop in Sri Lanka (Panabokke and Punyawardena, 2000). Except a few AERs in the Up country Wet and Intermediate Zones where nighttime minimum temperature limits the growth and development of rice plant, it is the most common land use in valley bottoms in all other AERs of the country. Rice production of Sri Lanka highly depends on the rainfall rhythm of respective regions. Out of three major rice eco-systems of the country, namely, Major irrigation, Minor irrigation and Rainfed, only the Major irrigation system is less sensitive to features of seasonal rains being highly dependent on trans-basin diversion of rivers originating from high rainfall areas of the island Wet Zone. However, time has come to motivate farmers in this rice eco-system as well as to use early rains of the season for land preparation as against to their conventional practice of waiting until the tank is filled to start the land preparation. Under such a situation, all three rice eco-systems of the country would be highly sensitive to features of seasonal rains such as onset, cessation or withdrawal, length and any mid season dry spells. Out of those four features of seasonal rains, onset of seasonal rains has a special significance in determination of cultivation calendar in an area, in particular to decide the date of water issue from the tanks in major and minor irrigation systems. It assures a minimum degree of staggered cultivation while saving the water to use in latter part of the season where rainfall is generally uncertain both in terms of amount and timing.

Conventionally, farmers and policy makers use early April and early October as the onset time of Yala and Maha rains, respectively with the gradual setting of intermonsoonal rains due to convectional activity and/or influence of the Inter Tropical Convergence Zone (ITCZ) over the island. However, many people in the agriculture sector are anxious whether onset time of rains in Yala and Maha seasons has been shifted due to climate change. Therefore, the main objective of this study was to determine, from a retrospective point of view, the onset of rains in Yala and Maha seasons in major rice growing regions of Sri Lanka, namely, Dry and Intermediate Zones with a simple procedure using only weekly rainfall data.

Materials and Methods

Study area

The Fig. 1 shows major rice growing areas of Sri Lanka. It clearly shows that within three major climatic zones, namely, Wet, Intermediate and Dry Zones major rice tracts have been concentrated only in the Dry and Intermediate Zones where all three aforesaid rice eco-systems are found in abundance. Even though the Fig. 1 along with the Table 1 suggest that considerable extent of rice lands spread across the Wet Zone, onset of seasonal rains in this region does not make significant impacts on the rice cultivation calendar owing to availability of phreatic water throughout the region. Meanwhile, recent studies have clearly shown that

farmers in the Wet Zone are weaning away from rice cultivation due to various physical and socio-economic reasons (Kendaragama and Bandara, 2000) in spite of its climatological conduciveness for rice cultivation in both seasons (Chithranayana *et al.*, 2004). Therefore, this study has limited its scope only to the Dry and Intermediate Zones of Sri Lanka where onset of rains in both Yala and Maha seasons would make a significant bearing on the rice cultivation calendar in respective regions and consequently the national rice production.

Data summary

Weekly rainfall data collected from 36 stations scattered throughout the Dry and Intermediate Zones were

taken for the study. In selecting these rainfall stations given in Table 2, major emphasis was given to the locations which are having long time series of data recorded during most recent times with a fair spatial coverage across all agro-ecological regions of both Dry and Intermediate Zones. In this study, the Yala season was considered as the period spanning from March to August while the period from September to February was chosen as the effective period of the Maha season. Consequently, any standard week after standard week No. 9 and 34 for Yala and Maha seasons, respectively, that satisfies the requirement of onset criteria was chosen as the start of respective seasons.

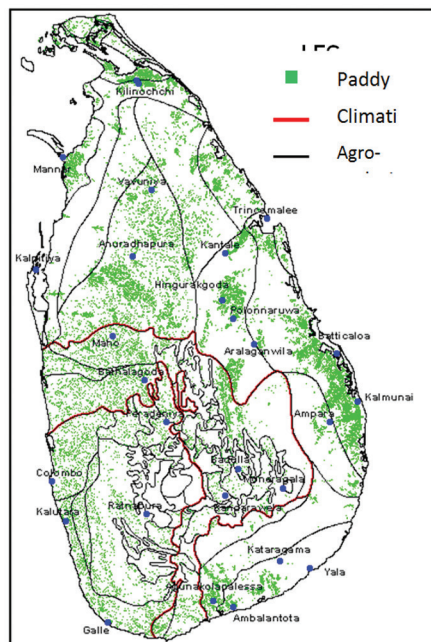


Figure1: Major rice growing areas of Sri Lanka.

Table 1: Extent of rice lands under each agro-ecological region of Sri Lanka.

Agro-ecological Region	Rice Extent (ha)	Agro-ecological Region	Rice Extent (ha)
WU1	36	IM1b	1,463
WU2a	0	IM1c	1,064
WU2b	145	IM2a	3,524
WU3	36	IM2b	2,793
WM1a	1,928	IM3a	1,130
WM1b	1,263	IM3b	1,463
WM2a	1,862	IM3c	2,793
WM2b	4,654	IL1a	47,997
WM3a	1,995	IL1b	22,454
WM3b	6,915	IL1c	5,465
WL1a	35,904	IL2	25,473
WL1b	12,204	IL3	36,827
WL2a	36,558	DL1a	4,231
WL2b	7,737	DL1b	1,81,067
WL3	31,491	DL1c	61,487
IU1	344	DL1d	25,135
IU2	1,374	DL1e	36,622
IU3a	109	DL1f	10,946
IU3b	109	DL2a	15,630
IU3c	2,142	DL2b	1,25,601
IU3d	399	DL3	34,006
IU3e	2,687	DL4	41,286
IM1a	5,053	DL5	19,940

Table 2: Selected rainfall stations for the study.

Station	Longitude	Latitude	Agro-ecological Region	Period
Hambegamuwa	6.53 ⁰	80.95 ⁰	DL1a	1979-1984
Okkampitiya	6.75 ⁰	81.30 ⁰	DL1a	1985-2000
Udawalawe	6.42 ⁰	80.82 ⁰	DL1a	1989-1999

Maha Illuppallama	8.12 ⁰	80.47 ⁰	DL1b	1976-2009
Angunakolapellessa	6.17 ⁰	80.88 ⁰	DL1b	1976-2009
Vavuniya	8.75 ⁰	80.50 ⁰	DL1b	1979-2009
Pelwehera	7.90 ⁰	80.68 ⁰	DL1b	1974-2003
Tanamalwila	6.47 ⁰	81.12 ⁰	DL1b	1977-1996
Medawachchiya	8.53 ⁰	80.48 ⁰	DL1b	1971-1985
Panama	6.75 ⁰	81.77 ⁰	DL1b	1970-1988
Minneriya	8.05 ⁰	80.90 ⁰	DL1c	1980-2009
Polonnaruwa	7.92 ⁰	81.03 ⁰	DL1c	1979-2009
Kantale	8.35 ⁰	80.98 ⁰	DL1c	1976-2009
Hingurakgoda	8.05 ⁰	80.95 ⁰	DL1c	2000-2007
Padawiya	8.83 ⁰	80.77 ⁰	DL1e	1972-1981
Horowpathana	8.53 ⁰	80.80 ⁰	DL1e	1960-1970
Batticaloa	7.72 ⁰	81.70 ⁰	DL2b	1979-2009
Aralaganwila	7.68 ⁰	81.18 ⁰	DL2b	1984-2009
Ampara	7.28 ⁰	81.67 ⁰	DL2b	1990-2009
Trincomalee	8.58 ⁰	81.25 ⁰	DL2b	1979-2009
Vanathavillu	8.27 ⁰	79.85 ⁰	DL3	1976-2009
Kalpitiya	8.25 ⁰	79.77 ⁰	DL3	1988-2002
Mannar	8.98 ⁰	79.92 ⁰	DL4	1979-2009
Paranthan	9.42 ⁰	80.42 ⁰	DL4	1975-1989
Weerawila	6.28 ⁰	81.23 ⁰	DL5	1976-2009
Hambantota	6.12 ⁰	81.13 ⁰	DL5	1979-2006
Ambalantota	6.12 ⁰	81.02 ⁰	DL5	1979-2009
Bandagiriya	6.23 ⁰	81.15 ⁰	DL5	1958-1988
Bathalagoda	7.52 ⁰	80.45 ⁰	IL1a	1976-2009
Makandura - NWP	7.32 ⁰	79.97 ⁰	IL1a	1982-2009
Lunuwila	7.33 ⁰	79.87 ⁰	IL1a	1977-2002
Moneragala	6.87 ⁰	81.35 ⁰	IL1c	1983-2009
Girandurukotte	7.45 ⁰	81.08 ⁰	IL2	1976-2009
Maho	7.82 ⁰	80.28 ⁰	IL3	1984-2009
Nalanda	7.67 ⁰	80.63 ⁰	IM3b	1976-2002
Hewaheta	7.07 ⁰	80.46 ⁰	IU2	1976-2001
Bandarawela	6.83 ⁰	80.98 ⁰	IU3c	1976-2009

Determination of onset of seasonal rains and mapping

If any consecutive three week period has a cumulative weekly rainfall of 20 mm or more in each week after a pre-specified week, week No. 9 and 34 for Yala and Maha seasons, respectively, then the first week of the said consecutive period was chosen as the onset of the seasonal rains. However, if the preceding week of the series has received a cumulative rainfall of greater than 10 mm or above, then it was considered as the onset of the seasonal rains taking in to consideration the fact that 10 mm of cumulative rainfall is adequate to soak the soil for land preparation activities. Meanwhile, if any consecutive three week period has a week with less than 20 mm of cumulative rainfall, but bounded either side by 20 mm or more than, the first week of the said period was chosen as the start of the seasonal rains. This relaxation was highly applicable to the Yala season where a true onset of seasonal rains cannot be easily discernible.

Based on the above criteria, surfaces depicting the spatial distribution of onset of each season were developed for the Dry and Intermediate Zones using ArcView GIS Software. Each surface was superimposed with the Dry and Intermediate Zones of the Agro-ecological Region Map of Sri Lanka (Punyawardena *et al.*, 2003) to ascertain the spatial relationship of onset of each season with the agro-ecology of major rice growing regions of Sri Lanka.

Results and Discussion

Onset of the Yala season

The analysis used in this study reveals that onset of the Yala season across Dry and Intermediate Zones of Sri Lanka, generally swings among standard week numbers 12 to 14, a period between 19th March to 2nd April. However, it should be noteworthy to put it in records that some AERs in the Dry and Intermediate Zones do not possess a distinct Yala season, but only a few weeks of rains with just fulfilling the requirements of the definition of onset criteria used in this study. These regions include DL2a, DL2b AERs in Digamadulla area and DL3, DL4, DL1c. DL1b (northern part), DL1e AERs of the Wanni region and DL5 and DL1b (southeastern Dry Zone) in Ruhuna area.

Fig. 2 shows spatial distribution of onset time of Yala season in Dry and Intermediate Zones of Sri Lanka. It is clear that apart from an area in the IL1a agro-ecological region, which is neighboring the WM3a and WM3b AERs in the central highlands of the Wet Zone, all other areas in the Low Country Intermediate Zone of the western half of Sri Lanka commence their Yala season in the standard week number 13, a period starting from 26th March onwards. However, the aforesaid area in the IL1a region, mainly the eastern part of the Kurunegala district commences its Yala rains a week earlier (19th March onwards). In fact, the most productive rice tracts of Kurunegala district are

located in this region including the experimental fields of Rice Research and Development Institute (RRDI). Meanwhile, the IL2 AER in the eastern part of the Low Country Intermediate Zone commences its Yala rains only on standard week number 14 (2nd April onwards). However, as mentioned before, these rains will last only for a few week period and hardly support any kind of crop growth. All other AERs of the Intermediate Zone located in Mid and Up Country regions begin to receive their expected Yala rains by standard week number 13, a period from last week of March (Fig. 2).

Onset of Yala season in the Dry Zone generally delays by one or two weeks than that of the Intermediate Zone. However, most parts of the Dry Zone commence their Yala rains by first week of April (standard week number 14). But, DL1a region of the Dry Zone commences its Yala season two weeks before the other areas of the Dry Zone (Fig. 2). This area stretches as a strip sandwiched between DL1b AER of southern Dry Zone and, foothills of southeastern edge of central hills and IL1c AER. In contrary, despite being able to identify a start of the Yala rains by the first week of April in the eastern and southeastern parts of the Dry Zone, the continuation of same rains to support a crop growth during the season cannot be guaranteed due to well established climatological reasons.

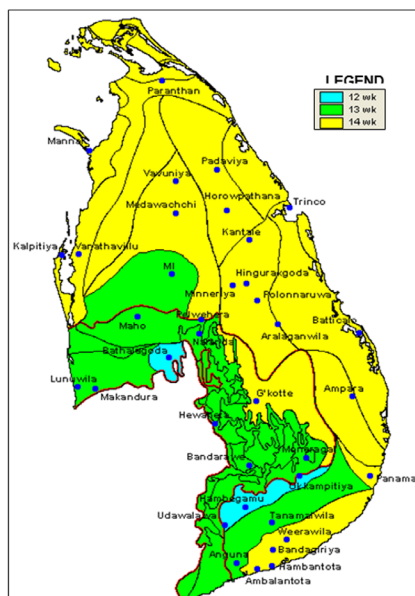


Figure 2: Onset of Yala season in Dry and Intermediate Zones of Sri Lanka.

Onset of the Maha season

Fig. 3 shows the spatial distribution of onset of Maha rains in the Dry and Intermediate Zones. Unlike the Yala season, the onset of seasonal rains in major rice growing regions of Sri Lanka is ranging from standard week number 37 to 41, a period from 10th September to 8th October with a five week span. The earliest onset of Maha rains on standard week number 37, 10th September onwards is recorded in IL1a region in the Low Country Intermediate Zone along with most of IU regions. The other parts of the Low Country Intermediate Zone, namely IL1b, IL1c, western part of the IL2

region and all IM regions begin to experience effective Maha rains a week later, a period from 17th September onwards. The rest of the IL2 region that adjoins the Dry Zone along with the IL3 region commences their Maha rains by standard week number 39, 24th September onwards.

Even though, it is customary to presume that onset of Maha season rains occurs only after early October in the Dry Zone, some regions of it such as DL1a region, southern part of the DL1b region and a part of the DL1c region that neighbors IL2 region begin to experience effective Maha rains from 24th September onwards (Standard week 34). Meanwhile, the rest of the Dry Zone receives its effective Maha rains between standard weeks 40 and 41, October 1st and 8th, respectively (Fig. 3). Late start of the Maha rains (Standard week 41) occurs in northern, northwestern, southeastern parts of the Dry Zone including the entire DL5 region. The Rajarata region of the Dry Zone along with the northern part of the DL2b region begins to receive its Maha rains at conventional onset time of Maha rains, the first of October. In fact, this is the area where our hydraulic civilization was most pronounced with cascades of tanks during ancient times. Nevertheless, within this region

a lense-shaped strip of area starting from the meeting point of the Dry, Wet and Intermediate Zones in the Matale district towards northeastern direction up to Kantale can be discernible where onset of the season is delayed by a week (Fig. 3).

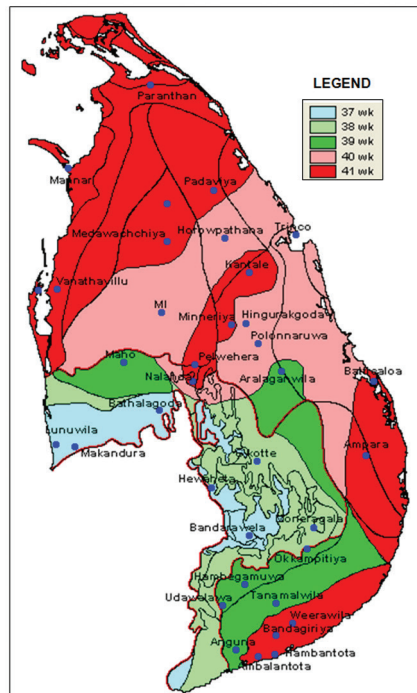


Figure 3: Onset of Maha season in the Dry and Intermediate Zones of Sri Lanka.

Conclusions and Recommendations

This study has shown that there is a high spatial variability of onset time of rains both in Yala and Maha seasons. However, it is also clear that conventional onset time of rains in the major rice growing areas of the country remain unchanged despite changes of rainfall regime attributed to climate change. Meanwhile, it is

a well known fact that having a precise knowledge on the onset of seasonal rains in a region would reduce the risk at crop establishment while it makes easier and less risky to take decisions on water management in agriculture. As the depicted probable onset times of both Yala and Maha rains have been based on the long series of quality controlled historical rainfall data, they can be used in decision making on the cropping calendar and water issues from the tanks in respective regions without wide margin of errors. A prudent user can also make use of them as a decision support tool in planning rainfed upland agricultural systems as well.

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Achievements and Strategies for Improving Water Productivity in Rice Cultivation

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Abstract

Water, apart from its critical functions in plant physiology, plays an important role in many of the practices in rice cultivation such as weed management, crop establishment, fertilizer application, and pest & disease control etc. Water is significant in determining quality and yield of rice grain, influences the environmental quality creating conditions such as salinity, alkalinity, iron toxicity etc. and affects the growth of rice plant differently at different phenological stages of the crop. As rice is a semi aquatic plant, it can grow well under a range of water regimes. However, soil water conditions affect the plant nutrition and growth and yield of the crop. Traditionally, rice is grown under irrigation keeping standing water in the dykes (Liyadi). This practice uses several folds of the water required to meet the evapotranspiration of the crop increasing the water demand and reducing the water productivity of the crop. Therefore, water management is crucial in rice cultivation and has to be based on scientific knowledge on soil-water- plant relation of the crop within practically feasible limits considering socio-economic factors too.

Studies on water management in rice have been initiated at the Agricultural Research Station in Mahalluppallama way back in 1966 and the research carried out since then has worked out components of water requirements under different soil and drainage conditions. Water requirement for land preparation, influence of water regimes and stress conditions on growth and yield and some varietal screening work to identify lines which tolerate moisture stress conditions have been studied. Early findings on water requirement for rice culture under different soil and drainage classes and the water requirement for land preparation have been communicated to the Mahaweli Authority by the DOA and have been used to make irrigation schedules. Apart from that the findings so far have very limited practical value but they will help us guiding ourselves on the future direction of research.

Rice water management research has been started at the Rice Research and Development Institute in 2004 and water saving irrigation technologies such as saturated soil culture, and alternate wetting & drying have been introduced recently. Present research is focused on aerobic rice culture which is a new concept to decrease water requirement in rice cultivation. Research is also being conducted to find out the performance of rice varieties under different moisture stress conditions during crop growth stages. Reducing seepage and percolation water losses and increasing the yield per unit volume of water are the two main aspects that need attention in order to increase the water productivity of rice.

This needs input from many disciplines such as soil science, irrigation agronomy, plant breeding and weed science. Research has been initiated on system level water management using new technologies such as Global Positioning System (GPS) and Geographic Information Systems (GIS) along with Remote Sensing (RS). It is expected to link on-farm water management with system level water management in the irrigation systems covering socio-economic aspects.

Key words: Water requirement, Water productivity, Water saving

Introduction

Rice is grown under a wide range of elevations in different climatic, soil and hydrological conditions in Sri Lanka. Rice lands are distributed over all agro-ecological regions of the country except in regions located at very high elevations. The performance of the crop is largely determined by the characteristics of micro environment of growing areas. There is a wide range of rainfall, temperature and soil types within these environments. Rice is grown under different water regimes in the country which are mainly determined by the irrigation systems. Based on these differences rice growing systems have been categorized in Sri Lanka as major irrigation schemes, minor irrigation schemes and rain-fed systems. Major irrigation schemes are those having a command area of more than 80 ha, where the water supply may be either from a major tank, a river or a major stream diversion system. Minor irrigation schemes are those that have a command area less than 80 ha and water supply is by a small tank or stream diversion. Rain-fed lands are dependent mostly on the rainfall for water supply. Paddy lands under minor and major irrigation schemes are mainly distributed within the Dry

and Intermediate Zones of the country while rain-fed lands spread more or less in the whole country. Rice yields are highly variable in rain-fed areas compared to irrigated areas showing that water is a critical factor for rice cultivation in those environments.

Water, apart from its critical functions in plant physiology, plays an important role in many of the practices in rice cultivation such as weed management, fertilizer application, and pest and disease control. Water is significant in determining grain quality and quantity of rice. Further, it influences the growing environmental quality creating conditions such as salinity, alkalinity, iron toxicity etc. Therefore, water management is crucial in rice cultivation at all the growth stages of the crops. It influences the growth of rice plant in different levels at different stages of the crop growth. As rice is a semi aquatic plant, it can grow well under a range of water regimes which may vary from 25% depletion level of soil moisture to 10 cm depth of standing water (Nayakekorala, quoted by Meththananda, 1990). Beyond these two critical levels, water stress as well as excess water create problems for growth of the crop. Conditions beyond the critical levels extremely affect the

growth of rice plant, nutrient status of soil and growth of weed.

The objective of this paper is to summarize the research studies conducted during the past 40 years on irrigation agronomy and on-farm water management of rice cultivation in Sri Lanka and to discuss the future direction and strategies of research in on-farm water management with a view of improving water productivity.

Progress of water management studies

Studies on rice water management have been initiated at the Agricultural Research Station at Mahailuppallama in 1966 and carried out until 1994. After re-structuring of the DOA and establishment of the crop specific research stations in 1994, rice research became the sole responsibility of the Rice Research and Development Institute (RRDI). However, after establishment of the RRDI there was no much emphasis given on water management research until 2004.

Early emphasis has been on water requirement research for land preparation and crop growth, rice cultivation under different water regimes, effects of water stress conditions and varietal screening for moisture stress. Components of water requirement namely, evapotranspiration, seepage and percolation rates of soils have been given priority in these studies. As water requirement data was required for planning and management of new irrigation projects

at that period much emphasis has been placed on that aspect. On-farm water management research conducted in early period has covered both upland and lowland rice cultures but more emphasis has been on lowland rice culture.

The Symposium on Research and Production of Rice in Ceylon conducted in 1966 has reported the work on water requirement of rice by Muarakami and Vignarajah (1966). Their work is the pioneering systematic research on water management conducted in Sri Lanka on paddy found in the literature. In the Rice Symposium conducted in 1980, Rasiyah (1980) summarized the research on water management conducted in Sri Lanka on paddy till then. In Rice Congress of 1990 water management research had been discussed under agronomy where some important findings of water management research conducted till then had been captured. Rice Congress of 2000 has reported some aspects of water management in system level. At the conclusion of the congress, a proposal has been made named “Gannoruwa proposal”, where improvement of water management has been identified as a mean of increasing cropping intensity of paddy in the irrigation schemes but has not reported how to do that or any direction for research. It has to be noted that review of work in previous rice symposia and congresses has not been very objective with respect to rice water management and not any specific recommendations have been made to follow up in the forth coming period.

Components of rice water requirement

Water requirement (WR) of rice depends on the evapo-transpiration (ET) and seepage and percolation (S&P) rates of water in the field. WR can be estimated by the following relationship.

$$\text{WR} = \text{ET} + \text{S\&P}$$

The ET rate is governed by the climatic factors and is kept constant in a specific climatic zone. However, S&P rates depend on the soil characteristics which may vary within short distance of a field. Hence, S&P rates on different soil types are required to determine the crop water requirements. Murakami and Vignarajah (1966) estimated by field studies the values for S&P rates to be 7 mm/day for LHG soils. Gunasekera and Wickham (1978) based on the field trials in Kagama colonization scheme reported value of 4-7 mm/day for the same soil group. Holmes *et al.* (1981) based on work conducted in Kaudulla Irrigation scheme reported values of 9, 22 and 43 mm/day, respectively for LHG, RBE moderately drained and RBE well drained soils. Hunting Technical Limited (1978) reported values of 3 and 10 mm/day for LHG and RBE soils, respectively. The data shows very wide variation of S&P rates in the field. Kitamura (1984) who studied consumptive use of water for paddy in the Rajangana irrigation scheme reported very high S&P rates averaging 20 mm/day. Nayakekorala (1985) after analyzing these data proposed that it would be justifiable

to use S&P rate of 3-4 mm/day for LHG soils and 7-10 mm/day for RBE soils. However, present knowledge on behavior of S&P rates in our soils is very poor. Kitamura (1984) reported that superficial layers of soil determine the percolation rates. He attributed this high percolation rates to the cracks formed in the soil during dry cycle in intermittent irrigation events in rotational water issues. Even if the puddling operation could reduce S&P rates in the paddy field the area under the dykes are not puddle and those areas could be the pathways for the high S&P rates in paddy fields (Lowe and Rushton, 1990). This is an area where future research has to be directed to improve the water use efficiency in paddy. This review indicates that we have not adequately captured the variability of the two main components of rice water requirement namely ET and S&P in various rice growing environments. If our aim is improving the water productivity we need to gather data on these aspects at least for major rice growing environments of the country.

Water requirement for land preparation

Methods of land preparation for highland and lowland paddy cultivations are different. In highland cultivation land is ploughed and harrowed when the soil is moist and allowed the weeds to dry up. In the lowland conditions soil is ploughed when the soil is wet and land is inundated with water to allow weeds to decay for some time. When the soil is soaked and weeds are

decayed there will be another plowing which enhances the decay of weeds. After this operation soil is kept again submerged for some time. Then the soil is puddled with water, excess water is drained and land is leveled for crop establishment. Puddling in this process is an important operation to reduce the percolation rates of water in soil. Water requirement during the land preparation period can be given by the following relationship.

$$LWR = SR + S\&P + E$$

Where, LWR is the land preparation water requirement, SR is the soil soaking requirement, E is the evaporation requirement and S&P is the seepage and percolation losses. From this relationship it appears that lengthening of this period increases the water use. Nayakekorala (1990) reported that paddy farmers in a field channel in Mahaweli system H took nearly 35 days to complete the land preparation (wetland) from the date of water issue. Limitations on availability of machinery and labor for

land preparation were the reasons for comparatively a longer duration.

Good land preparation facilitates easy water management while availability of sufficient water is important for good land preparation. Water requirement for land preparation varies with soil type, cultivation practice and time and duration of land preparation. Despite the fact that rice is grown on different soil types, studies on land preparation water requirement have been carried out only in RBE, LHG and NCB soils. Several workers have reported water requirement for land preparation for lowland paddy by using different experimental approaches. Their findings are presented in Table 1. Data from the different workers cannot be compared because of differences in the variables used. However, the data show that there are differences in water requirement in different soil types and drainage classes. It is important to note that there are differences in length of the period used by different researchers.

Table 1: Land preparation water requirement for paddy reported by various workers.

Reference	Soil type	Length of land preparation (days)	Quantity of water used (mm)		Remarks
			Maha	Yala	
Murakami and Vignarajah, 1966	LHG	25	170	170	Experiment has been conducted in isolated plots
Lewis, 1976	RBE/ WD	Not reported	270	-	Research conducted in large farmers' fields
	RBE/MD		168		
	LHG		96		

Rasiah, 1980	RBE/WD	7	231	Research conducted in large fields
	RBE/MD	12	104	
	LHG	12	157	
Nayakekorala, 1983	RBE/MD	14	204	Research conducted in large fields
	LHG	14	163	

The data in Table 1 indicate that well drained RBE soils require more water for land preparation when compared to poorly drained LHG soils. As the water requirement varies with the soil type, methods of land preparation also affect the water required. According to Alles (1967) water requirement for land preparation was very low in minimum tillage method (185 mm) compared to conventional method (450 mm), although the yields were similar in both cases.

The water required for land preparation depends on the length of the period of operation. Lengthening of the period will increase the water requirement. On the other hand shortening of the period will affect the quality of land preparation and increase weed population after crop establishment. Nayakekorala and Hemakumara (1991) showed that lengthening of land preparation period beyond 14 days has no beneficial effect on the yield increase but increase the water requirement in LHG soils. However, lengthening of period beyond 14 days helped reduce the weed population after the crop establishment.

According to the above review it is clear that the length of land

preparation period determines the water requirement in a given soil type. As the length of land preparation period depends on many socio economic factors, the optimum water requirement has to be studied taking these aspects into consideration using interdisciplinary and participatory research approaches.

Crop water requirement during growth stages

Crop water requirement depends on the characteristics of climate (evaporation), soil (seepage and percolation) and the plant (transpiration) as stated before. Murakami and Vignarajah (1966) showed by pot experiments that transpiration of lowland paddy gradually increased from crop establishment till heading stage and decreased afterwards while evaporation component gradually decreased until the time of harvesting. The trend of ET variation was gradual. The peak ET was about 1.8 times the initial evapo-transpiration.

Alles (1967) found that the total water requirement of paddy cultivation in feet is numerically equal to the age of the paddy in months. He further observed that the amount of water

needed during growing period vary with the drainage class of soil due to differences of seepage and percolation losses. The 4-4½ month paddy crop used 713 mm of irrigation water and 202 mm of effective precipitation to produce grain yield of 4.64 t/ha at 14 % moisture in the poorly drained soil. In the imperfectly drained soils 3-3½ month paddy crop used 838 mm of

effective precipitation to produce yield of 1.9 t/ha at 14% moisture. In the well drained soils 3-3½ month paddy crop used 965 mm of irrigation water and 67 mm of effective precipitation to yield 3.29 t/ha at 14% moisture. The results of water requirement of paddy during growth period recorded by various researchers are given in Table 2.

Table 2: Water requirement during crop growth period as reported by various researchers.

Reference	Soil Type	Crop Duration (months)	Quantity of water used (mm)		Remarks
			Maha	Yala	
Murakami and Vignarajah, 1966	LHG	4	1545	1910	Experiment has been conducted in isolated plots
Lewis, 1976	RBE/ WD	3½	1745	-	Research conducted in large farmers' fields. Values include effective rainfall
	RBE/MD		1177		
	LHG		401		
Nayakekorala, 1983	RBE/MD	3½	853		Research conducted in large fields. Values include effective rainfall.
		4½	1028		
	LHG	3½	948		
		4½	1128		

The very low value in LHG soils for 3½ month paddy as reported by Lewis (Table 2) is attributed to water gain as seepage from upper terraces. The high variability of the data reported by the different researchers could be due to variations in S&P rates and run off losses. The researchers have reported comparatively higher yields under these experiments. The data have a

limited value due to lack of breakdown of different components such as ET and S&P. However, the values can be considered for practical purposes.

Experimentally determined ET rates for paddy in Sri Lanka are very rare in the literature. Murakami and Vignarajah (1966) have determined ET of paddy using transpiration ratios (grain produced per unit of water

used). Nayakekorala (unpublished data) estimated the ET using long term average climatic data of Mahailuppallama by Penman method. The data are reported in Table 3.

Values for Maha season of 3½ month paddy closely compared with that of Murakami and Vignarajah (1966) while there were some discrepancies in the case of 4½ month maturity duration.

Table 3: Experimentally determined and estimated ET rates (mm) of paddy.

Reference	Maturity duration (months)	Yala Season (mm)	Maha Season (mm)
Murakami and Vignarajah, 1966	3½	-	455
	4½	-	830
Nayakekorala, 1983 Unpublished data	3½	665	463
	4½	768	520

Rathnayake (unpublished data) has reported the calculated ET rates for the Dry, Intermediate and Wet Zones as 4.7, 3.6 and 3.0 mm/day in Yala season and 3.0, 2.7 and 2.6 mm/day in Maha season, respectively. Rathnayake (unpublished) estimated

water requirement of different maturity durations of rice for three different climatic zones in Yala and Maha seasons (Table 4). Estimation of ET was done using long term evaporation data all over the country and S&P were estimated using values from literature.

Table 4: Water requirement of different maturity durations for three different climatic zones in Yala and Maha seasons.

Climatic zone	Water Requirement (mm)					
	4 Months		3½ Months		3 Months	
	Yala	Maha	Yala	Maha	Yala	Maha
Dry	1550	1350	1475	1325	1425	1275
Intermediate	1425	1325	1375	1300	1325	1250
Wet	1350	1325	1300	1275	1275	1250

The review of data indicates that the present knowledge on water requirement of paddy in Sri Lanka is very poor. More studies are required to gather basic data on the various components of water requirement

such as ET, seepage and percolation rates in different agro ecological zones on different soils and drainage classes. It has been reported earlier that the seepage rates are very high on some soils. It is important to study

the percolation in our soils. Literature shows that in some soils minimum percolation rates are required for growth of paddy. We need to study this aspect as well and determine the minimum percolation rates if required. Research also has to be conducted to develop ways and means to reduce percolation rates in the soil.

Irrigation methods

Lewis (1976) studied the irrigation methods for rice crop and concluded that rice can be grown successfully with intermittent irrigation even in the well drained soils. The rice crop needs frequent and light irrigation. If water has to be supplied by surface method, the land must be well graded. Rasiah (1980) found that the physical behavior of the well and imperfectly drained RBE soils were similar during Yala season, hence the crop performance and field irrigation requirements of crops were similar on these two soils. Rice yield on RBE soils was about 80% of that obtained on LHG soils.

Rice cultivation under different water regimes

Different water regimes in the field influence the growth of rice plant and nutrient uptake at different growth stages. Therefore, most of the studies on water regimes have been carried out under standing water condition and saturated condition. Alles (1966) reported that the yield differences between the 20 and 10 day periods of standing water is greatest at the primordial initiation stage and least at

the effective tillering stage. A 20 day period of ponded water at any critical stage is as good as or better than having continuous ponded water. Alles (1966) further reported that the treatments where there were periods without standing water during non critical stages of growth were far better than those with continuous standing water.

A comparison between standing water and saturated conditions in the soil from 3 weeks after flower initiation until late booting of 3½ month variety revealed that the grams of paddy produced per kg of water applied increased from 1.077 to 1.455 and 2.829 to 12.65 during Yala and Maha seasons, respectively (RRDI, 2008). The highest yield of 8 t/ha from Bg 359, 7 t/ha from Bg 300 and 4.6 t/ha from Bg 250 were recorded under saturated condition (RRDI, 2009). Experiments conducted at RRDI, Batalagoda reconfirmed that there is no difference in grain yield between both conditions suggesting that there is room for conserving irrigation water through better water management practices. However, negative results under saturation condition could be observed in the experiments conducted on NCB soils at Aralaganwila. Silva and Bandaranayake (1987) reported that the maintenance of 5 cm water level recorded the highest yield (12 t/ha) while saturated water level recorded the lowest yield (7.67 t/ha) in NCB soils. But the lowest water requirement was recorded in saturated condition as 58 cm of water while maintenance of 15 cm water level recorded the highest water requirement of 161 cm. Chandrapala (2006) reported that

continuous irrigation up to maximum tillering recorded the highest growth parameters and grain yield of paddy while maintenance of saturated water level recorded the lowest growth parameters and yield during Yala season.

Effects of water stress

Prolonged irrigation interval may subject the rice plants to water stress resulting in low yield. Hence, irrigation interval has to be based on the plant-soil-water relations of the rice plant and aim at preventing water stress in the plant. This requires the daily water requirement data including water losses. Response of the rice plant to water stress varies with its growth stage and other agronomic practices. Information on this aspect under local conditions is scarce. The preliminary studies conducted in Sri Lanka by Murakami and Vignarajah (1966) indicated that when plants were subjected to moisture stress during the 15 day period before and 15 day period after the heading, yield was severely affected.

The degree of yield reduction depends on the growth stage, length of the stress period and the ability of the variety to withstand moisture stress. Degree and duration of stress imposed by different irrigation intervals depend on the soil type, drainage class and the maturity duration of paddy (FCRDI, 1982). The impact of soil moisture at different growth stages of rice of different maturity durations were studied at

RRDI, Batalagoda (RRDI, 2009). Bg 359, Bg 300 and Bg 250 varieties were used to represent 3½, 3 and 2½ month maturity durations, respectively in the experiment. The highest yields of 8 t/ha (Bg 359), 7 t/ha (Bg 300) and 4.6 t/ha (Bg 250) were recorded under saturated soil condition. Rice yield was reduced by 50% when water stress was imposed from panicle initiation (PI) to maturity. In comparison to yield at saturated condition or no stress condition, 30% yield loss was observed when the stress was introduced from PI to 50% flowering stage and also from flowering to maturity stage. When the stress was continued up to PI, yield loss was about 20% (RRDI, 2009).

Severity of moisture stress as observed in rice crop mainly depends on soil water characteristics of the soil profile. Therefore, the studies should be focused on behavior of water movements and water retention in different paddy growing soils. Climatic conditions such as rainfall, temperature and relative humidity also affect water stress and studies are needed covering all these aspects.

Varietal screening for moisture stress

Rice varieties respond differently to drought conditions. Alles (1966) indicated that H-4 and H-8 varieties were as good as other drought resistant varieties like Pinulot and Dikwee for restrictively irrigated fields. Bg 250, an ultra short duration variety, performed well when recommended management

practices for a 3 month age rice crop were used. This variety is specially recommended for unfavorable rainfed areas where rainfall is not adequate for a successful 3 month rice crop (RRDI, 2004). Variety screening for moisture stress was initiated in 2006 at RRDI. Over 160 rice lines were screened for drought tolerance and the varieties were identified from traditional germplasms, recommended improved varieties and from INGER program (RRDI, 2008). Altogether 100 rice lines including 61 rice lines received through the INGER program were tested under aerobic, saturated and flooded soil conditions. Seven lines with very promising phenotypic acceptability and 3 lines (line numbers) with moderate acceptability were selected from the aerobic nursery. In addition under aerobic conditions, 5 lines were selected based on grain yield. Seven lines were selected under saturated conditions. These lines will be used in the future breeding programs to develop drought tolerance rice varieties (RRDI, 2008).

Future research needs

Reviews of past research reveal that future research has to be directed to improve the water use efficiency in paddy. This review indicates that we have not adequately captured the variability of the two main components of rice water requirement namely ET and S&P in various rice growing environments. If our aim is improving the water productivity we need to gather data on these aspects at least for

the major rice growing environments of the country. As the length of land preparation period depends on many socio economic factors, the optimum water requirement has to be studied taking these aspects into consideration using interdisciplinary and participatory research approaches. Research also has to be conducted for development of ways and means to reduce water losses as seepage and percolation rates in paddy soils are high. This requires study of effectiveness of paddy dykes and land preparation on these losses. Severity of moisture stress as observed in rice crop mainly depends on soil water characteristics of the soil profile. Therefore, the studies should be focused on behavior of water movements and water retention in different paddy growing soils. Climatic conditions such as rainfall, temperature and relative humidity also affect water stress and studies are needed covering all these aspects. Varietal development for water stress conditions is also important and it should be included in future rice improvement programs.

The most suitable land areas where traditional wetland rice culture could be practiced and technologies to increase water productivity should be identified. For the identification of land areas, new technologies such as Geo-informatics, Global Positioning Systems technology, Geographic Information Systems and Satellite images from Remote Sensing can be utilized. They also can be used for system level water management approaches.

Linking on-farm water management with system level irrigation management based on land and crop requirement is an important aspect in conducting future research. In this regard, all the above efforts have to be focused in an interdisciplinary manner. Water management research should be carried out with the collaboration of other sectors such as agronomy, irrigation, soil science, social science and economics.

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Productivity Improvement of Rice Lands through Soil Fertility Management

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Abstract

Many studies have been conducted to improve the fertility and productivity of rice soils during the past decade starting from year 2000. New fertilizer recommendation based on target yield was introduced in year 2001. Use of chemical fertilizers along with organic manures was recommended in the new fertilizer recommendation and afterward use of organic manure in rice fields was encouraged through various programs. Zinc sulphate was included as a fertilizer in the new fertilizer recommendation. Rice growing soils were characterized up to series level and efforts were made to develop site specific fertilizer recommendations for rice using the above information in 2007. To encourage farmers on site specific fertilizer recommendation, biological test kit was introduced as a low cost and easy method to realize the fertilizer needs in farmer fields. Target yield and water availability in a particular location were taken into consideration for issuing fertilizer for rice at subsidiary rates. To identify highly suitable rice lands, feasibility studies were initiated to prepare thematic maps using soil fertility as parameters. Productivity declining in rice fields was taken into consideration and farmer participatory studies were conducted at low fertility sites at Ambanganga and salinity affected areas at Kalingaella and Angithamkulama and beneficial management practices to increase productivity were identified. Introduction of granular urea in place of prilled urea and application of N based on leaf color increased N use efficiency by 10 -15% so that nitrogen fertilizer use could be reduced by 25 - 30%. Single super phosphate (SSP) produced from ERP was introduced as a P source for rice in 2008. Soil fertility studies in the future should be directed to find out ways to enhance utilization of naturally available plant nutrients in rice fields to reduce chemical fertilizer usage. In the mean time cost effective measures should be studied to improve productivity of marginal rice fields. Suitable rice lands should be identified using geographic information systems and they should be mapped to identify high potential rice growing soils.

Introduction

Average rice yield was 3.8 tha⁻¹ in the year 2000 and it increased up to 4.3 tha⁻¹ in 2010. There are many reasons behind this success but management of soil fertility is the main reason. Based

on the proposals made at the 3rd Rice Congress held in year 2000, the site specific nutrient management to reduce the gap between farmer yield and potential yield has been taken as the theme during the last ten years. Number of studies were under taken at Rice

Research and Development Institute (RRDI), Batalagoda and all the seven substations to find ways and means to improve production and productivity of rice soils. To increase fertilizer use in the country fertilizer subsidy was introduced. Even with the subsidized fertilizer, some farmers were unable to improve productivity in their rice fields. In the mean time depletion of soil fertility and profitability of rice farming could be observed. Thus the objective of this article is to compile the findings made during the last 10 years on productivity improvement of rice lands.

Rice soils in Sri Lanka

Rice growing soils in Sri Lanka range from waterlogged and poorly drained to well drained and rice is also grown under many different climatic and hydrologic conditions. Consequently, there is considerable variability in pedogenetic and morphological characteristics of rice-growing soils in Sri Lanka (Panabokke, 1996). As a result of the above fertility characteristics, productivity, risks and limitations vary among rice fields through out the country. Therefore, rice soils of Sri Lanka have been classified up to series level during last ten years (Mapa *et al.*, 1999; Mapa *et al.*, 2005; Mapa *et al.*, 2010). Most of the soil series in the Dry Zone viz. Elayapattuwa, Hurathgama, Kahatagasdeliya, Alawakumbura, Nawagattegama, Galwewa, Kuda Oya, Katagal Ara, Akkaraipattuwa, Damana, Ilupayadichennai, Muthukandiya, Pallegama, Omadiamadu and Seruwila except Puttalam and Siyambala series

have favorable soil properties for growth of rice crop. Soil fertility in terms of base saturation of these soils is conducive for higher yield.

Rice growing soil series in the Wet Zone have low fertility characteristics. Therefore, rice yields in the Wet Zone are low compared to that of Dry Zone soils. Madabokka, Palatuwa, Wagura and Gampaha soil series are found in the Low Country Wet Zone (LCWZ) while Gampola and Kiribathkumbura soil series are found in the Mid Country Wet Zone (Mapa *et al.*, 1999).

Depending on the extent of cultivation, major and minor rice growing soil series are found in the Intermediate Zone of Sri Lanka. Batalagoda, Balalla, Kudaoya, Muthukandiya and Ulhitiya series come under major rice growing soils while Hebarawa, Wariyapola, Kurunegala, Bibila, Badulla, Bandarawela and Welimada are minor rice growing soils in the Intermediate Zone soils in Sri Lanka (Mapa *et al.*, 2005). All these soil series are superior to that of Wet Zone soils but inferior to that of Dry Zone soils.

Inorganic fertilizer usage in rice

Fertilizer use in rice sector has increased during the past ten years. Present fertilizer statistics show that rice consumed around 0.28 m mt of urea, 0.09 m mt of triple super phosphate (TSP) and 0.07 m mt of muriate of potash (MOP) during Maha 2009/10 and Yala 2010 seasons (National Fertilizer Secretariat, 2010). Of the

total imports of urea, MOP and TSP, about 63%, 36% and 70%, respectively are used for rice cultivation in Sri Lanka. Ninety one percent of the rice farmers in Sri Lanka use recommended level of urea. However, MOP and TSP use was far below the recommended levels until the introduction of subsidy for P and K in 2005. Following the subsidy, with the use of P and K fertilizers, rice yield started to increase with the increased rate of urea application.

Fertilizer response studies

Many experiments have been conducted in the Wet Zone soils during the last ten years to find out the optimum levels of NPK for efficient rice production under Wet Zone conditions. Response to P application up to 40 $\text{P}_2\text{O}_5 \text{ ha}^{-1}$, N application up to 50 kg N ha^{-1} and K application up to 60 $\text{kg K}_2\text{O ha}^{-1}$ in alluvial soils at Bombuwala (RRRDC/Bombuwela, 2004). On the other hand, no response was observed for NPK fertilizer application in ill drained saline soils in the coastal flood plains in the LCWZ (RRRDC/Bombuwela, 2001). No response to K application was observed in mineral soils at Labuduwa. Application of Mg up to 20 kg ha^{-1} and application of dolomite up to 1.5 tha^{-1} increased the grain yield in LCWZ significantly (RRRDC/Bombuwela 2004). Bandara and Silva (2000) reported that most of the rice soils of Sri Lanka are low in secondary and micro nutrients, particularly S, Zn and Cu. Bandara *et al.* (2003) reported that N use efficiency is increased by applied Zn. Nutrient response studies

for Zn in LCWZ have shown that no response to Zn application in LHG soils found in the LCWZ (RRRDC/Bombuwela, 2004). Details of the effect of micronutrients on grain yield of rice in the Low Country Dry and Intermediate Zones are presented in another review article in this publication.

For different soils of the Wet Zone, N application ranged from 50 to 60 kg N ha^{-1} . N application up to 50 kg N ha^{-1} increased yield in low humic gley soils in the LCWZ (RRRDC/Bombuwela 2004). However, further increase in yield may be achieved if other nutrient deficiencies are corrected. Leaf-N content is closely related to photosynthetic rate and biomass production (Peng *et al.*, 1996). It is a sensitive indicator of the dynamic changes in crop-N demand within a growing season. Sirisena *et al.* (2002) found that leaf color measurement with SPAD meter can be effectively used to determine the N responsiveness of rice. Sirisena and Wickramasinghe (2003) further reported that SPAD meter results can also be used to reduce the N application by $\frac{1}{4}$ of the recommended N level. However, high price of SPAD meter limits its use by farmers. Therefore, leaf color chart (LCC) which measures leaf greenness related to the leaf N content in leaf has been tested to determine N requirement. Sirisena *et al.* (2006a) reported that irrespective of varieties, LCC readings linearly increased with application of N fertilizer but at high level of nitrogen there is a reduction in both leaf color as well as the grain yield of rice. They

pointed out that LCC reading is a good indicator to determine N requirement of rice varieties under Sri Lankan condition. The critical levels of LCC reading change according to the growth stage of the crop. At early and latter stages, it should be around 3.5 but at middle it should be around 4.5. LCC scores at 4 weeks after sowing can be used to screen rice variety response to added N. Experiments conducted in 3 agro-ecological zones revealed that amount of N application for rice crop can be cut down by this method thus increasing N use efficiency. Recovery of applied N to wetland rice in Sri Lanka is around 30% and agronomic N use efficiency is as low as 10 kg grain yield per 1 kg N (Sirisena *et al.*, 2001). Thus, new methodologies were tested at RRDI, Batalagoda to increase N use efficiency. In these experiments, granular urea having granule size of 3 mm, prilled urea coated with neem oil and prilled urea mixed with soil (Urea:Soil - 1:2 and kept at room temperature over night) have been tested against normal prilled urea with normal application. Between the two urea sources, granular urea produced higher yield as well as higher agronomic N use efficiency compared to prilled urea in both Yala and Maha seasons. The highest agronomic N use efficiency value of 29 kg grain yield per 1 kg N was recorded for granular urea and followed by soil mixed urea and neem coated urea. Agronomic N use efficiency of soil mixed prilled urea, neem coated urea and normal prilled urea was found to be 25.2, 22.8 and 20.4 kg grain yield per 1 kg N, respectively (Sirisena and Wickramasinghe, 2003).

Studies on phosphorous and potassium requirement for rice

MOP and TSP were subsidized since 2005, thus, the use of these two fertilizers increased with the increased use of urea from 2005. Alluvial soils at Bombuwela responded to P application up to 40 P_2O_5 ha⁻¹ (RRRDC/Bombuwela, 2004) and K application up to 60 kg K₂O ha⁻¹ (RRRDC/Bombuwela, 2004). When nutrients are supplied at a balance, P responded up to 30 kg P ha⁻¹ and K responded up to 60 kg K ha⁻¹ (Bandara, *et al.*, 2005; Bandara, *et al.*, 2006b; Bandara, *et al.*, 2007a). Use of P and K for rice has created nutrient balance for the crop. Response studies for P and K were conducted throughout the island during the past ten years. Accordingly, P fertilizer at a rate of 15-25 kg P ha⁻¹ and K fertilizer at a rate of 30 - 56 kg ha⁻¹ were recommended for the Low Country Dry and Intermediate Zones. Bandara *et al.* (2004) reported that high P application reduced Zn availability and consequently reduced the rice yield. Experimental results have proved that single super phosphate can be used as the P fertilizer source in both Low Country Dry and Intermediate Zones and in the LCWZ as well (RRRDC/Bombuwela 2004; RRDI, 2008).

When 75% of the required P supplied as TSP and 25% as high grade Epplawela rock phosphate (ERP) gave same yield as the yield given by TSP alone in soils of the LCWZ (RRRDC/Bombuwela, 2004). However, a similar

trend was not observed in the soils of the Low Country Intermediate Zone (RRDI, 2008). Experiments conducted at RRDI, Batalagoda have shown that Eppawela single super phosphate, a product of ERP, can be used as the source of P fertilizer in the Low Country Wet, Dry and Intermediate Zones (Sirisena *et al.*, 2007; RRRDC/Bombuwela, 2004). Bandara *et al.* (2005) have found that rice straw at the rate of 5 t ha⁻¹ can supply K requirement of rice at early stage of crop growth. Therefore, they suggested the possibility of reducing present dosage of K fertilizer for rice using K fertilizer only at later growth stages.

Farmers in the Dry Zone do not use micronutrients for their rice fields even though responses of rice to some micronutrients are observed. Therefore, farmers do not always achieve the full benefit of applied P and K fertilizers. Series of long term experiments were started at RRDI, Batalagoda in year 2009 and continued to find out ways and means to maximize the residual P fertilizer and for better use of K nutrients in rice straw and irrigation water. In these studies application of P fertilizer in alternative seasons and application of K fertilizer only at PI were tested against DOA recommended N, P and K application (RRDI, 2010). These studies should be conducted in multi-locations in the future to identify P and K requirement for rice fields in different soil series.

Studies on micro-nutrient requirement for rice

Nutrient imbalances in rice soils are the major reason for low response to application of fertilizer under farmer condition. They further reported that nutrient imbalance in rice fields leads to low fertilizer use efficiency, low grain yields and low farmer profit. Once a nutrient declines to its critical level, yields fall dramatically, even when large amounts of other nutrients are applied (PPIC, 2000). Bandara *et al.* (2007a) have reported the importance of balance fertilization in order to increase crop yields. Widespread deficiency of Zn has been observed throughout the country and soil analysis have shown that the level of Zn in the soil is lower than the critical Zn level for rice. As a result, a number of experiments have been conducted in the island and application of ZnSO₄ at the rate of 5 kg ha⁻¹ for rice has been recommended by the fertilizer recommendation committee (Bandara and Silva, 2001; DOA, 2001).

Studies on soil fertility improvement through organic manure

Long term field trials conducted at the RRDI, Batalagoda have found that application of chemical fertilizer along with organic manure consistently increased rice yield over that of application of organic manure or chemical fertilizer alone (RRDI, 2008).

Soil organic matter of all rice fields appears poor and this has been the main reason for low rice yields in Sri Lanka. Hence, the Fertilizer Recommendation Committee of the DOA met in 2001 decided to recommend application of organic manure to all rice fields. Further it was decided that no reduction should be made in the amount of chemical fertilizer added when organic manure is applied. Application of all the rice straw produced by the previous crop back to the field at first ploughing and application of 5 tha^{-1} of cattle manure or 2.5 tha^{-1} of poultry manure during the second ploughing are recommended. Application of green manures at the rate of 1 tha^{-1} is also recommended (DOA, 2001). Bandara *et al.* (2006a) reported that use of poultry manure and cattle manure provided most of the micronutrients particularly Zn to the rice crop. Further, sulfur nutrient can also be supplied by the use of poultry manure and cattle manure at recommended level. Therefore, it proves that application of micro nutrients is not required if these organic manures are added.

Site specific fertilizer application

Site specific fertilizer application has been the major theme during the last ten years. Therefore, site specific fertilizer recommendation for Kurunegala, Elayapattuwa and Galwewa soil series in the Dry and Intermediate Zones were formulated using initial soil analysis with ASI method of analysis and fixation studies. Fertilizer

rates calculated based on initial soil analysis, fixation curves and optimal nutrients ratios performed better in the field giving high rice yields. Accordingly, the tested technique gave better results and gave a balanced fertilizer recommendation for the rice crop. Hence, suggested nutrients rates to be applied to rice crop grown in Kurunegala soil series were 150 kg N, 50 kg P_2O_5 , 75 kg K_2O , 25 kg S, 15 kg Mg, 1 kg B and 1 kg Zn per hectare to obtain high grain yield (1 t more yield per hectare) (Bandara *et al.*, 2005). Suggested nutrients rates to be applied to rice crop grown in Elayapaththuwa soil series were 160 kg N, 35 kg P, 60 kg K, 25 kg S, 10 kg Mg, and 2 kg Zn per hectare to obtain high grain yield. This nutrient application out yielded by 1.5 t grain yield per hectare in comparison to the present DOA recommendation (Bandara *et al.*, 2006b). Similarly site specific fertilizer recommendation using site specific procedure for Galwewa series in the Low Country Dry Zone were formulated (160 kg N, 40 kg P, 60 kg K, 10 kg Mg, 15 kg S, 0.4 kg B, 0.5 kg Cu, 2 kg Zn) and this formulation produced 25% more yield than that of current recommendation (140 kg N, 22 kg P, 42 kg K and 1kg Zn) (Bandara *et al.*, 2007a).

Biological soil nutrient test kit has been developed by RRDI for the identification of deficiency, sufficiency or toxicity of nutrients in soils (Bandara *et al.*, 2007b). Missing element technique has been used in this test kit. They reported that this method was effective for soils of differing

characteristics such as pH, CEC and texture. This method has been tested in other countries as well (Kumaragamage and Indraratne, 2003).

Productivity improvement of rice growing in salinity affected soils

Accumulation of excess quantities of Na, Ca and Mg in soil is identified as salinity condition which makes the soils unsuitable for rice cultivation. Accumulation of excess amounts of Ca, Mg and Na in soil increases soil pH and electrical conductivity and hinder the availability of important plant nutrients such as P and Zn. Soil salinity has been observed in rice soils distributed throughout the island. Saline soils occur mostly in major irrigated inland areas is largely the result of insufficient drainage facilities (Jeganathan and Adam Pain, 1980) while the source of salt in the area is weathered rocks and irrigation water (Handawala, 1982). Salinity levels are particularly severe in areas where evaporation exceeds precipitation. Sikurajapathy *et al.* (1983) reported that about 5% of the total extent of poorly drained rice soils in Mahaweli H area is affected by inland salinity while Punyawardena *et al.* (1989) reported that about 30% of poorly drained soils in newly developed rice fields in Kirindi Oya project area are affected by salinity. Sirisena *et al.* (2008) have reported accumulation of Ca, Mg and Na throughout the catena in the feeding area of Girithale irrigation scheme. More than 12% of the rice extent of the above area has electrical conductivity

values more than 0.125 dSm⁻¹ (1:5 soil:water) due to accumulation of salts. Accumulation is higher in the lower part of the feeding area than that in the upper or middle parts.

Sirisena *et al.* (2010) have found that there are some rice fields where electrical conductivity is even above 15 dSm⁻¹ in rice fields at Angiththamkulam yaya situated in the Puttalam district. Electrical conductivity values measured at the time of land preparation were much higher than the critical electrical conductivity level of 2 dS m⁻¹ required for optimal rice growth as given by Ikehashi and Ponnampuruma (1978). The most prominent cations in the area were Ca, Mg and Na. Soils of the Angiththamkulam yaya come under the Nawagaththegama soil series which has an extremely high content of salt-containing minerals (Dulanjalee and Pitawala, 2008). The electrical conductivity of the irrigation water received by these lands is as high as 1.5 dSm⁻¹. Exchangeable Ca levels in the study area were higher than that reported previously by Mapa *et al.* (2010) for rice-growing soils in the same area.

Irrespective of climatic zone, rice growing soils in the coastal belt of Sri Lanka are salinized periodically, making rice growing extremely difficult (Dimantha, 1977). The area affected by coastal salinity was estimated at around 0.112 million ha (De Alwis and Panabokke, 1972). Electrical conductivity of the extract of saturated soil paste exceeding 4 dSm⁻¹ is not uncommon in these areas (Panabokke,

1978). Moreover, approximately 4000 ha of coastal rice fields in Sri Lanka were flooded with salt water during 2004 becoming these lands unsuitable for rice cultivation.

A study conducted at the Angiththamkulama revealed that introduction of cultural practices such as land leveling, application of organic manure, maintenance of stagnate water at 2-5 cm throughout the growing season and cleaning of drainage canals helped to reduce salt affecting the rice plants (Sirisena *et al.*, 2010). As a result, yield levels of 3.5 tha^{-1} could be obtained from the salt sensitive rice variety Bg 352. The highest yield of 7 tha^{-1} could be obtained from the new rice line Bg11-139, followed by Bg 5-110 (6.4 tha^{-1}) and Bg 4-91 (6.1 tha^{-1}) which have been developed at the RREDI recently. Sirisena *et al.* (2009) have observed that cleaning of drainage canals helps to reduce Mg and Na contents in soils in studies conducted at Kalingaela in Polonnaruwa district. As a result, soil pH and electrical conductivity values were also reduced. As a short-term measure to reduce salinity conditions in the Dry Zone rice soils, they have recommended that farmers should be encouraged to improve drainage and to control excessive irrigation to prevent the water table from rising.

Four soil amendments (gypsum, dolomite, cinnamon ash and rice-husk-charcoal) have been tested for their influence on improving the yield parameters of rice grown in tsunami-affected and non-affected areas. Yield

parameters were compared with an untreated control using salt susceptible (At 362) and salt resistant (At 354) rice varieties. The salt resistant variety had the highest grain yield. The two amendments, gypsum and rice-husk-charcoal, led to an increase in grain yield compared to the untreated control, whereas dolomite and cinnamon ash gave no significant yield increase.

Productivity improvement of rice growing in Fe toxicity affected soils

Fe toxicity is becoming a newly emerged problem in the Dry Zone areas. Studies conducted in Kalingaela in Polonnaruwa have shown that Fe toxicity is a newly emerged soil problem in the lower part of the catena in Maha season. These studies further revealed that iron toxicity problem is becoming severe in some areas in the Dry Zone in Maha season due to water stagnation in rice fields in the lower part of the catena caused by blockage of drainage canals. A range of agronomic management interventions have been advocated to reduce Fe^{2+} concentration in the soil or to foster the rice plants' ability to cope with excess iron in either soil or in the plant. However, none of those options is universally applicable or efficient under diverse environmental conditions where Fe toxicity is expressed. Amelioration measures such as liming, mid season drying, interception of inter flow, good fertilizer management (application of P and K) and use of tolerant varieties are considered effective (Bandara and Gunathilake, 1994; Sahrawart, 2003).

Additions of extra 15 K₂O kg ha⁻¹ to rice fields have been recommended to withstand iron toxicity in K deficient soils (DOA, 2001). Weerasinghe (1994) suggested that maintenance of flowing water regime keeping soil continuously submerged and application of lime to neutralize the acidity were effective in alleviating iron toxicity. Continuous water flow in rice fields is helpful to reduce iron toxicity incidence in soil by helping to supply the nutrients required by the plant and to flush the iron from root zone. These cultural measures are uneconomical due to extra cost incurred to farmers. However, several rice varieties which are tolerant to iron toxicity have been developed and introduced by the DOA (RRDI, 2008).

Natural nutrient supply

Rice varieties developed at RRDI were screened in low fertility rice fields at RRDI, Batalagoda where rice has been cultivated for last 30 years over two seasons per year without addition of any form of fertilizer or manure. Results of these studies revealed that average yield level of many rice varieties ranged from 1.0 to 1.5 tha⁻¹. This suggested that natural nutrient supply under the conditions at RRDI was sufficient only for a grain yield of about 1-1.5 tha⁻¹ in rice.

Identification of land suitability for rice cultivation

Most of the studies conducted so far have considered the macro level in different agro-ecological regions. It is only recently that some attention has been given to take into account the variability of the soil fertility of rice growing soils. Variability in soil fertility created by farmer practices was reported by Sirisina *et al.* (2008) in their experiments in Parakrama Samudraya and Girithale irrigation schemes. Attempts were made by Rathnayake *et al.* (2009) and Nayanaka *et al.* (2009) to map the suitability of soils on different soil fertility factors. Taking into consideration of pH, EC, P, K and Zn, soils have been categorized to suitable, marginal and unsuitable categories for rice cultivation. Though intensive rice cultivation is practiced for past few decades in this particular area, only 25% of the area is highly suitable for rice cultivation. A similar study was conducted for Kurunegala district and found that only 66% of the area is suitable for rice cultivation (Rathnayake *et al.*, 2008). Accordingly all the rice fields in Sri Lanka can be mapped and suitable rice lands in Sri Lanka and their extent can be estimated in the future so that maximum effort can be concentrated on suitable lands and other lands can be taken for other profitable ventures.

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